

CYP2D6 DDGI Qualification

Version	main-OSP12.2
Qualification Plan Release	https://github.com/Open-Systems-Pharmacology/Qualification-DDGI-CYP2D6/releases/tag/vmain
OSP Version	12.2
Qualification Framework Version	3.5

This qualification report is filed at:

<https://github.com/Open-Systems-Pharmacology/OSP-Qualification-Reports>

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1 Introduction

1.1 Objective

This qualification report evaluates the developed physiologically based pharmacokinetic (PBPK) cytochrome P450 (CYP) 2D6 drug-drug-gene interaction (DDGI) network for the ability to perform simulations with the intended purpose to predict CYP2D6 (CYP2D6)-mediated DD(G)Is.

The respective *qualification plan* to produce this *qualification report* is transparently documented and provided open-source (<https://github.com/Open-Systems-Pharmacology/OSP-Qualification-Reports>). The same applies for all presented PBPK models including *evaluation reports* on model building and evaluation of each model (<https://github.com/Open-Systems-Pharmacology/OSP-PBPK-Model-Library>).

Evaluation reports including descriptions on model building and detailed evaluations of the included models are documented separately (see [Section 1.2](#)).

Please refer to the [Appendix](#) to learn more details:

- An overview over the Open Systems Pharmacology (OSP) Suite is given in chapter [Section 5.1](#)
- [Section 5.2](#) shows the implementation of the underlying mathematical equations for DDIs in the OSP suite.
- A detailed general description of the performed qualification workflow (*qualification plan*, *qualification report*, etc.) can be found in chapter [Section 5.3](#).

1.2 CYP2D6 DDGI Network

Figure 1 shows the developed CYP2D6 DDGI network of interacting perpetrator and victim drugs. (a) Schematic illustration of the modeled interactions of CYP2D6 perpetrator and victim drugs. Black arrows indicate metabolism or transport, green arrows indicate induction, red solid lines indicate competitive inhibition, red dashed lines down-regulation (bupropion), noncompetitive inhibition (verapamil P-gp inhibition) or mechanism-based inactivation (others). (b-c) Drug-drug-(gene) interaction matrix for modeled interactions mediated by (a) CYP2D6 and (b) CYP3A4 and P-gp. Colors indicate categories according to the FDA's Examples of Drugs that Interact with CYP Enzymes and Transporter Systems.⁴² Height of the grey ribbons indicates the number of clinical studies for the respective interaction covered by the network, numbers in brackets indicate the number of clinical interaction studies for the corresponding compound. CYP: cytochrome P450, P-gp: P-glycoprotein.

Figure 1: Quinidine DDGI modeling network

The following victim and/or perpetrator PBPK models were applied:

- **Alprazolam** Model snapshot: <https://github.com/Open-Systems-Pharmacology/Alprazolam-Model/blob/master/Alprazolam-Model.json>
- **Atomoxetine** Model snapshot: <https://github.com/Open-Systems-Pharmacology/Atomoxetine-Model/blob/main/atomoxetine-model.json>
- **Bupropion** Model snapshot: <https://github.com/Open-Systems-Pharmacology/Bupropion-Model/blob/main/Bupropion-Model.json>

- **Carbamazepine** Model snapshot: <https://github.com/Open-Systems-Pharmacology/Carbamazepine-Model/blob/main/Carbamazepine-Model.json>
- **Cimetidine** Model snapshot: <https://github.com/Open-Systems-Pharmacology/Cimetidine-Model/blob/master/Cimetidine-Model.json>
- **Clarithromycin** Model snapshot: <https://github.com/Open-Systems-Pharmacology/Clarithromycin-Model/blob/master/Clarithromycin-Model.json>
- **(E)-Clomiphene** Model snapshot: [https://github.com/Open-Systems-Pharmacology/Clomiphene-Model/blob/main/\(E\)-clomiphene-DGI-Model.json](https://github.com/Open-Systems-Pharmacology/Clomiphene-Model/blob/main/(E)-clomiphene-DGI-Model.json)
- **Desipramine** Model snapshot: <https://github.com/Open-Systems-Pharmacology/Desipramine-Model/blob/main/Desipramine-Model.json>
- **Dextromethorphan** Model snapshot: https://github.com/Open-Systems-Pharmacology/Dextromethorphan-Model/blob/main/dextromethorphan_aggregated_simulations.json
- **Digoxin** Model snapshot: <https://github.com/Open-Systems-Pharmacology/Digoxin-Model/blob/master/Digoxin.json>
- **Erythromycin** Model snapshot: <https://github.com/Open-Systems-Pharmacology/Erythromycin-Model/blob/master/Erythromycin-Model.json>
- **Fluvoxamine** Model snapshot: <https://github.com/Open-Systems-Pharmacology/Fluvoxamine-Model/blob/master/Fluvoxamine-Model.json>
- **Itraconazole** Model snapshot: <https://github.com/Open-Systems-Pharmacology/Itraconazole-Model/blob/master/Itraconazole-Model.json>
- **Ketoconazole** Model snapshot: <https://github.com/Open-Systems-Pharmacology/Ketoconazole-Model/blob/main/Ketoconazole-Model.json>
- **Metoprolol** Model file: <https://github.com/Open-Systems-Pharmacology/Metoprolol-Model/blob/main/Metoprolol-Model.pksim5>
- **Mexiletine** Model snapshot: <https://github.com/Open-Systems-Pharmacology/Mexiletine-Model/blob/main/Mexiletine-Model.json>
- **Omeprazole** Model snapshot: <https://github.com/Open-Systems-Pharmacology/Omeprazole-Model/blob/main/Omeprazole-Model.json>
- **Paroxetine** Model snapshot: <https://github.com/Open-Systems-Pharmacology/Paroxetine-Model/blob/main/paroxetine-model.json>
- **Quinidine** Model snapshot: <https://github.com/Open-Systems-Pharmacology/Quinidine-Model/blob/main/Quinidine-Model.json>
- **Rifampicin** Model snapshot: <https://github.com/Open-Systems-Pharmacology/Rifampicin-Model/blob/master/Rifampicin-Model.json>
- **Risperidone** Model snapshot: <https://github.com/Open-Systems-Pharmacology/Risperidone-Model/blob/main/risperidone-model.json>
- **Verapamil** Model snapshot: <https://github.com/Open-Systems-Pharmacology/Verapamil-Model/blob/master/Verapamil-Model.json>

The following interaction scenarios were predicted and used to qualify the final DDGI network:

- Alprazolam as victim:
 - Paroxetine-alprazolam-DDI
- Atomoxetine as victim:
 - Bupropion-atomoxetine-DDGI
 - Fluvoxamine-atomoxetine-DDI
 - Paroxetine-atomoxetine-DDGI
- (E)-Clomiphene as victim:
 - Clarithromycin-clomiphene-DDGI
 - Paroxetine-clomiphene-DDGI
- Desipramine as victim:

- Atomoxetine-desipramine-DDI
- Bupropion-desipramine-DDI
- Paroxetine-desipramine-DDGI
- Quinidine-desipramine-DDI
- Dextromethorphan as victim:
 - Paroxetine-dextromethorphan-DDGI
 - Quindine-dextromethorphan-DDI
- Digoxin as victim:
 - Quinidine-digoxin-DDI
- Metoprolol as victim:
 - Cimetidine-metoprolol-DDI
 - Paroxetine-metoprolol-DDI
 - Quinidine-metoprolol-DDGI
 - Rifampicin-metoprolol-DDI
- Mexiletine as victim:
 - Quinidine-mexiletine-DDGI
- Midazolam as victim:
 - Atomoxetine-midazolam-DDI
- Paroxetine as victim:
 - Itraconazole-paroxetine-DDI
 - Quinidine-paroxetine-DDI
- Quinidine as victim:
 - Carbamazepine-quinidine-DDI
 - Cimetidine-quinidine-DDI
 - Erythromycin-quinidine-DDI
 - Fluvoxamine-quinidine-DDI
 - Itraconazole-quinidine-DDI
 - Omeprazole-quinidine-DDI
 - Rifampicin-quinidine-DDI
 - Verapamil-quinidine-DDI
- Risperidone as victim:
 - Ketoconazole-risperidone-DDI
 - Rifampicin-risperidone-DDI
 - Verapamil-risperidone-DDI

The published DD(G)I studies between the respective perpetrators and victim drugs were simulated and compared to observed data. The following sections give an overview of the clinical studies being part of this qualification report.

1.2.1 Paroxetine-Alprazolam-DDI

The paroxetine-alprazolam interaction was evaluated using clinical DDI studies listed in [Table 1](#).

Source	Route	Schedule	Pop.	Sex	N	Perpetrator
Calvo 2004	po	1 mg q.d.	European	f	22	+/- PAR, 20 mg q.d. po

Table 7: f: female, N: number of study participants, PAR: paroxetine, po: oral, pop.: population used in simulations, q.d.: once daily. If perpetrator or victim drugs were applied in form of salts, the respective dose of base was calculated and incorporated in simulations.

1.2.2 Fluvoxamine-Atomoxetine-DDI

The fluvoxamine-atomoxetine interaction was evaluated using clinical DDI studies listed in [Table 3](#).

Source	Route	Schedule	Pop.	Sex	N	Perpetrator
Todor 2017	po	25 mg s.d.	European	m	18	+/- FLV, 50/100 mg q.d. po

Table 7: FLV: fluvoxamine, m: male, N: number of study participants, po: oral, pop.: population used in simulations, q.d.: once daily, s.d.: single dose. If perpetrator or victim drugs were applied in form of salts, the respective dose of base was calculated and incorporated in simulations.

1.2.3 Paroxetine-Atomoxetine-DDGI

The paroxetine-atomoxetine interaction was evaluated using clinical DDGI studies listed in [Table 4](#).

Source	Route	Schedule	Pop.	Sex	N	Perpetrator
Jung 2020	po	20 mg s.d.	Asian	m	7 AS0	+/- PAR, 20 mg q.d. po
Jung 2020	po	20 mg s.d.	Asian	m	9 AS1.25	+/- PAR, 20 mg q.d. po
Jung 2020	po	20 mg s.d.	Asian	m	10 AS2	+/- PAR, 20 mg q.d. po
Todor 2015	po	25 mg s.d.	European	m	22	+/- PAR, 20 mg b.i.d./q.d. po
Belle 2002	po	20 mg b.i.d.	American	m	22	+/- PAR, 20 mg q.d. po

Table 7: AS: CYP2D6 activity score, b.i.d.: twice daily, m: male, N: number of study participants, PAR: paroxetine, po: oral, pop.: population used in simulations, q.d.: once daily, s.d.: single dose. If perpetrator or victim drugs were applied in form of salts, the respective dose of base was calculated and incorporated in simulations.

1.2.4 Clarithromycin-Clomiphene-DDGI

The clarithromycin-clomiphene interaction was evaluated using clinical DDGI studies listed in [Table 5](#).

Source	Route	Schedule	Pop.	Sex	N	Perpetrator
Mürdter 2016	po	42 mg s.d.	European	f	5 AS0	+/- CLA, 500 mg b.i.d. po
Mürdter 2016	po	42 mg s.d.	European	f	4 AS0.5	+/- CLA, 500 mg b.i.d. po
Mürdter 2016	po	42 mg s.d.	European	f	2 AS1	+/- CLA, 500 mg b.i.d. po
Mürdter 2016	po	42 mg s.d.	European	f	2 AS2	+/- CLA, 500 mg b.i.d. po
Mürdter 2016	po	42 mg s.d.	European	f	2 AS3	+/- CLA, 500 mg b.i.d. po

Table 7: AS: CYP2D6 activity score, b.i.d.: twice daily, CLA: clarithromycin, f: female, N: number of study participants, po: oral, pop.: population used in simulations, s.d.: single dose. If perpetrator or victim drugs were applied in form of salts, the respective dose of base was calculated and incorporated in simulations.

1.2.5 Paroxetine-Clomiphene-DDGI

The paroxetine-clomiphene interaction was evaluated using clinical DDGI studies listed in [Table 6](#).

Source	Route	Schedule	Pop.	Sex	N	Perpetrator
Mürdter 2016	po	42 mg s.d.	European	f	4 AS0	+/- PAR, 40 mg q.d. po
Mürdter 2016	po	42 mg s.d.	European	f	4 AS0.5	+/- PAR, 40 mg q.d. po
Mürdter 2016	po	42 mg s.d.	European	f	2 AS1	+/- PAR, 40 mg q.d. po
Mürdter 2016	po	42 mg s.d.	European	f	2 AS2	+/- PAR, 40 mg q.d. po
Mürdter 2016	po	42 mg s.d.	European	f	2 AS3	+/- PAR, 40 mg q.d. po

Table 7: AS: CYP2D6 activity score, b.i.d.: twice daily, f: female, N: number of study participants, PAR: paroxetine, po: oral, pop.: population used in simulations, q.d.: once daily, s.d.: single dose. If perpetrator or victim drugs were applied in form of salts, the respective dose of base was calculated and incorporated in simulations.

1.2.6 Atomoxetine-Desipramine-DDI

The atomoxetine-desipramine interaction was evaluated using clinical DDI studies listed in [Table 7](#).

Source	Route	Schedule	Pop.	Sex	N	Perpetrator
Sauer 2004	po	50 mg s.d.	American	m	22	+/- ATO, 60 mg b.i.d. po

Table 7: ATO: atomoxetine, b.i.d.: twice daily, m: male, N: number of study participants, po: oral, pop.: population used in simulations, s.d.: single dose. If perpetrator or victim drugs were applied in form of salts, the respective dose of base was calculated and incorporated in simulations.

1.2.7 Paroxetine-Desipramine-DDGI

The paroxetine-desipramine interaction was evaluated using clinical DDGI studies listed in [Table 9](#).

Source	Route	Schedule	Pop.	Sex	N	Perpetrator
Nichols 2009	po	50 mg s.d.	American	m	20	+/- PAR, 20 mg q.d. po
Brøsen 1993	po	100 mg s.d.	European	m	8 PM	+/- PAR, 20 mg q.d. po
Brøsen 1993	po	100 mg s.d.	European	m	5 NM	+/- PAR, 20 mg q.d. po
Brøsen 1993	po	100 mg s.d.	European	m	4 fast NM	+/- PAR, 20 mg q.d. po
Aldermann 1997	po	50 mg q.d.	American	m	6	+/- PAR, 20/30 mg q.d. po

Table 7: m: male, N: number of study participants, PAR: paroxetine, NM: CYP2D6 normal metabolizer, PM: CYP2D6 poor metabolizer, po: oral, pop.: population used in simulations, q.d.: once daily, s.d.: single dose. If perpetrator or victim drugs were applied in form of salts, the respective dose of base was calculated and incorporated in simulations.

1.2.8 Quinidine-Desipramine-DDI

The quinidine-desipramine interaction was evaluated using clinical DDI studies listed in [Table 10](#).

Source	Route	Schedule	Pop.	Sex	N	Perpetrator
Brøsen 1989	po	100 mg s.d.	European	m	6	+/- QUI, 200 mg q.d. po

Table 7: m: male, N: number of study participants, po: oral, pop.: population used in simulations, q.d.: once daily, QUI: quinidine, s.d.: single dose. If perpetrator or victim drugs were applied in form of salts, the respective dose of base was calculated and incorporated in simulations.

1.2.9 Paroxetine-Dextromethorphan-DDGI

The paroxetine-dextromethorphan interaction was evaluated using clinical DDGI studies listed in [Table 11](#).

Source	Route	Schedule	Pop.	Sex	N	Perpetrator
Storelli 2018	po	5 mg s.d.	European	f	16 IM	+/- PAR, 20 mg b.i.d. po
Storelli 2018	po	5 mg s.d.	European	f	17 NM	+/- PAR, 20 mg b.i.d. po
Schoedel 2012	po	30 mg b.i.d.	American	m	13	+/- PAR, 30 mg q.d. po

Table 1: b.i.d.: twice daily, f: female, IM: CYP2D6 intermediate metabolizer, m: male, N: number of study participants, NM: CYP2D6 normal metabolizer, po: oral, pop.: population used in simulations, QUI: quinidine, q.d.: once daily, s.d.: single dose. If perpetrator or victim drugs were applied in form of salts, the respective dose of base was calculated and incorporated in simulations.

1.2.10 Quinidine-Dextromethorphan-DDI

The quinidine-dextromethorphan interaction was evaluated using clinical DDI studies listed in [Table 12](#).

Source	Route	Schedule	Pop.	Sex	N	Perpetrator
Capon 1996	po	30 mg s.d.	European	m	6 NM	+/- QUI, 50 mg s.d. po
Schadel 1995	po	30 mg s.d.	American	f	5 NM	+/- QUI, 100 mg s.d. po

Table 1: f: female, m: male, N: number of study participants, NM: CYP2D6 normal metabolizer, po: oral, pop.: population used in simulations, QUI: quinidine, s.d.: single dose. If perpetrator or victim drugs were applied in form of salts, the respective dose of base was calculated and incorporated in simulations.

1.2.11 Quinidine-Digoxin-DDI

The quinidine-digoxin interaction was evaluated using clinical DDI studies listed in [Table 13](#).

Source	Route	Schedule	Pop.	Sex	N	Perpetrator
Steiness 1980	iv	10 µg/kg s.d.	European	m	6	+/- QUI, 200 mg b.i.d. po
Ochs 1981	iv	1 mg s.d.	European	m	7	+/- QUI, 200 mg q.i.d. po

Table 2: b.i.d.: twice daily, m: male, iv: intravenous, N: number of study participants, po: oral, pop.: population used in simulations, q.i.d.: four times daily, QUI: quinidine, s.d.: single dose. If perpetrator or victim drugs were applied in form of salts, the respective dose of base was calculated and incorporated in simulations.

1.2.12 Cimetidine-Metoprolol-DDI

The cimetidine-metoprolol interaction was evaluated using clinical DDI studies listed in [Table 14](#).

Source	Route	Schedule	Pop.	Sex	N	Perpetrator
Chellingsworth 1988	po	100 mg s.d.	European	m	12	+/- CIM, 800 mg q.d. po
Chellingsworth 1988	po	100 mg b.i.d.	European	m	12	+/- CIM, 800 mg q.d. po
Toon 1988	po	100 mg b.i.d.	European	m	12	+/- CIM, 800 mg q.d. po
Kirch 1982	po	100 mg b.i.d.	European	m	6	+/- CIM, 200/400 mg q.i.d. po

Table 7: b.i.d.: twice daily, CIM: cimetidine, m: male, N: number of study participants, po: oral, pop.: population used in simulations, q.d.: once daily, q.i.d.: four times daily, s.d.: single dose. If perpetrator or victim drugs were applied in form of salts, the respective dose of base was calculated and incorporated in simulations.

1.2.13 Paroxetine-Metoprolol-DDI

The paroxetine-metoprolol interaction was evaluated using clinical DDI studies listed in [Table 15](#).

Source	Route	Schedule	Pop.	Sex	N	Perpetrator
Stout 2011	po	50 mg s.d.	American	m	10	+/- PAR, 10 mg b.i.d. po
Stout 2011	po	100 mg s.d.	American	m	10	+/- PAR, 10 mg b.i.d. po
Parker 2011	po	100 mg s.d.	American	m	12	+/- PAR, 20 mg q.d. po
Parker 2011	po	100 mg b.i.d.	American	m	12	+/- PAR, 20 mg q.d. po
Parker 2011	po	200 mg s.d.	American	m	12	+/- PAR, 20 mg q.d. po
Hemeryck 2000	po	100 mg s.d.	European	m	8	+/- PAR, 10 mg b.i.d. po

Table 7: b.i.d.: twice daily, m: male, N: number of study participants, PAR: paroxetine, po: oral, pop.: population used in simulations, q.d.: once daily, s.d.: single dose. If perpetrator or victim drugs were applied in form of salts, the respective dose of base was calculated and incorporated in simulations.

1.2.14 Quinidine-Metoprolol-DDGI

The quinidine-metoprolol interaction was evaluated using clinical DDGI studies listed in [Table 16](#).

Source	Route	Schedule	Pop.	Sex	N	Perpetrator
Leemann 1993	iv	20 mg s.d.	European	m	3 PM	+/- QUI, 50 mg s.d. po
Leemann 1993	iv	20 mg s.d.	European	m	4 NM	+/- QUI, 50 mg s.d. po
Leemann 1993	iv	20 mg s.d.	European	m	3 PM	+/- QUI, 250 mg b.i.d. po
Leemann 1993	iv	20 mg s.d.	European	m	4 NM	+/- QUI, 250 mg b.i.d. po
Johnson 1996	po	200 mg s.d.	American	m	10	+/- QUI, 100 mg q.d. po
Johnson 1996	po	200 mg s.d.	American	m	10	+/- QUI, 100 mg q.d. po

Table 7: b.i.d.: twice daily, iv: intravenous, m: male, N: number of study participants, NM: CYP2D6 normal metabolizer, PM: CYP2D6 poor metabolizer, po: oral, pop.: population used in simulations, q.d.: once daily, QUI: quinidine, s.d.: single dose. If perpetrator or victim drugs were applied in form of salts, the respective dose of base was calculated and incorporated in simulations.

1.2.15 Rifampicin-Metoprolol-DDI

The rifampicin-metoprolol interaction was evaluated using clinical DDI studies listed in [Table 17](#).

Source	Route	Schedule	Pop.	Sex	N	Perpetrator
Bennett 1982	po	100 mg s.d.	European	m	12	+/- RIF, 600 mg q.d. po

Table 7: m: male, N: number of study participants, po: oral, pop.: population used in simulations, q.d.: once daily, RIF: rifampicin, s.d.: single dose. If perpetrator or victim drugs were applied in form of salts, the respective dose of base was calculated and incorporated in simulations.

1.2.16 Quinidine-Mexiletine-DDGI

The quinidine-mexiletine interaction was evaluated using clinical DDGI studies listed in [Table 18](#).

Source	Route	Schedule	Pop.	Sex	N	Perpetrator
Abolfathi 1993	po	200 mg s.d.	American	m	4 PM	+/- QUI, 50 mg q.i.d. po
Abolfathi 1993	po	200 mg s.d.	American	m	10 NM	+/- QUI, 50 mg q.i.d. po

Table 3: m: male, N: number of study participants, NM: CYP2D6 normal metabolizer, PM: CYP2D6 poor metabolizer, po: oral, pop.: population used in simulations, q.i.d.: four times daily, QUI: quinidine, s.d.: single dose. If perpetrator or victim drugs were applied in form of salts, the respective dose of base was calculated and incorporated in simulations.

1.2.17 Atomoxetine-Midazolam-DDI

The atomoxetine-midazolam interaction was evaluated using clinical DDI studies listed in [Table 19](#).

Source	Route	Schedule	Pop.	Sex	N	Perpetrator
Sauer 2004	po	5 mg s.d.	American	f	8	+/- ATO, 60 mg b.i.d. po

Table 7: ATO: atomoxetine, b.i.d.: twice daily, f: female, N: number of study participants, po: oral, pop.: population used in simulations, q.d.: once daily, s.d.: single dose. If perpetrator or victim drugs were applied in form of salts, the respective dose of base was calculated and incorporated in simulations.

1.2.18 Itraconazole-Paroxetine-DDI

The itraconazole-paroxetine interaction was evaluated using clinical DDI studies listed in [Table 20](#).

Source	Route	Schedule	Pop.	Sex	N	Perpetrator
Yasui-Furukori 2007	po	20 mg s.d.	Japanese	f	13	+/- ITR, 100 mg b.i.d. po

Table 7: b.i.d.: twice daily, f: female, ITR: itraconazole, N: number of study participants, po: oral, pop.: population used in simulations, s.d.: single dose. If perpetrator or victim drugs were applied in form of salts, the respective dose of base was calculated and incorporated in simulations.

1.2.19 Quinidine-Paroxetine-DDI

The quinidine-paroxetine interaction was evaluated using clinical DDI studies listed in [Table 21](#).

Source	Route	Schedule	Pop.	Sex	N	Perpetrator
Schoedel 2012	po	20 mg q.d.	American	m	14 NM	+/- QUI, 30 mg b.i.d. po

Table 4: m: male, N: number of study participants, NM: CYP2D6 normal metabolizer, po: oral, pop.: population used in simulations, q.d.: once daily, QUI: quinidine. If perpetrator or victim drugs were applied in form of salts, the respective dose of base was calculated and incorporated in simulations.

1.2.20 Carbamazepine-Quinidine-DDI

The carbamazepine-quinidine interaction was evaluated using clinical DDI studies listed in [Table 22](#).

Source	Route	Schedule	Pop.	Sex	N	Perpetrator
Andreassen 2007	po	200 mg s.d.	European	m	10	+/- CBZ, 200/400 mg b.i.d. po

Table 5: b.i.d.: twice daily, CBZ: carbamazepine, m: male, N: number of study participants, po: oral, pop.: population used in simulations, s.d.: single dose. If perpetrator or victim drugs were applied in form of salts, the respective dose of base was calculated and incorporated in simulations.

1.2.21 Cimetidine-Quinidine-DDI

The cimetidine-quinidine interaction was evaluated using clinical DDI studies listed in [Table 23](#).

Source	Route	Schedule	Pop.	Sex	N	Perpetrator
Kolb 1984	po	400 mg s.d.	American	m	9	+/- CIM, 300 mg q.d. po
Hardy 1983	po	400 mg s.d.	American	m	9	+/- CIM, 300 mg q.i.d. po

Table 6: CIM: cimetidine, m: male, N: number of study participants, po: oral, pop.: population used in simulations, q.d.: once daily, q.i.d.: four times daily, s.d.: single dose. If perpetrator or victim drugs were applied in form of salts, the respective dose of base

was calculated and incorporated in simulations.

1.2.22 Erythromycin-Quinidine-DDI

The erythromycin-quinidine interaction was evaluated using clinical DDI studies listed in [Table 24](#).

Source	Route	Schedule	Pop.	Sex	N	Perpetrator
Damkier 1999b	po	200 mg s.d.	European	m	6	+/- ERY, 250 mg q.i.d. po

Table 7: ERY: erythromycin, m: male, N: number of study participants, po: oral, pop.: population used in simulations, q.i.d.: four times daily, s.d.: single dose. If perpetrator or victim drugs were applied in form of salts, the respective dose of base was calculated and incorporated in simulations.

1.2.23 Fluvoxamine-Quinidine-DDI

The fluvoxamine-quinidine interaction was evaluated using clinical DDI studies listed in [Table 25](#).

Source	Route	Schedule	Pop.	Sex	N	Perpetrator
Damkier 1999a	po	200 mg s.d.	American	m	6	+/- FLV, 100 mg q.d. po

Table 7: FLV: fluvoxamine, m: male, N: number of study participants, po: oral, pop.: population used in simulations, q.d.: once daily, s.d.: single dose. If perpetrator or victim drugs were applied in form of salts, the respective dose of base was calculated and incorporated in simulations.

1.2.24 Itraconazole-Quinidine-DDI

The itraconazole-quinidine interaction was evaluated using clinical DDI studies listed in [Table 26](#).

Source	Route	Schedule	Pop.	Sex	N	Perpetrator
Kaukonen 1997	po	100 mg s.d.	European	f	9	+/- ITR, 200 mg q.d. po
Damkier 1999b	po	200 mg s.d.	European	m	6	+/- ITR, 100 mg q.d. po

Table 7: f: female, ITR: itraconazole, m: male, N: number of study participants, po: oral, pop.: population used in simulations, q.d.: once daily, s.d.: single dose. If perpetrator or victim drugs were applied in form of salts, the respective dose of base was calculated and incorporated in simulations.

1.2.25 Omeprazole-Quinidine-DDI

The omeprazole-quinidine interaction was evaluated using clinical DDI studies listed in [Table 27](#).

Source	Route	Schedule	Pop.	Sex	N	Perpetrator
Ching 1991	po	400 mg s.d.	European	m	8	+/- OME, 40 mg q.d. po

Table 8: m: male, N: number of study participants, OME: omeprazole, po: oral, pop.: population used in simulations, q.d.: once daily, s.d.: single dose. If perpetrator or victim drugs were applied in form of salts, the respective dose of base was calculated and incorporated in simulations.

1.2.26 Rifampicin-Quinidine-DDI

The rifampicin-quinidine interaction was evaluated using clinical DDI studies listed in [Table 28](#).

Source	Route	Schedule	Pop.	Sex	N	Perpetrator
Damkier 1999	po	200 mg s.d.	European	m	6	+/- RIF, 600 mg q.d. po

Table 9: m: male, N: number of study participants, po: oral, pop.: population used in simulations, q.d.: once daily, RIF: rifampicin, s.d.: single dose. If perpetrator or victim drugs were applied in form of salts, the respective dose of base was calculated and incorporated in simulations.

1.2.27 Verapamil-Quinidine-DDI

The verapamil-quinidine interaction was evaluated using clinical DDI studies listed in [Table 29](#).

Source	Route	Schedule	Pop.	Sex	N	Perpetrator
Edwards 1987	po	400 mg s.d.	European	m	6	+/- VER, 80 mg t.i.d. po
Edwards 1987	po	400 mg s.d.	European	m	6	+/- VER, 120 mg t.i.d. po

Table 10: m: male, N: number of study participants, po: oral, pop.: population used in simulations, s.d.: single dose, t.i.d.: three times daily, VER: verapamil. If perpetrator or victim drugs were applied in form of salts, the respective dose of base was calculated and incorporated in simulations.

1.2.28 Ketoconazole-Risperidone-DDI

The ketoconazole-risperidone interaction was evaluated using clinical DDI studies listed in [Table 30](#).

Source	Route	Schedule	Pop.	Sex	N	Perpetrator
Mahatthanatrakul 2012	po	2 mg s.d.	Asian	m	10	+/- KET, 200 mg q.d. po

Table 7: KET: ketoconazole, m: male, N: number of study participants, po: oral, pop.: population used in simulations, q.d.: once daily, s.d.: single dose. If perpetrator or victim drugs were applied in form of salts, the respective dose of base was calculated and incorporated in simulations.

1.2.29 Rifampicin-Risperidone-DDI

The rifampicin-risperidone interaction was evaluated using clinical DDI studies listed in [Table 31](#).

Source	Route	Schedule	Pop.	Sex	N	Perpetrator
Kim 2018	po	1 mg s.d.	Asian	m	10	+/- RIF, 600 mg q.d. po
Mahatthanatrakul 2007	po	4 mg s.d.	Asian	m	10	+/- RIF, 600 mg q.d. po

Table 7: m: male, N: number of study participants, po: oral, pop.: population used in simulations, q.d.: once daily, RIF: rifampicin, s.d.: single dose. If perpetrator or victim drugs were applied in form of salts, the respective dose of base was calculated and incorporated in simulations.

1.2.30 Verapamil-Risperidone-DDI

The verapamil-risperidone interaction was evaluated using clinical DDI studies listed in [Table 32](#).

Source	Route	Schedule	Pop.	Sex	N	Perpetrator
Nakagami 2005	po	1 mg s.d.	Japanese	m	12	+/- VER, 80 mg t.i.d. po

Table 7: m: male, N: number of study participants, po: oral, pop.: population used in simulations, s.d.: single dose, t.i.d.: three times daily, VER: verapamil. If perpetrator or victim drugs were applied in form of salts, the respective dose of base was calculated and incorporated in simulations.

2 Qualification of Use Case CYP2D6-mediated DDIs

The following section shows the correlations between observed and model-predicted AUC_{last} and C_{max} ratios, respectively.

Specifically, the PBPK model performance for the PK parameters **AUC ratio (AUCR)** and **C_{max} ratio (CmaxR)** is assessed via:

- predicted (*Pred*) vs. observed (*Obs*) plots
- *Pred/Obs* vs. *Obs* plots
- geometric mean fold error (GMFE):

$$10^{\frac{\sum |\log(\frac{Pred}{Obs})|}{n}}$$

- number of AUCR and CMAXR falling within 2-fold error range and within the limits suggested by [Guest et al. 2011](#)
- detailed table of results for each study

In the plots,

- the dotted lines denote 0.50–2.00 (2-fold) criterion,
- the solid lines denote the limits suggested by [Guest et al. 2011](#),
- the bold solid line denotes the unity line,
- each color represents one combination of perpetrator and victim (parent drug or metabolite)

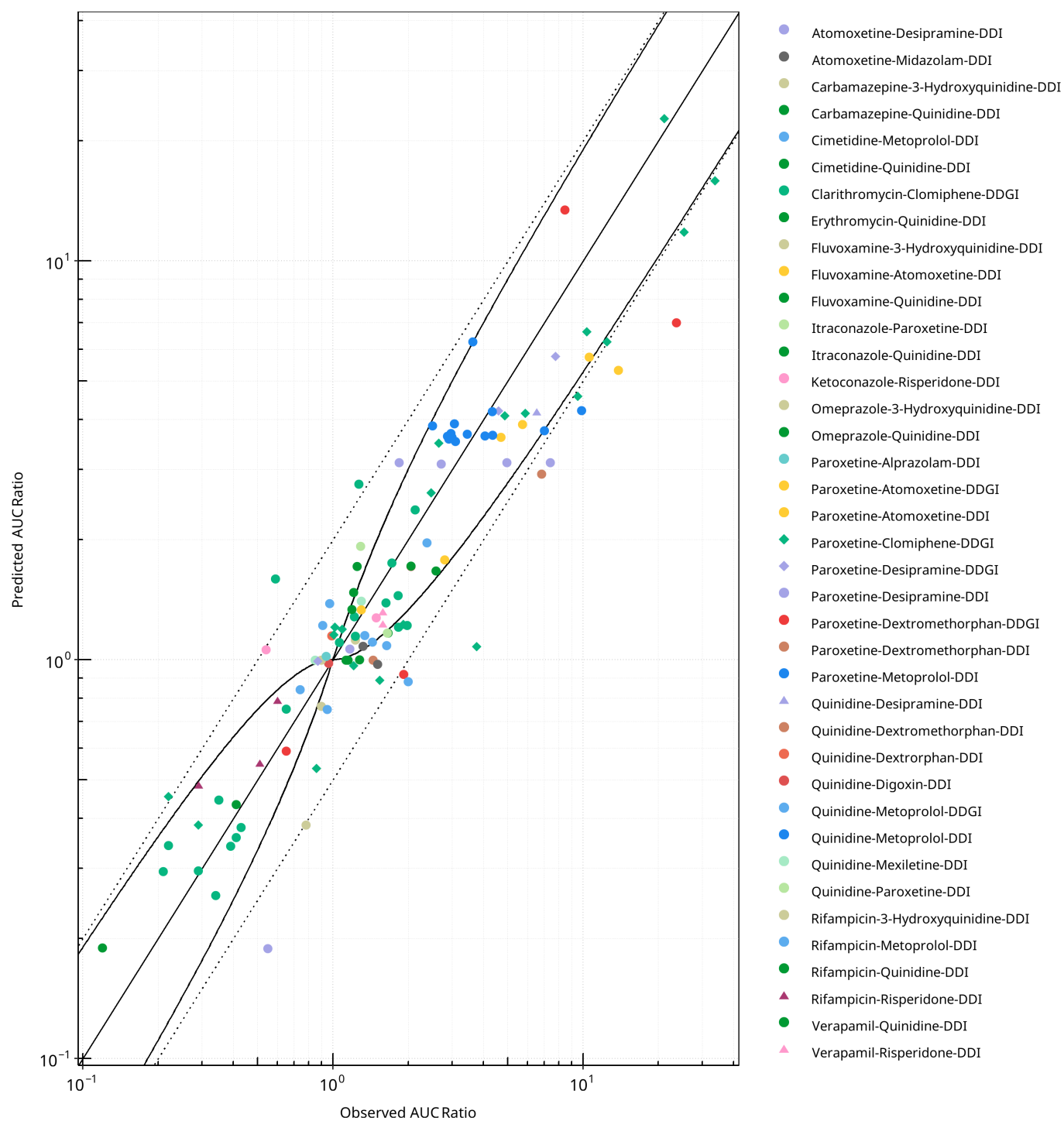


Figure 2-1: CYP2D6 DDIs. Predicted vs. Observed AUC Ratio. ($\delta = 1$ in Guest *et al.* formula)

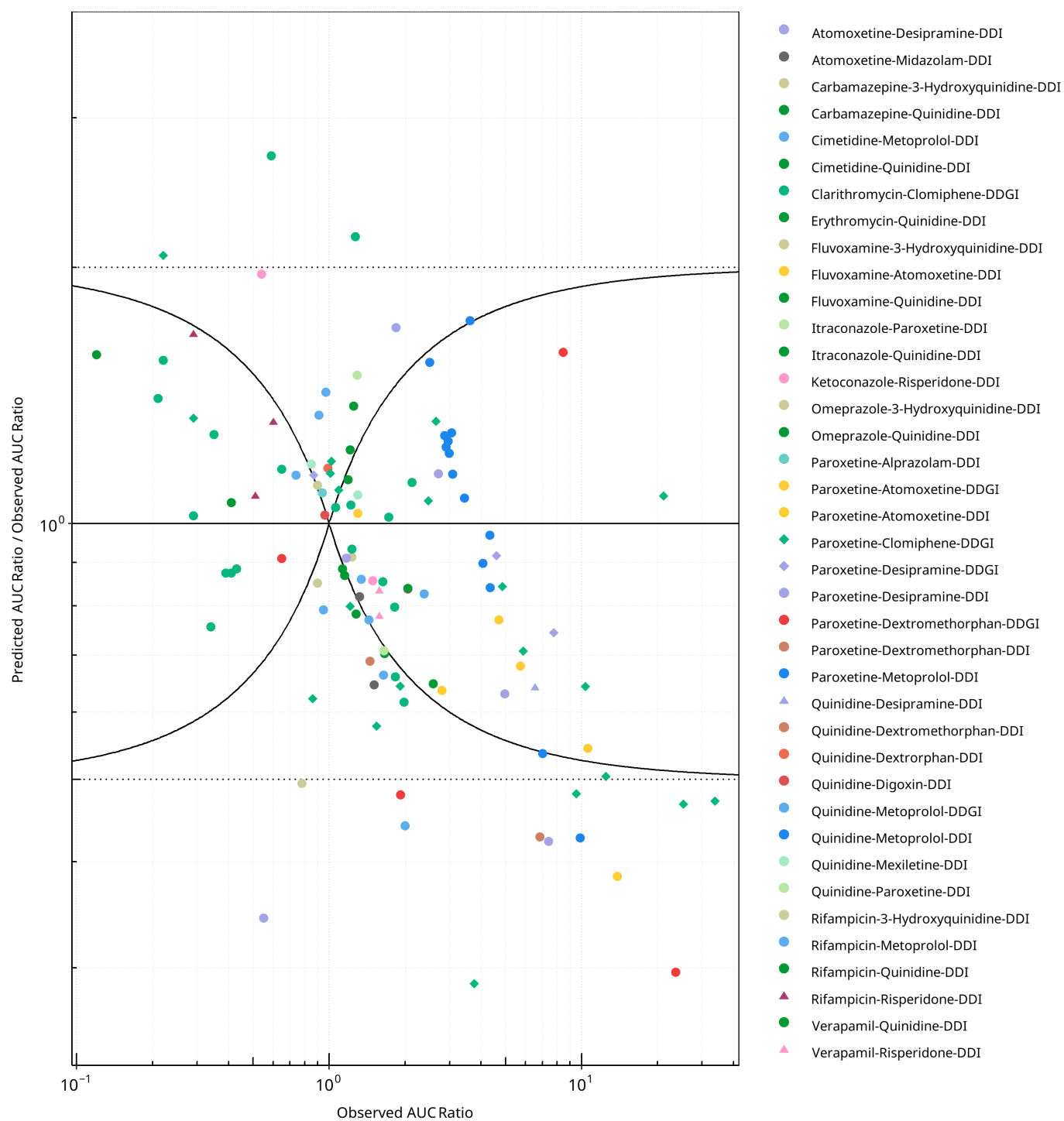


Figure 2-2: CYP2D6 DDIs. Predicted/Observed vs. Observed AUC Ratio. ($\delta = 1$ in Guest *et al.* formula)

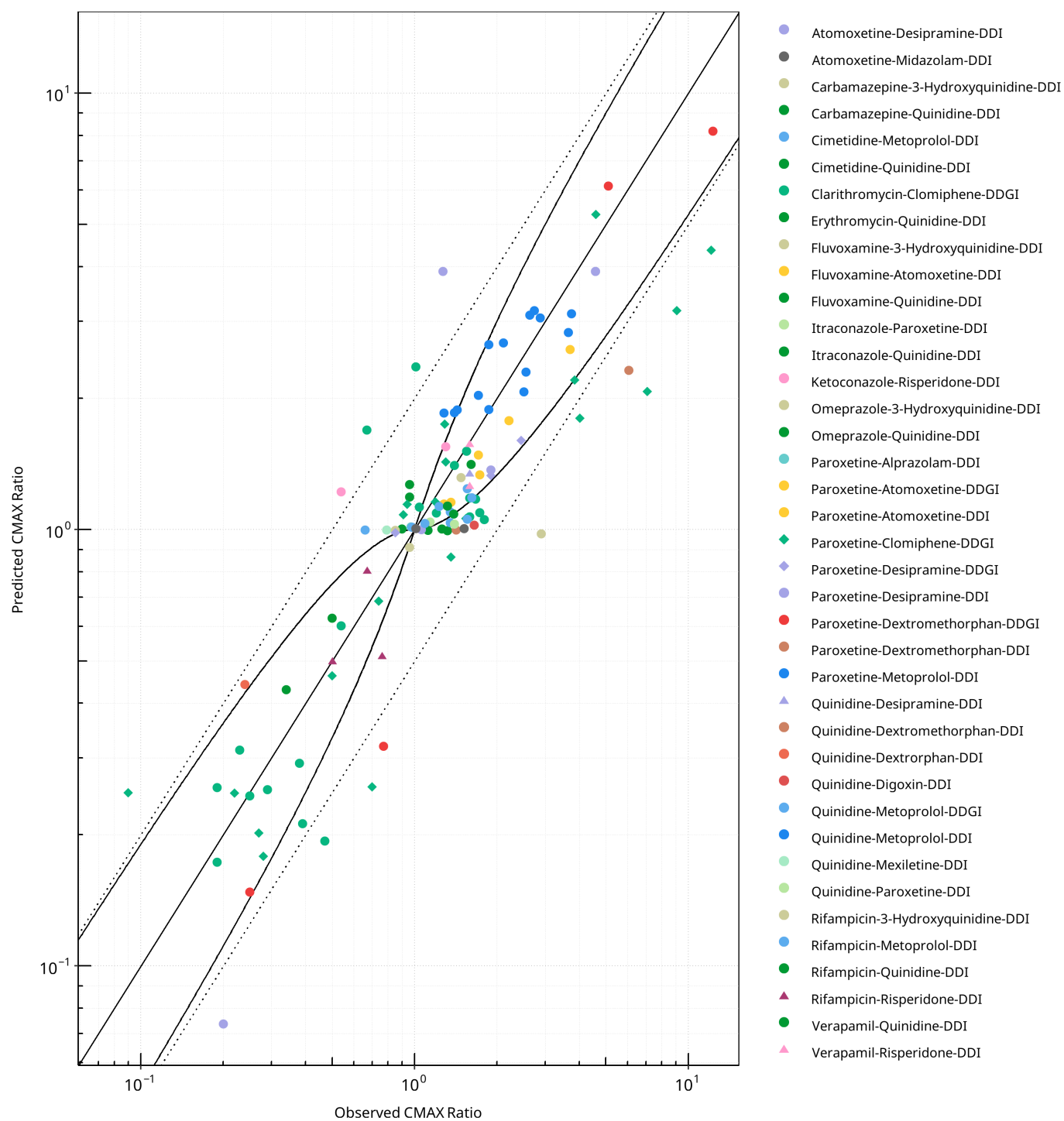


Figure 2-3: CYP2D6 DDIs. Predicted vs. Observed CMAX Ratio. ($\delta = 1$ in Guest *et al.* formula)

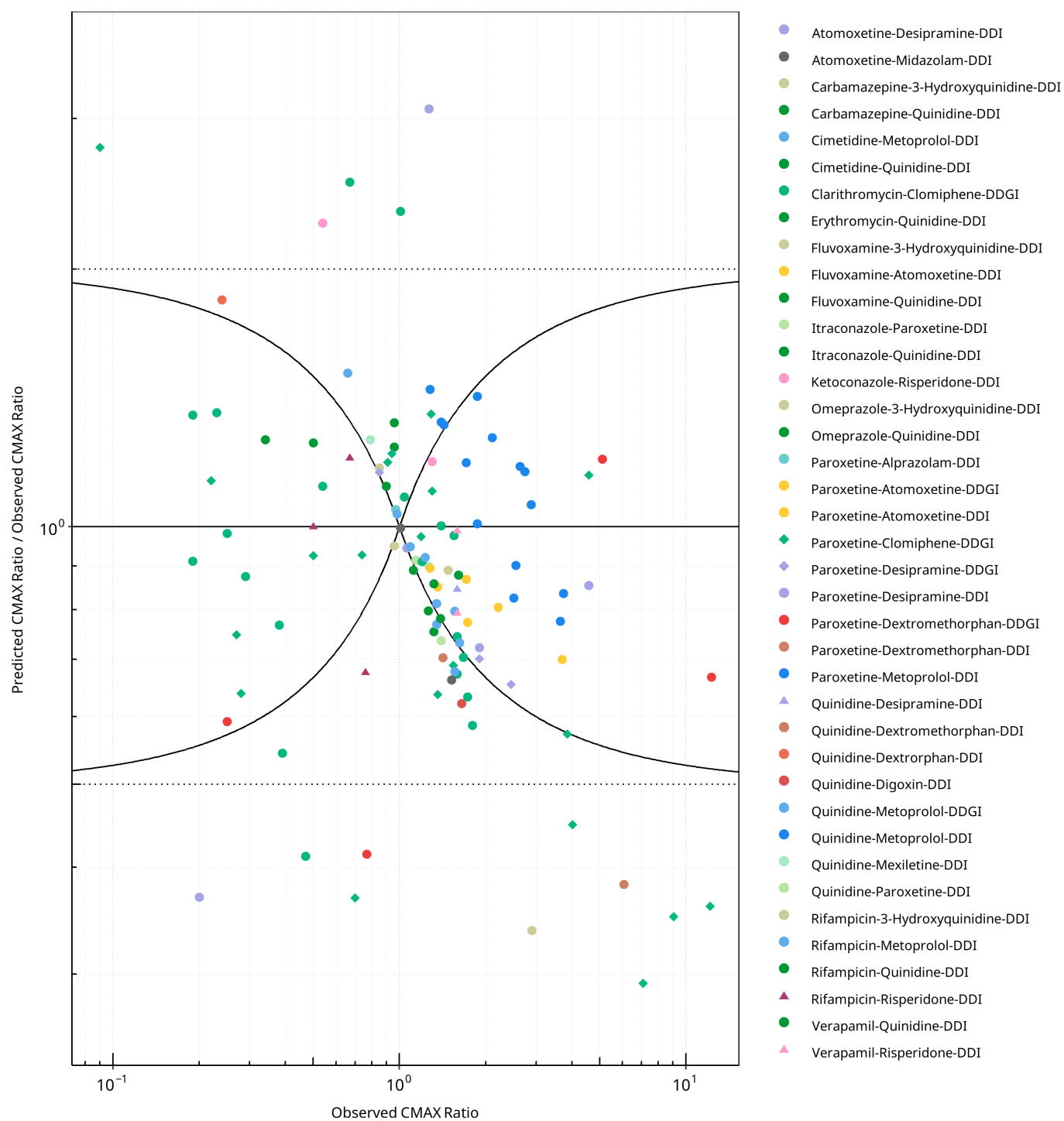


Figure 2-4: CYP2D6 DDIs. Predicted/Observed vs. Observed CMAX Ratio. ($\delta = 1$ in Guest *et al.* formula)

Table 2-1: GMFE for CYP2D6 DDIs Ratio

PK parameter	GMFE
AUC	1.40
CMAX	1.38

Table 2-2: Summary table for CYP2D6 DDIs - AUC Ratio. ($\delta = 1$ in Guest *et al.* formula)

AUC	Number	Ratio [%]
Points total	117	-
Points within Guest <i>et al.</i>	67	57.26
Points within 2 fold	101	86.32

Table 2-3: Summary table for CYP2D6 DDIs - CMAX Ratio. ($\delta = 1$ in Guest *et al.* formula)

CMAX	Number	Ratio [%]
Points total	116	-
Points within Guest <i>et al.</i>	64	55.17
Points within 2 fold	101	87.07

Table 2-4: Summary table for CYP2D6 DDIs

DataID	Perpetrator	Victim	Predicted AUC Ratio	Observed AUC Ratio	Pred/Obs AUC Ratio	Predicted C _{MAX} Ratio	Observed C _{MAX} Ratio	Pred/Obs C _{MAX} Ratio	Reference
16940	Quinidine, 50 mg, PO, SD: 1h before DEX	Dextromethorphan, PO	2.93	6.84	0.43	2.32	6.07	0.38	Capon 1996
16942	Quinidine, 50 mg, PO, SD: 1h before DEX	Total dextrothorphan, PO	1.15	0.99	1.16	0.44	0.24	1.84	Capon 1996
16946	Quinidine, 200 mg, PO, BID	Digoxin, IV	0.98	0.96	1.02	0.68	-	-	Steiness 1980
16948	Quinidine, 200 mg, PO, QID	Digoxin, IV	1.72	2.05	0.84	1.03	1.65	0.62	Ochs 1981
16950	Quinidine, 50 mg, PO, QID	Mexiletine, PO	1.00	0.85	1.18	1.00	0.79	1.27	Abolfathi 1993
16952	Quinidine, 50 mg, PO, QID	Mexiletine, PO	1.41	1.30	1.08	1.14	1.27	0.90	Abolfathi 1993
16954	Quinidine, 30 mg, PO, BID	Paroxetine, PO	1.93	1.29	1.50	1.04	1.14	0.92	Schoedel 2012
16956	Carbamazepine, 200/400 mg, PO, BID	Quinidine, PO	0.43	0.41	1.06	0.63	0.50	1.26	Andreassen 2007
16958	Carbamazepine, 200/400 mg, PO, BID	3-Hydroxyquinidine, PO	0.77	0.90	0.85	1.32	1.48	0.89	Andreassen 2007
16960	Cimetidine, 300 mg, PO, QD	Quinidine, PO	1.00	1.13	0.89	1.01	0.90	1.12	Kolb 1984
16962	Cimetidine, 300 mg, PO, QID	Quinidine, PO	1.00	1.28	0.78	1.01	1.26	0.80	Hardy 1983
16964	Fluvoxamine, 100 mg, PO, QD	Quinidine, PO	1.17	1.66	0.70	1.00	1.32	0.76	Damkier 1999a
16966	Fluvoxamine, 100 mg, PO, QD	3-Hydroxyquinidine, PO	1.12	1.23	0.91	0.91	0.96	0.95	Damkier 1999a
16968	Omeprazole, 40 mg, PO, QD	Quinidine, PO	1.00	1.15	0.87	1.00	1.12	0.89	Ching 1991
16970	Omeprazole, 40 mg, PO, QD	3-Hydroxyquinidine, PO	1.00	0.90	1.11	1.00	0.85	1.17	Ching 1991
16972	Rifampicin, 600 mg, PO, QD	Quinidine, PO	0.19	0.12	1.58	0.43	0.34	1.27	Damkier 1999
16974	Rifampicin, 600 mg, PO, QD	3-Hydroxyquinidine, PO	0.39	0.78	0.50	0.98	2.90	0.34	Damkier 1999

DataID	Perpetrator	Victim	Predicted AUC Ratio	Observed AUC Ratio	Pred/Obs AUC Ratio	Predicted C _{MAX} Ratio	Observed C _{MAX} Ratio	Pred/Obs C _{MAX} Ratio	Reference
16976	Verapamil, 80 mg, PO, TID	Quinidine, PO	1.48	1.21	1.22	1.19	0.96	1.24	Edwards 1987
16978	Verapamil, 120 mg, PO, TID	Quinidine, PO	1.72	1.25	1.38	1.27	0.96	1.32	Edwards 1987
16980	Paroxetine, 20 mg, PO, QD	Alprazolam, PO	1.02	0.94	1.09	1.02	0.97	1.05	Calvo 2004
16986	Fluvoxamine, 50/100 mg, PO, QD/QD	Atomoxetine, PO	1.34	1.30	1.03	1.16	1.36	0.85	Todor 2017
16988	Paroxetine, 20 mg, PO, QD	Atomoxetine, PO	1.78	2.80	0.64	1.15	1.28	0.90	Jung 2020
16990	Paroxetine, 20 mg, PO, QD	Atomoxetine, PO	3.62	4.70	0.77	1.49	1.71	0.87	Jung 2020
16992	Paroxetine, 20 mg, PO, QD	Atomoxetine, PO	5.76	10.57	0.54	1.78	2.21	0.81	Jung 2020
16994	Paroxetine, 20/20 mg, PO, BID/QD	Atomoxetine, PO	3.90	5.73	0.68	1.34	1.73	0.77	Todor 2015
16996	Paroxetine, 20 mg, PO, QD	Atomoxetine, PO	5.34	13.87	0.38	2.59	3.70	0.70	Belle 2002
16998	Clarithromycin, 500 mg, PO, BID	(E)-Clomiphene, PO	2.38	2.13	1.12	1.52	1.55	0.98	Mürdter 2016
17000	Clarithromycin, 500 mg, PO, BID	(E)-4-Hydroxyclophene, PO	2.77	1.27	2.18	2.36	1.01	2.34	Mürdter 2016
17002	Clarithromycin, 500 mg, PO, BID	(E)-N-Desethylclomiphene, PO	0.75	0.65	1.16	0.60	0.54	1.12	Mürdter 2016
17004	Clarithromycin, 500 mg, PO, BID	(E)-4-Hydroxy-N-desethylclomiphene, PO	1.60	0.59	2.71	1.70	0.67	2.53	Mürdter 2016
17006	Clarithromycin, 500 mg, PO, BID	(E)-Clomiphene, PO	1.45	1.82	0.80	1.18	1.59	0.75	Mürdter 2016
17008	Clarithromycin, 500 mg, PO, BID	(E)-4-Hydroxyclophene, PO	1.75	1.72	1.02	1.41	1.40	1.00	Mürdter 2016
17010	Clarithromycin, 500 mg, PO, BID	(E)-N-Desethylclomiphene, PO	0.30	0.29	1.02	0.17	0.19	0.91	Mürdter 2016

DataID	Perpetrator	Victim	Predicted AUC Ratio	Observed AUC Ratio	Pred/Obs AUC Ratio	Predicted C _{MAX} Ratio	Observed C _{MAX} Ratio	Pred/Obs C _{MAX} Ratio	Reference
17012	Clarithromycin, 500 mg, PO, BID	(E)-4-Hydroxy-N- desethylclomiphene, PO	0.45	0.35	1.27	0.25	0.25	0.98	Mürdter 2016
17014	Clarithromycin, 500 mg, PO, BID	(E)-Clomiphene, PO	1.22	1.98	0.62	1.10	1.73	0.63	Mürdter 2016
17016	Clarithromycin, 500 mg, PO, BID	(E)-4- Hydroxyclophene, PO	1.39	1.63	0.86	1.18	1.67	0.70	Mürdter 2016
17018	Clarithromycin, 500 mg, PO, BID	(E)-N- Desethylclomiphene, PO	0.26	0.34	0.76	0.19	0.47	0.41	Mürdter 2016
17020	Clarithromycin, 500 mg, PO, BID	(E)-4-Hydroxy-N- desethylclomiphene, PO	0.38	0.43	0.89	0.21	0.39	0.54	Mürdter 2016
17022	Clarithromycin, 500 mg, PO, BID	(E)-Clomiphene, PO	1.15	1.23	0.93	1.07	1.59	0.67	Mürdter 2016
17024	Clarithromycin, 500 mg, PO, BID	(E)-4- Hydroxyclophene, PO	1.28	1.22	1.05	1.13	1.04	1.09	Mürdter 2016
17026	Clarithromycin, 500 mg, PO, BID	(E)-N- Desethylclomiphene, PO	0.30	0.21	1.41	0.26	0.19	1.35	Mürdter 2016
17028	Clarithromycin, 500 mg, PO, BID	(E)-4-Hydroxy-N- desethylclomiphene, PO	0.34	0.22	1.56	0.25	0.29	0.88	Mürdter 2016
17030	Clarithromycin, 500 mg, PO, BID	(E)-Clomiphene, PO	1.11	1.06	1.05	1.06	1.80	0.59	Mürdter 2016
17032	Clarithromycin, 500 mg, PO, BID	(E)-4- Hydroxyclophene, PO	1.21	1.83	0.66	1.09	1.20	0.91	Mürdter 2016
17034	Clarithromycin, 500 mg, PO, BID	(E)-N- Desethylclomiphene, PO	0.34	0.39	0.88	0.31	0.23	1.36	Mürdter 2016
17036	Clarithromycin, 500 mg, PO, BID	(E)-4-Hydroxy-N- desethylclomiphene, PO	0.36	0.41	0.88	0.29	0.38	0.77	Mürdter 2016
17038	Paroxetine, 40 mg, PO, QD	(E)-Clomiphene, PO	1.19	1.09	1.10	1.08	0.91	1.19	Mürdter 2016
17040	Paroxetine, 40 mg, PO, QD	(E)-4- Hydroxyclophene, PO	1.21	1.02	1.19	1.16	1.19	0.98	Mürdter 2016

DataID	Perpetrator	Victim	Predicted AUC Ratio	Observed AUC Ratio	Pred/Obs AUC Ratio	Predicted C _{MAX} Ratio	Observed C _{MAX} Ratio	Pred/Obs C _{MAX} Ratio	Reference
17042	Paroxetine, 40 mg, PO, QD	(E)-N- Desethylclomiphenes, PO	0.97	1.21	0.80	0.87	1.36	0.64	Mürdter 2016
17044	Paroxetine, 40 mg, PO, QD	(E)-4-Hydroxy-N- desethylclomiphenes, PO	1.16	1.01	1.15	1.15	0.94	1.22	Mürdter 2016
17046	Paroxetine, 40 mg, PO, QD	(E)-Clomiphenes, PO	2.63	2.47	1.06	1.43	1.30	1.10	Mürdter 2016
17048	Paroxetine, 40 mg, PO, QD	(E)-4- Hydroxycyclophens, PO	0.46	0.22	2.07	0.25	0.09	2.78	Mürdter 2016
17050	Paroxetine, 40 mg, PO, QD	(E)-N- Desethylclomiphenes, PO	4.10	4.86	0.84	1.75	1.29	1.36	Mürdter 2016
17052	Paroxetine, 40 mg, PO, QD	(E)-4-Hydroxy-N- desethylclomiphenes, PO	0.39	0.29	1.33	0.25	0.22	1.13	Mürdter 2016
17054	Paroxetine, 40 mg, PO, QD	(E)-Clomiphenes, PO	4.59	9.53	0.48	1.81	4.02	0.45	Mürdter 2016
17056	Paroxetine, 40 mg, PO, QD	(E)-4- Hydroxycyclophens, PO	0.54	0.86	0.62	0.18	0.28	0.64	Mürdter 2016
17058	Paroxetine, 40 mg, PO, QD	(E)-N- Desethylclomiphenes, PO	11.85	25.32	0.47	3.18	9.07	0.35	Mürdter 2016
17060	Paroxetine, 40 mg, PO, QD	(E)-4-Hydroxy-N- desethylclomiphenes, PO	1.23	1.91	0.64	0.46	0.50	0.93	Mürdter 2016
17062	Paroxetine, 40 mg, PO, QD	(E)-Clomiphenes, PO	6.66	10.35	0.64	2.21	3.85	0.57	Mürdter 2016
17064	Paroxetine, 40 mg, PO, QD	(E)-4- Hydroxycyclophens, PO	0.89	1.54	0.58	0.20	0.27	0.75	Mürdter 2016
17066	Paroxetine, 40 mg, PO, QD	(E)-N- Desethylclomiphenes, PO	22.80	21.14	1.08	5.28	4.59	1.15	Mürdter 2016
17068	Paroxetine, 40 mg, PO, QD	(E)-4-Hydroxy-N- desethylclomiphenes, PO	3.50	2.65	1.32	0.69	0.74	0.93	Mürdter 2016
17070	Paroxetine, 40 mg, PO, QD	(E)-Clomiphenes, PO	6.29	12.47	0.50	2.08	7.08	0.29	Mürdter 2016

DataID	Perpetrator	Victim	Predicted AUC Ratio	Observed AUC Ratio	Pred/Obs AUC Ratio	Predicted C _{MAX} Ratio	Observed C _{MAX} Ratio	Pred/Obs C _{MAX} Ratio	Reference
17072	Paroxetine, 40 mg, PO, QD	(E)-4- Hydroxyclophene, PO	1.08	3.76	0.29	0.26	0.70	0.37	Mürdter 2016
17074	Paroxetine, 40 mg, PO, QD	(E)-N- Desethylclomiphene, PO	15.92	33.72	0.47	4.38	12.14	0.36	Mürdter 2016
17076	Paroxetine, 40 mg, PO, QD	(E)-4-Hydroxy-N- desethylclomiphene, PO	4.16	5.87	0.71	1.06	1.54	0.69	Mürdter 2016
17078	Atomoxetine, 60 mg, PO, BID	Desipramine, PO	1.07	1.17	0.91	1.00	1.06	0.95	Sauer 2004
17082	Paroxetine, 20 mg, PO, QD	Desipramine, PO	3.11	2.71	1.15	1.37	1.90	0.72	Nichols 2009
17084	Paroxetine, 20 mg, PO, QD	2- Hydroxydesipramine, PO	0.19	0.55	0.34	0.07	0.20	0.37	Nichols 2009
17086	Paroxetine, 20 mg, PO, QD	Desipramine, PO	0.99	0.87	1.14	0.99	0.85	1.16	Brøsen 1993
17088	Paroxetine, 20 mg, PO, QD	Desipramine, PO	4.22	4.60	0.92	1.33	1.90	0.70	Brøsen 1993
17090	Paroxetine, 20 mg, PO, QD	Desipramine, PO	5.78	7.76	0.74	1.61	2.45	0.66	Brøsen 1993
17092	Paroxetine, 20/30 mg, PO, QD/QD	Desipramine, PO	3.13	1.84	1.70	3.91	1.27	3.08	Aldermann 1997
17093	Paroxetine, 20/30 mg, PO, QD/QD	Desipramine, PO	3.13	4.96	0.63	3.91	4.58	0.85	Aldermann 1997
17094	Paroxetine, 20/30 mg, PO, QD/QD	Desipramine, PO	3.13	7.40	0.42	3.91	4.58	0.85	Aldermann 1997
17096	Quinidine, 200 mg, PO, QD	Desipramine, PO	4.19	6.52	0.64	1.35	1.59	0.85	Brøsen 1989
17098	Paroxetine, 20 mg, PO,	Dextromethorphan, PO	13.46	8.45	1.59	6.14	5.11	1.20	Storelli 2018
17100	Paroxetine, 20 mg, PO,	Dextrophan, PO	0.59	0.65	0.91	0.15	0.25	0.59	Storelli 2018
17102	Paroxetine, 20 mg, PO,	Dextromethorphan, PO	7.01	23.61	0.30	8.21	12.29	0.67	Storelli 2018

DataID	Perpetrator	Victim	Predicted AUC Ratio	Observed AUC Ratio	Pred/Obs AUC Ratio	Predicted C _{MAX} Ratio	Observed C _{MAX} Ratio	Pred/Obs C _{MAX} Ratio	Reference
17104	Paroxetine, 20 mg, PO,	Dextrophan, PO	0.92	1.92	0.48	0.32	0.77	0.42	Storelli 2018
17106	Paroxetine, 20 mg, PO, QD	Dextromethorphan, PO	1.00	1.45	0.69	1.00	1.42	0.70	Schoedel 2012
17108	Cimetidine, 800 mg, PO, QD	Metoprolol, PO	1.22	0.91	1.34	1.24	1.56	0.80	Chellingworth 1988
17110	Cimetidine, 800 mg, PO, QD	Metoprolol, PO	1.15	1.34	0.86	1.19	1.62	0.73	Chellingworth 1988
17112	Cimetidine, 800 mg, PO, QD	Metoprolol, PO	1.11	1.44	0.77	1.13	1.23	0.92	Toon 1988
17114	Cimetidine, 200/200/200/400 mg, PO, QD	Metoprolol, PO	1.09	1.64	0.66	1.06	1.56	0.68	Kirch 1982
17116	Paroxetine, 10.10.2010 mg, PO, QD/BID	Metoprolol, PO	4.20	4.33	0.97	2.07	2.51	0.83	Stout 2011
17118	Paroxetine, 10.10.2010 mg, PO, QD/BID	Metoprolol, PO	6.28	3.62	1.73	2.84	3.65	0.78	Stout 2011
17120	Paroxetine, 20 mg, PO, QD	R-Metoprolol, PO	3.70	2.96	1.25	3.18	2.74	1.16	Parker 2011
17122	Paroxetine, 20 mg, PO, QD	S-Metoprolol, PO	3.59	2.91	1.23	3.11	2.64	1.18	Parker 2011
17124	Paroxetine, 20 mg, PO, QD	R-Metoprolol, PO	3.92	3.06	1.28	2.69	2.11	1.27	Parker 2011
17126	Paroxetine, 20 mg, PO, QD	S-Metoprolol, PO	3.87	2.50	1.55	2.66	1.87	1.42	Parker 2011
17128	Paroxetine, 20 mg, PO, QD	R-Metoprolol, PO	3.65	4.06	0.90	3.13	3.74	0.84	Parker 2011
17130	Paroxetine, 20 mg, PO, QD	S-Metoprolol, PO	3.54	3.09	1.14	3.06	2.88	1.06	Parker 2011
17132	Paroxetine, 10 mg, PO, BID	R-Metoprolol, PO	4.22	9.88	0.43	2.30	2.55	0.90	Hemeryck 2000
17134	Paroxetine, 10 mg, PO, BID	S-Metoprolol, PO	3.77	7.01	0.54	2.04	1.71	1.19	Hemeryck 2000
17136	Quinidine, 50 mg, PO, SD	Metoprolol, IV	0.75	0.95	0.79	1.02	0.98	1.04	Leemann 1993
17138	Quinidine, 50 mg, PO, SD	Metoprolol, IV	0.88	2.00	0.44	1.04	1.35	0.77	Leemann 1993

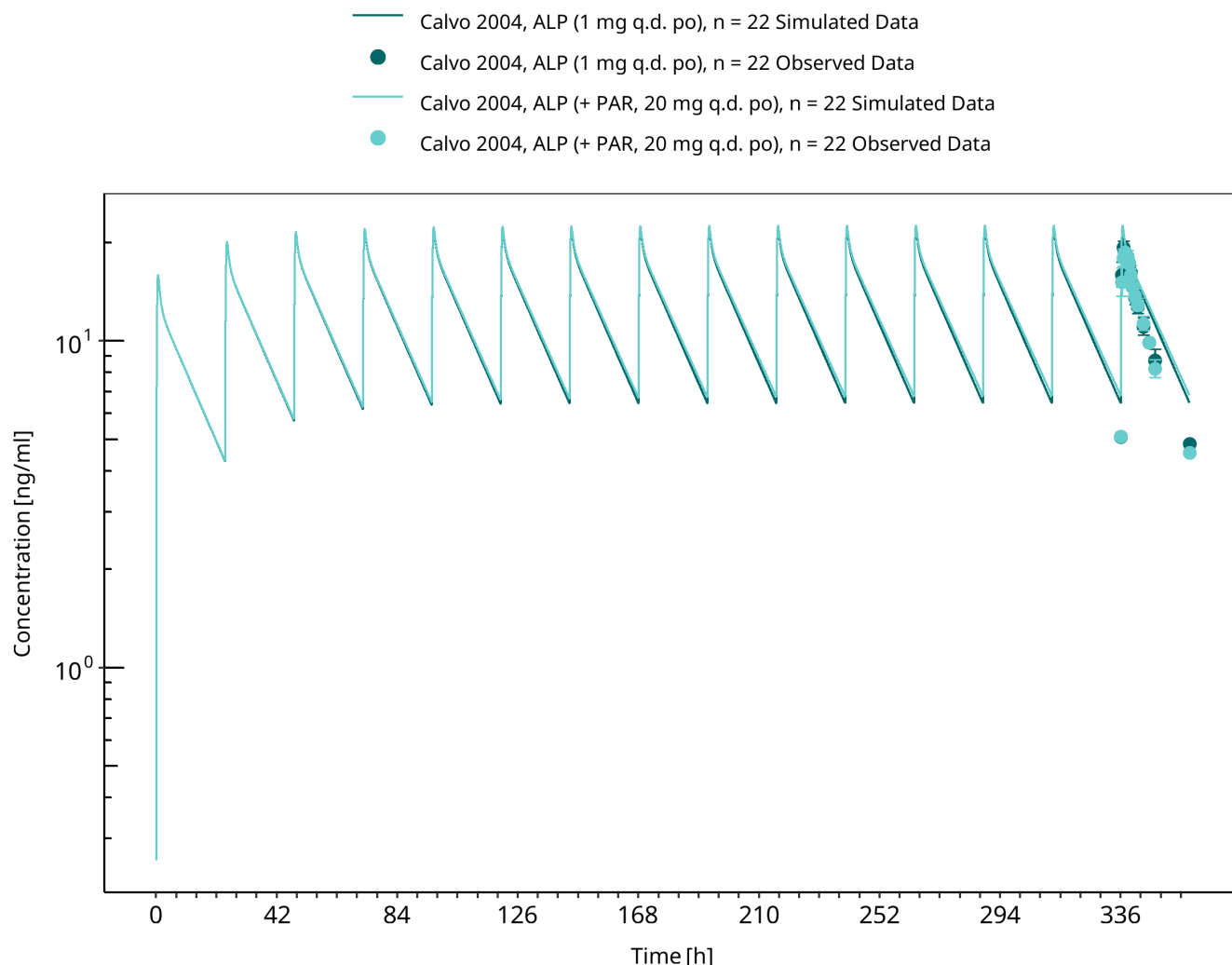
DataID	Perpetrator	Victim	Predicted AUC Ratio	Observed AUC Ratio	Pred/Obs AUC Ratio	Predicted C _{MAX} Ratio	Observed C _{MAX} Ratio	Pred/Obs C _{MAX} Ratio	Reference
17140	Quinidine, 250 mg, PO, BID	Metoprolol, IV	1.39	0.97	1.43	1.04	1.09	0.95	Leemann 1993
17142	Quinidine, 250 mg, PO, BID	Metoprolol, IV	1.97	2.38	0.83	1.10	1.35	0.81	Leemann 1993
17144	Quinidine, 100 mg, PO, QD	R-Metoprolol, PO	3.69	3.44	1.07	1.88	1.43	1.32	Johnson 1996
17146	Quinidine, 100 mg, PO, QD	S-Metoprolol, PO	3.65	2.87	1.27	1.86	1.28	1.45	Johnson 1996
17148	Quinidine, 100 mg, PO, QD	R-Metoprolol, PO	3.66	4.35	0.84	1.89	1.87	1.01	Johnson 1996
17150	Quinidine, 100 mg, PO, QD	S-Metoprolol, PO	3.62	2.99	1.21	1.86	1.40	1.33	Johnson 1996
17152	Rifampicin, 600 mg, PO, QD	Metoprolol, PO	0.84	0.74	1.14	1.00	0.66	1.52	Bennett 1982
17154	Atomoxetine, 60 mg, PO, BID	Midazolam, PO	1.08	1.32	0.82	1.01	1.01	1.00	Sauer 2004
17156	Atomoxetine, 60 mg, PO, BID	Midazolam, PO	0.98	1.51	0.65	1.01	1.52	0.66	Sauer 2004
17158	Paroxetine, 100 mg, PO, BID	Paroxetine, PO	1.17	1.65	0.71	1.03	1.40	0.74	Yasui-Furukori 2007
17160	Erythromycin, 250/250/250/250 mg, PO, QID	Quinidine, PO	1.34	1.19	1.13	1.09	1.39	0.78	Damkier 1999b
17162	Itraconazole, 200 mg, PO, QD	Quinidine, PO	1.72	2.05	0.84	1.42	1.61	0.88	Kaukonen 1997
17164	Itraconazole, 100 mg, PO, QD	Quinidine, PO	1.67	2.58	0.65	1.13	1.32	0.86	Damkier 1999b
17166	Ketoconazole, 200 mg, PO, QD	Risperidone, PO	1.28	1.49	0.86	1.55	1.30	1.19	Mahatthanatrakul 2012
17168	Ketoconazole, 200 mg, PO, QD	9-Hydroxyrisperidone, PO	1.06	0.54	1.97	1.22	0.54	2.27	Mahatthanatrakul 2012
17170	Rifampicin, 600 mg, PO, QD	Risperidone, PO	0.55	0.51	1.08	0.52	0.76	0.68	Kim 2018
17172	Rifampicin, 600 mg, PO, QD	9-Hydroxyrisperidone, PO	0.79	0.60	1.32	0.81	0.67	1.21	Kim 2018
17174	Rifampicin, 600 mg, PO, QD	Risperidone, PO	0.49	0.29	1.68	0.50	0.50	1.00	Mahatthanatrakul 2007

DataID	Perpetrator	Victim	Predicted AUC Ratio	Observed AUC Ratio	Pred/Obs AUC Ratio	Predicted C _{MAX} Ratio	Observed C _{MAX} Ratio	Pred/Obs C _{MAX} Ratio	Reference
17176	Verapamil, 80/80/80 mg, PO, TID	Risperidone, PO	1.32	1.58	0.84	1.58	1.59	0.99	Nakagami 2005
17178	Verapamil, 80/80/80 mg, PO, TID	9- Hydroxyrisperidone, PO	1.23	1.58	0.78	1.27	1.59	0.80	Nakagami 2005

3 Concentration-Time Profiles

The following section shows concentration-time profiles of the victim drugs of the simulated DD(G)I studies in comparison to observed data.

3.1 Paroxetine-Alprazolam-DDI



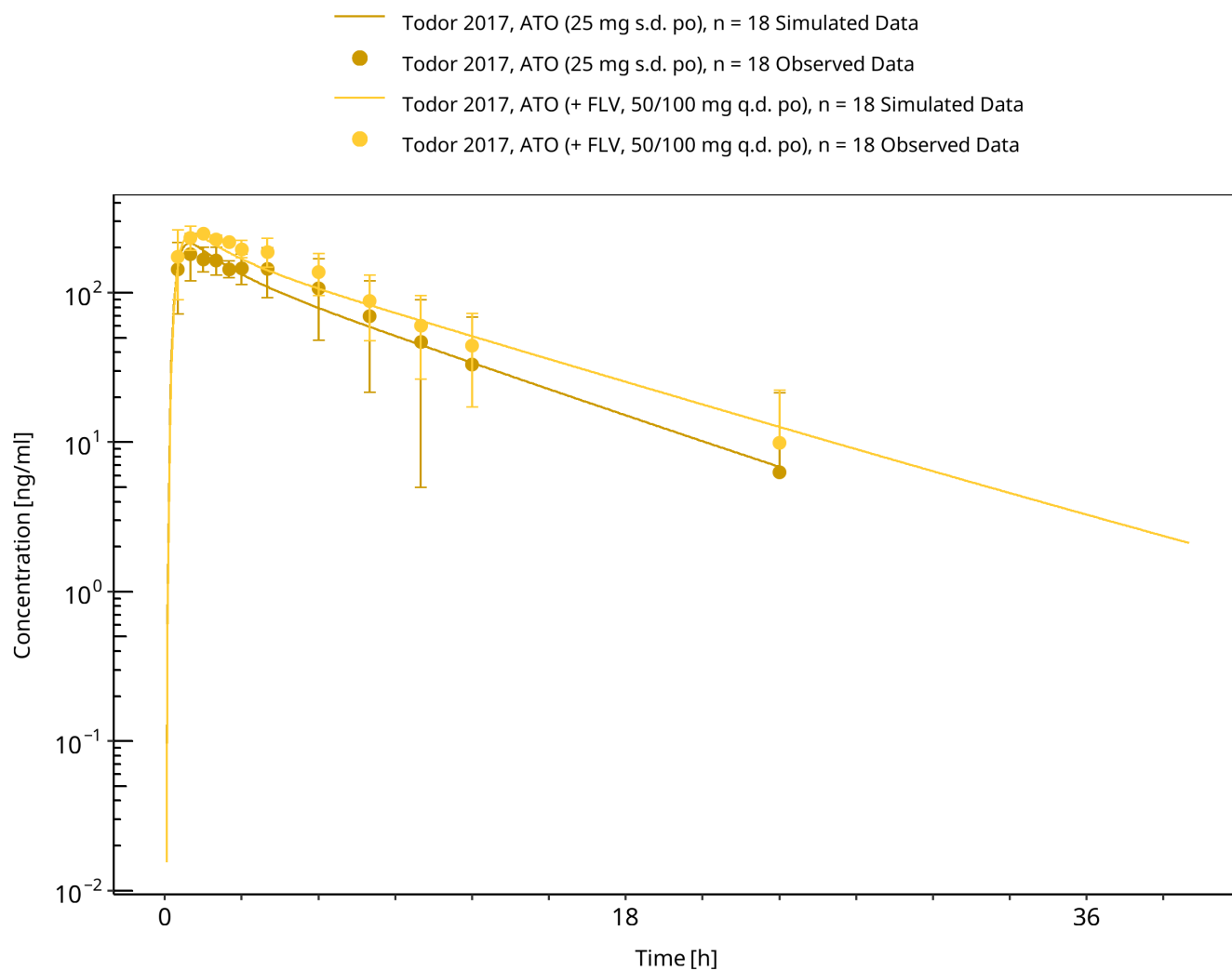


Figure 3-2: Fluvoxamine-Atomoxetine-DDI

3.3 Paroxetine-Atomoxetine-DDGI

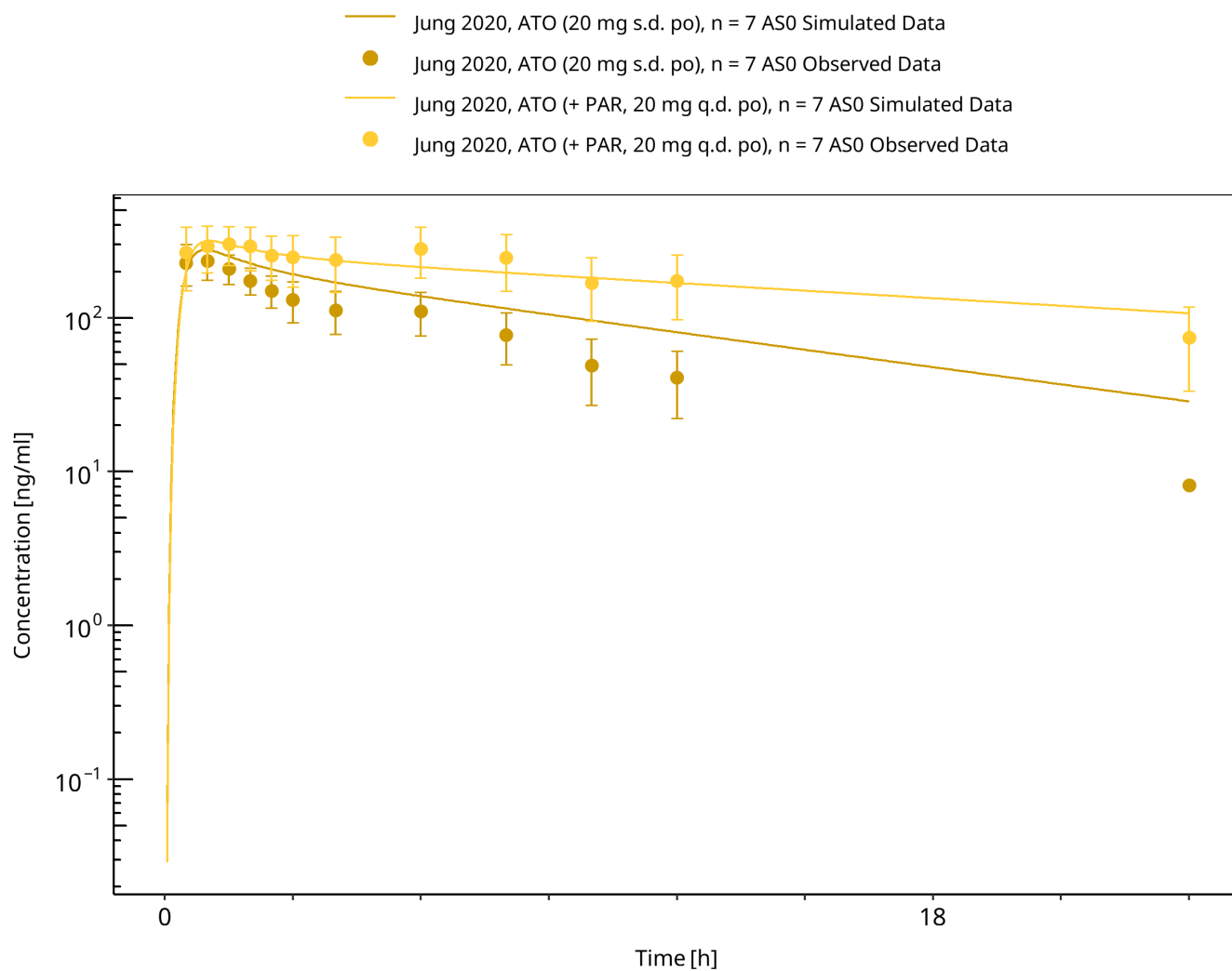


Figure 3-3: Paroxetine-Atomoxetine-DDGI

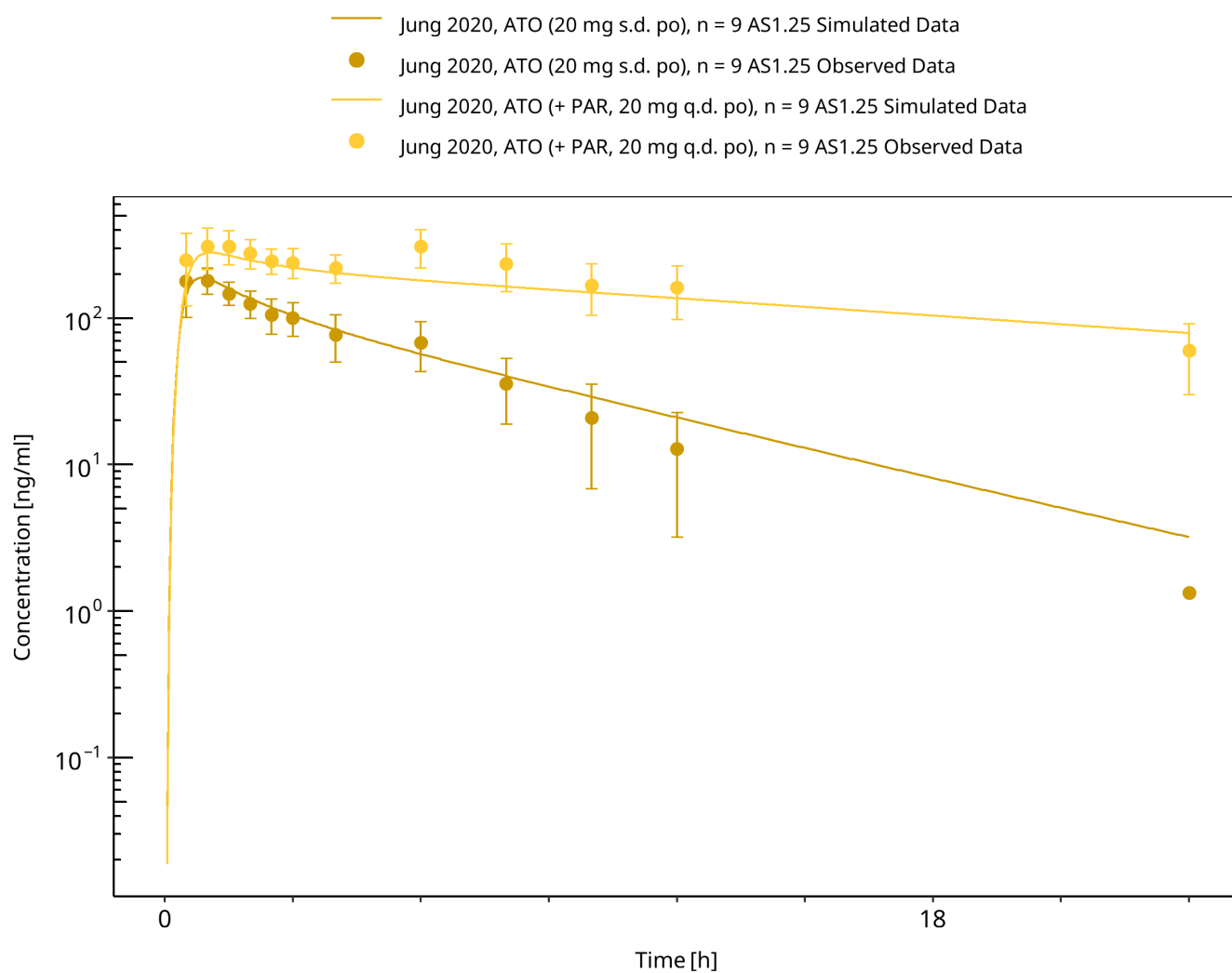


Figure 3-4: Paroxetine-Atomoxetine-DDGI

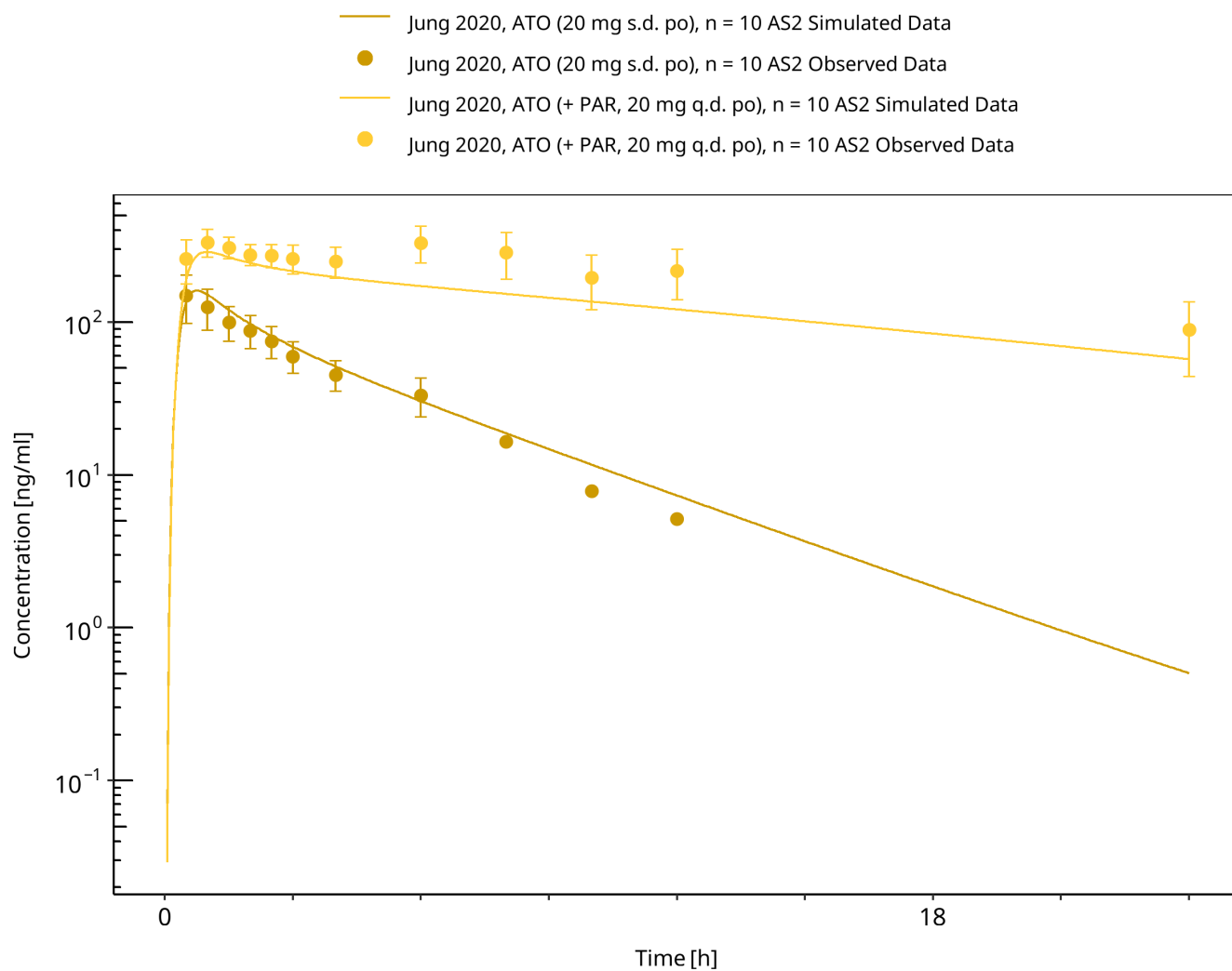


Figure 3-5: Paroxetine-Atomoxetine-DDGI

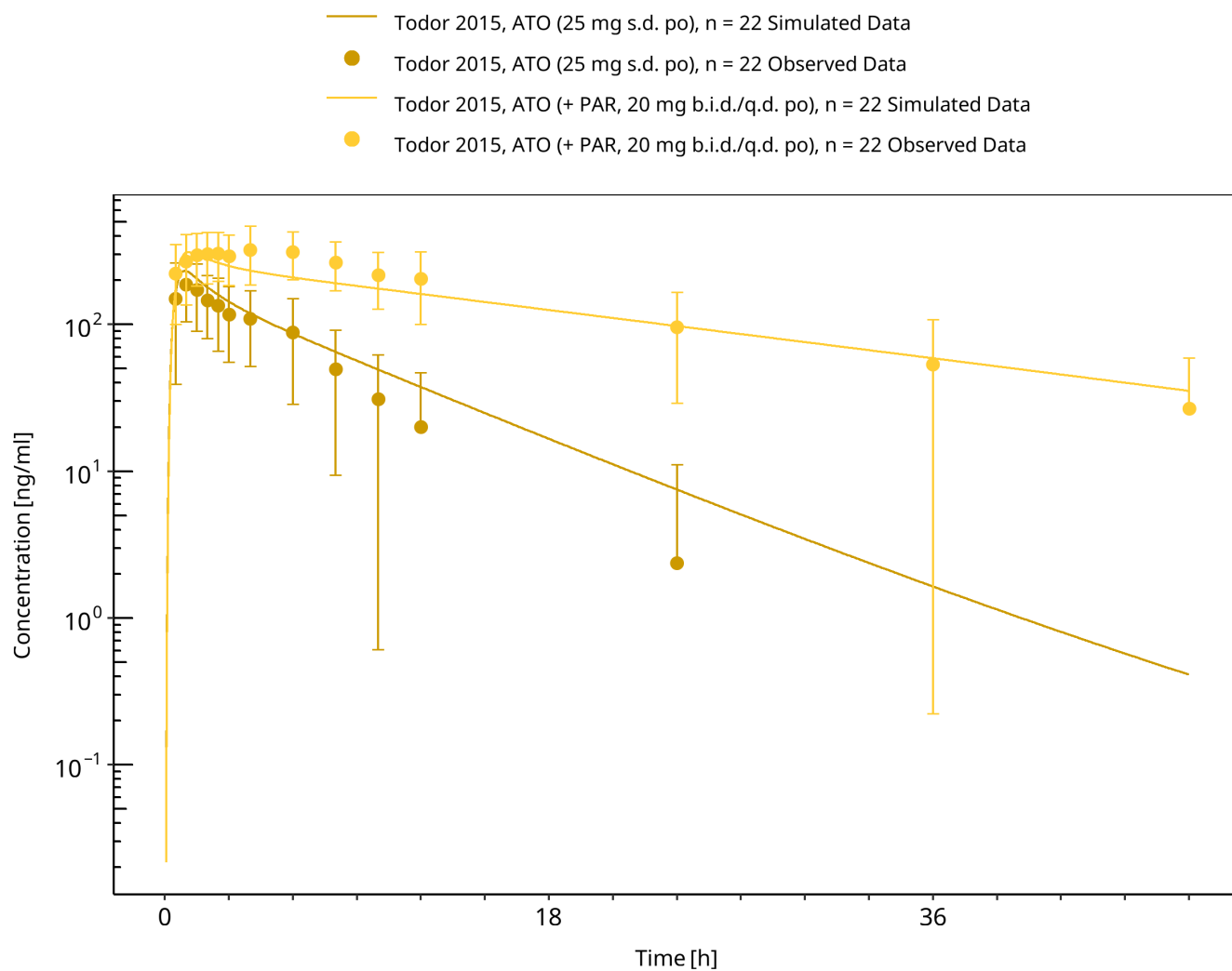


Figure 3-6: Paroxetine-Atomoxetine-DDI

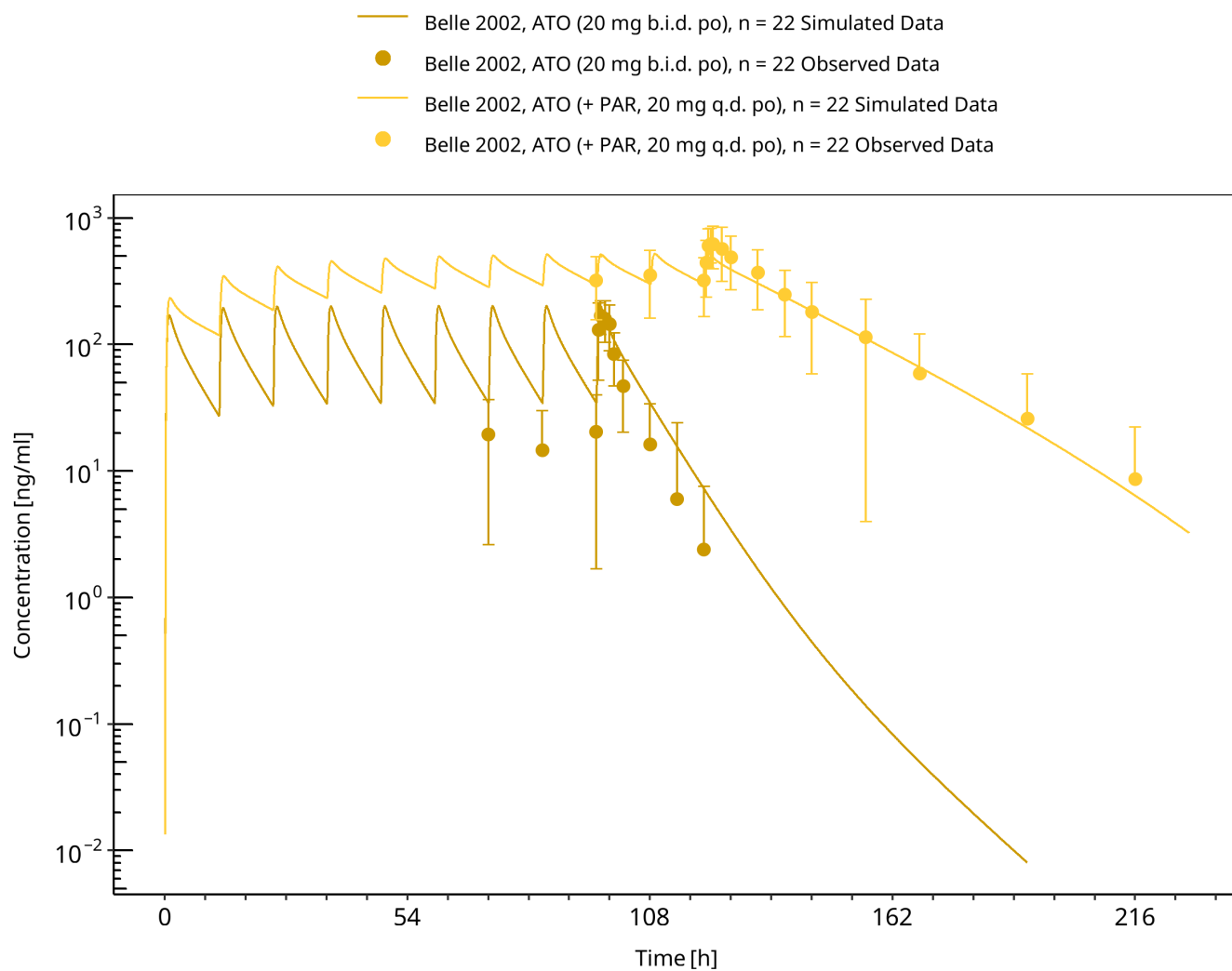


Figure 3-7: Paroxetine-Atomoxetine-DDI

3.4 Clarithromycin-Clomiphene-DDGI

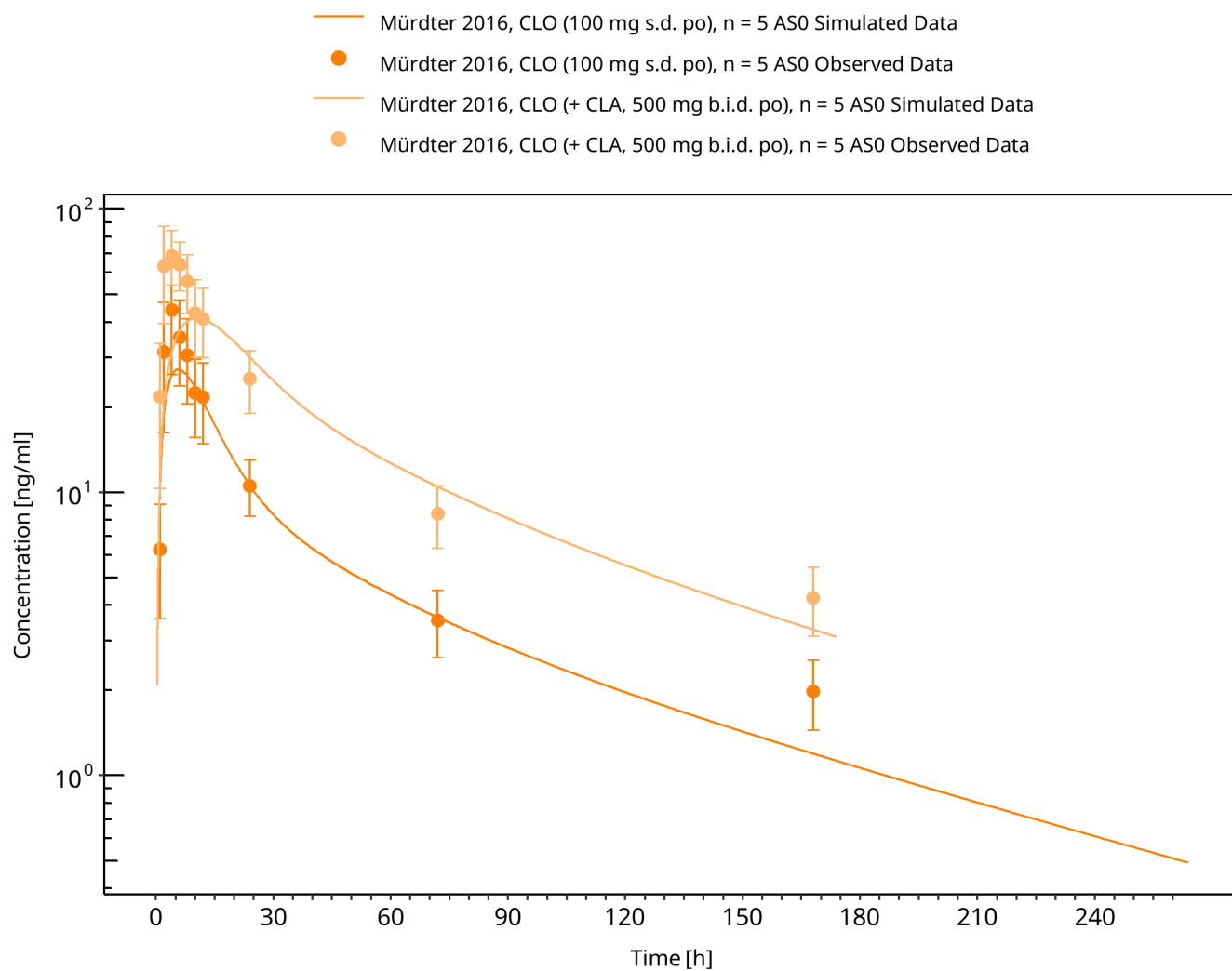


Figure 3-8: Clarithromycin-Clomiphene-DDGI

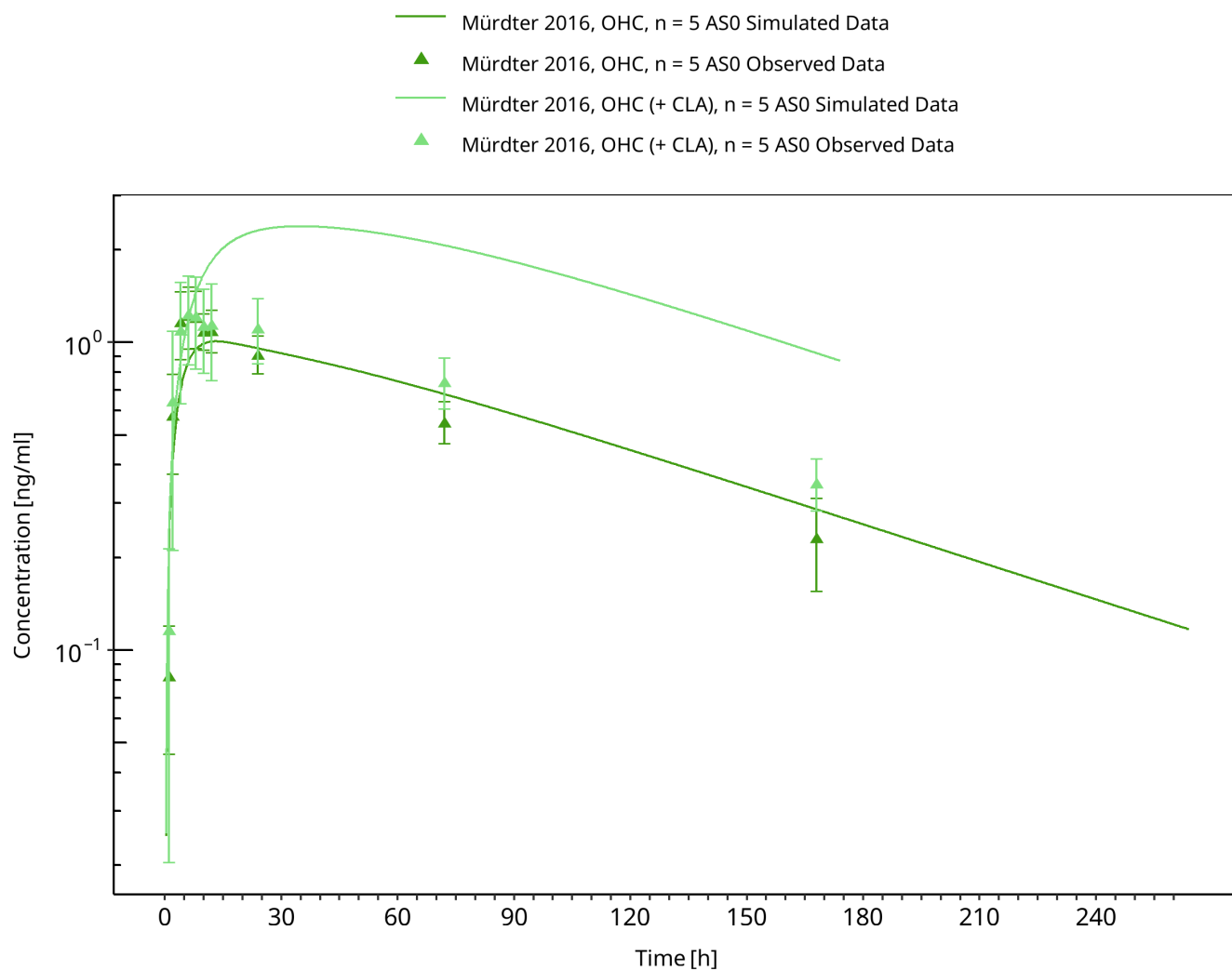


Figure 3-9: Clarithromycin-Clomiphene-DDGI

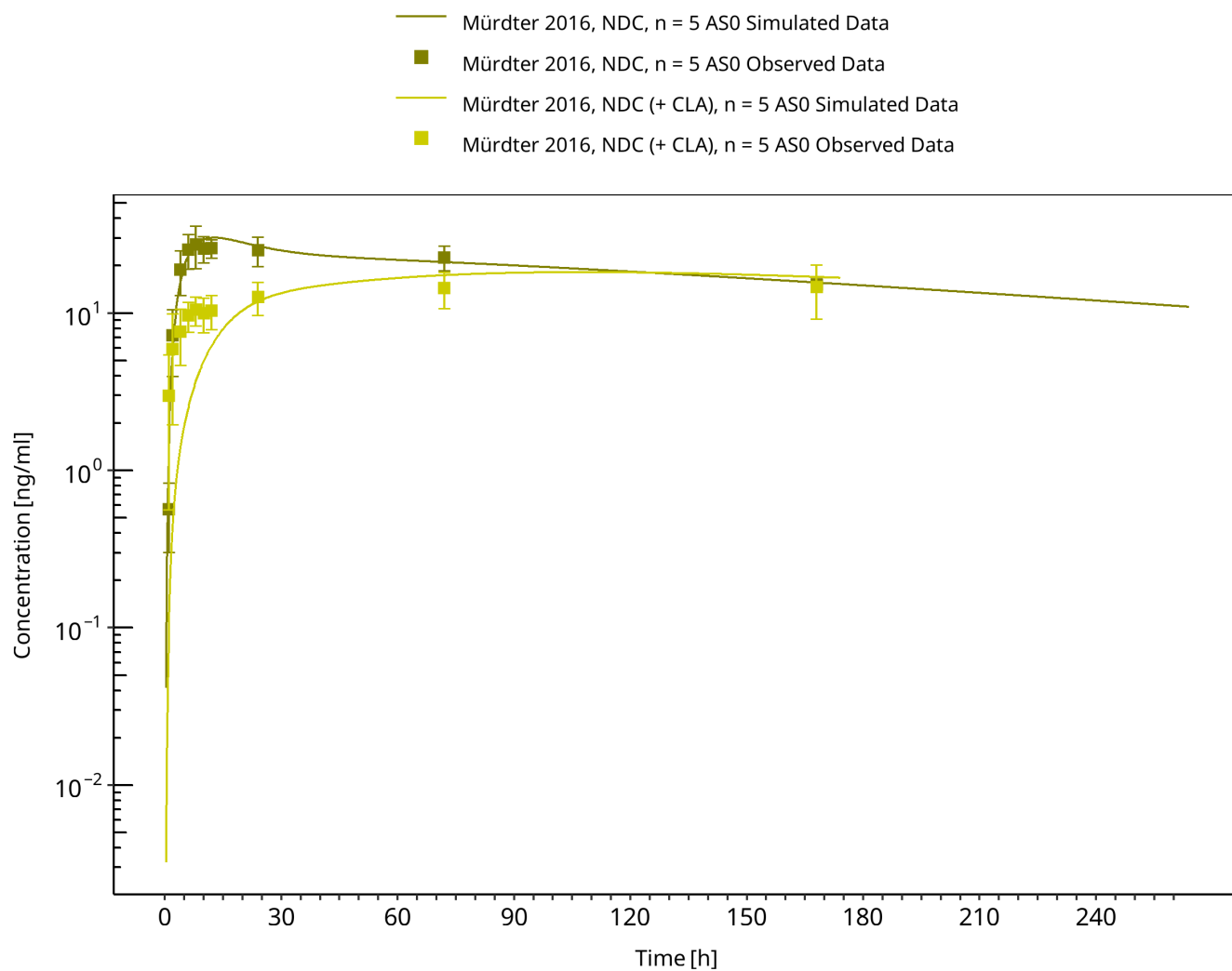


Figure 3-10: Clarithromycin-Clomiphene-DDGI

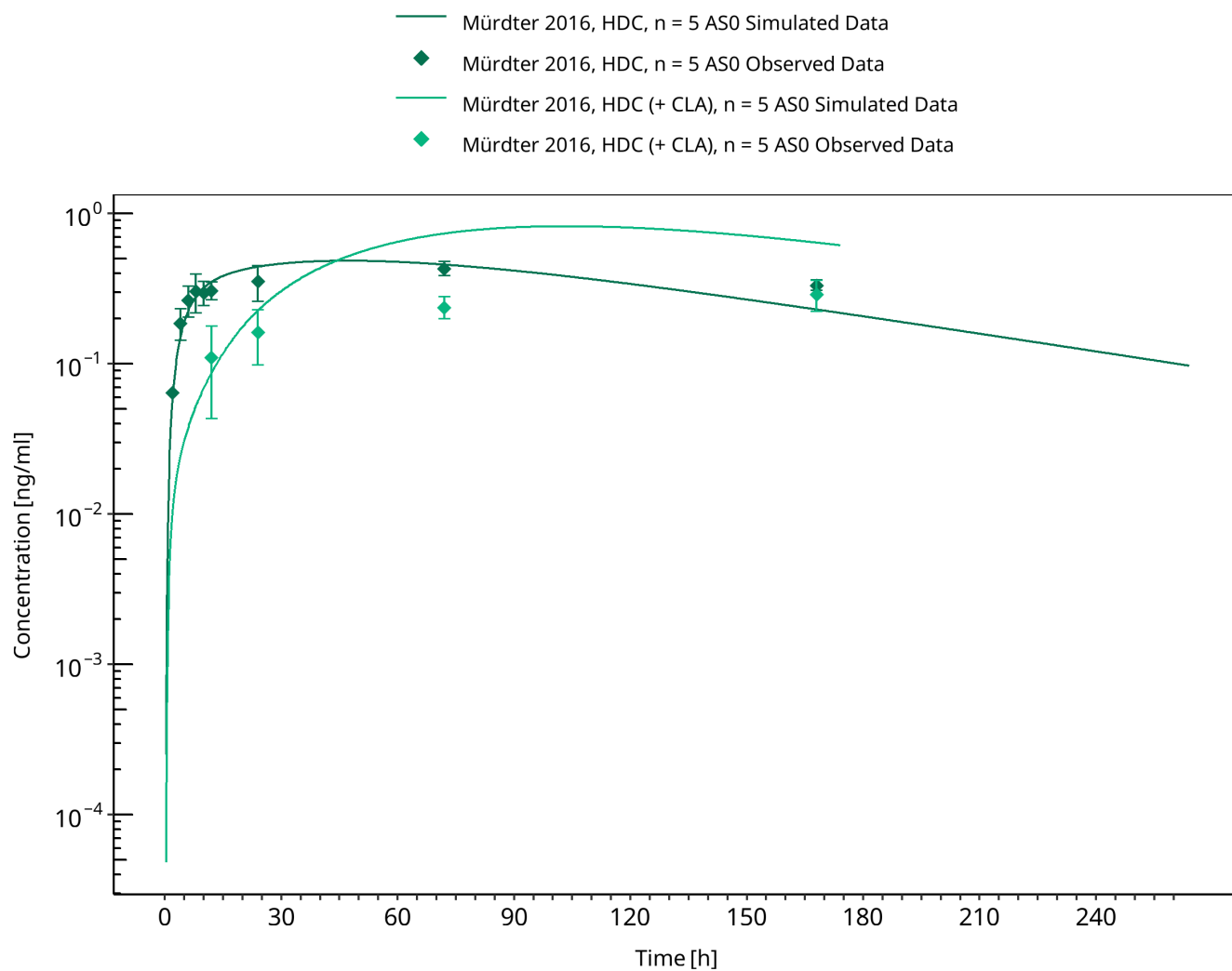


Figure 3-11: Clarithromycin-Clomiphene-DDGI

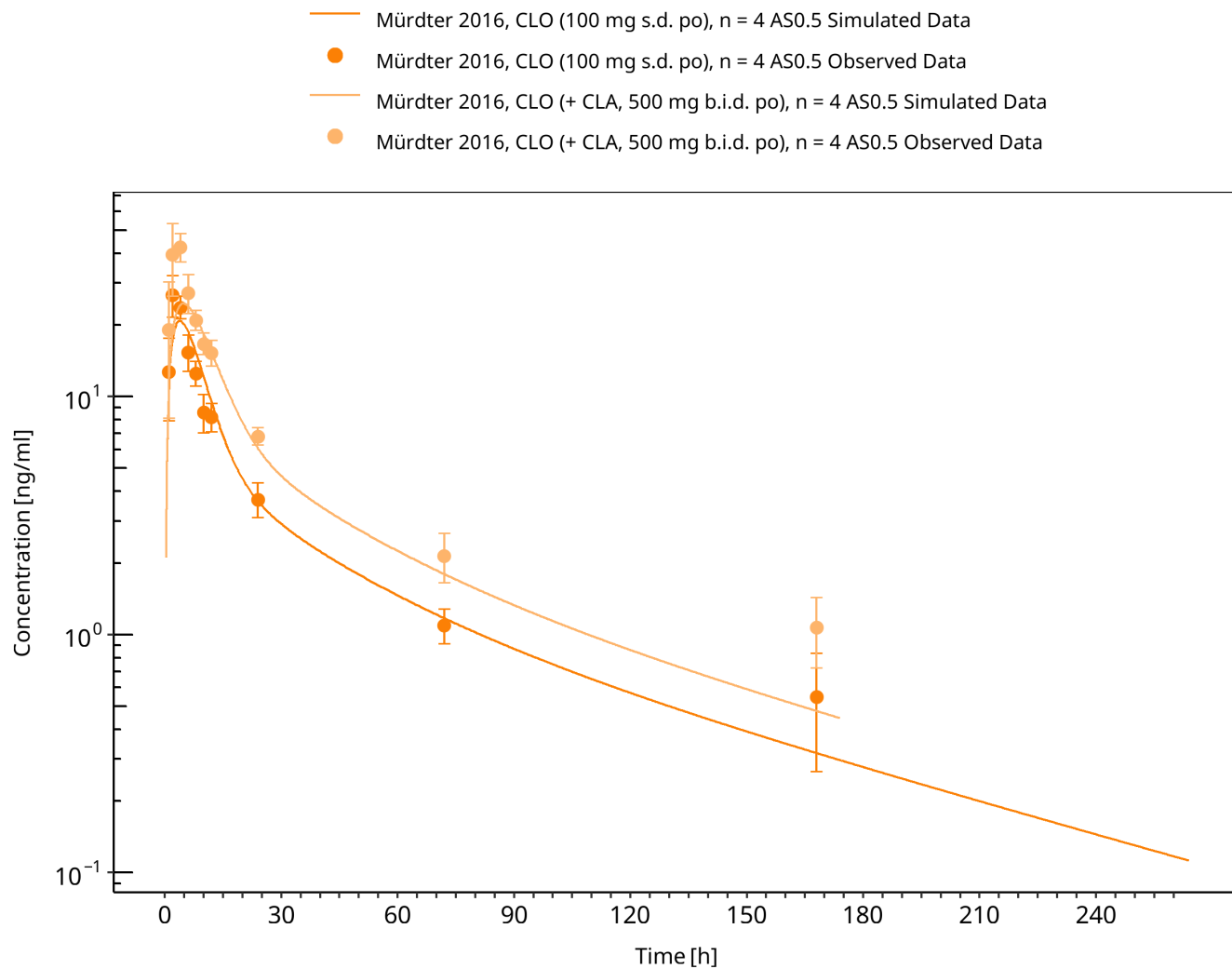


Figure 3-12: Clarithromycin-Clomiphene-DDGI

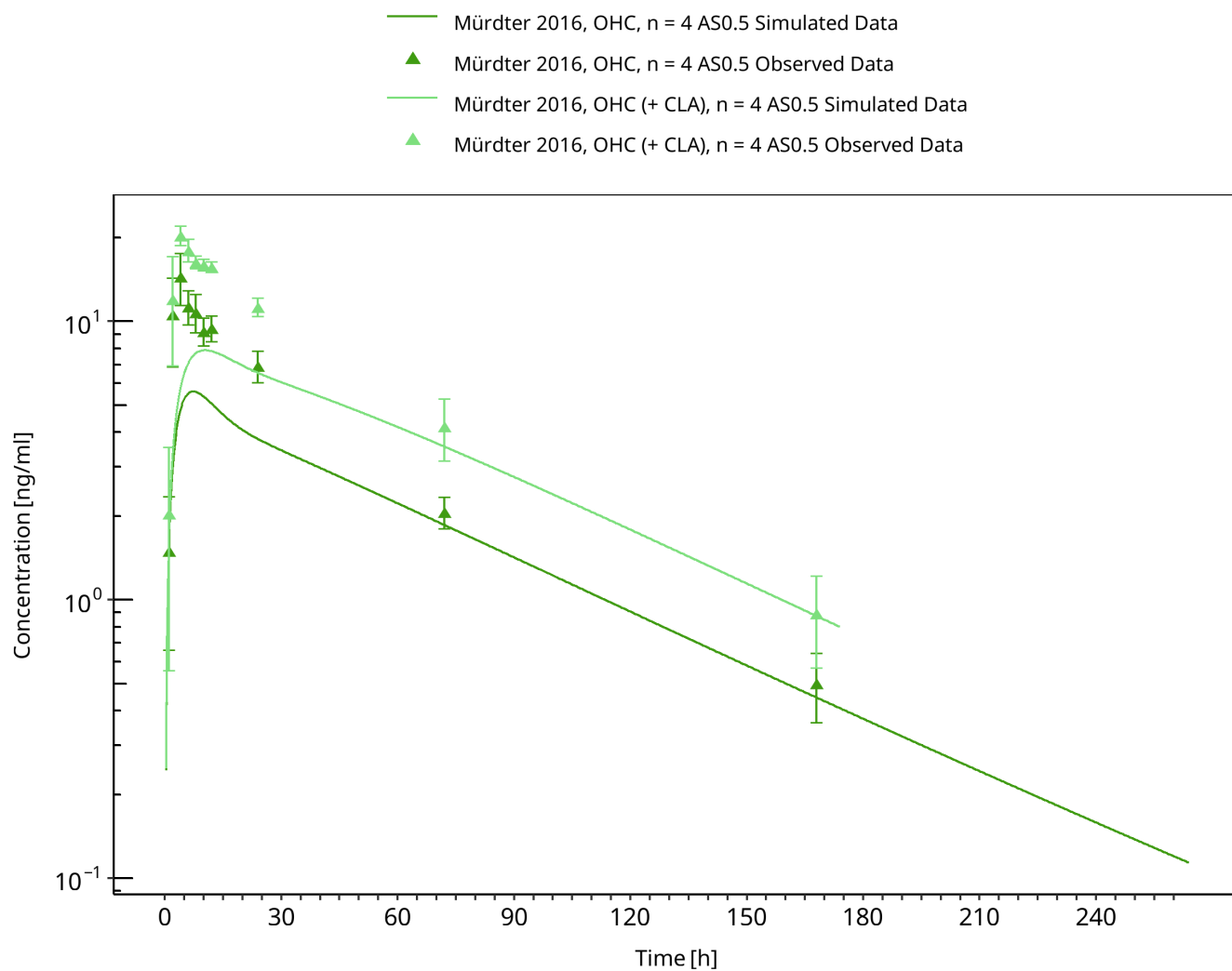


Figure 3-13: Clarithromycin-Clomiphene-DDGI

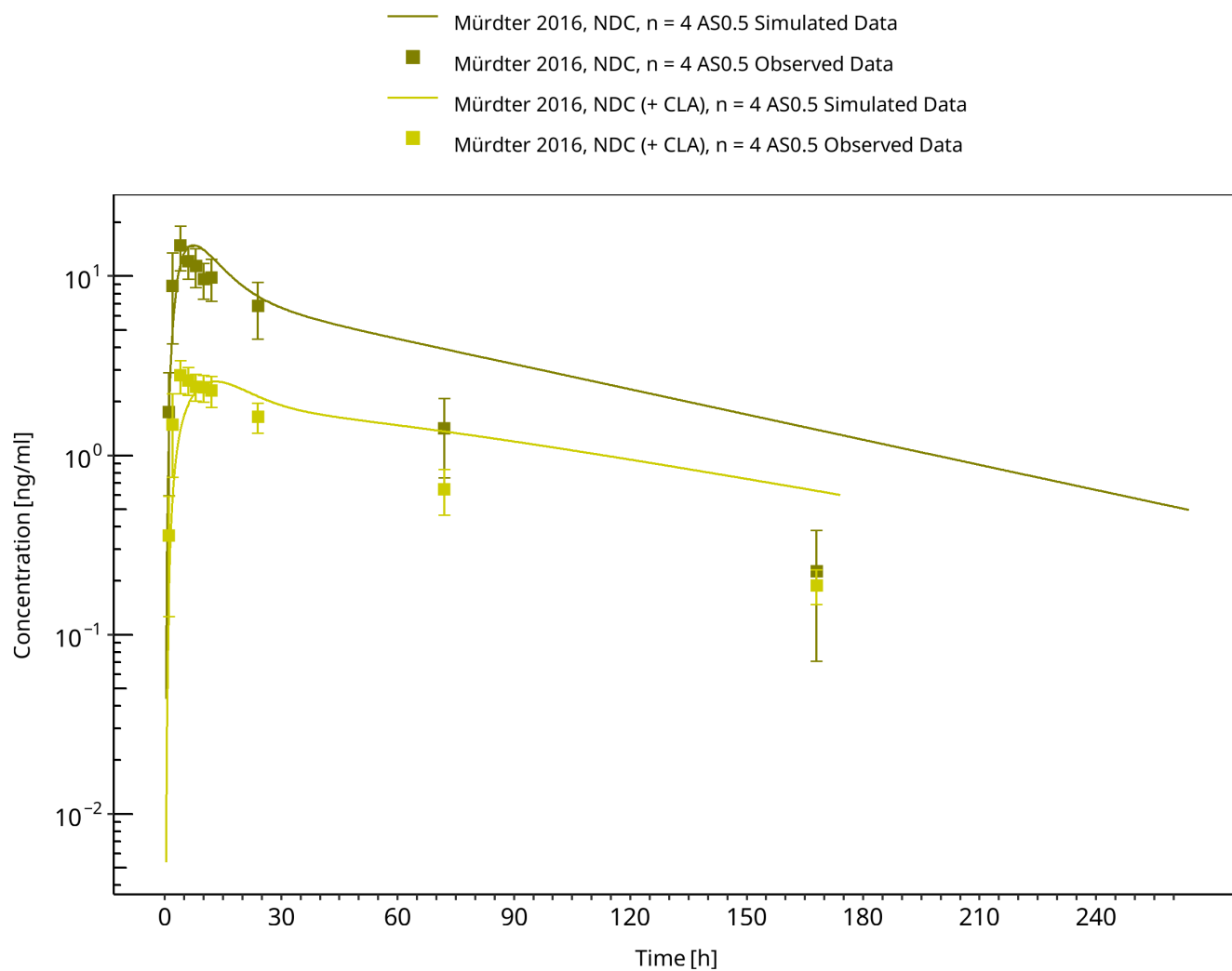


Figure 3-14: Clarithromycin-Clomiphene-DDGI

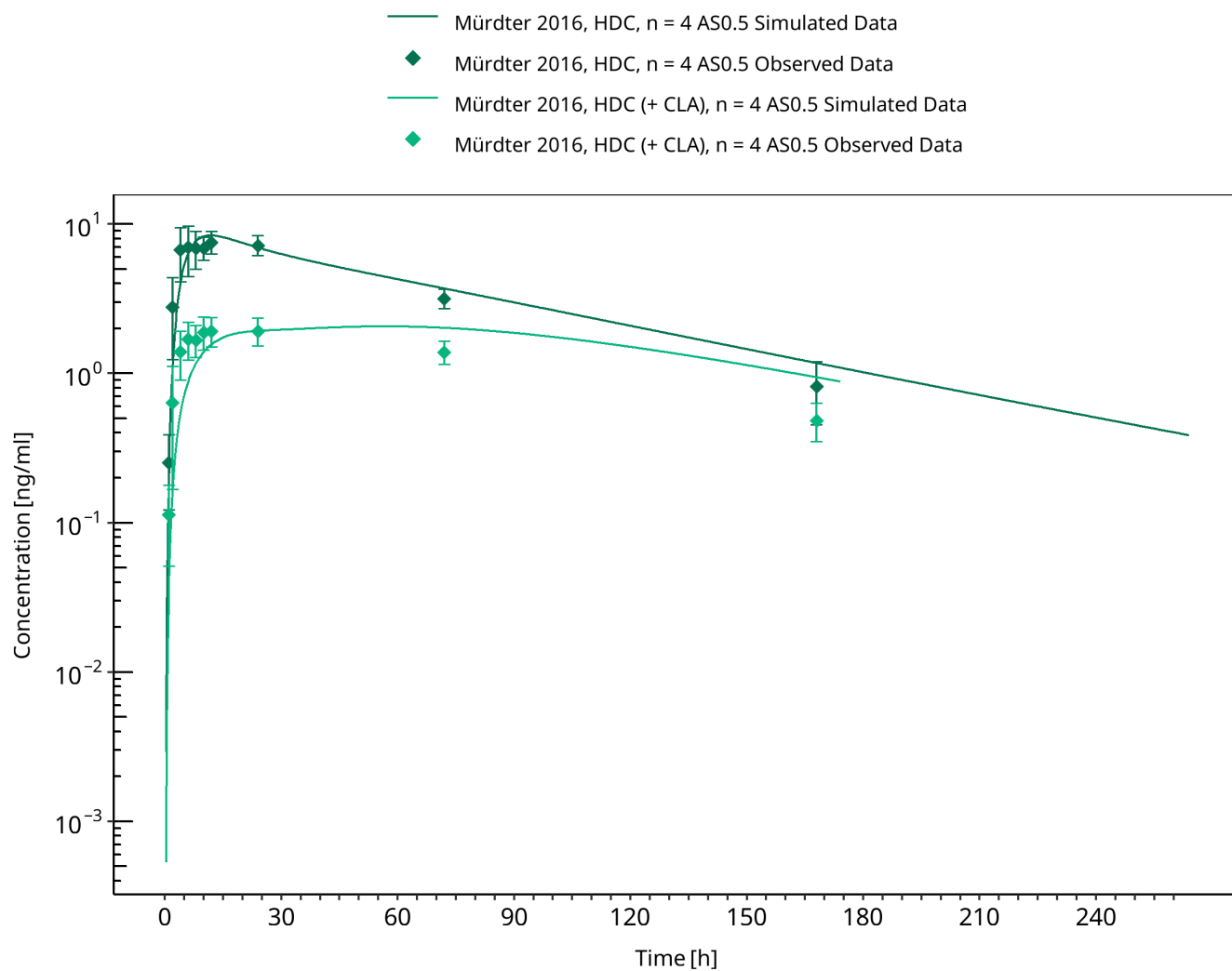


Figure 3-15: Clarithromycin-Clomiphene-DDGI

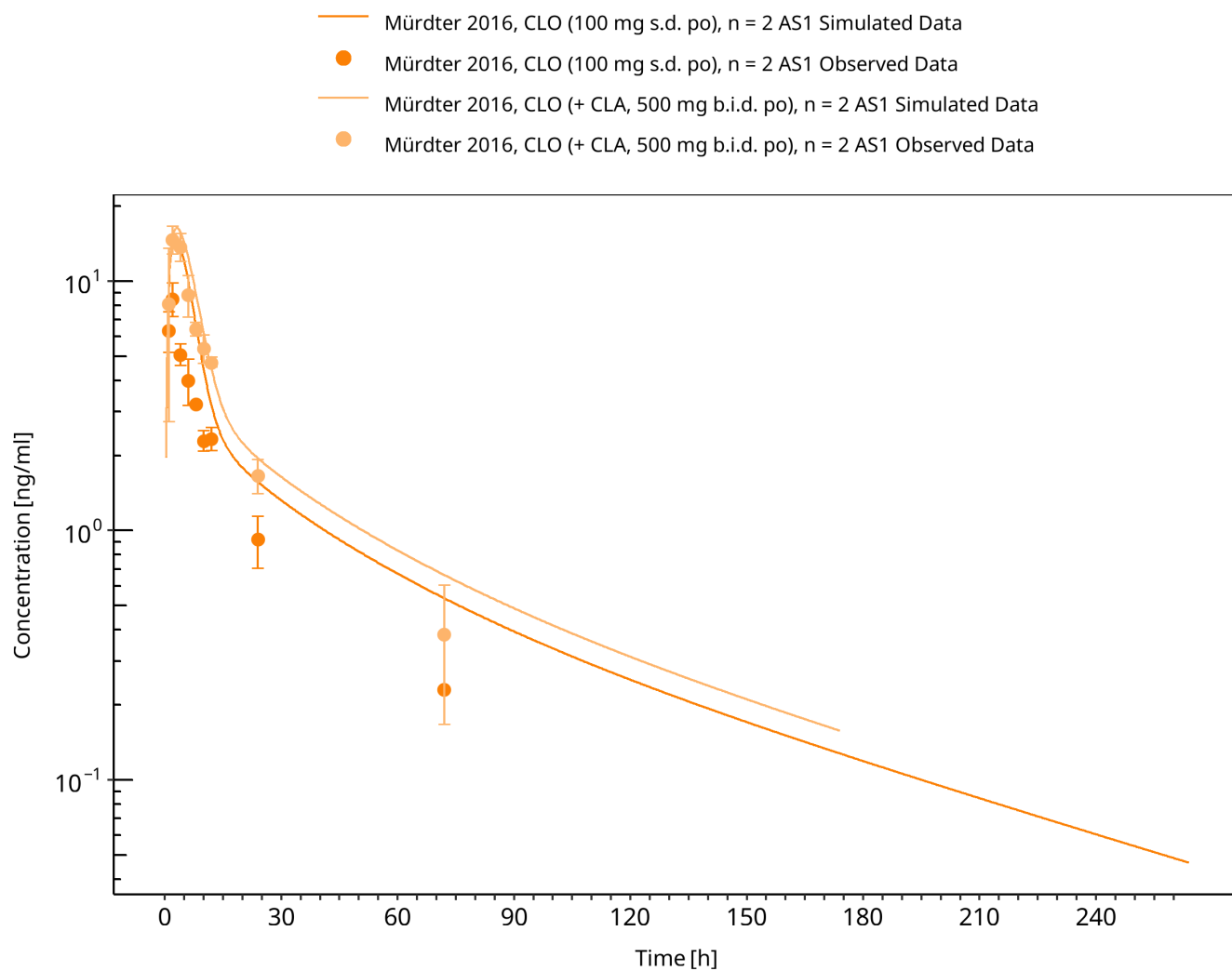


Figure 3-16: Clarithromycin-Clomiphene-DDGI

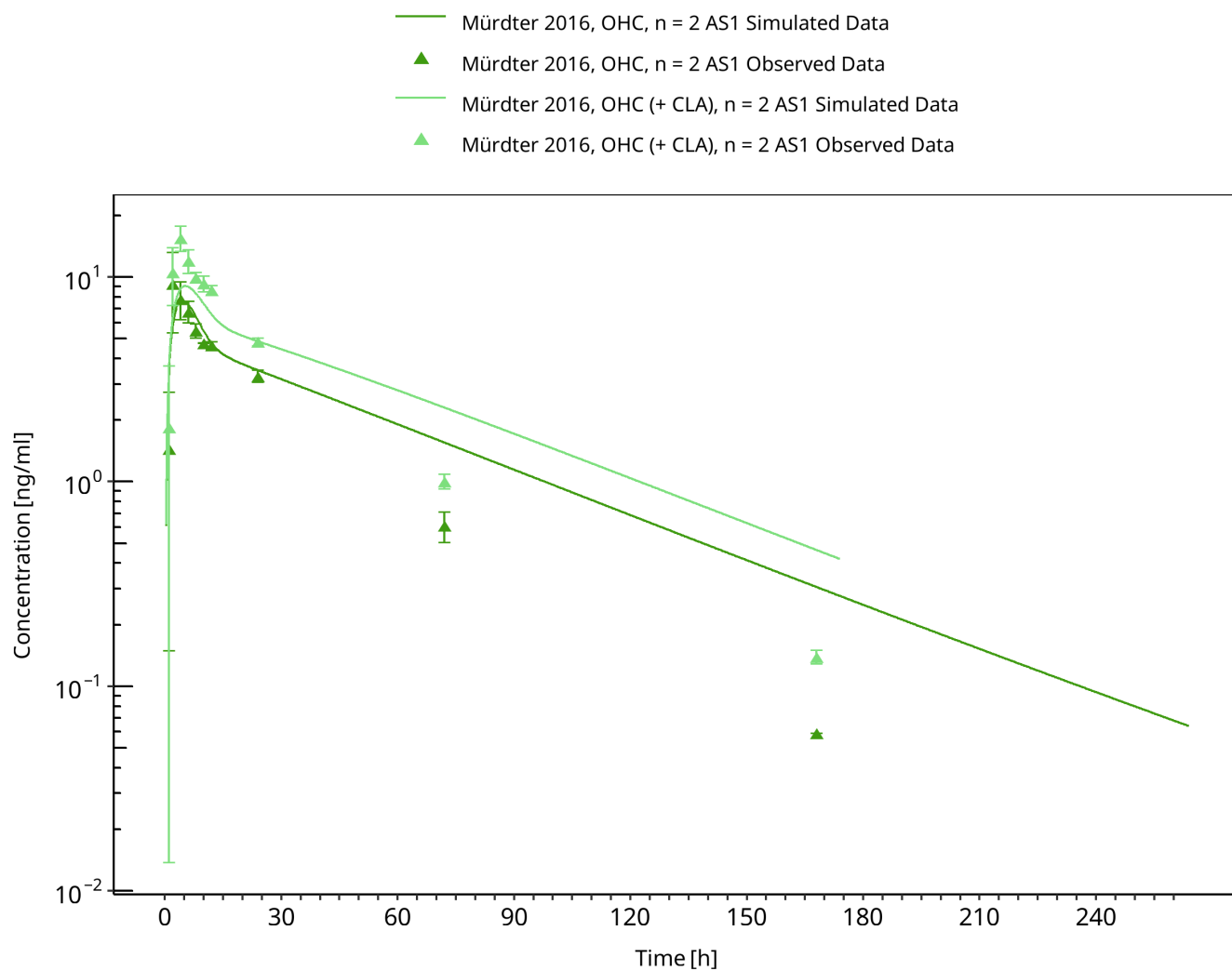


Figure 3-17: Clarithromycin-Clomiphene-DDGI

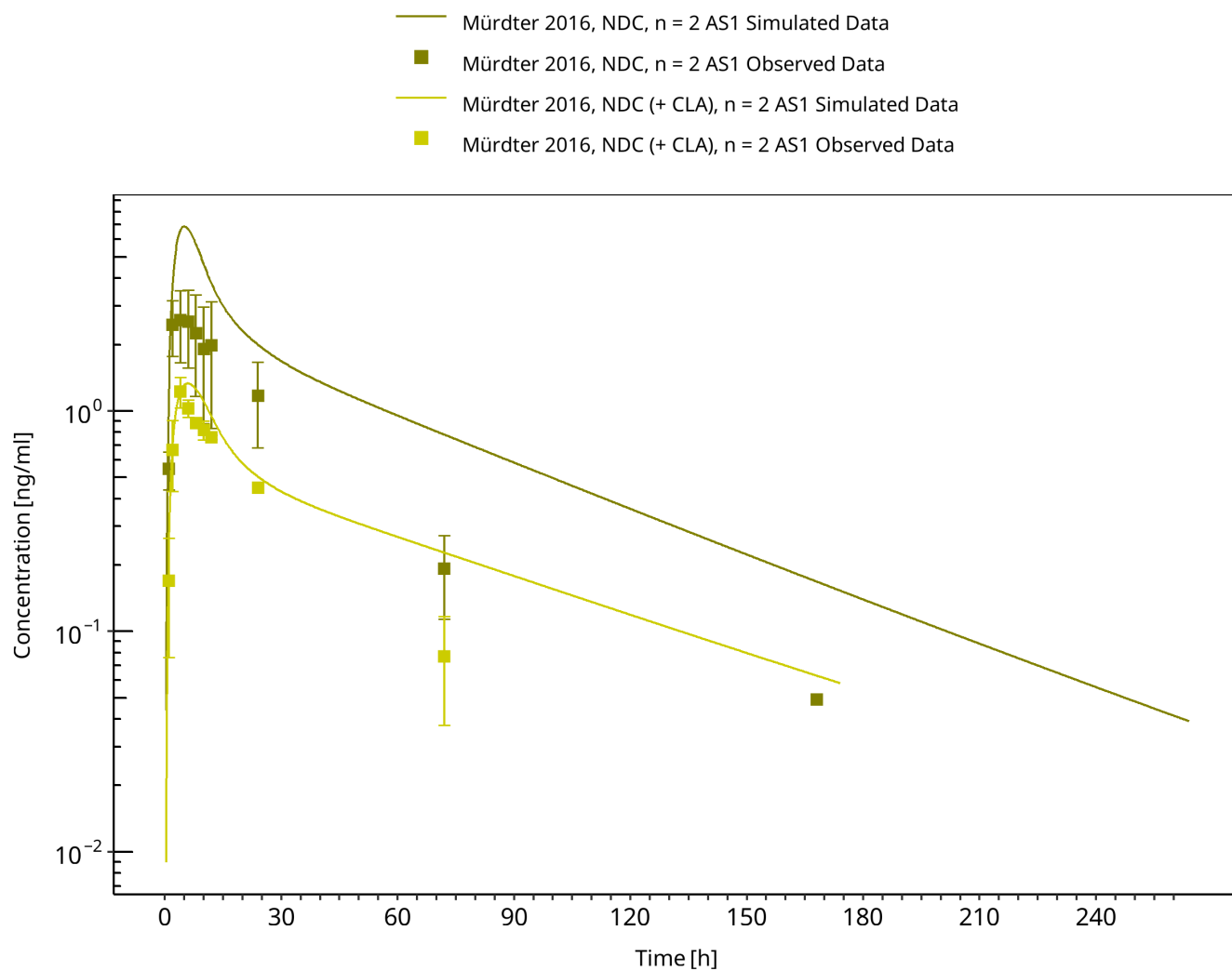


Figure 3-18: Clarithromycin-Clomiphene-DDGI

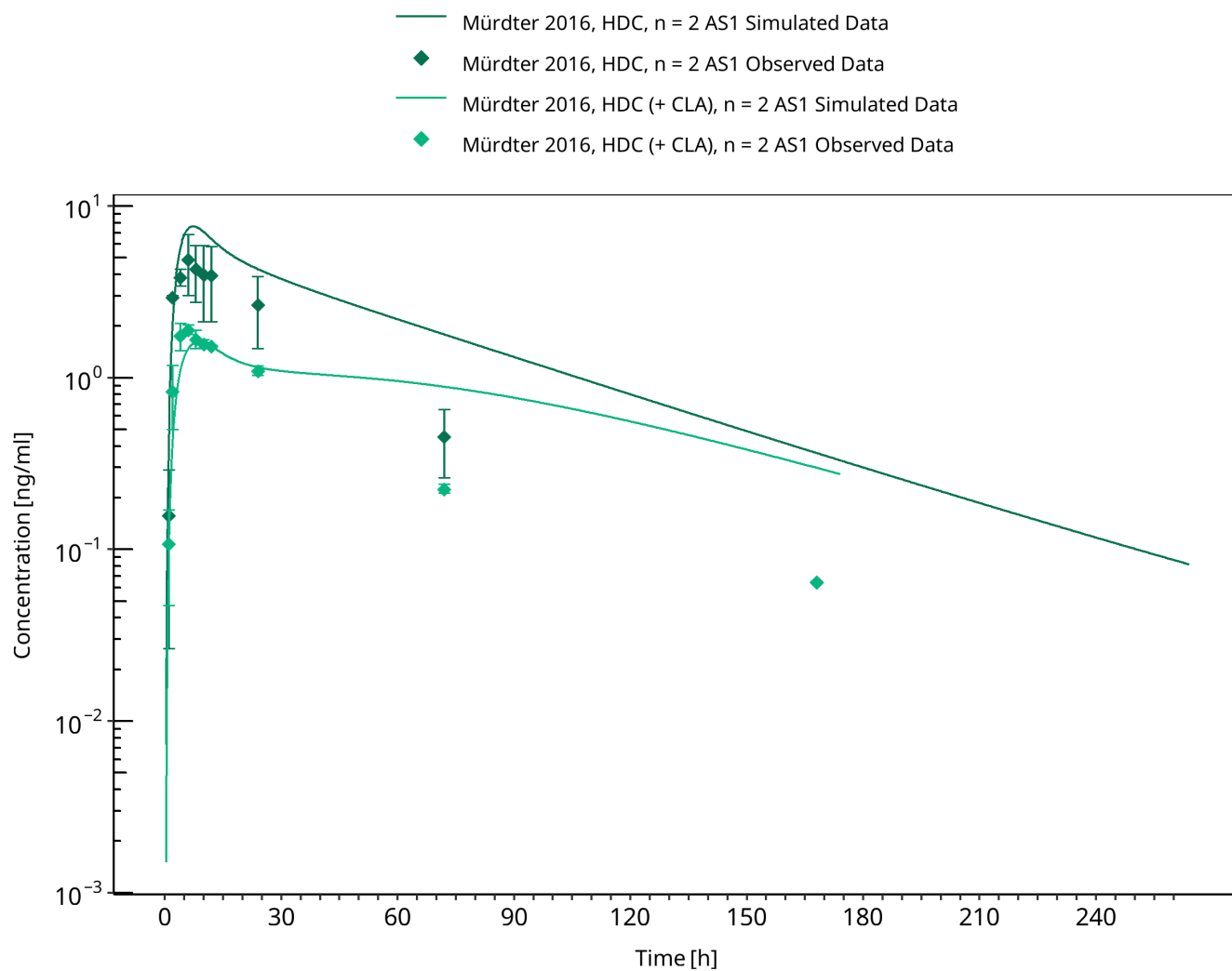


Figure 3-19: Clarithromycin-Clomiphene-DDGI

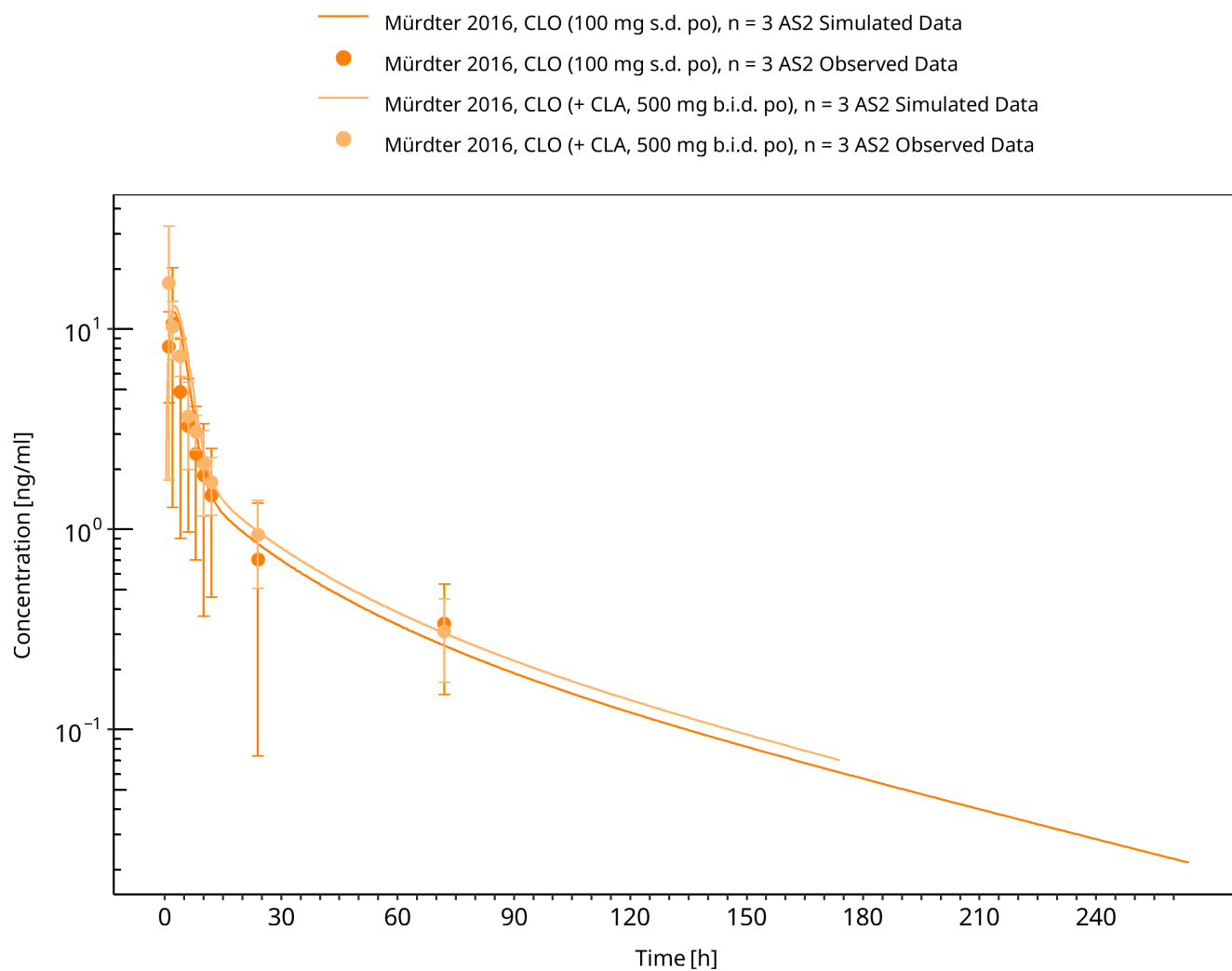


Figure 3-20: Clarithromycin-Clomiphene-DDGI

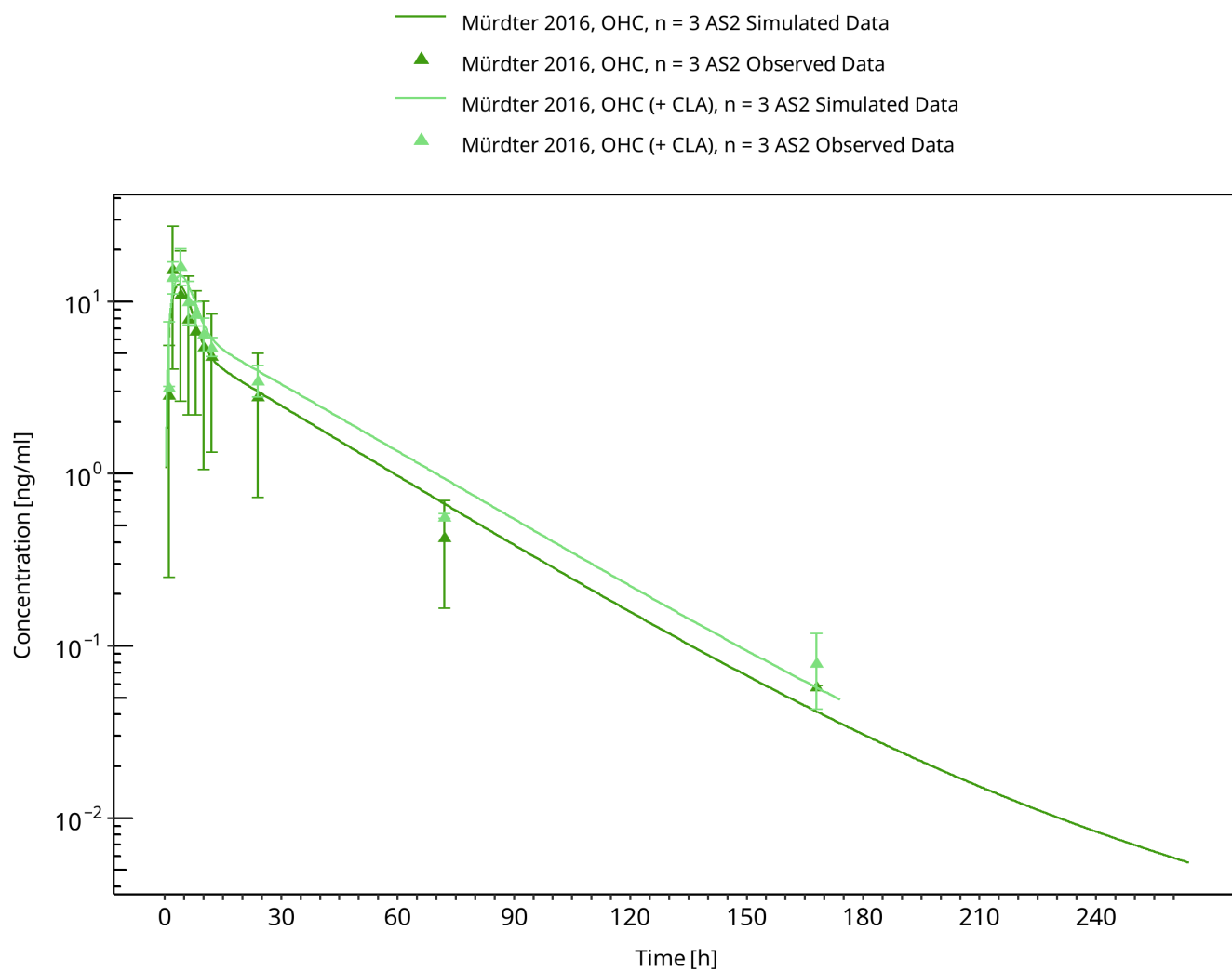


Figure 3-21: Clarithromycin-Clomiphene-DDGI

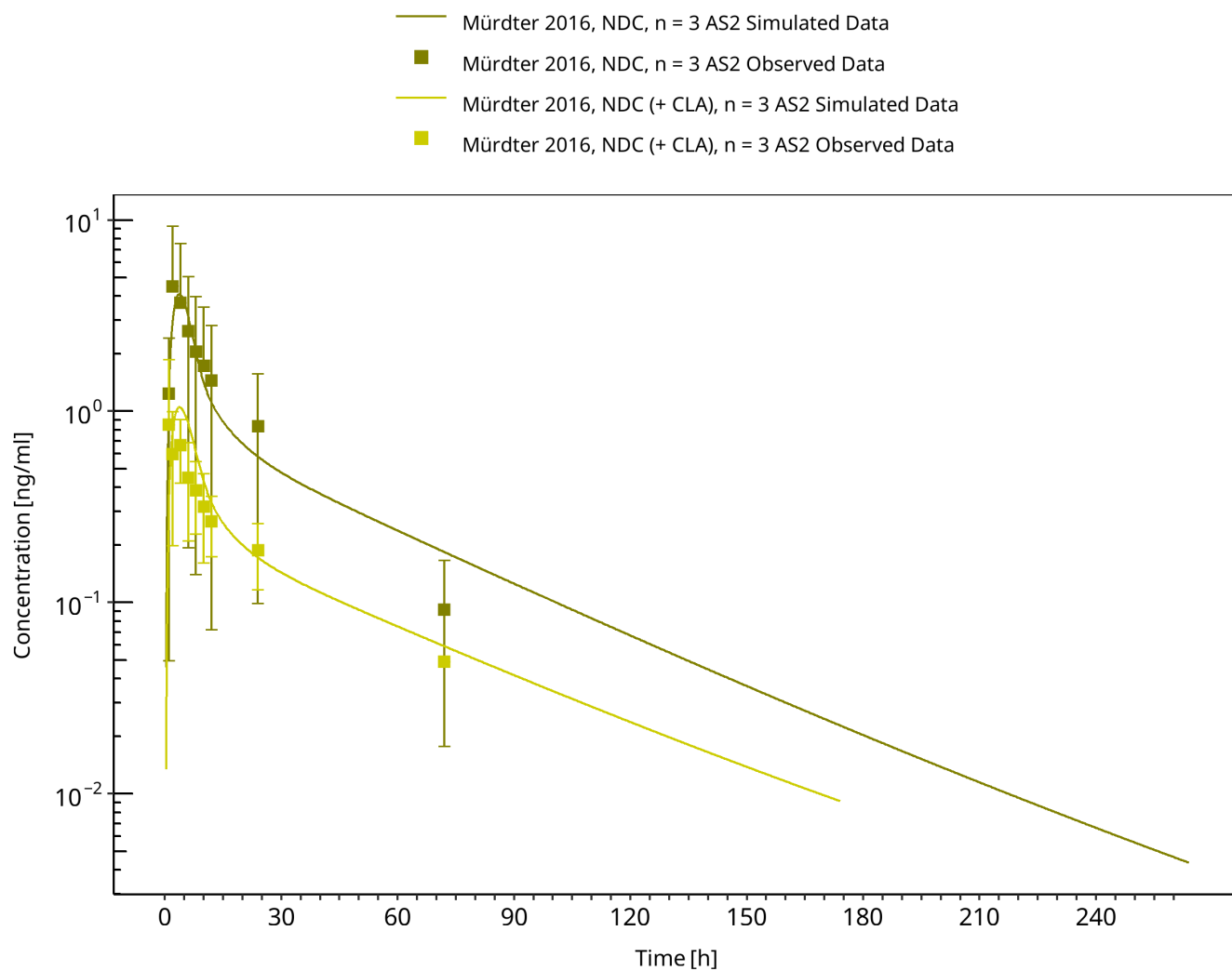


Figure 3-22: Clarithromycin-Clomiphene-DDGI

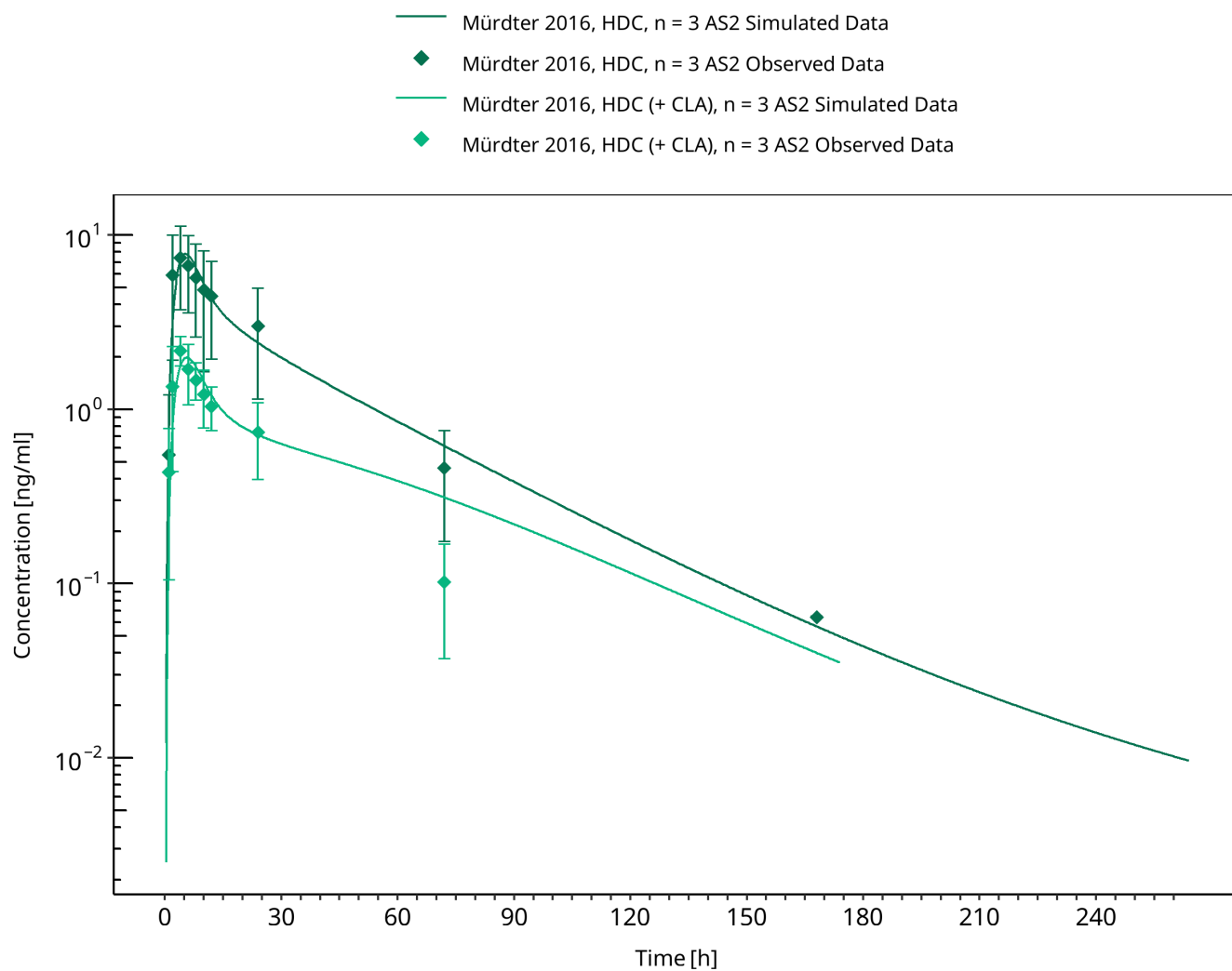


Figure 3-23: Clarithromycin-Clomiphene-DDGI

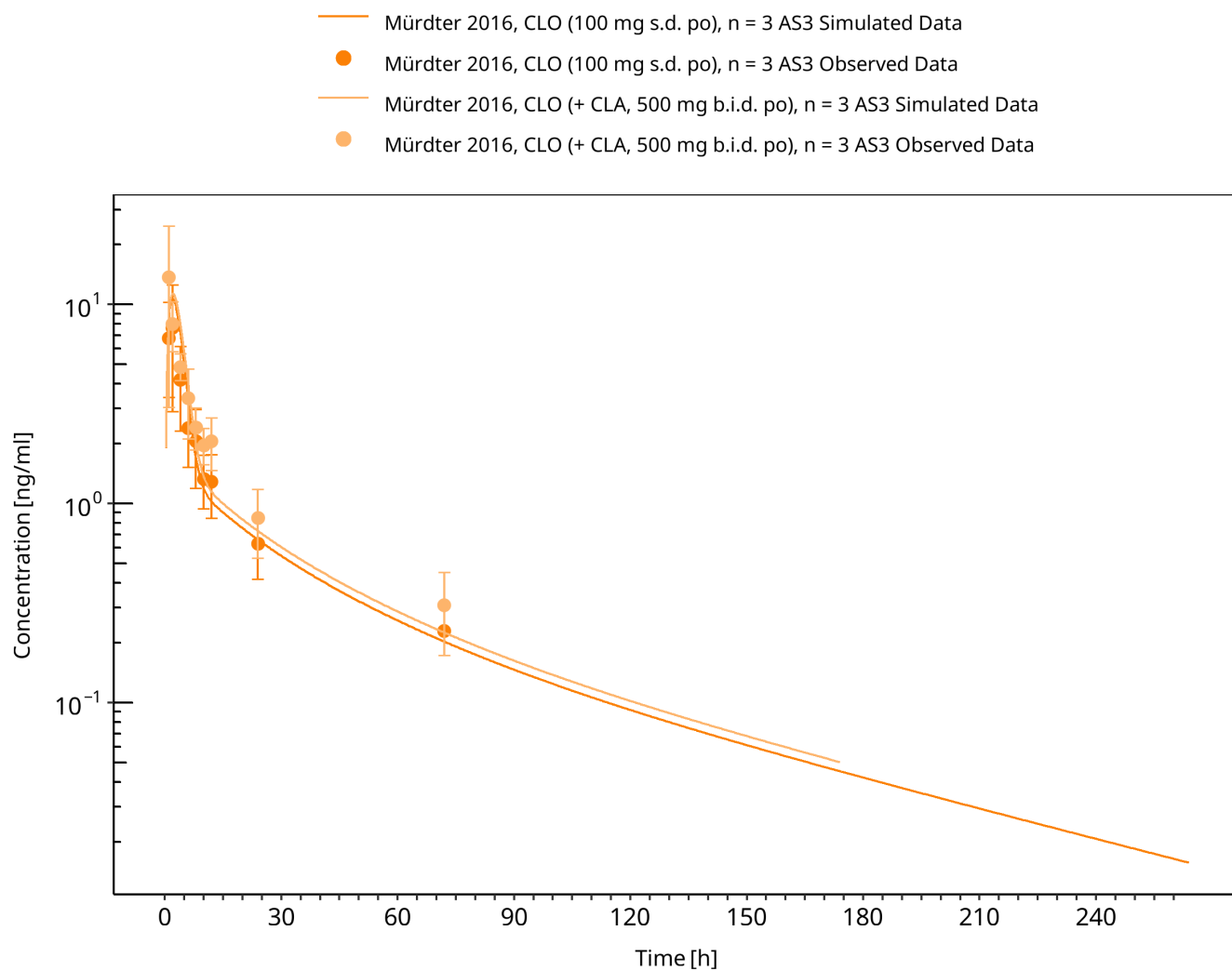


Figure 3-24: Clarithromycin-Clomiphene-DDGI

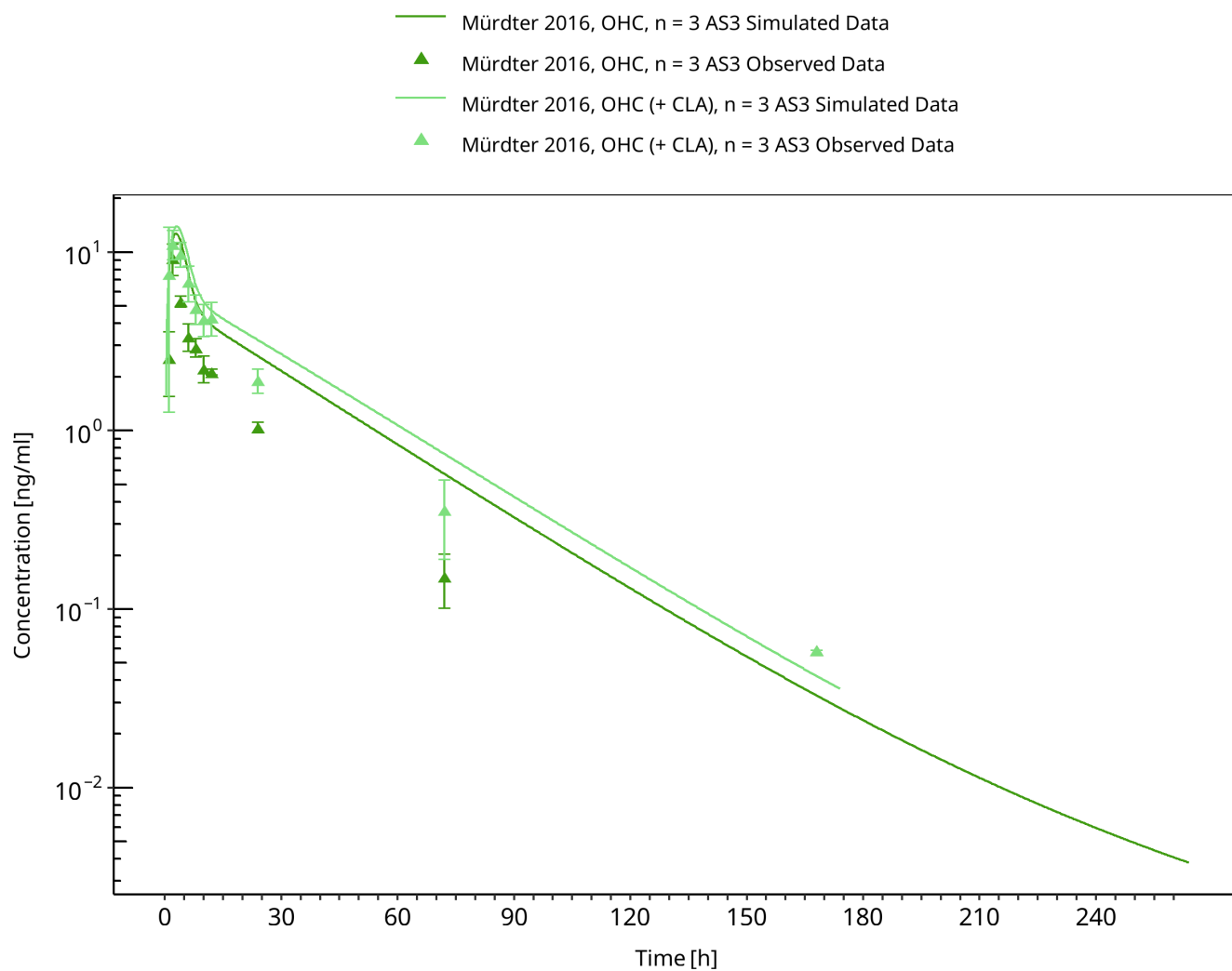


Figure 3-25: Clarithromycin-Clomiphene-DDGI

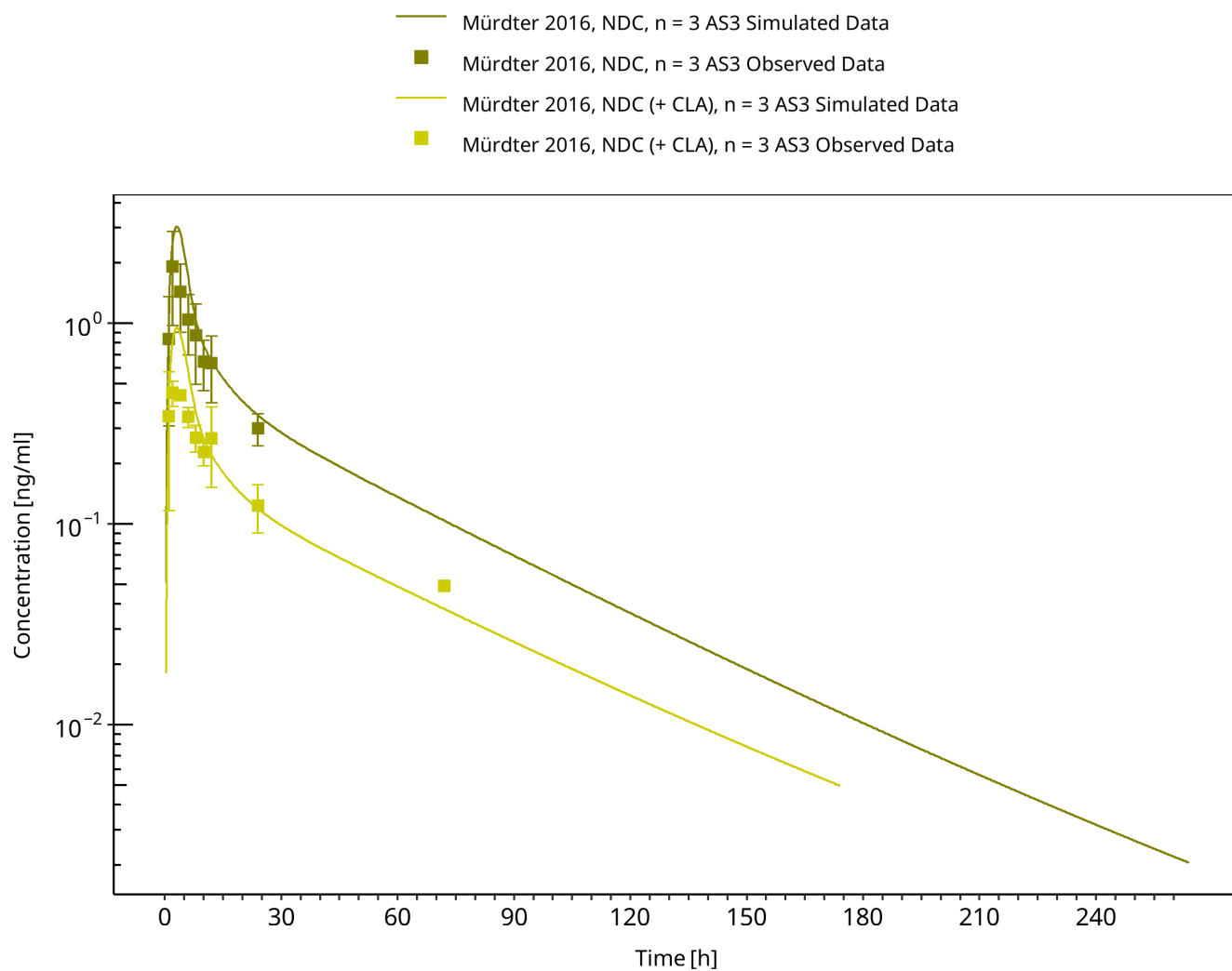


Figure 3-26: Clarithromycin-Clomiphene-DDGI

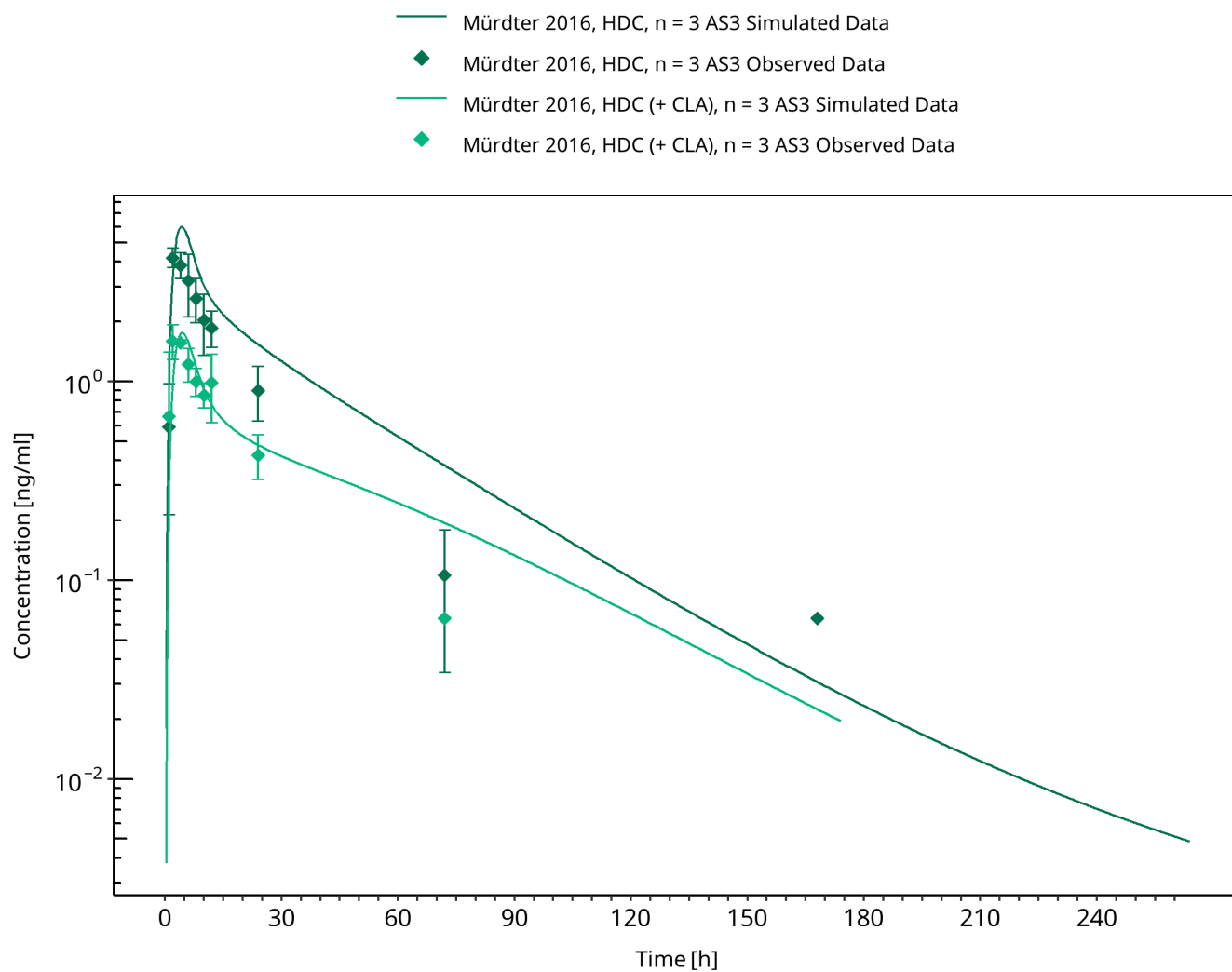


Figure 3-27: Clarithromycin-Clomiphene-DDGI

3.5 Paroxetine-Clomiphene-DDGI

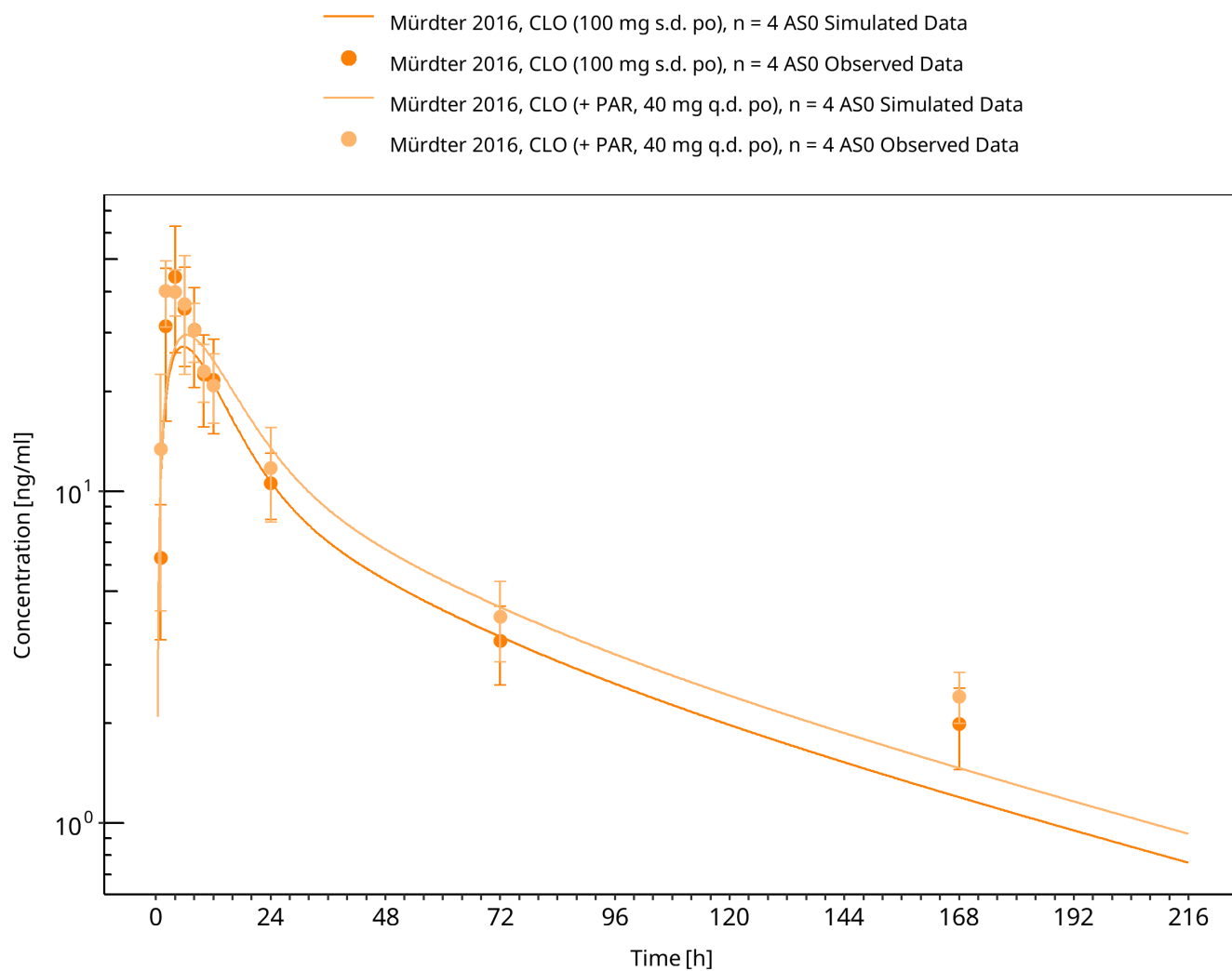


Figure 3-28: Paroxetine-Clomiphene-DDGI

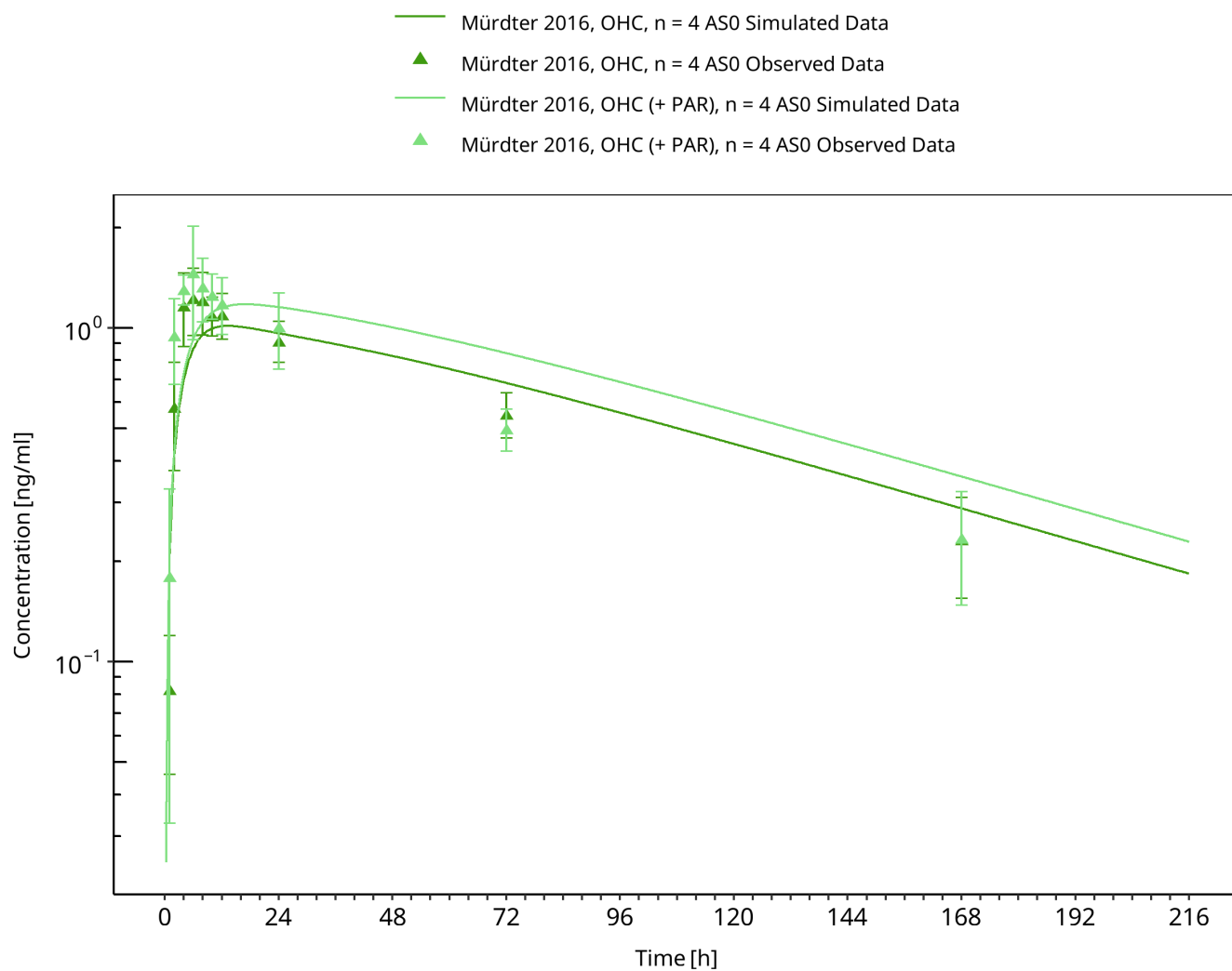


Figure 3-29: Paroxetine-Clomiphene-DDGI

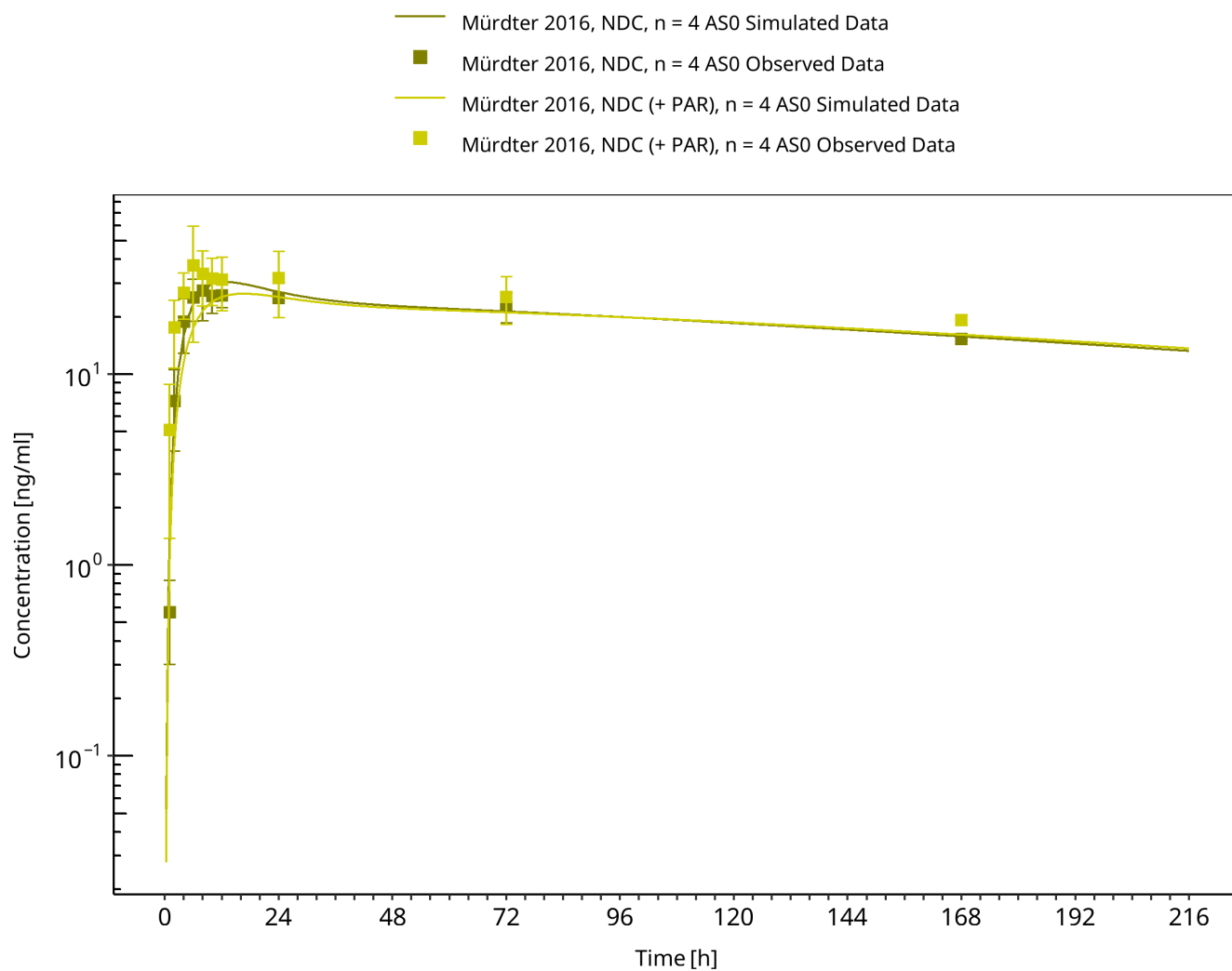


Figure 3-30: Paroxetine-Clomiphene-DDGI

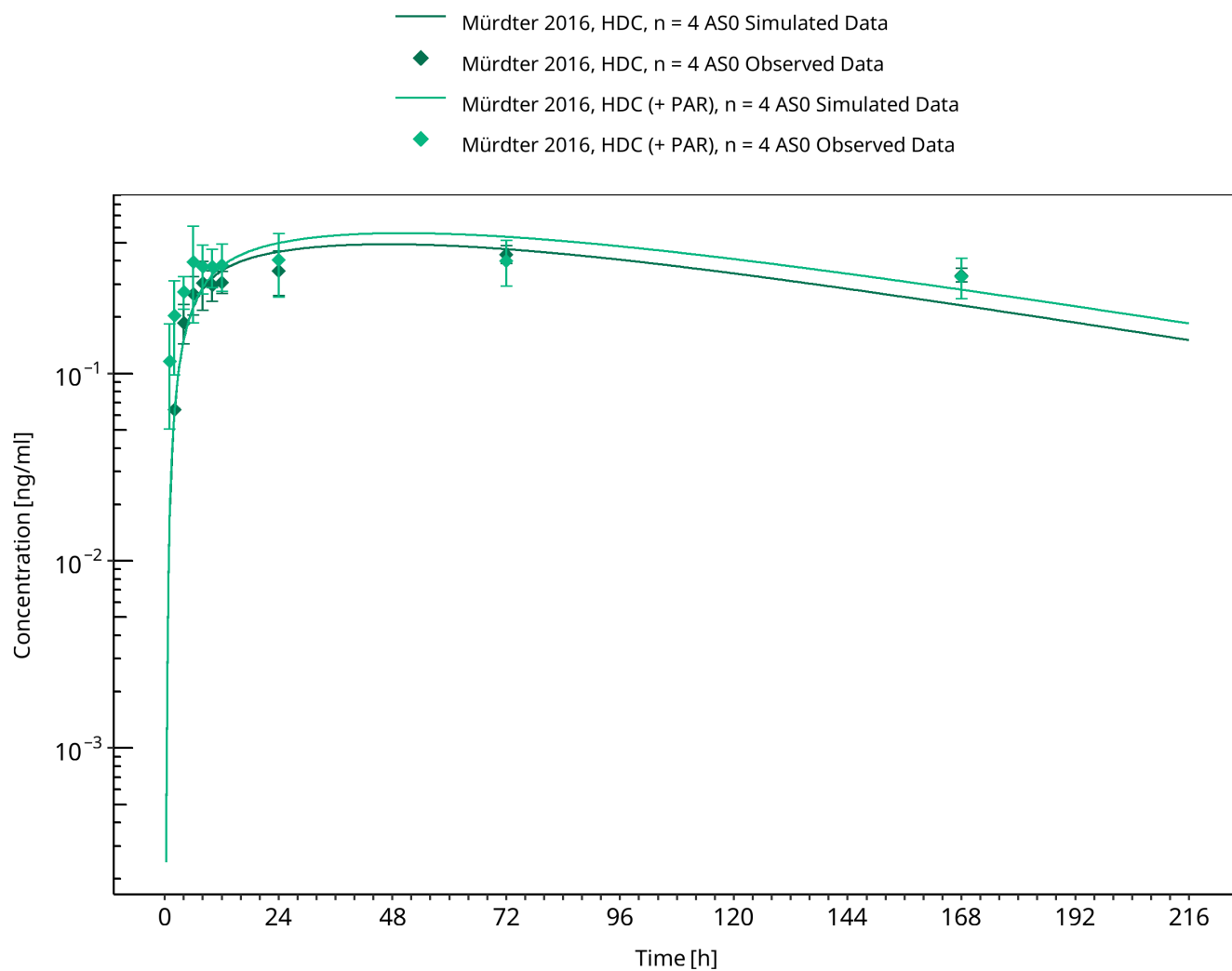


Figure 3-31: Paroxetine-Clomiphene-DDGI

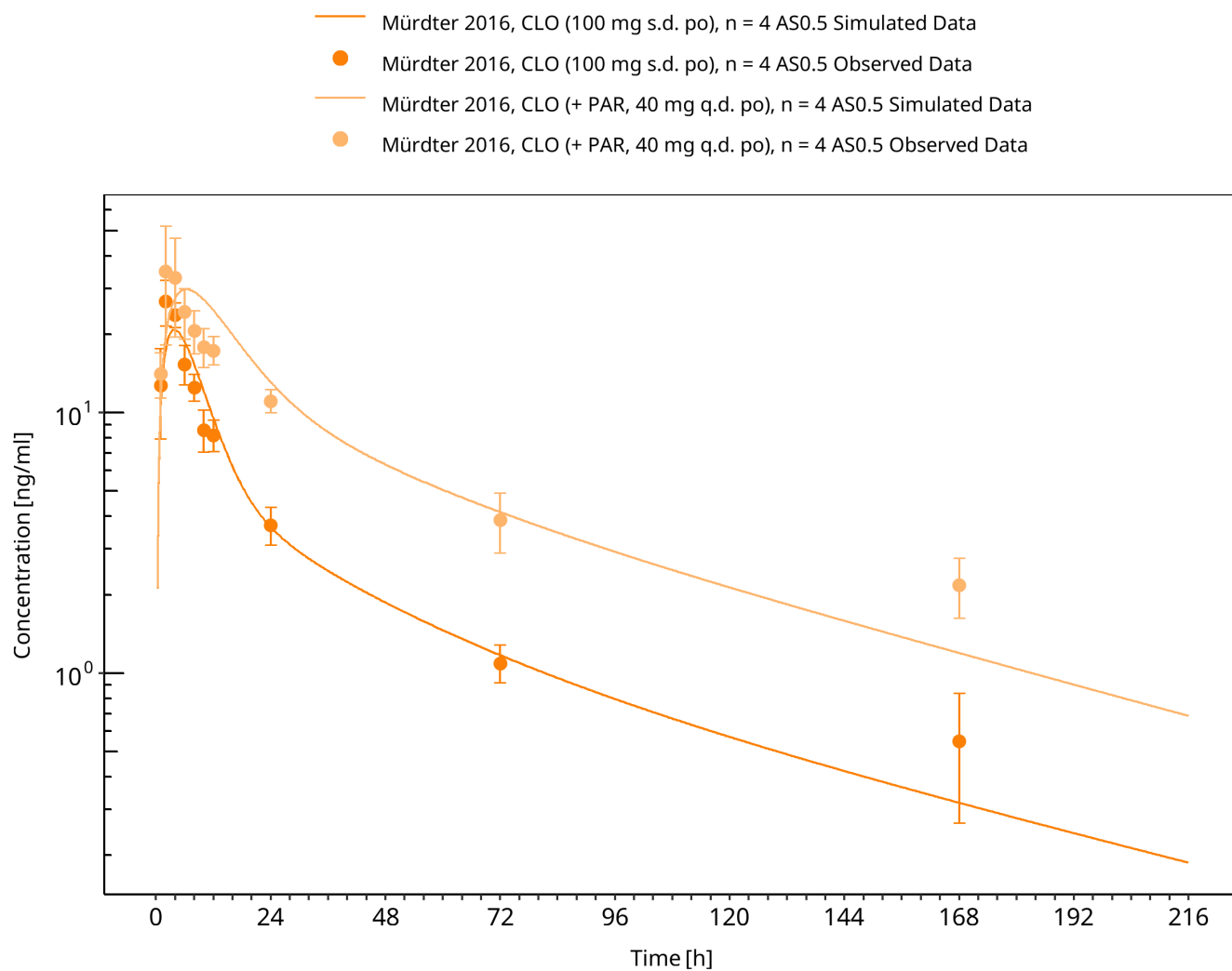


Figure 3-32: Paroxetine-Clomiphene-DDGI

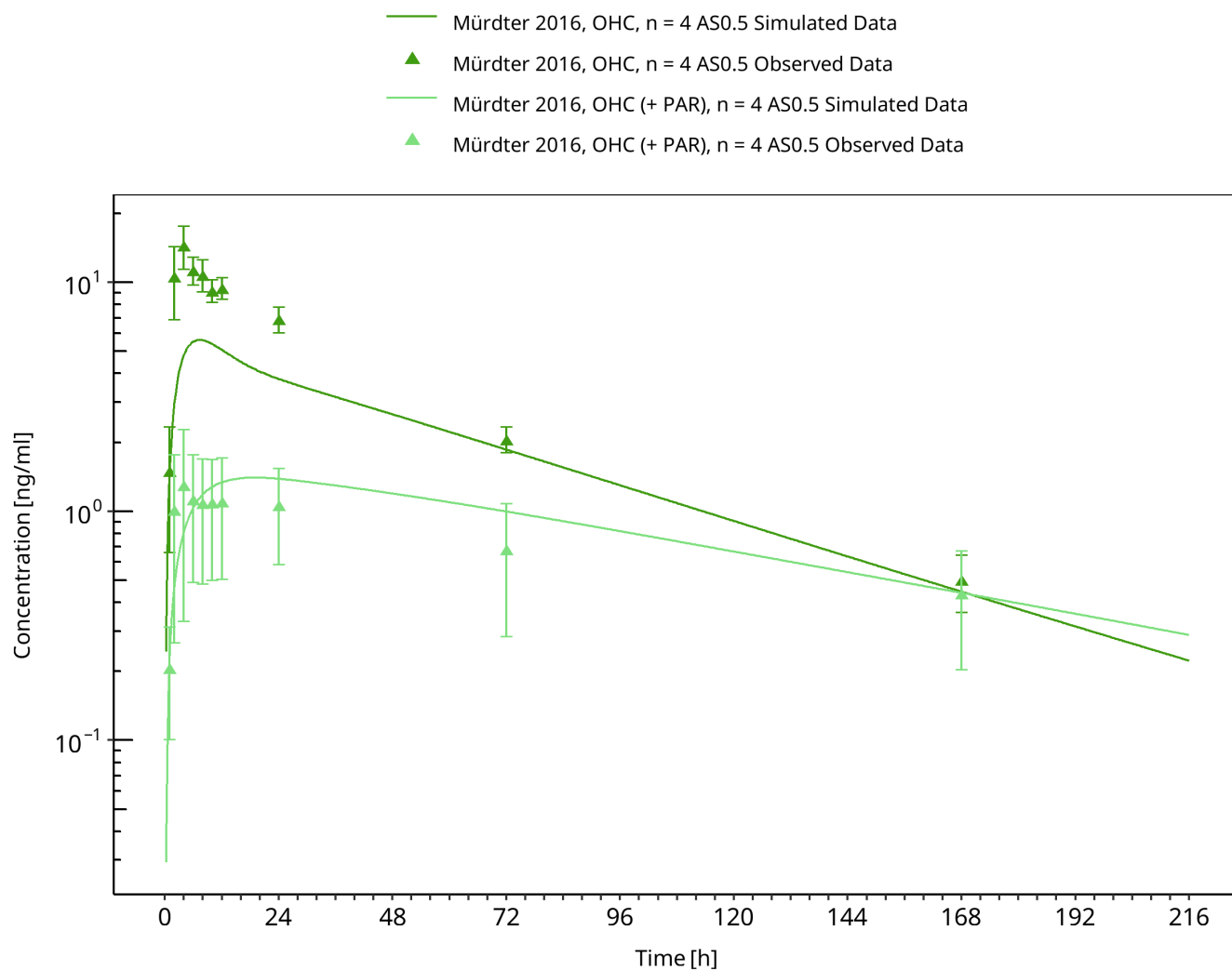


Figure 3-33: Paroxetine-Clomiphene-DDGI

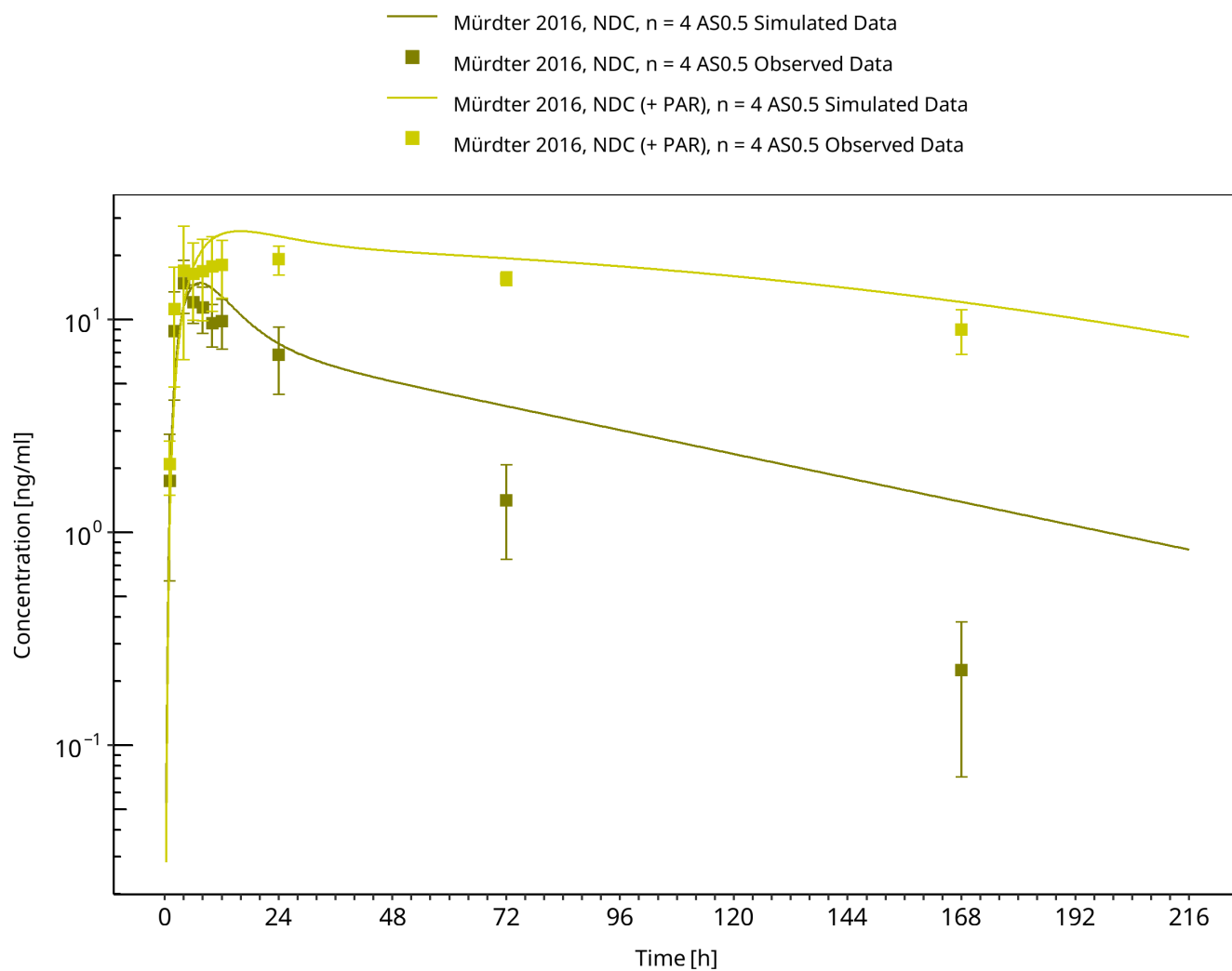


Figure 3-34: Paroxetine-Clomiphene-DDGI

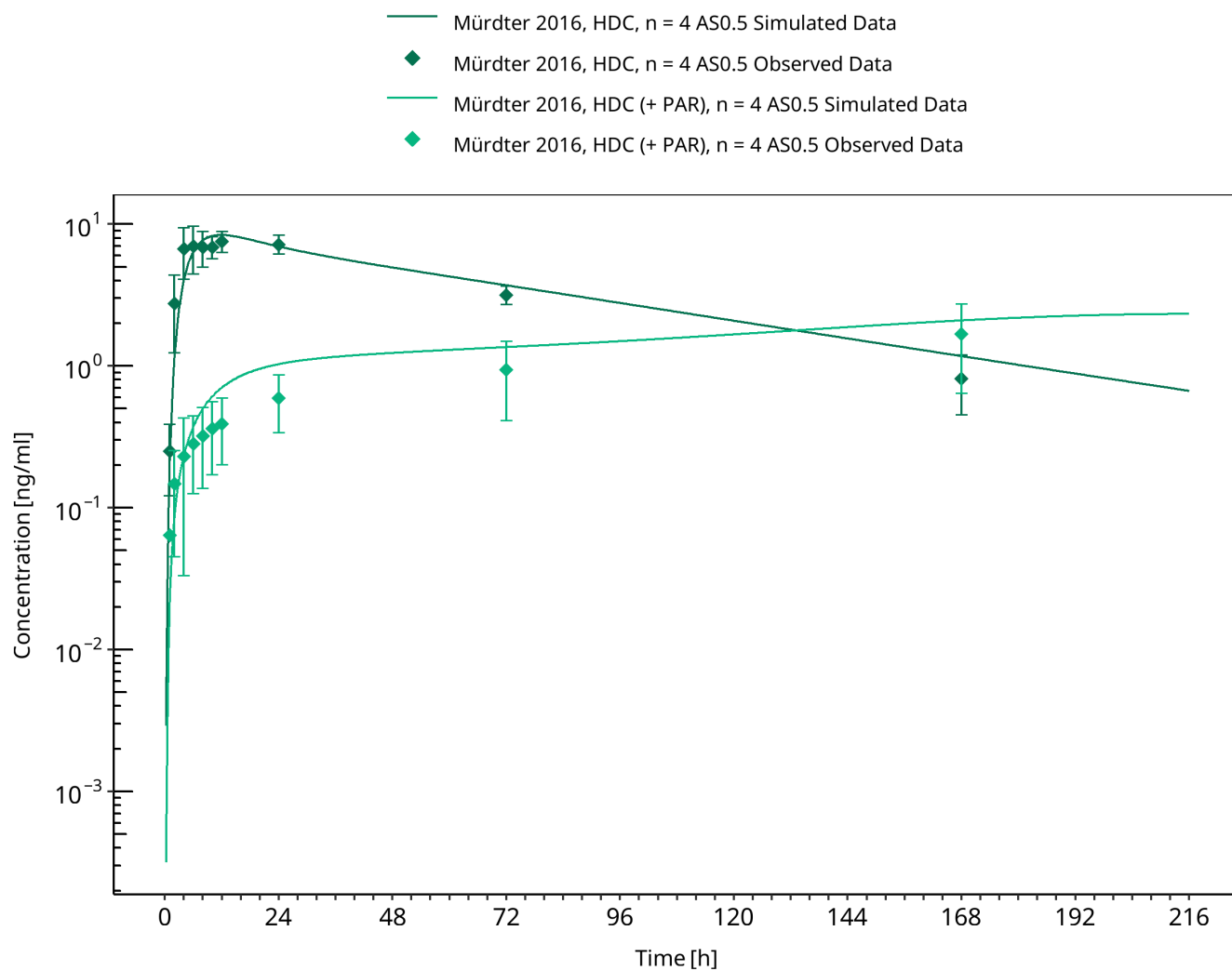


Figure 3-35: Paroxetine-Clomiphen-DDGI

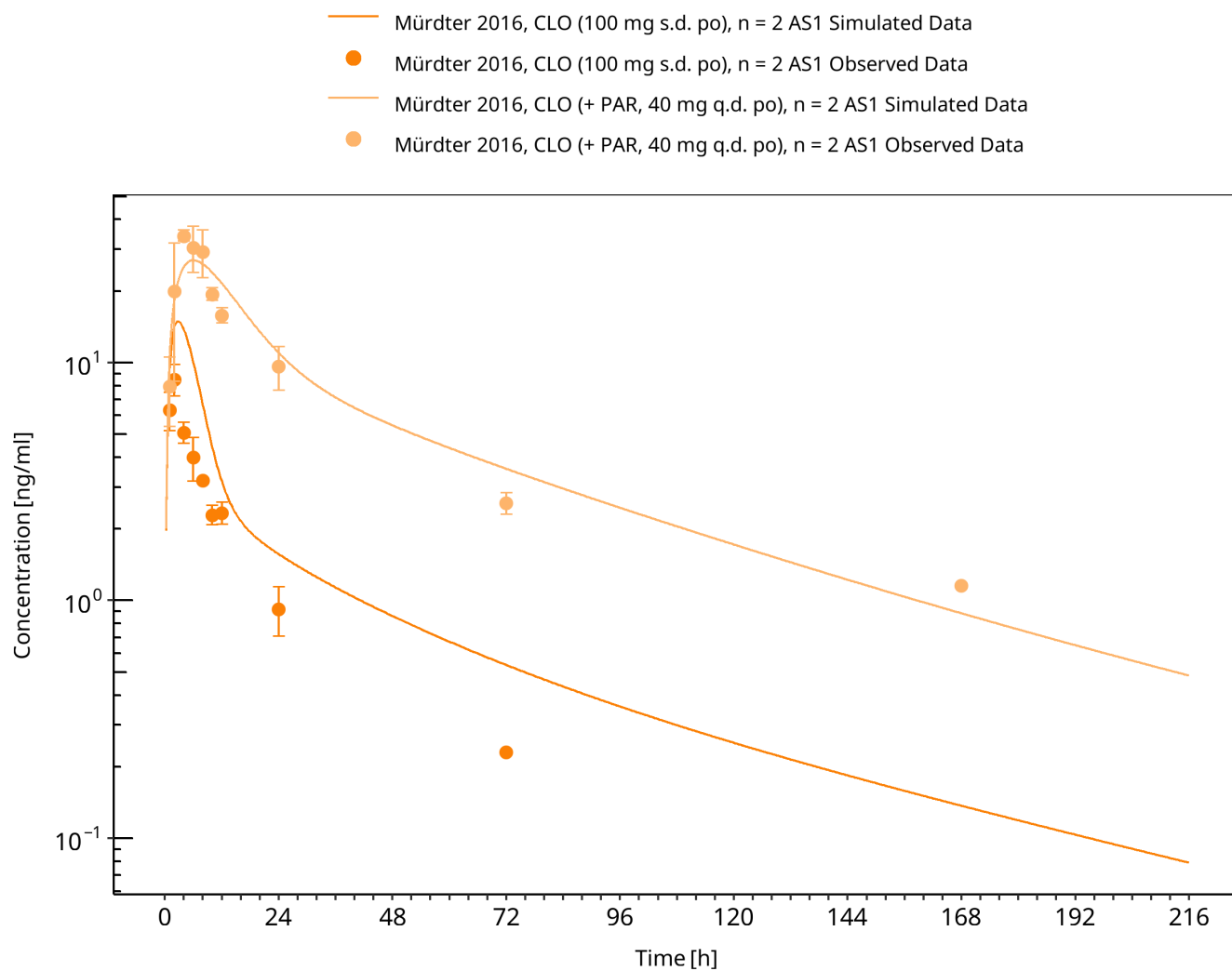


Figure 3-36: Paroxetine-Clomiphene-DDGI

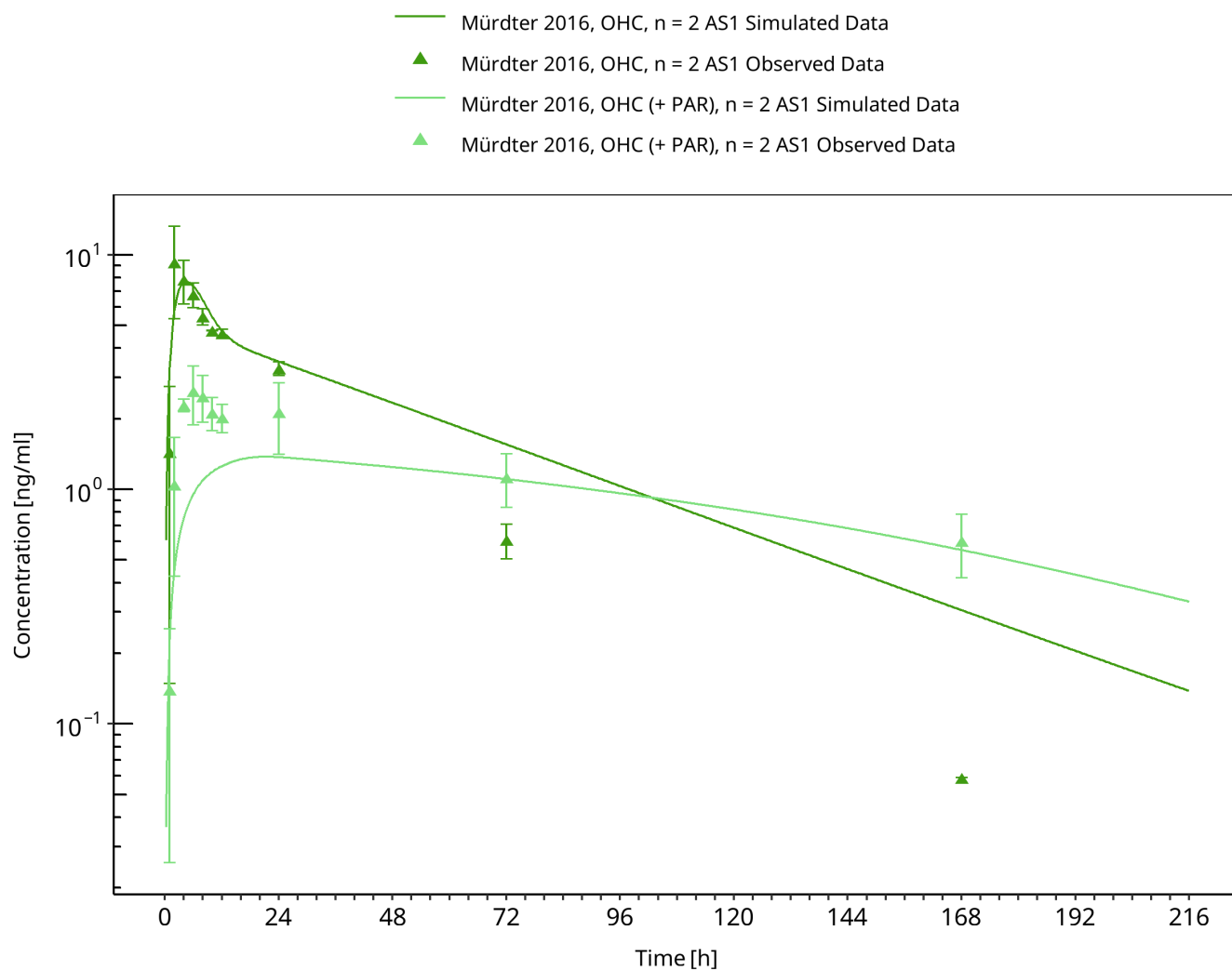


Figure 3-37: Paroxetine-Clomiphene-DDGI

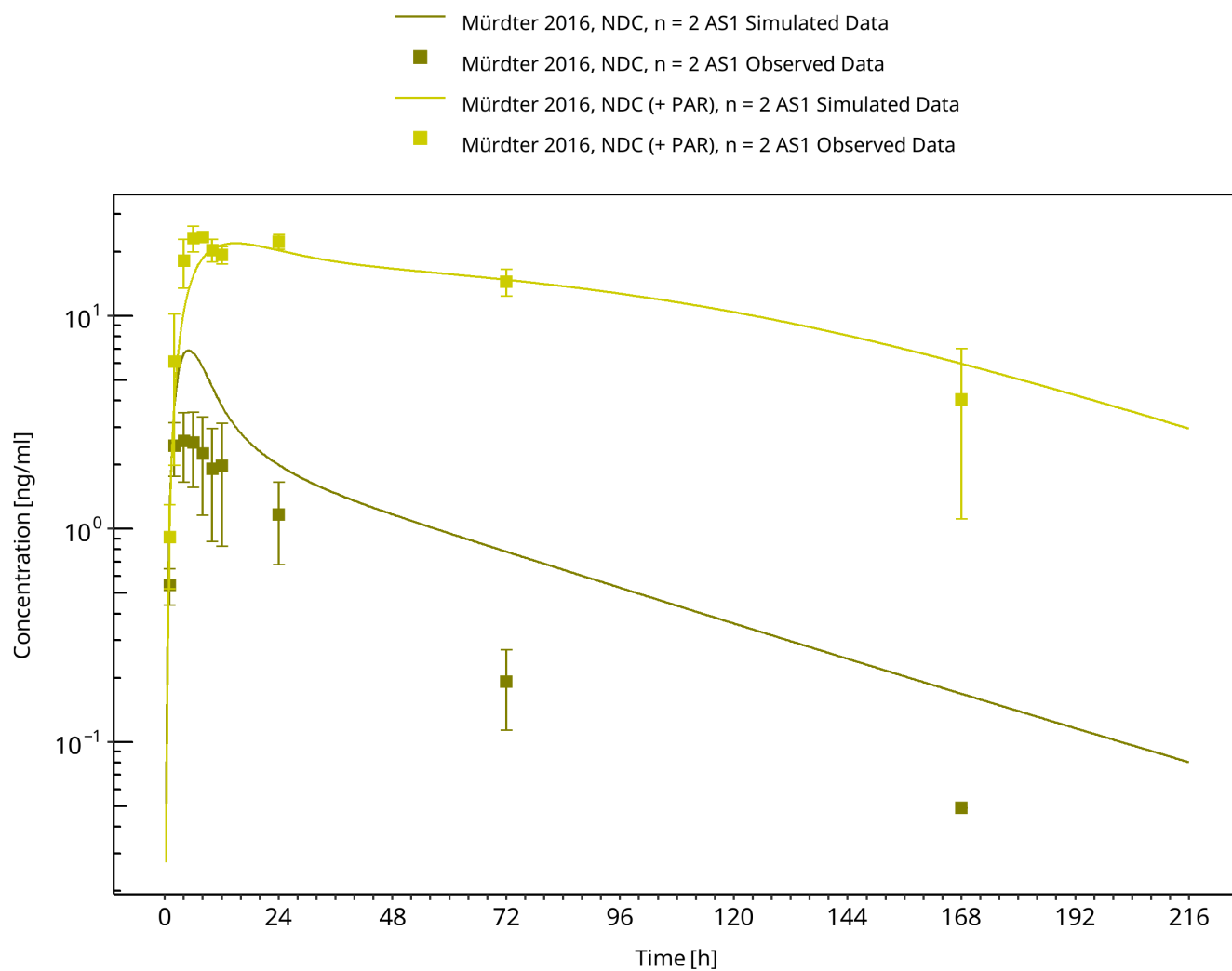


Figure 3-38: Paroxetine-Clomiphene-DDGI

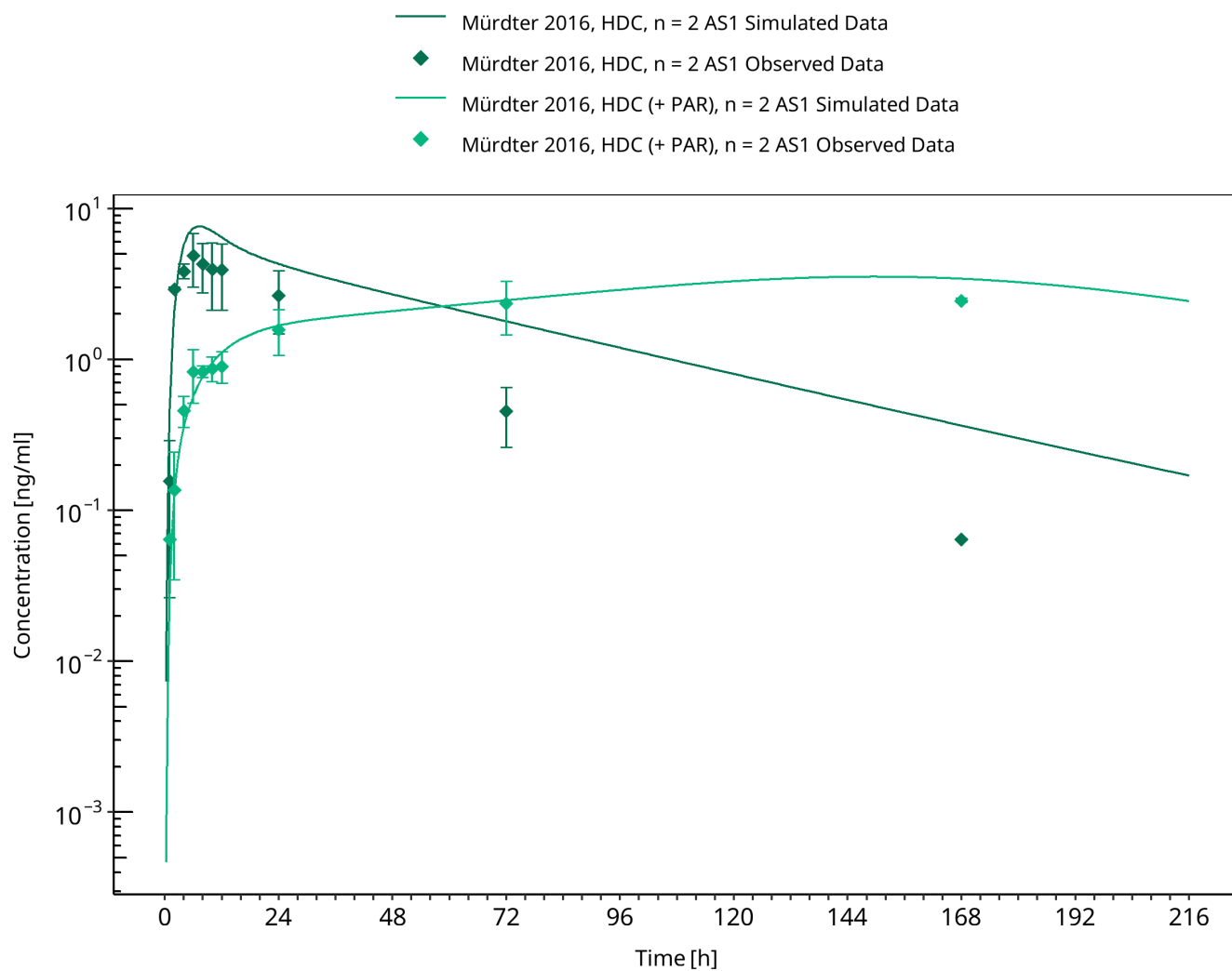


Figure 3-39: Paroxetine-Clomiphene-DDGI

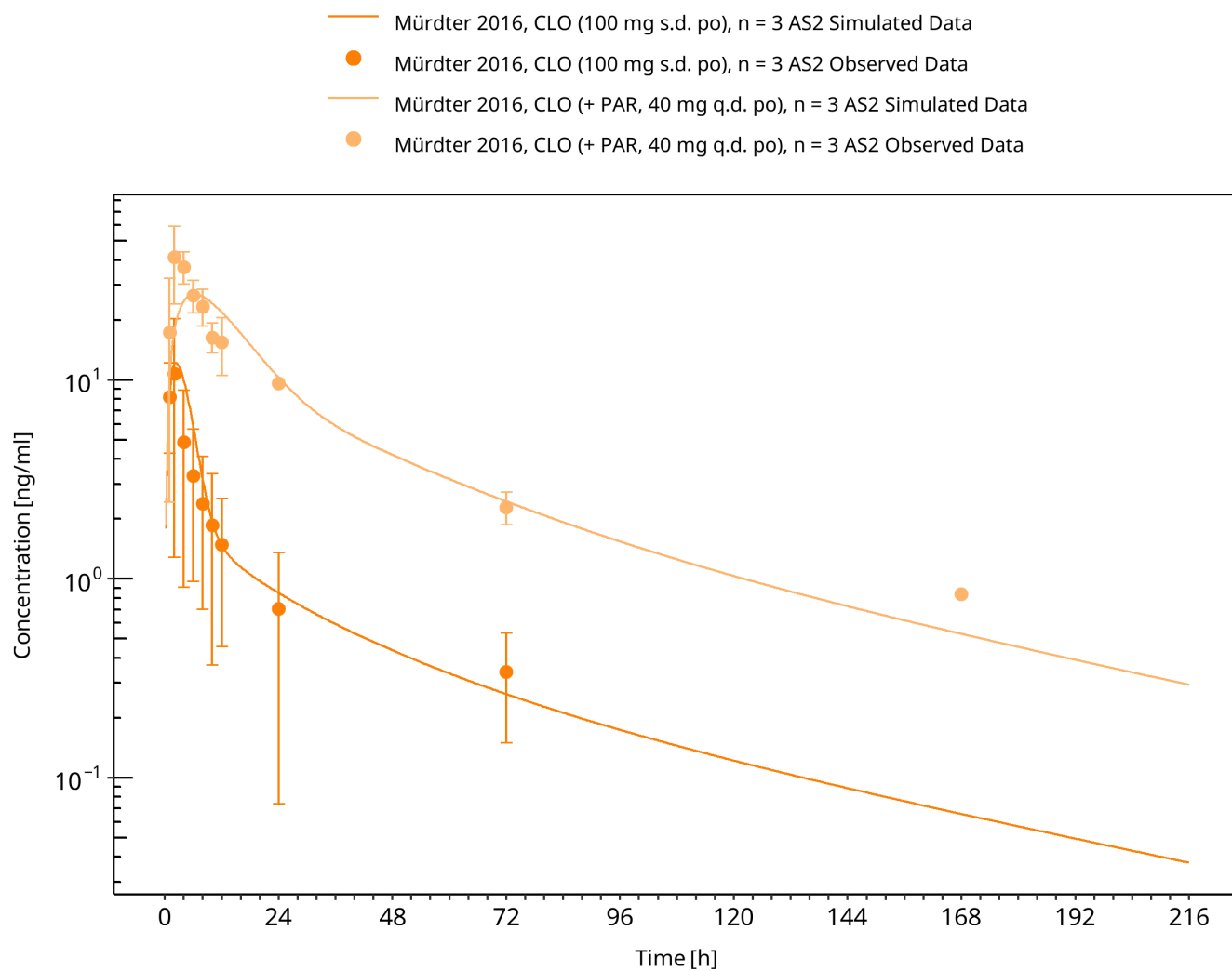


Figure 3-40: Paroxetine-Clomiphene-DDGI

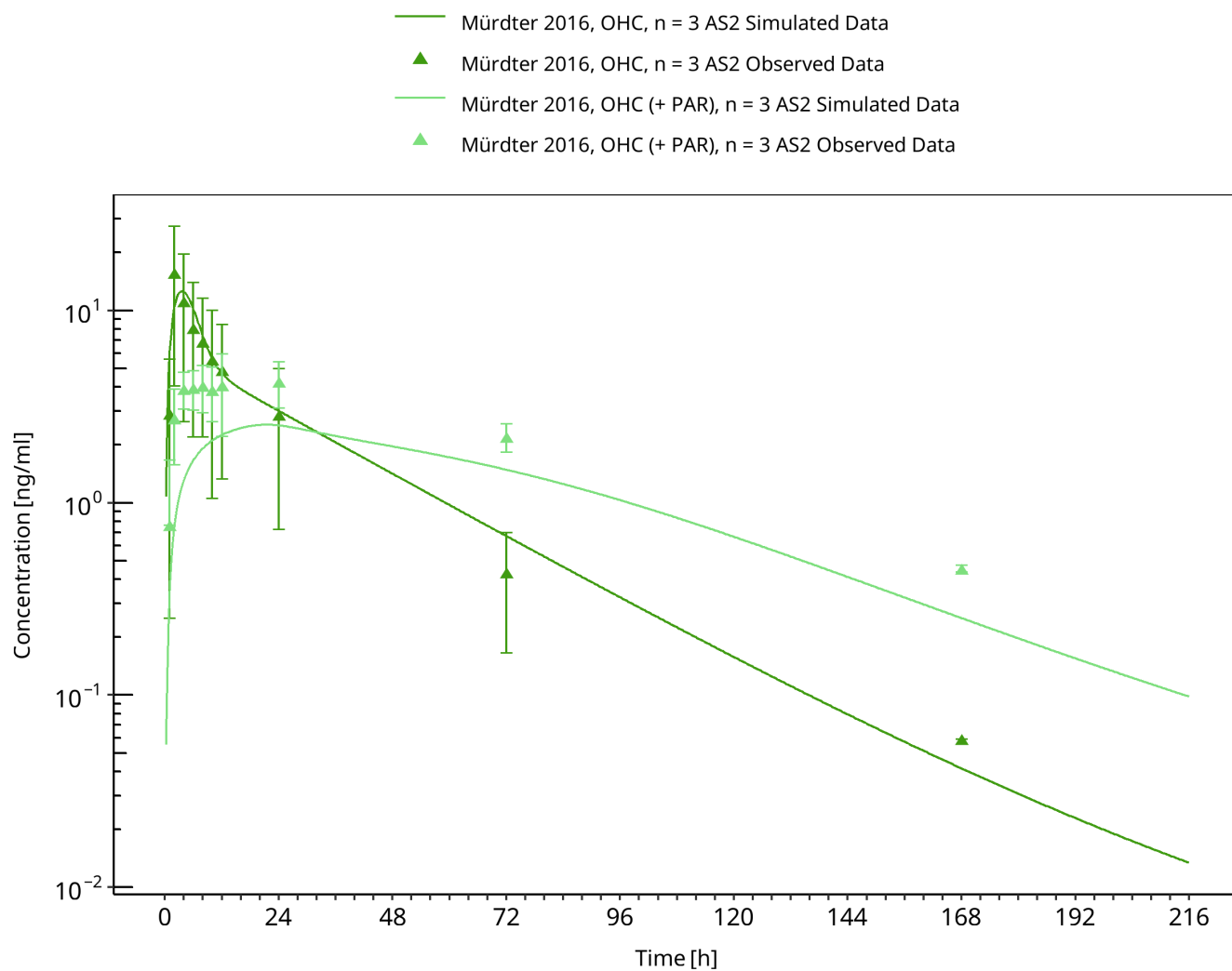


Figure 3-41: Paroxetine-Clomiphene-DDGI

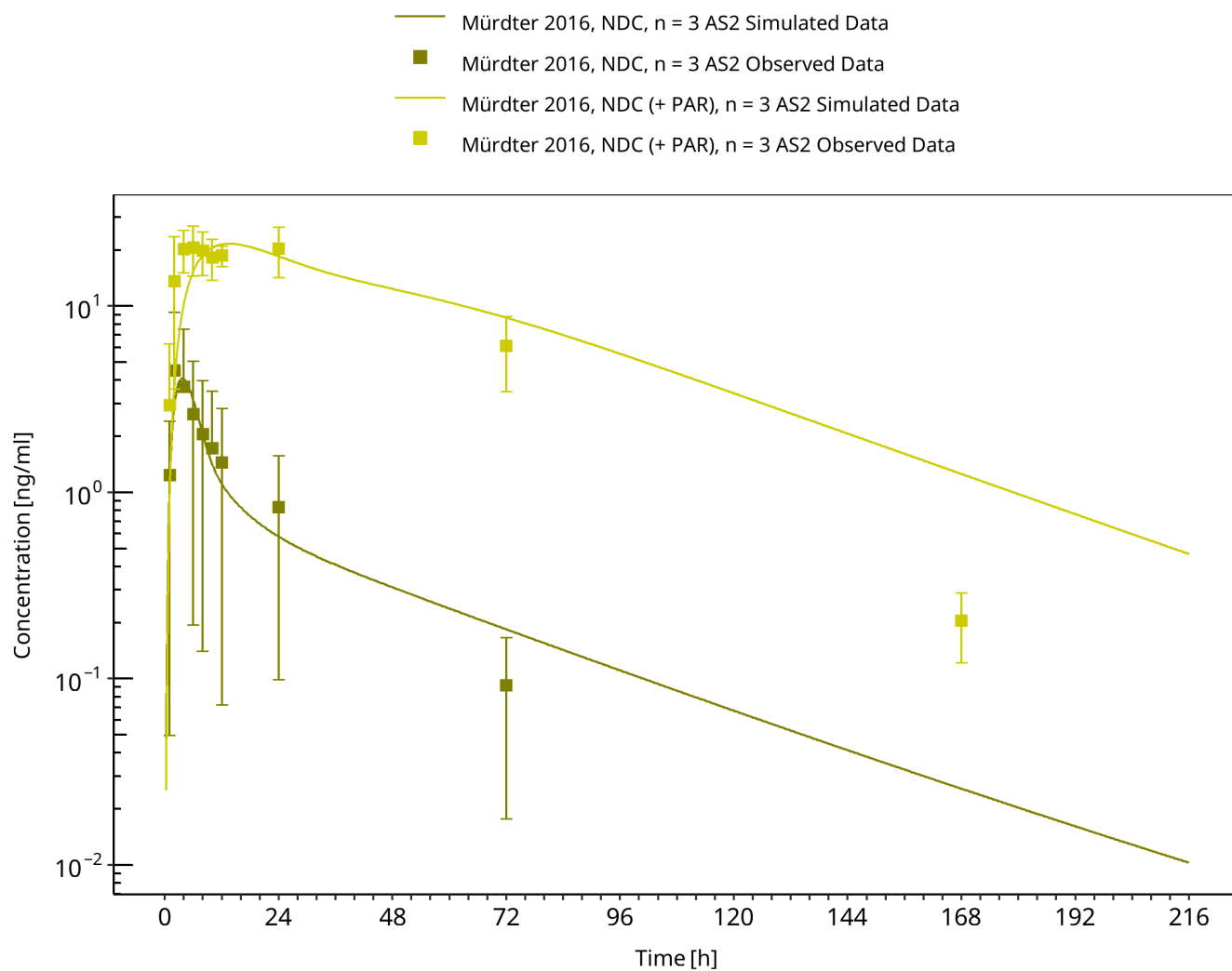


Figure 3-42: Paroxetine-Clomiphene-DDGI

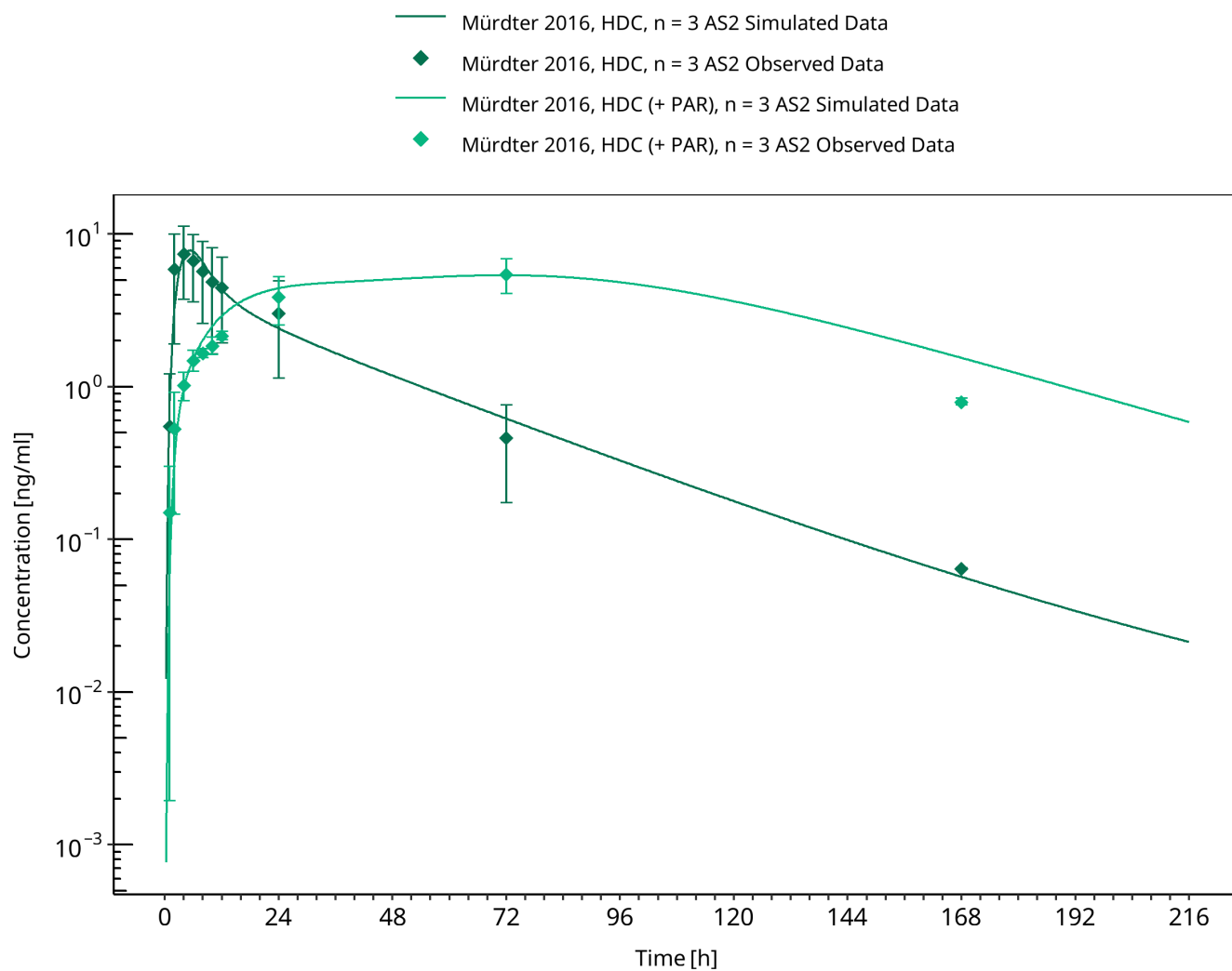


Figure 3-43: Paroxetine-Clomiphene-DDGI

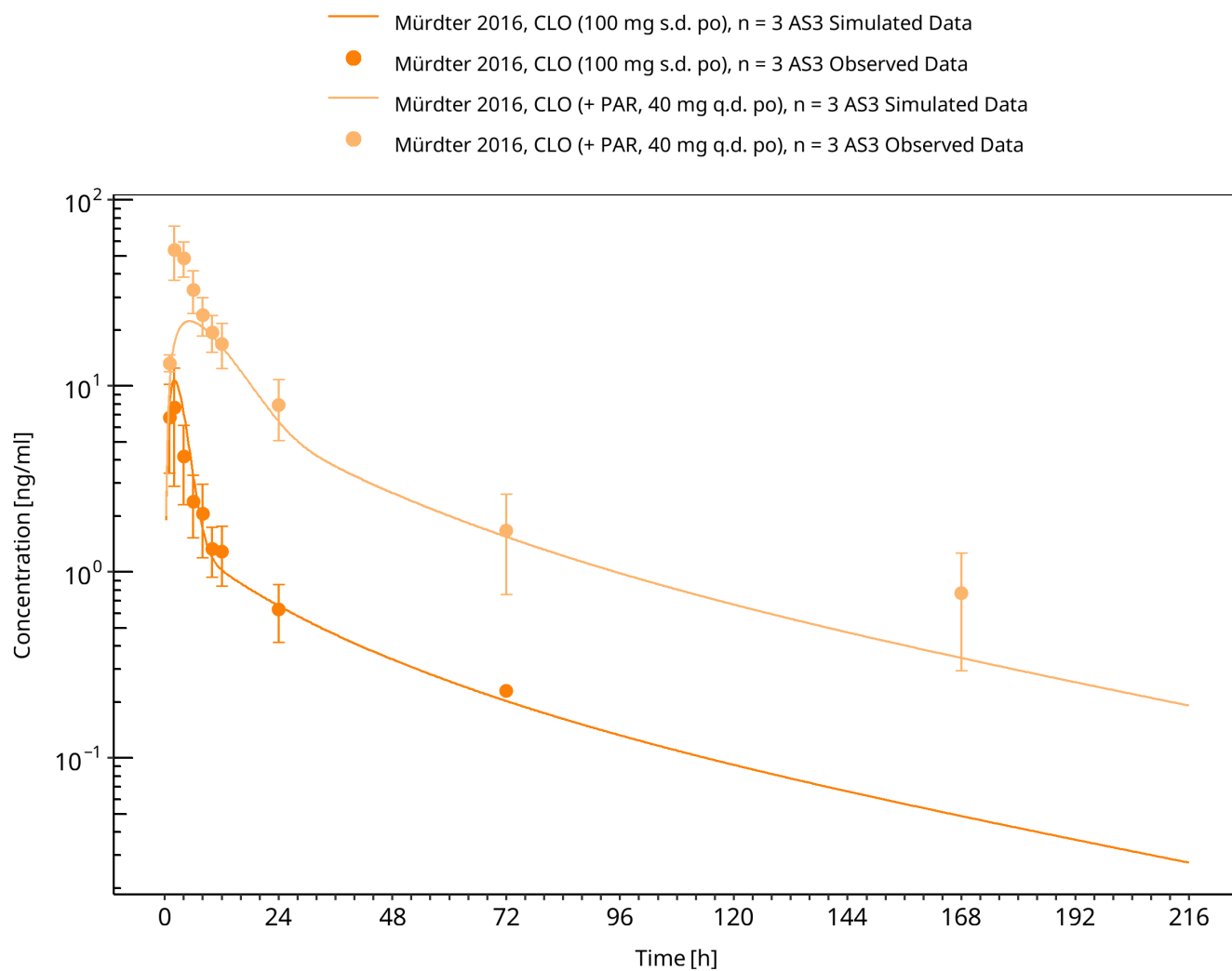


Figure 3-44: Paroxetine-Clomiphene-DDGI

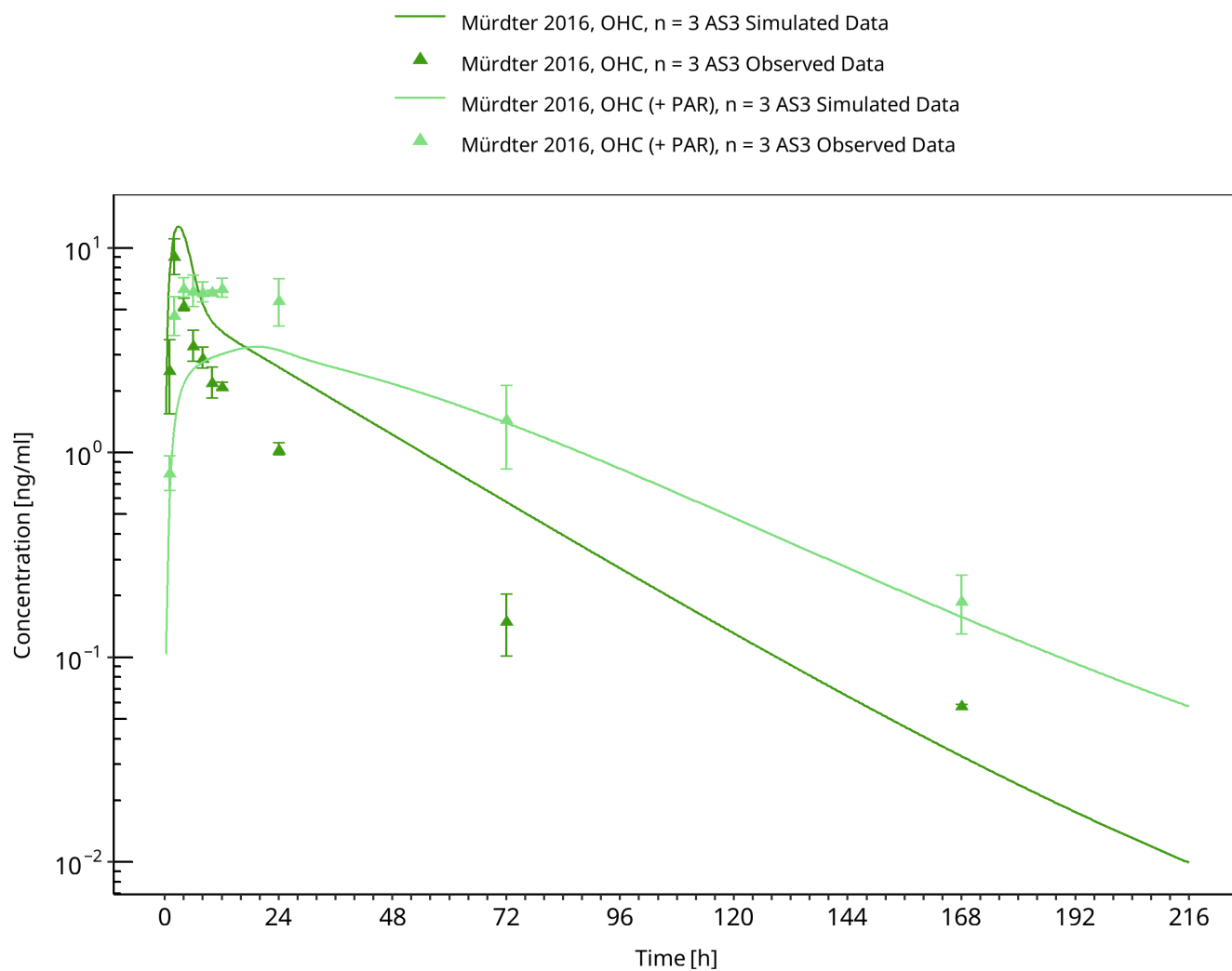


Figure 3-45: Paroxetine-Clomiphene-DDGI

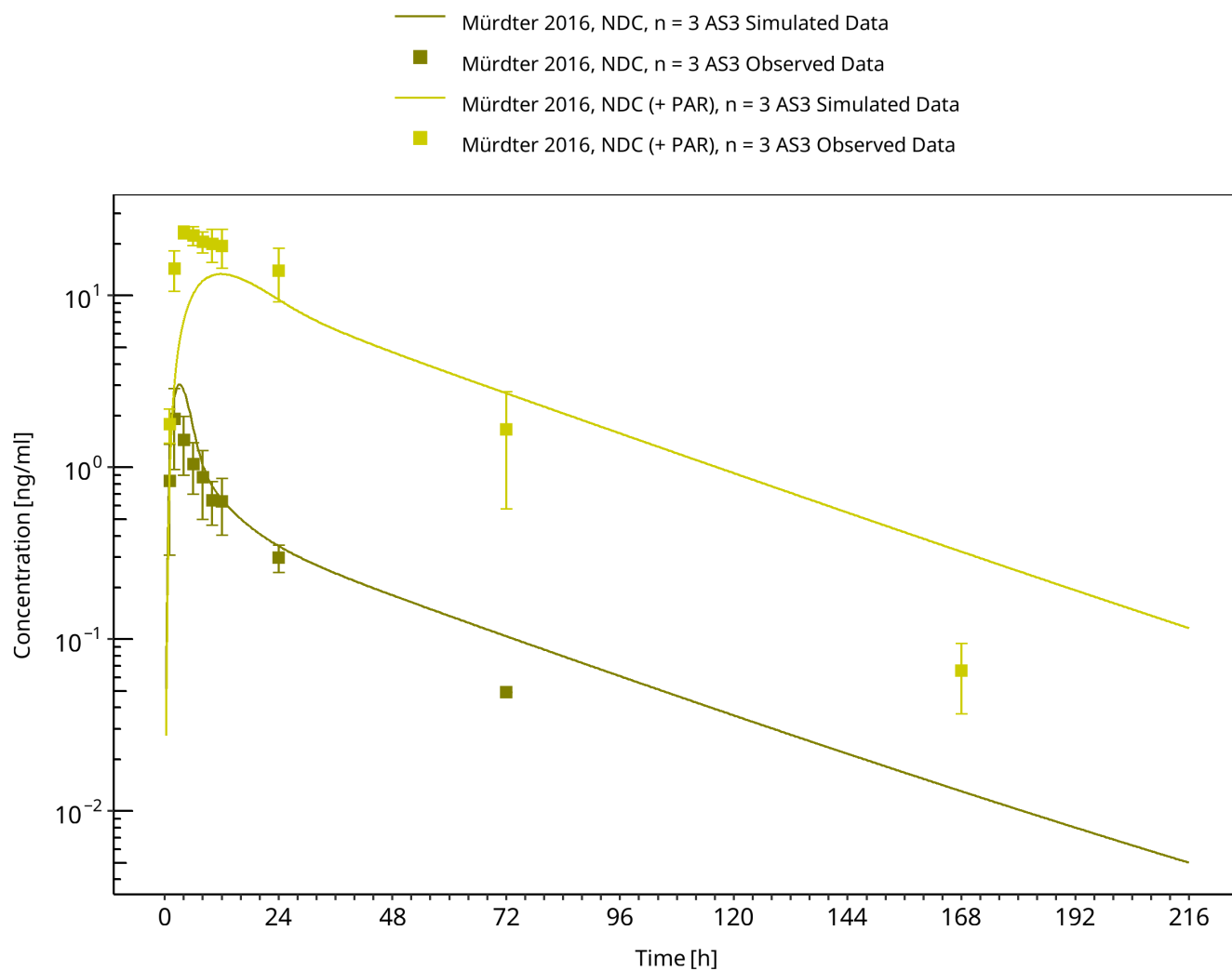


Figure 3-46: Paroxetine-Clomiphene-DDGI

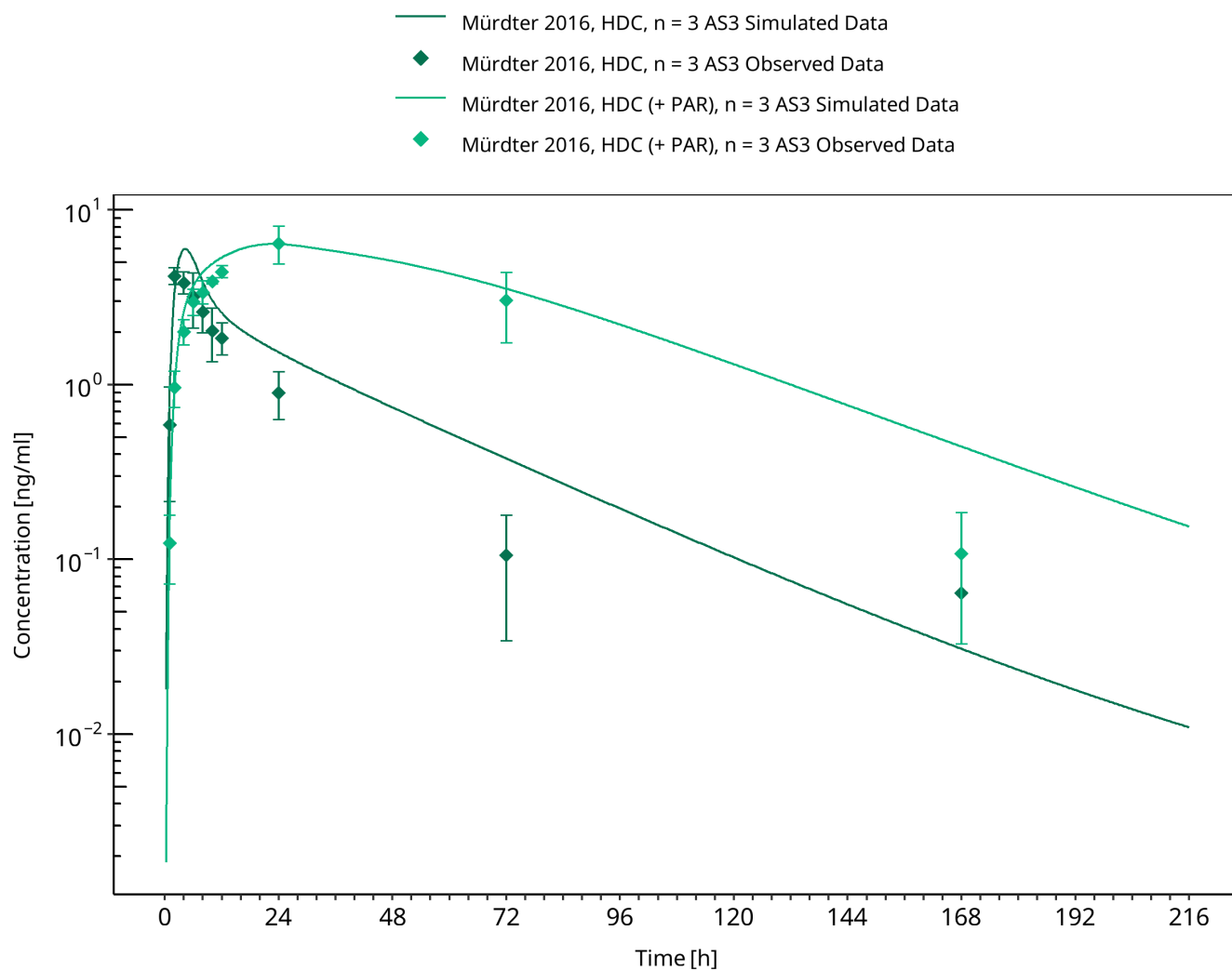


Figure 3-47: Paroxetine-Clomiphene-DDGI

3.6 Atomoxetine-Desipramine-DDI

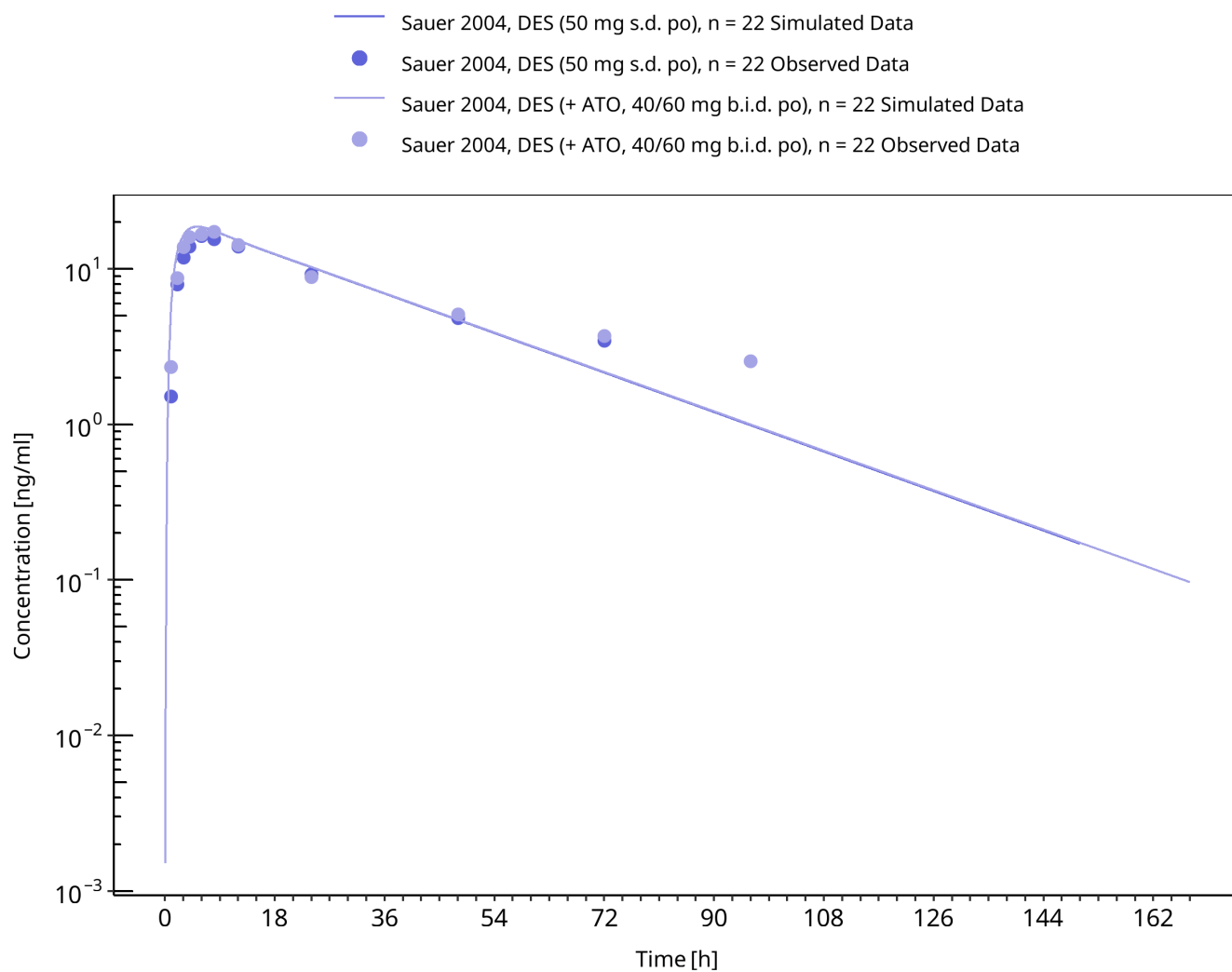


Figure 3-48: Atomoxetine-Desipramine-DDI

3.7 Paroxetine-Desipramine-DDGI

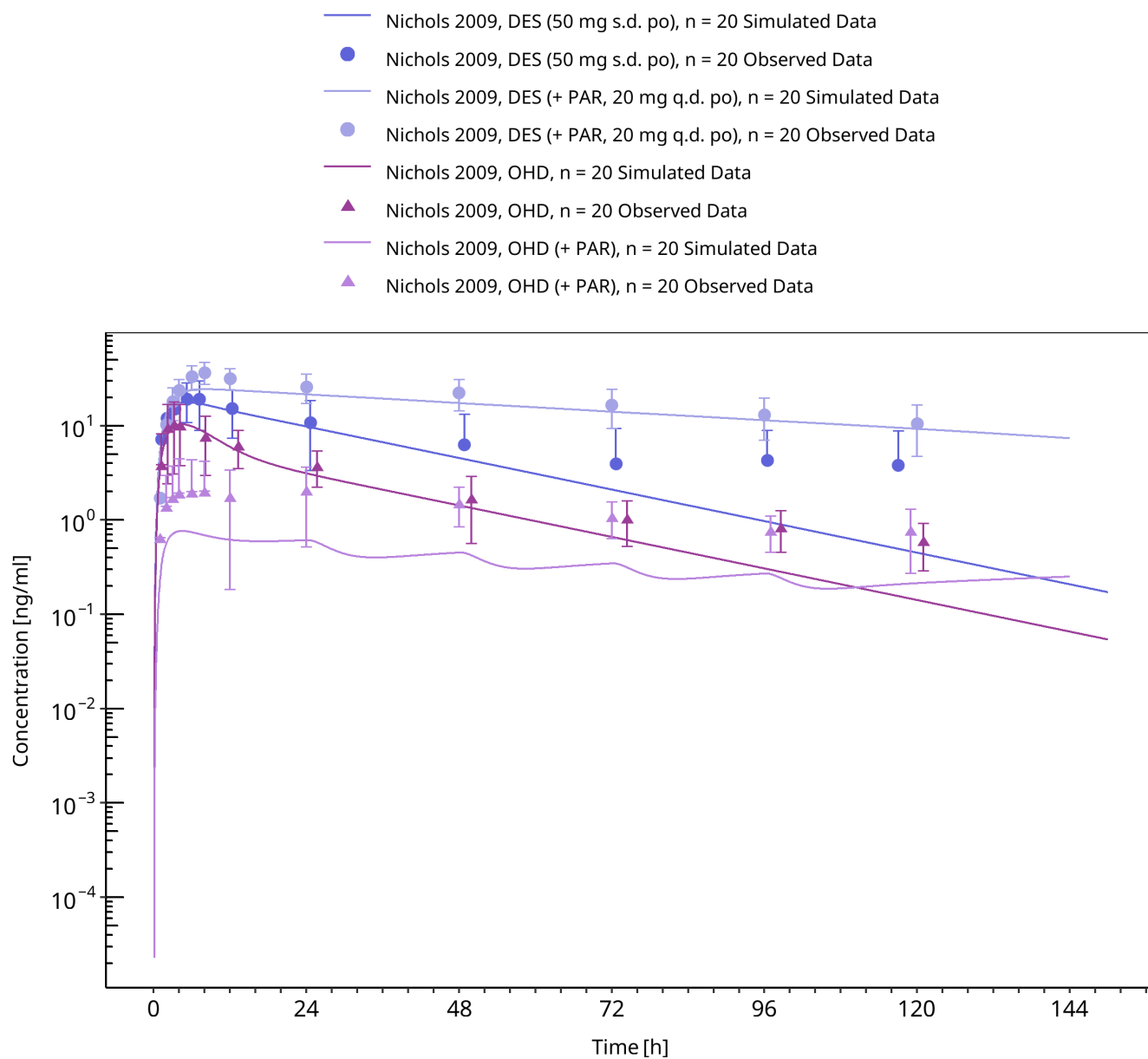


Figure 3-49: Paroxetine-Desipramine-DDI

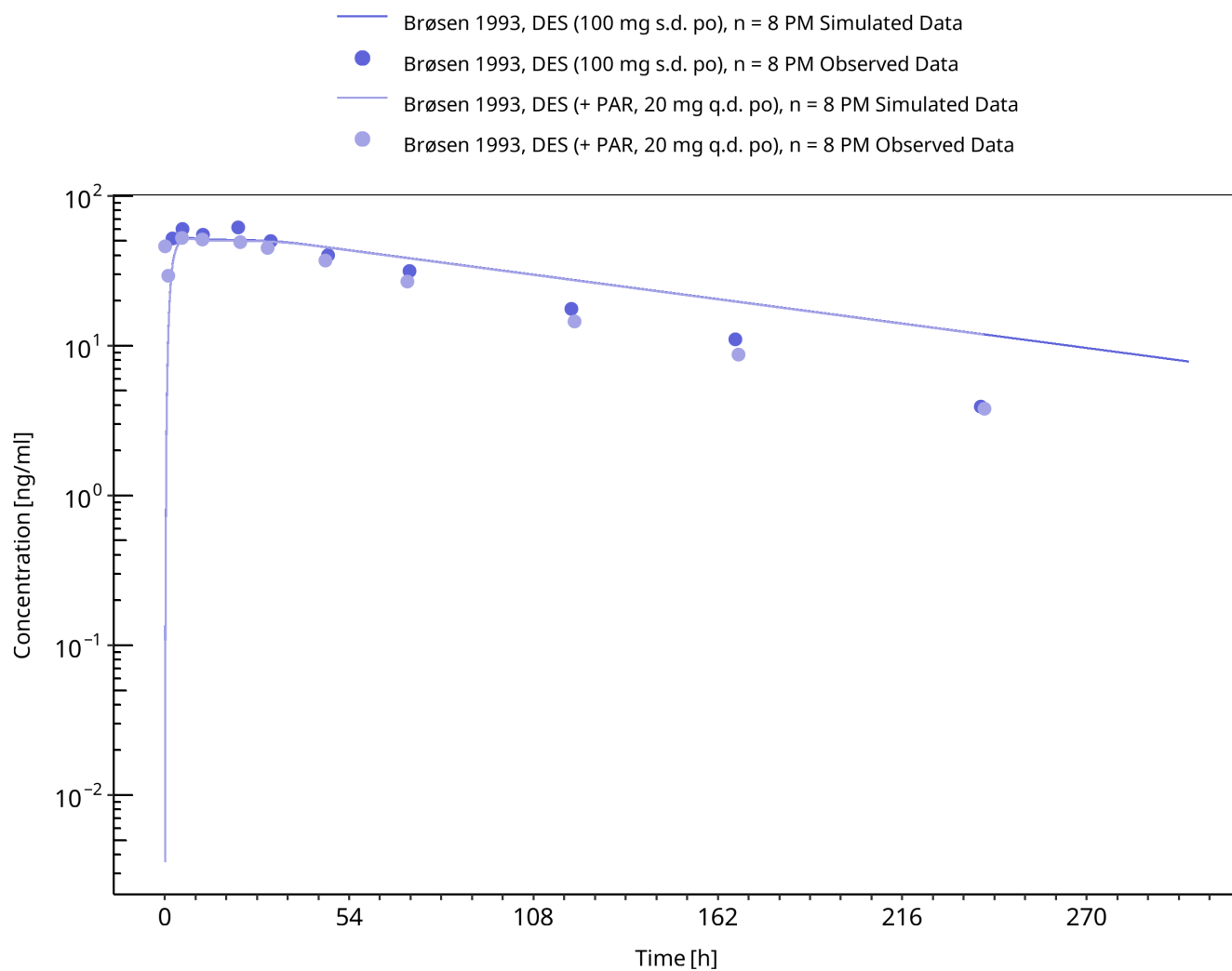


Figure 3-50: Paroxetine-Desipramine-DDGI

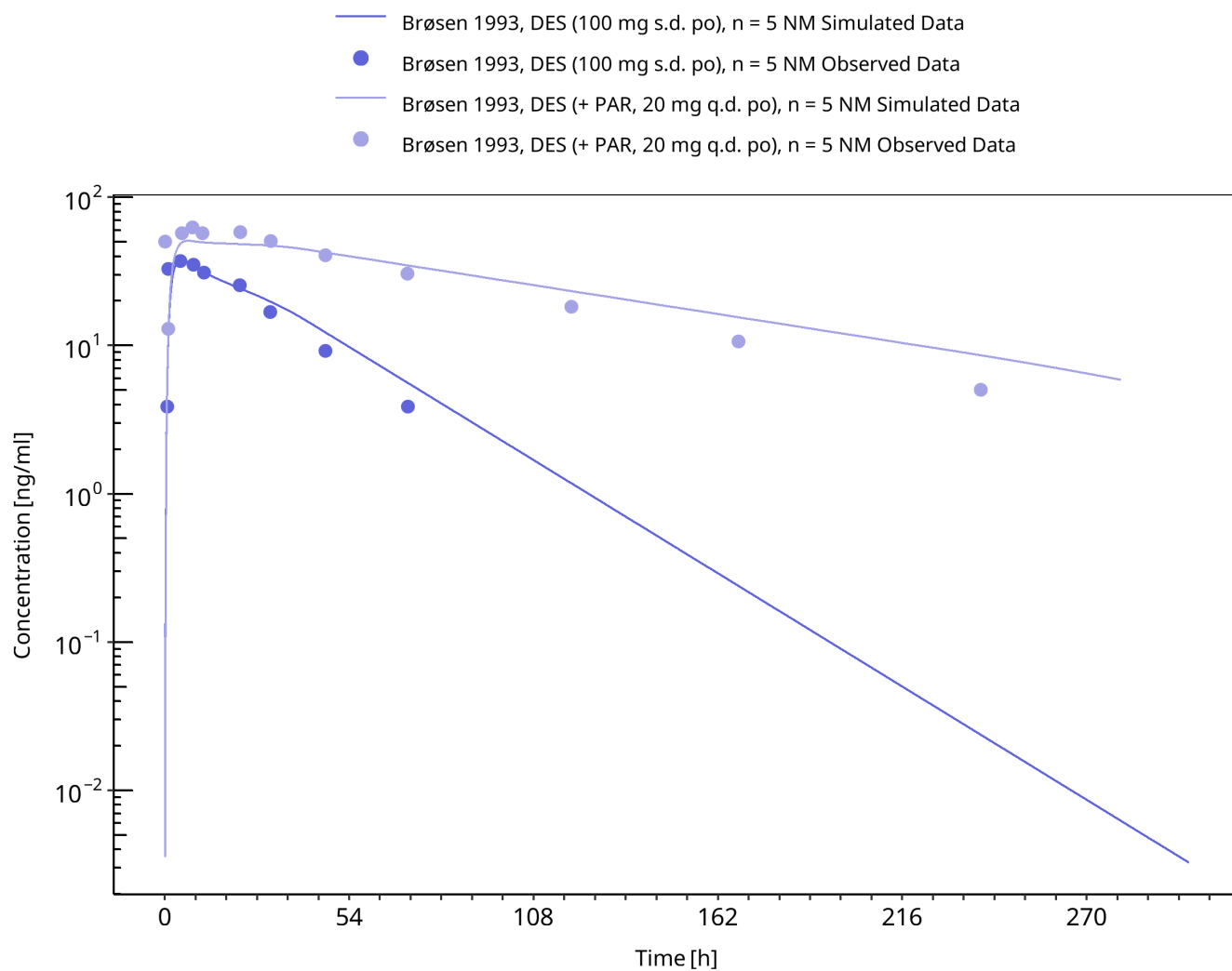


Figure 3-51: Paroxetine-Desipramine-DDGI

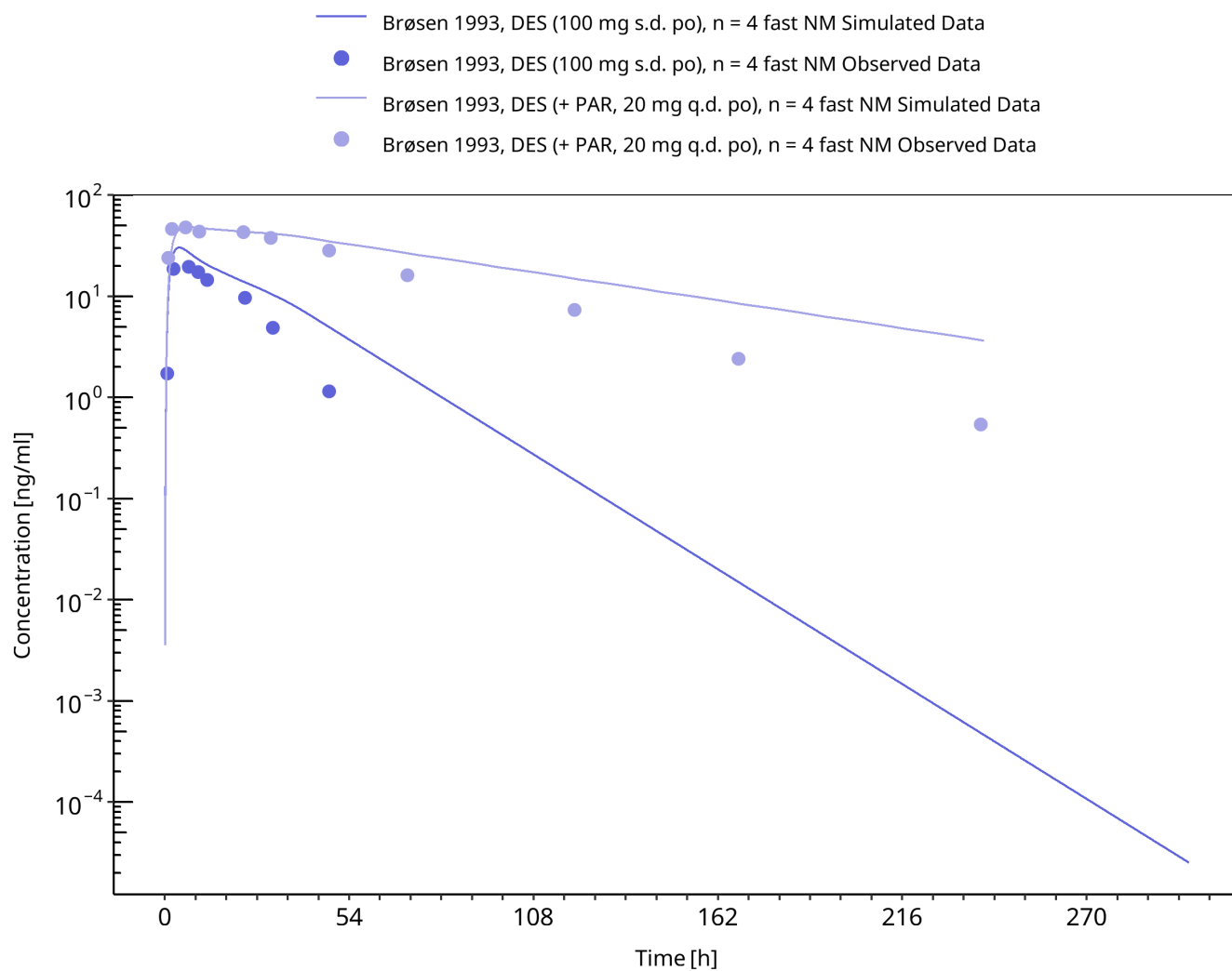


Figure 3-52: Paroxetine-Desipramine-DDGI

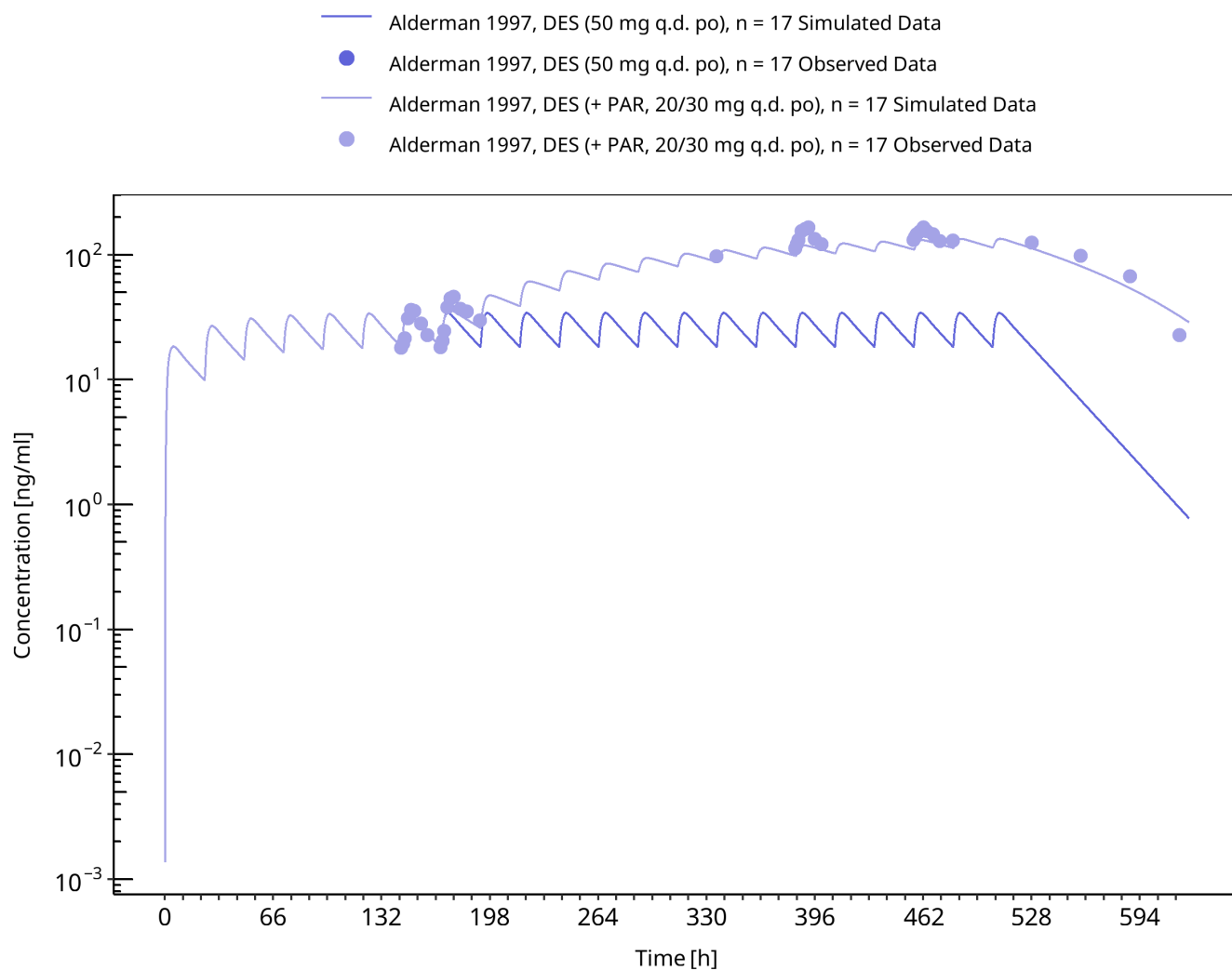


Figure 3-53: Paroxetine-Desipramine-DDI

3.8 Quinidine-Desipramine-DDI

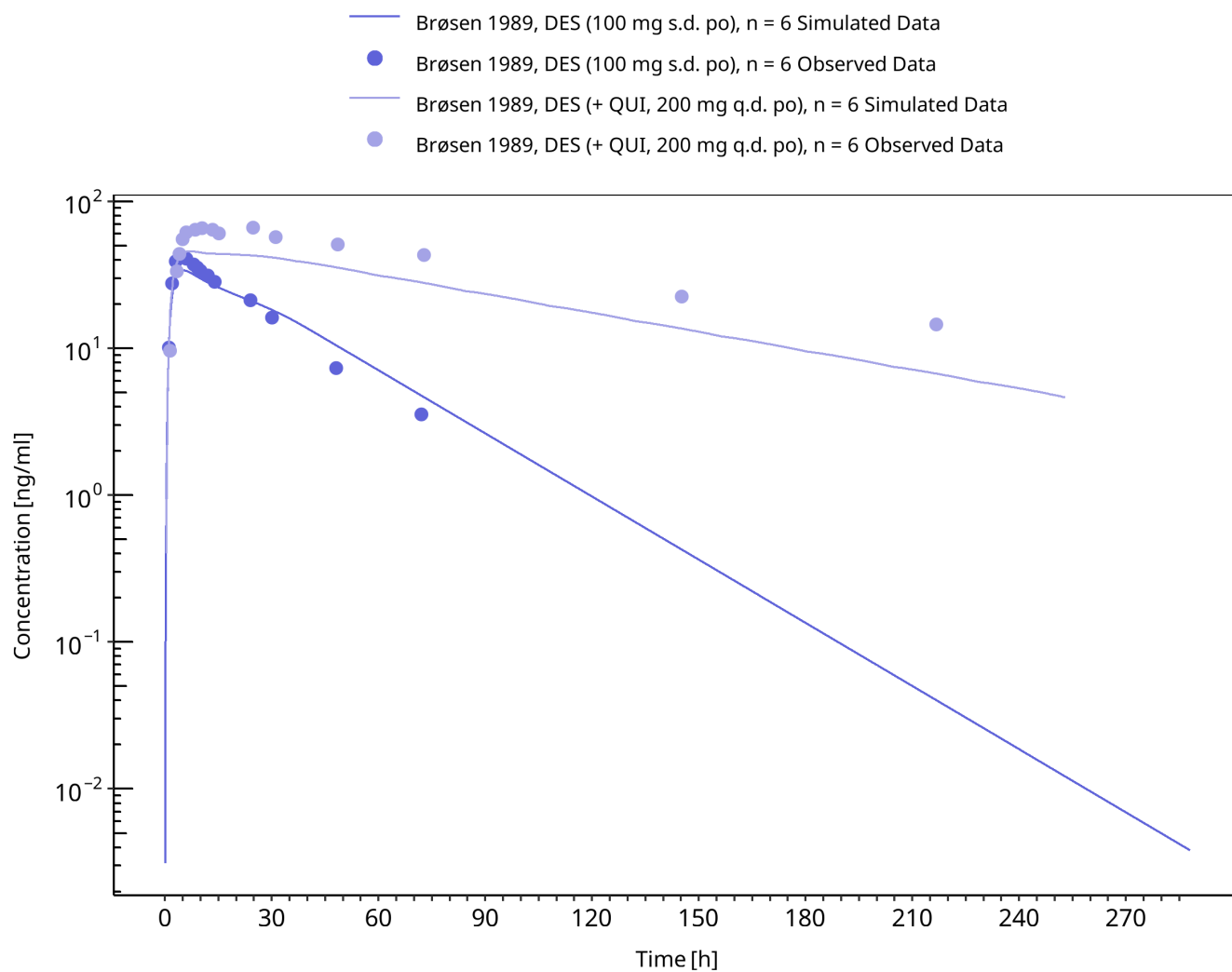


Figure 3-54: Quinidine-Desipramine-DDI

3.9 Paroxetine-Dextromethorphan-DDGI

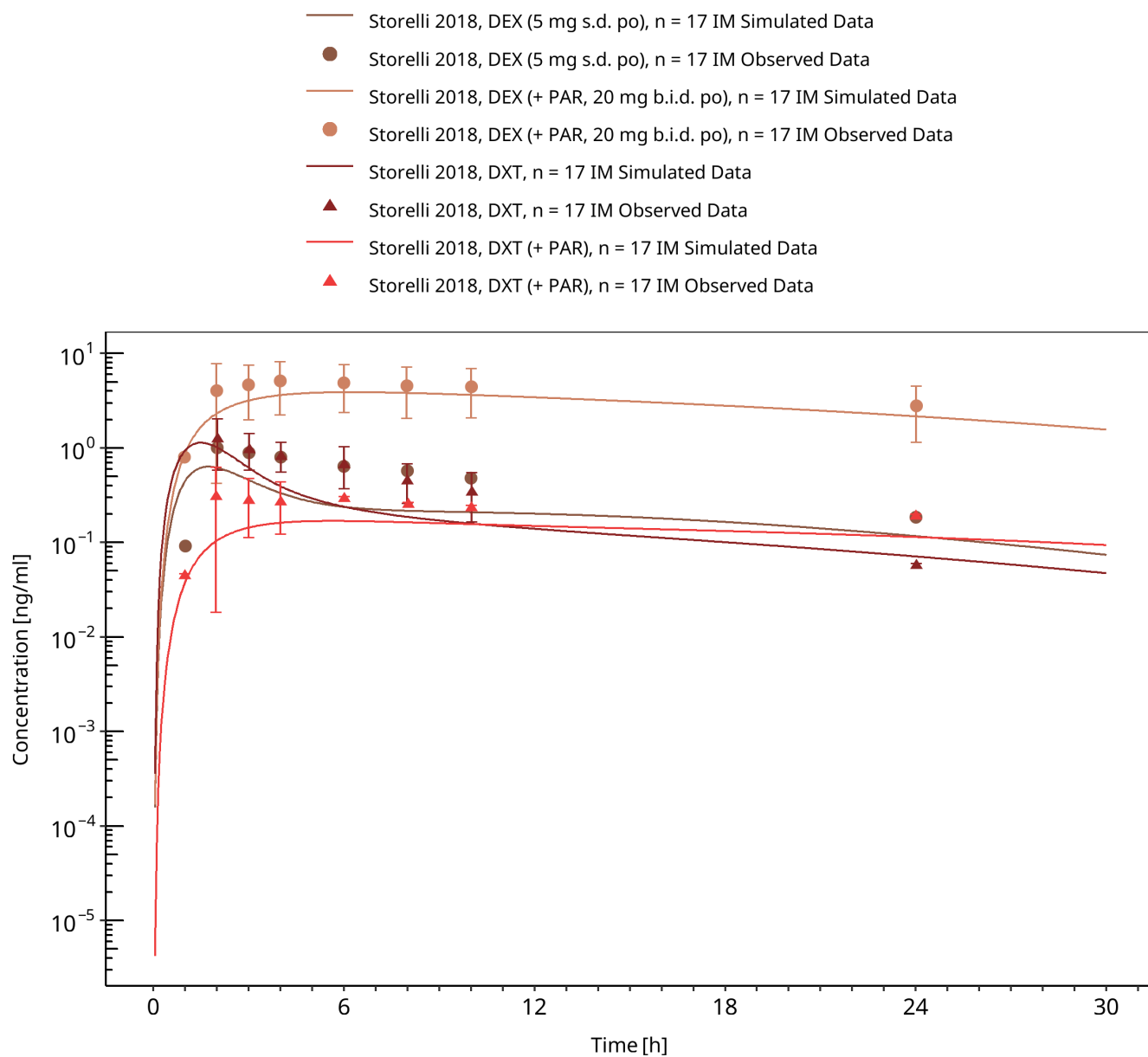


Figure 3-55: Paroxetine-Dextromethorphan-DDGI

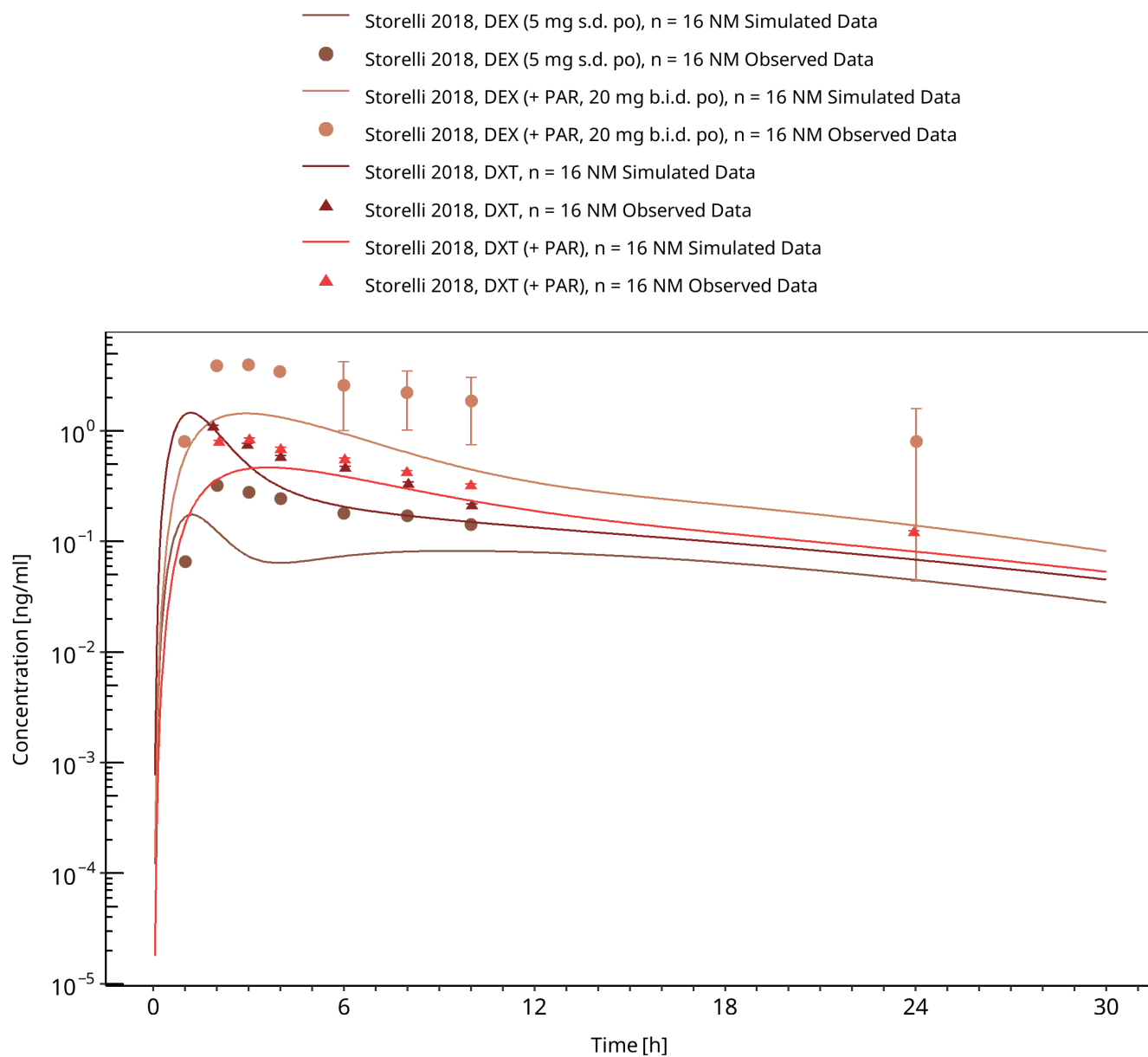


Figure 3-56: Paroxetine-Dextromethorphan-DDGI

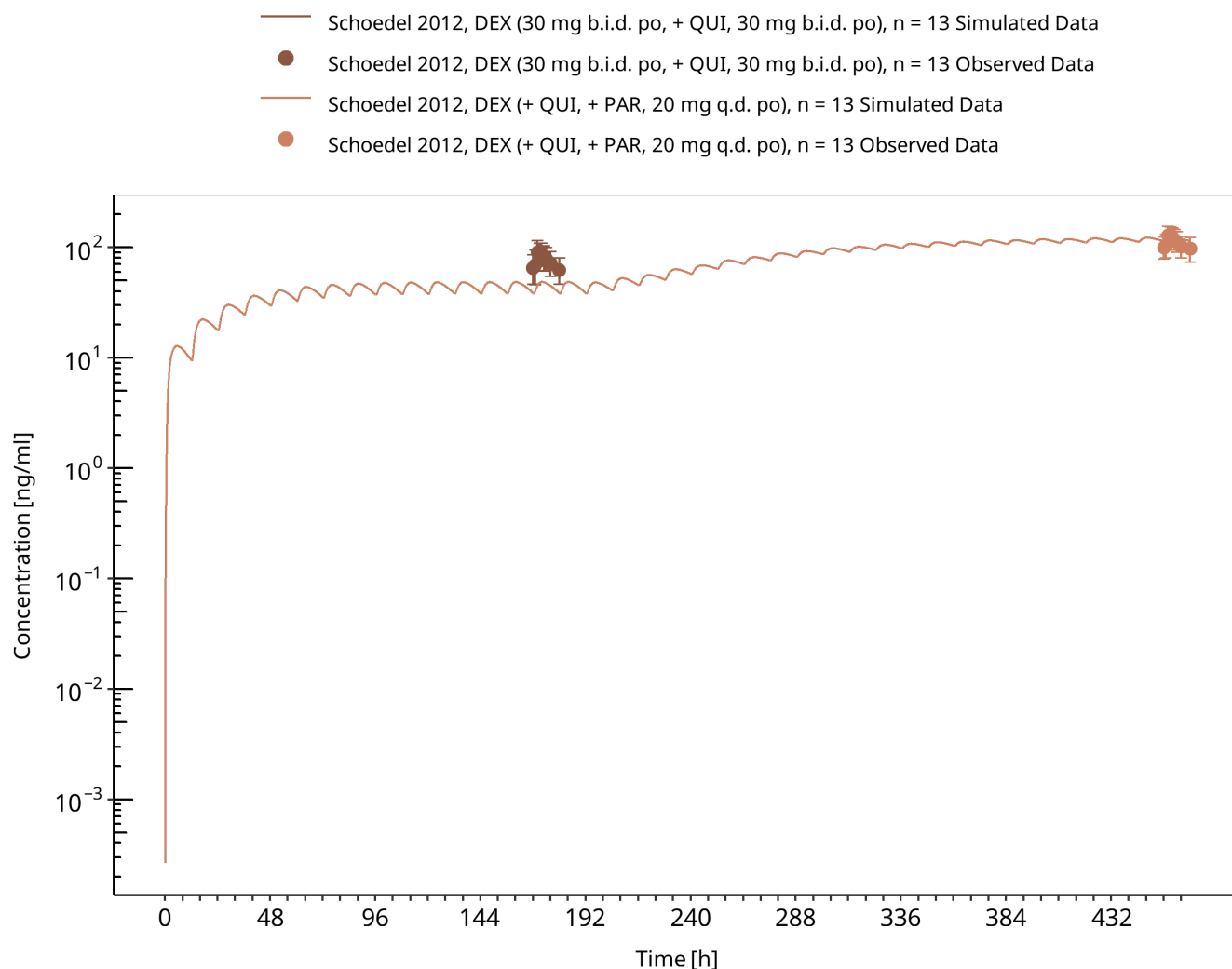


Figure 3-57: Paroxetine-Dextromethorphan-DDI

3.10 Quinidine-Dextromethorphan-DDI

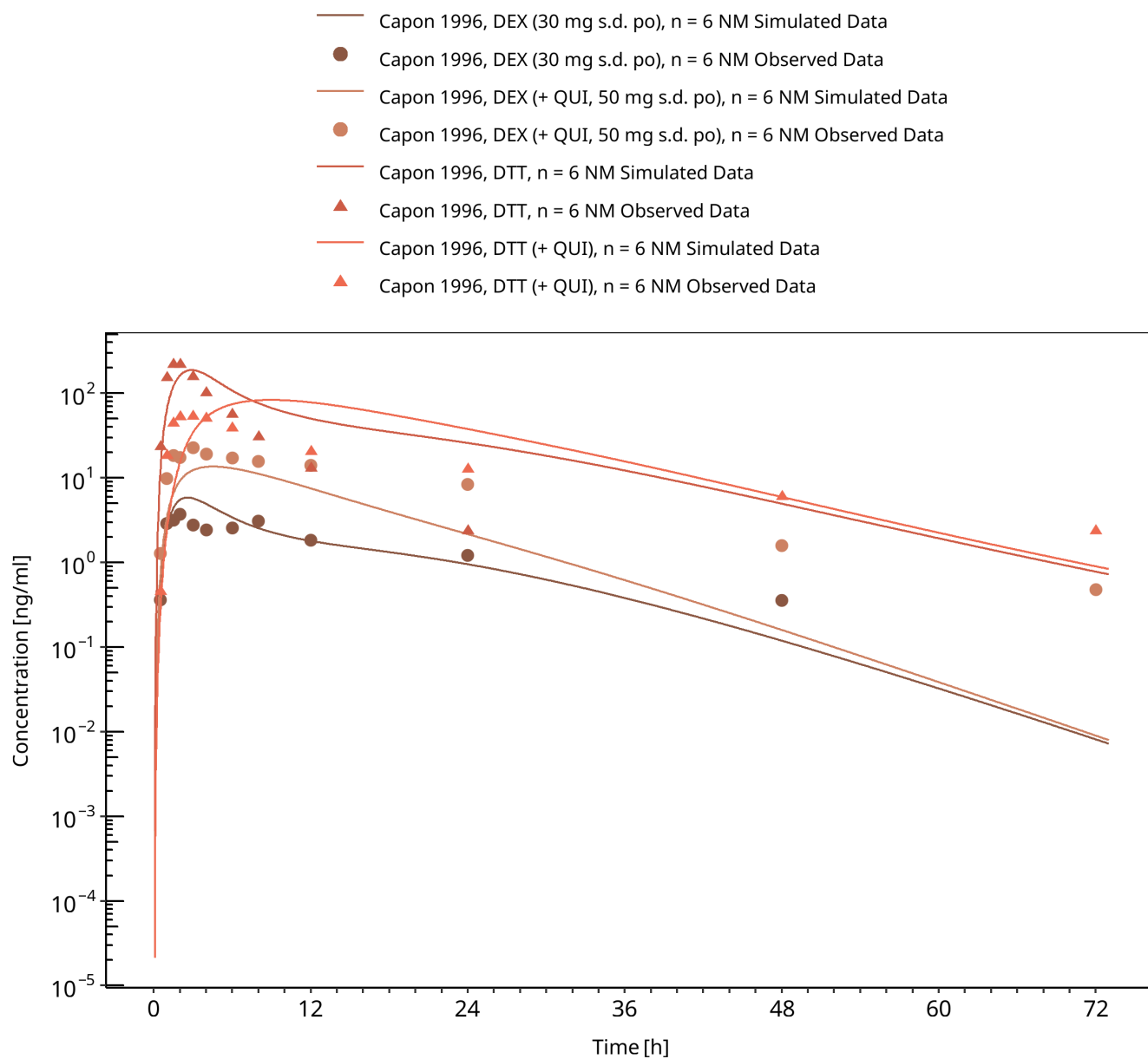


Figure 3-58: Quinidine-Dextromethorphan-DDI

3.11 Quinidine-Digoxin-DDI

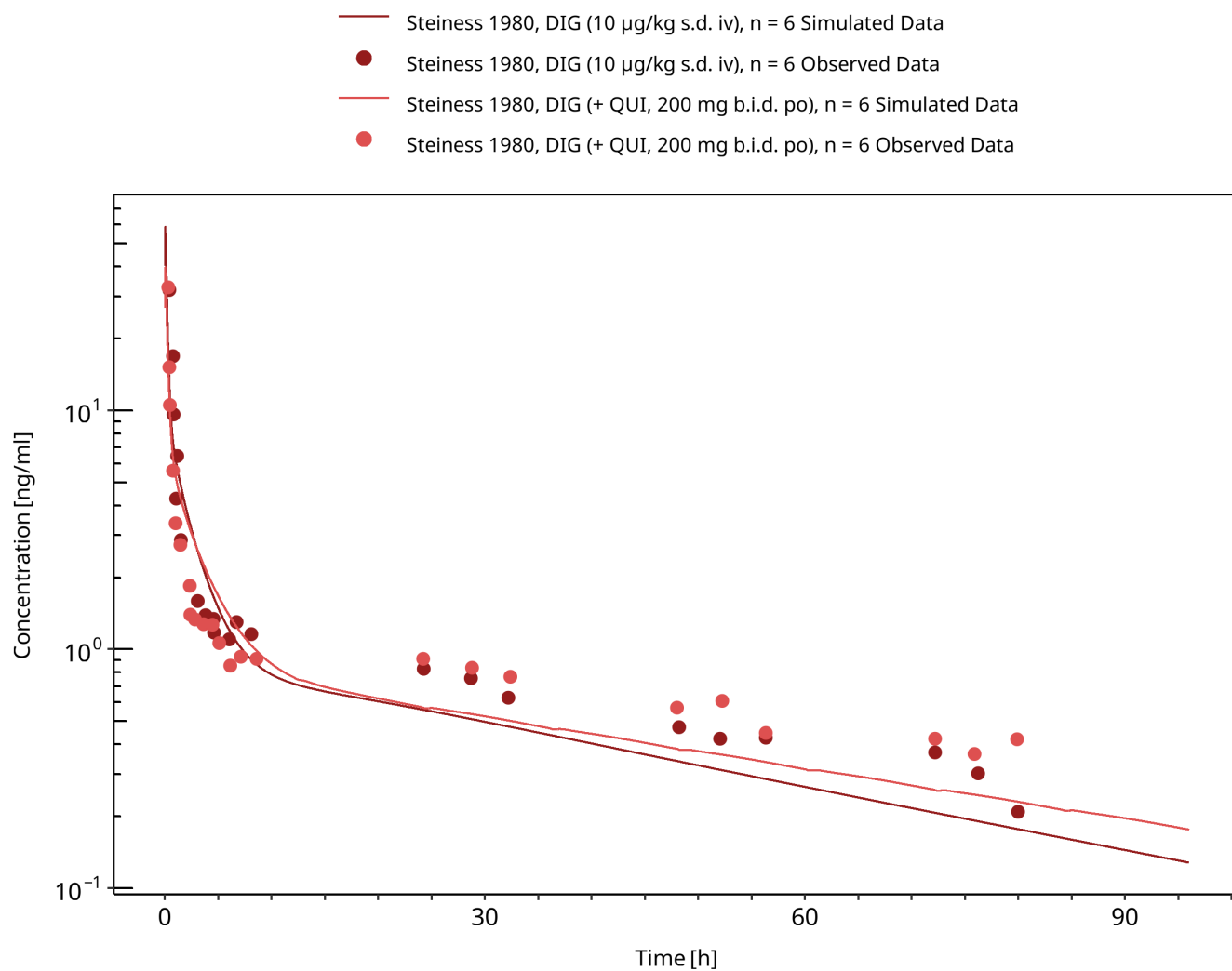


Figure 3-59: Quinidine-Digoxin-DDI

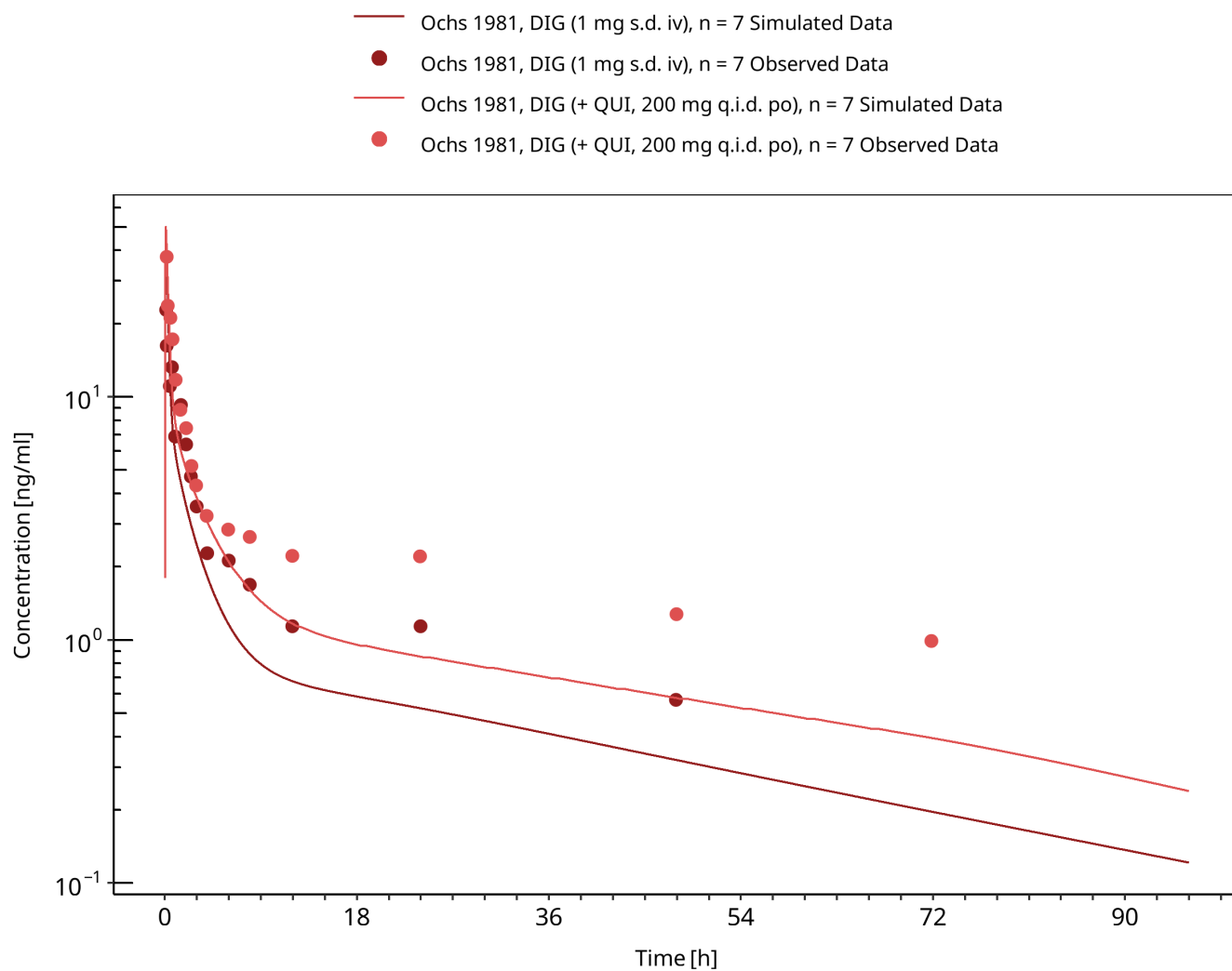


Figure 3-60: Quinidine-Digoxin-DDI

3.12 Cimetidine-Metoprolol-DDI

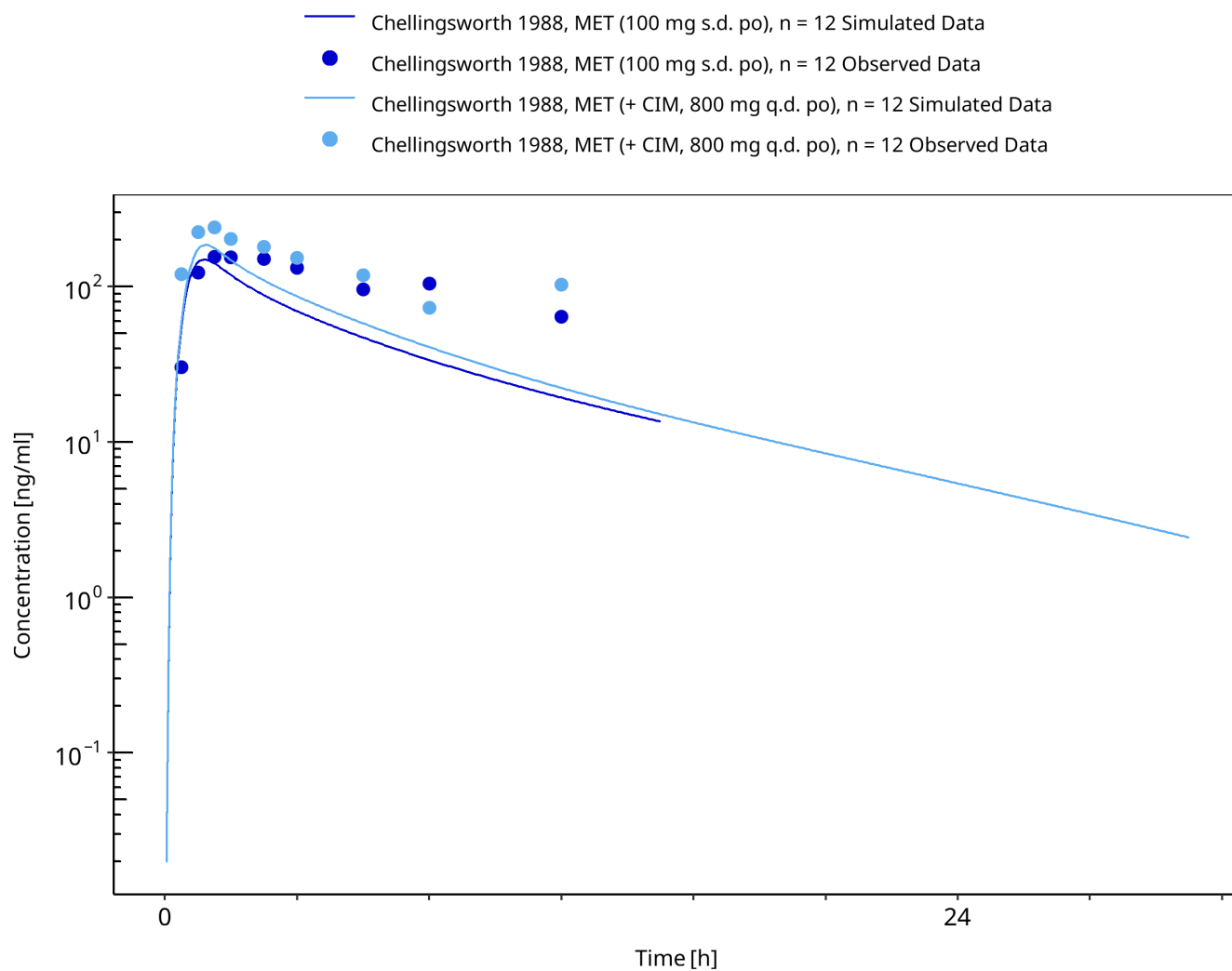


Figure 3-61: Cimetidine-Metoprolol-DDI

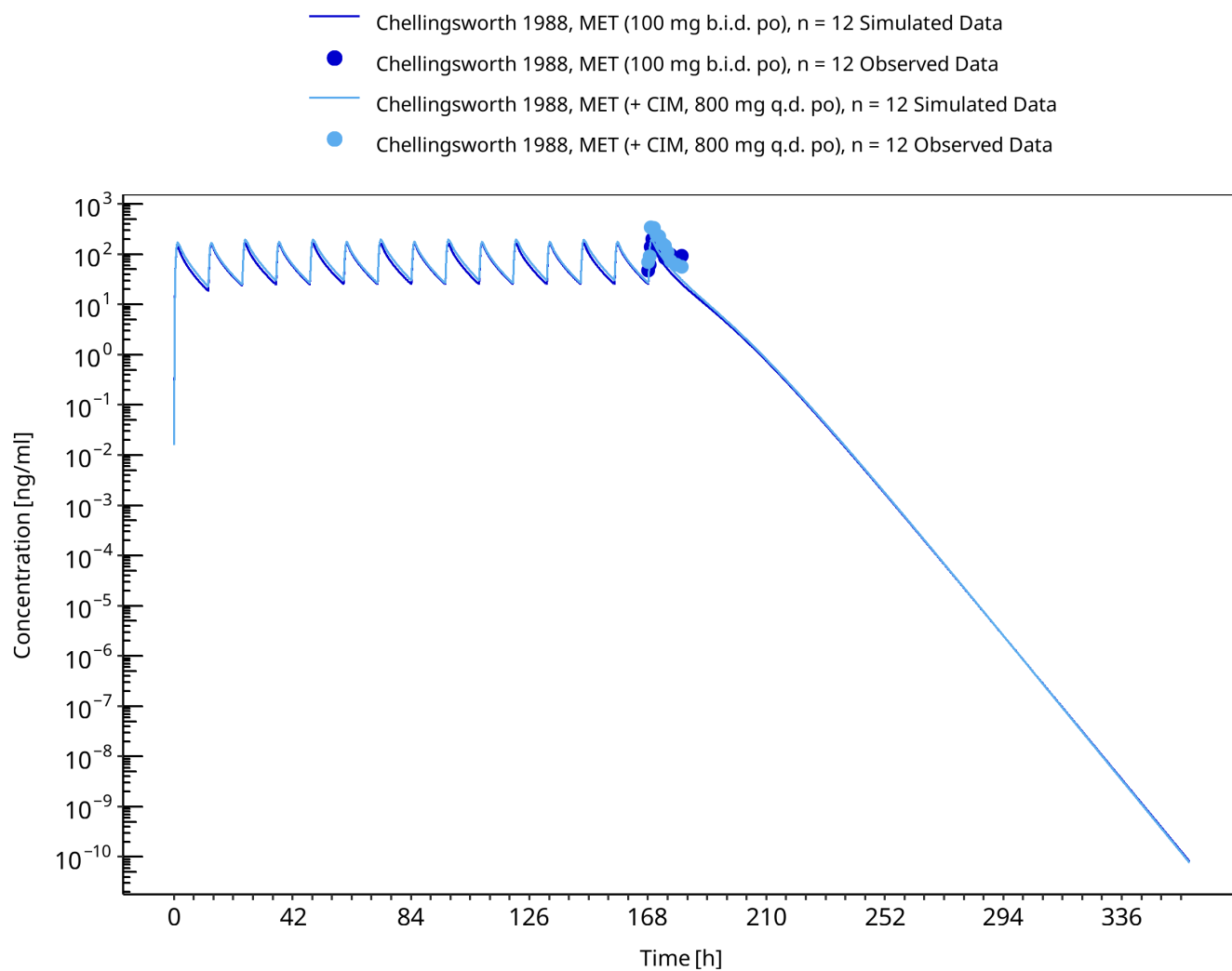


Figure 3-62: Cimetidine-Metoprolol-DDI

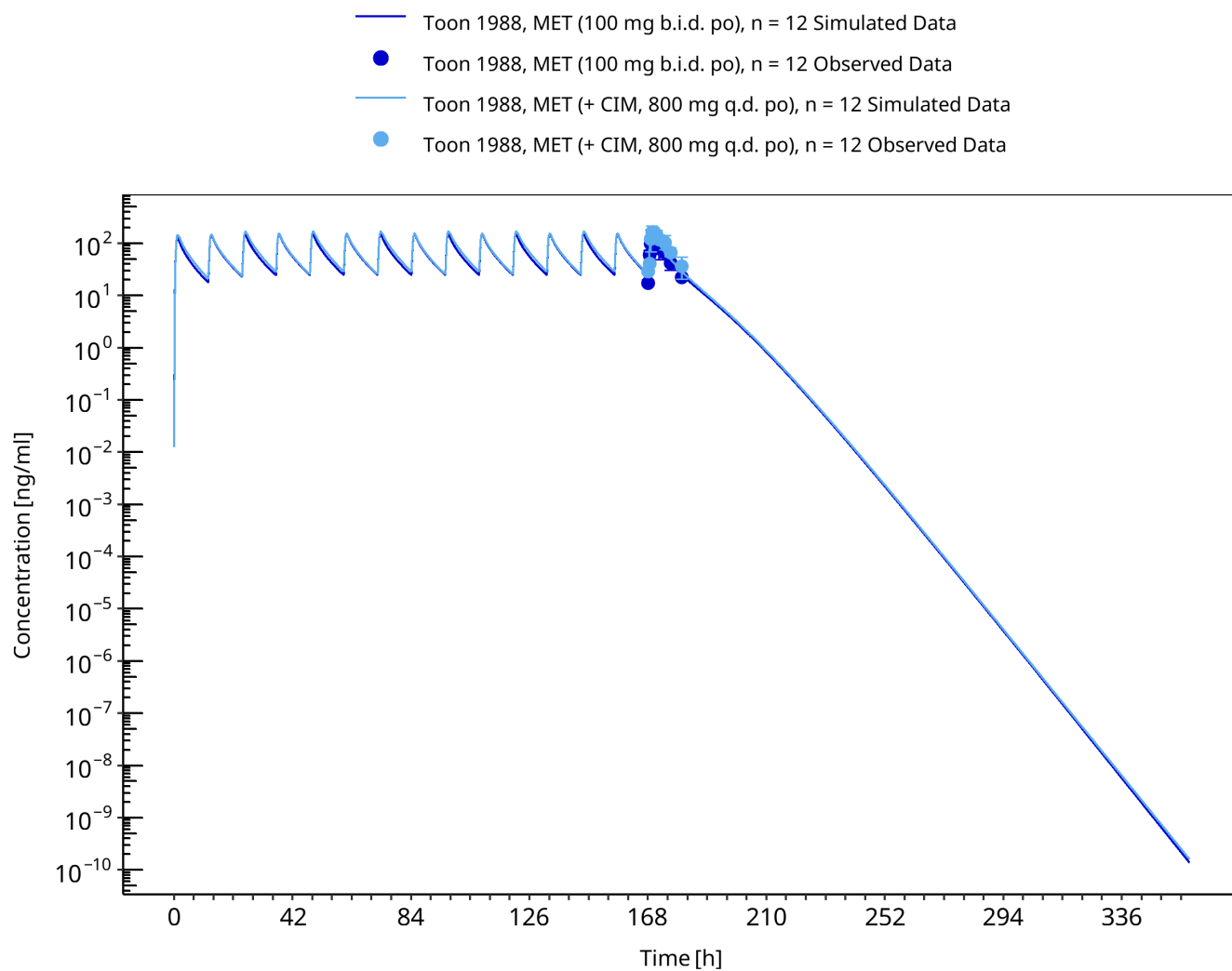


Figure 3-63: Cimetidine-Metoprolol-DDI

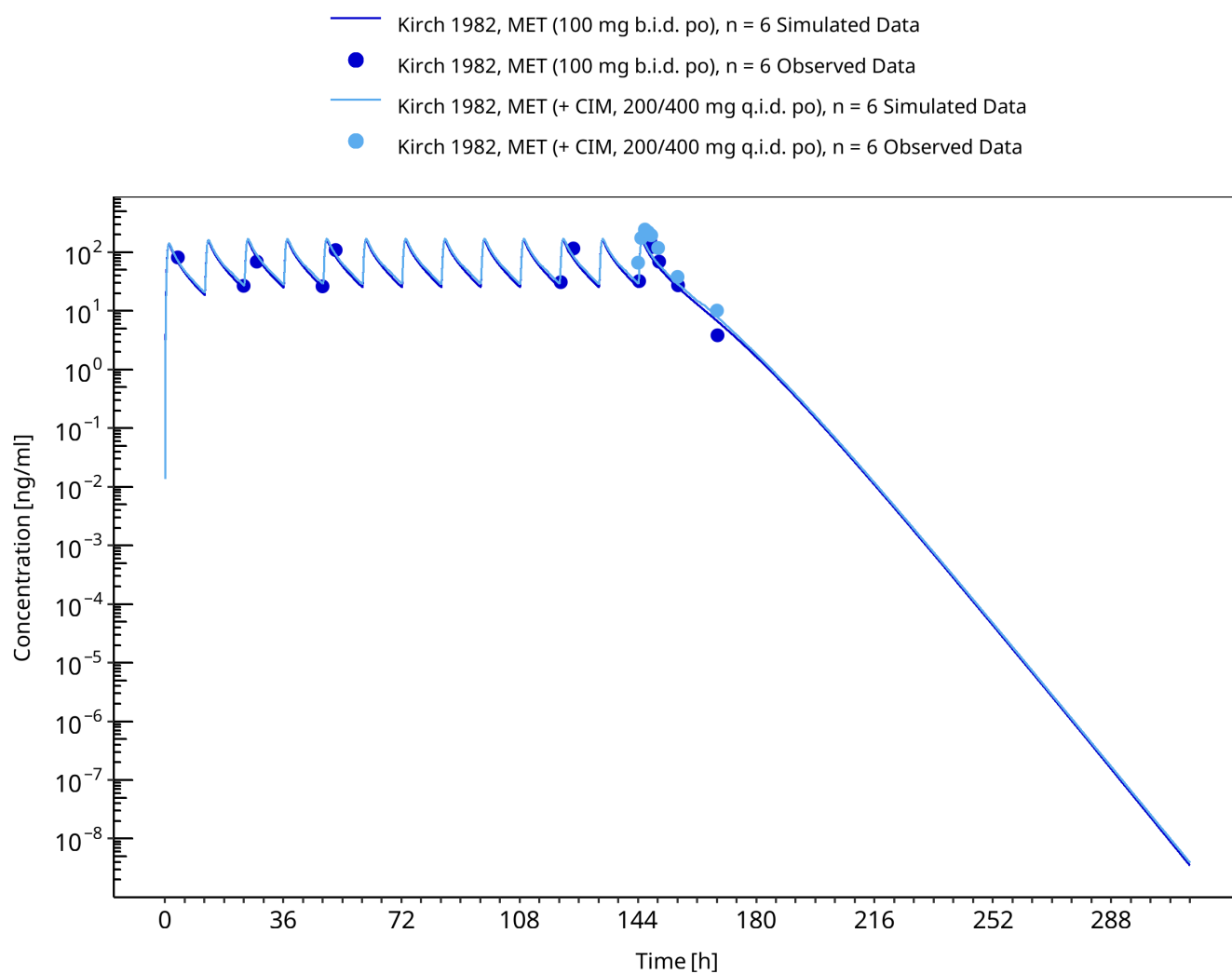


Figure 3-64: Cimetidine-Metoprolol-DDI

3.13 Paroxetine-Metoprolol-DDI

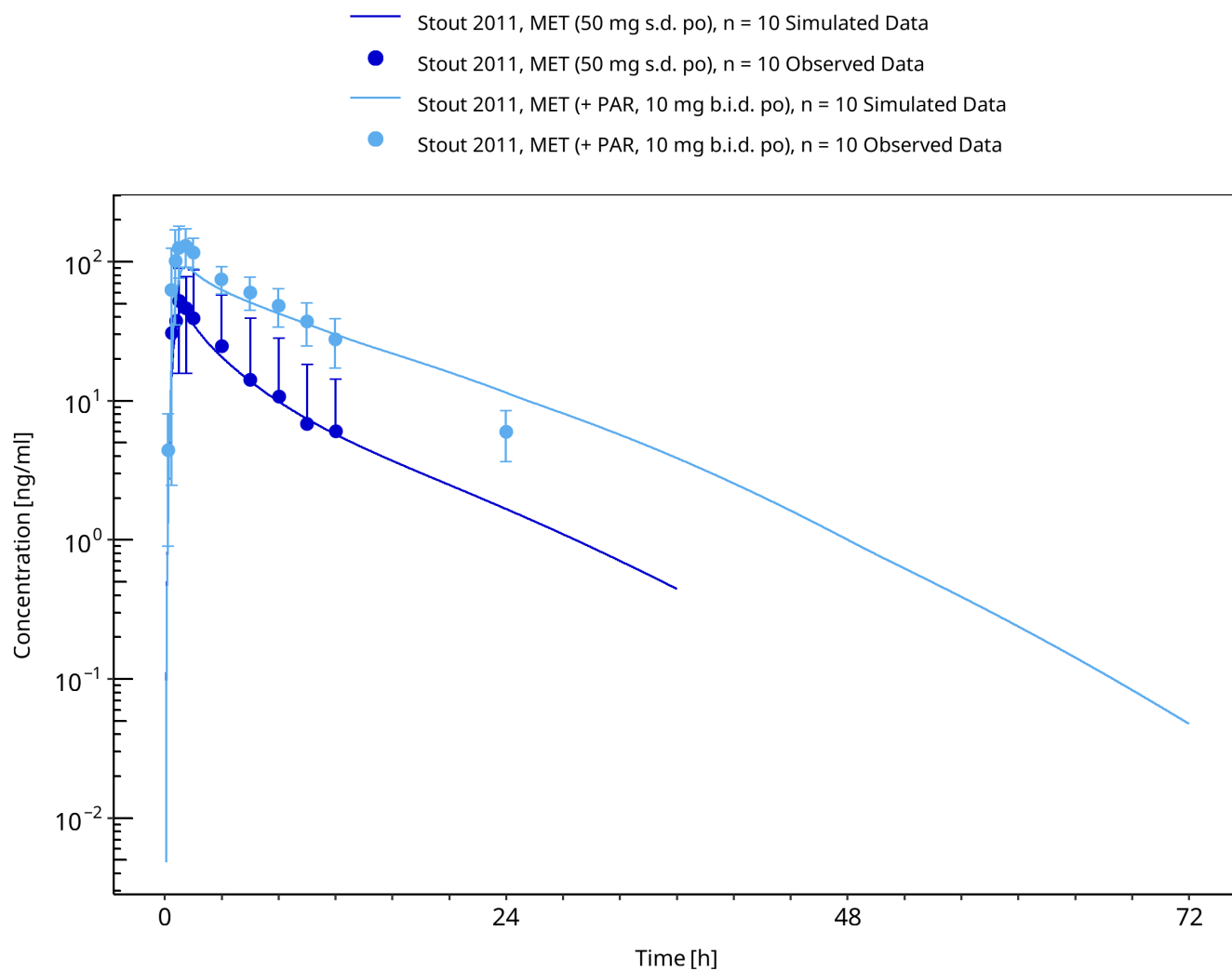


Figure 3-65: Paroxetine-Metoprolol-DDI

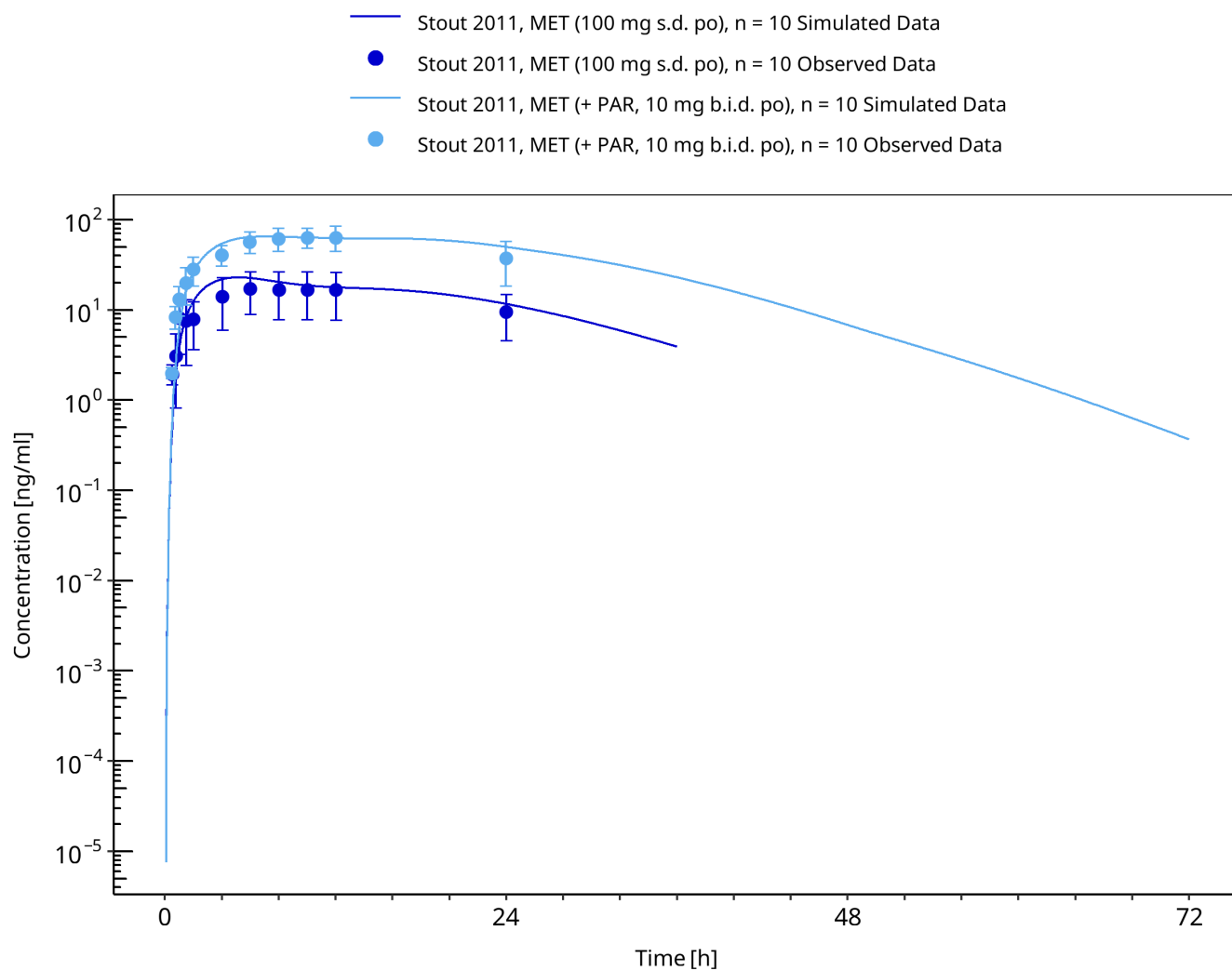


Figure 3-66: Paroxetine-Metoprolol-DDI

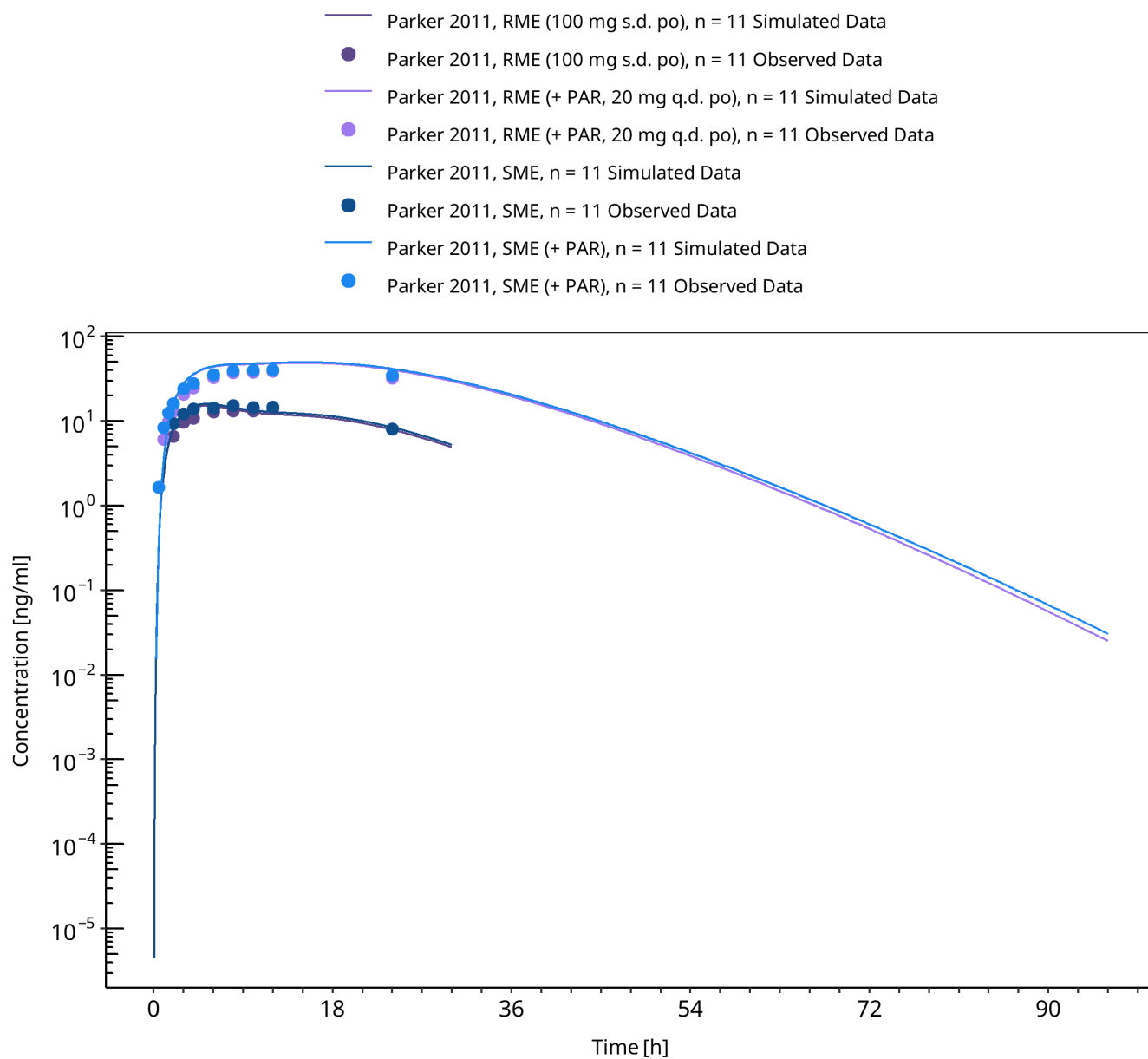


Figure 3-67: Paroxetine-Metoprolol-DDI

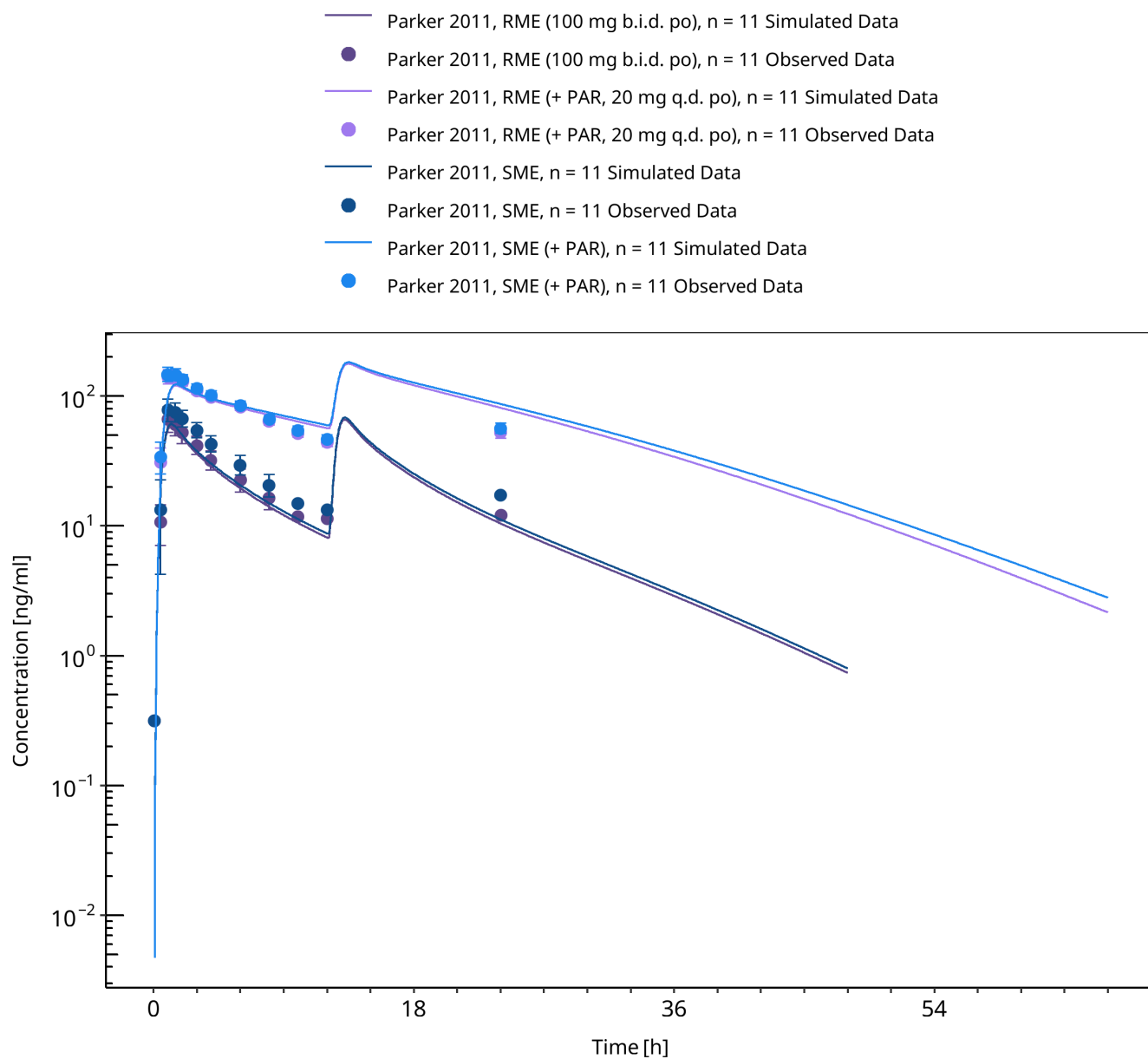


Figure 3-68: Paroxetine-Metoprolol-DDI

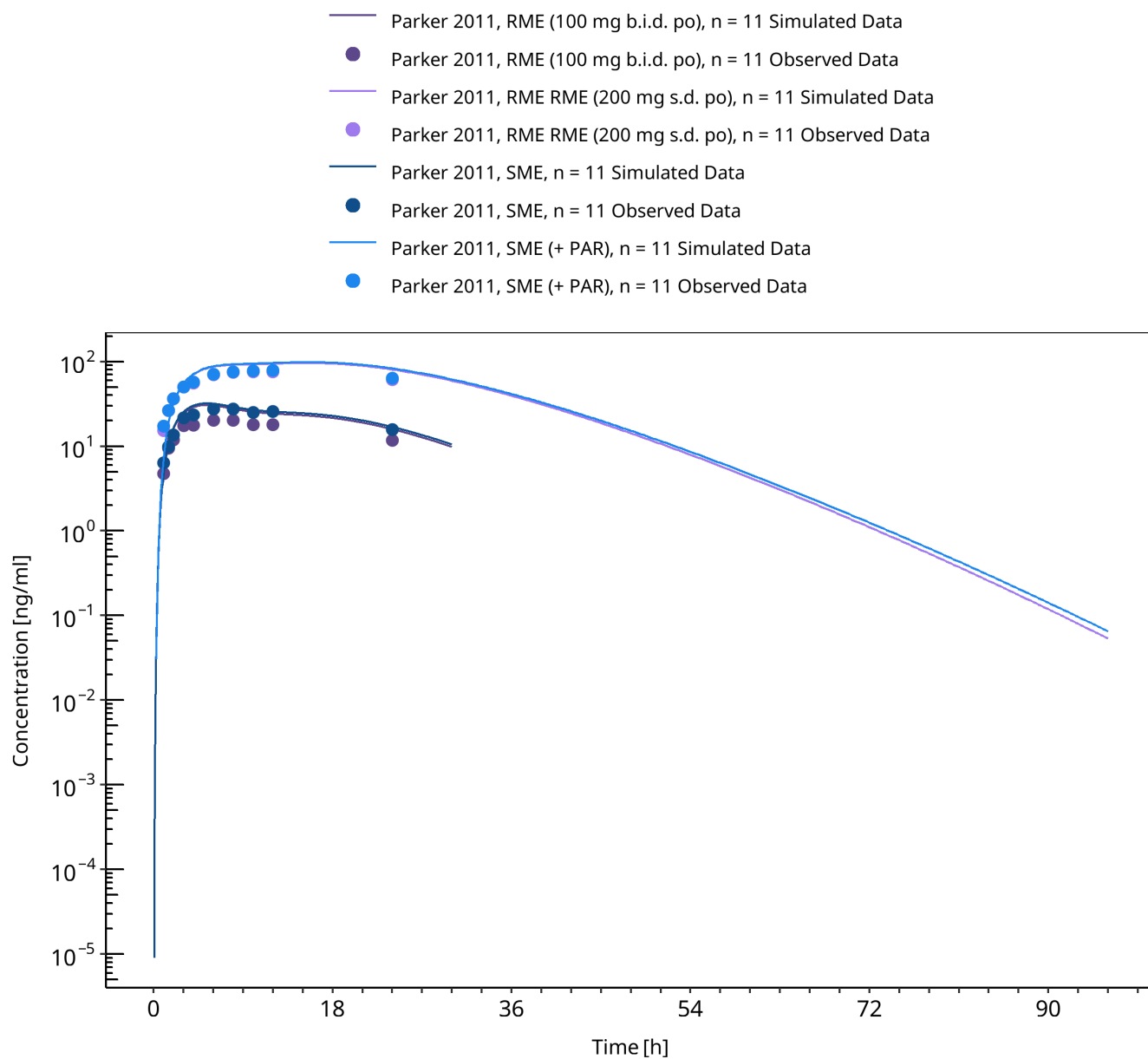


Figure 3-69: Paroxetine-Metoprolol-DDI

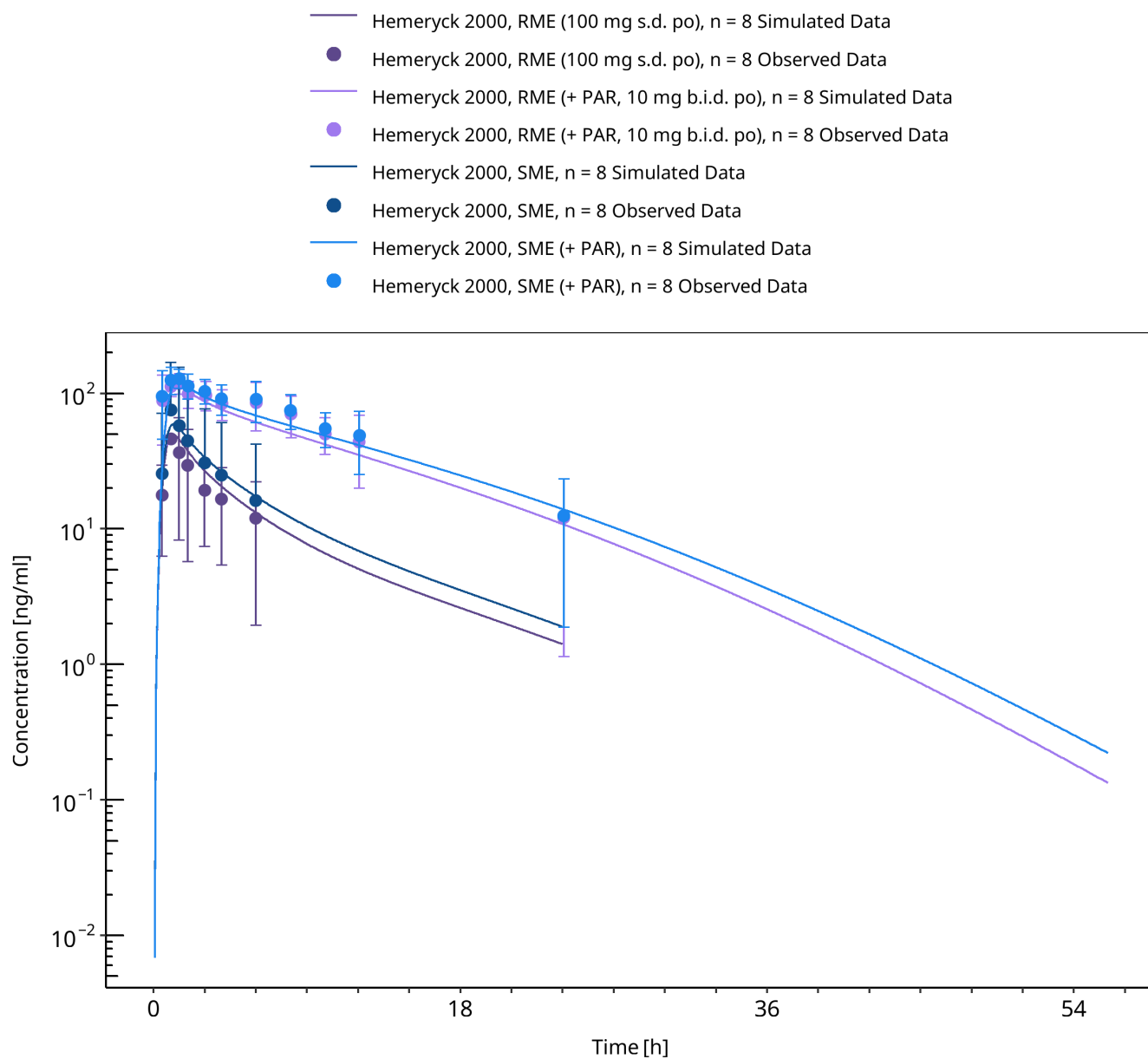


Figure 3-70: Paroxetine-Metoprolol-DDI

3.14 Quinidine-Metoprolol-DDGI

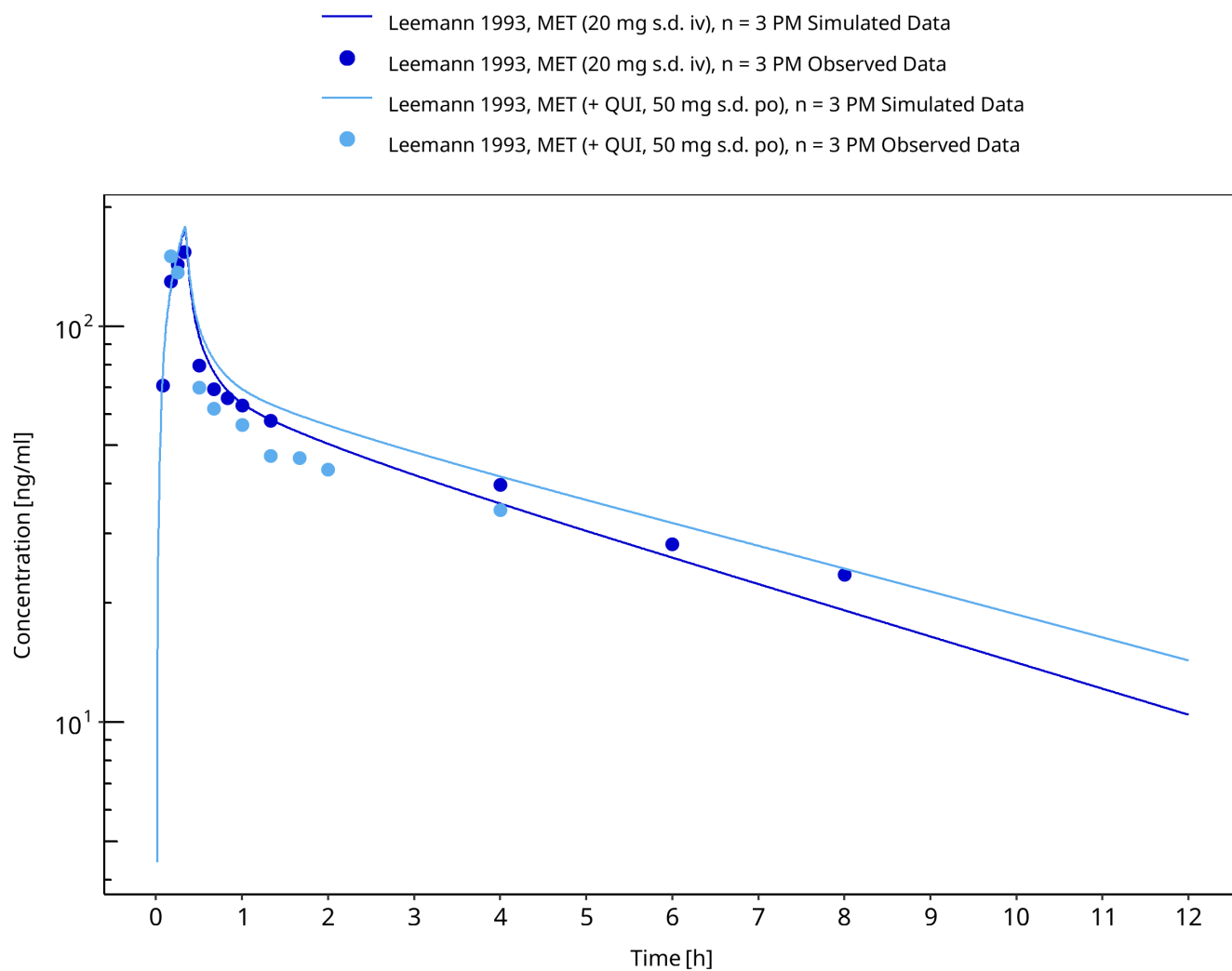


Figure 3-71: Quinidine-Metoprolol-DDGI

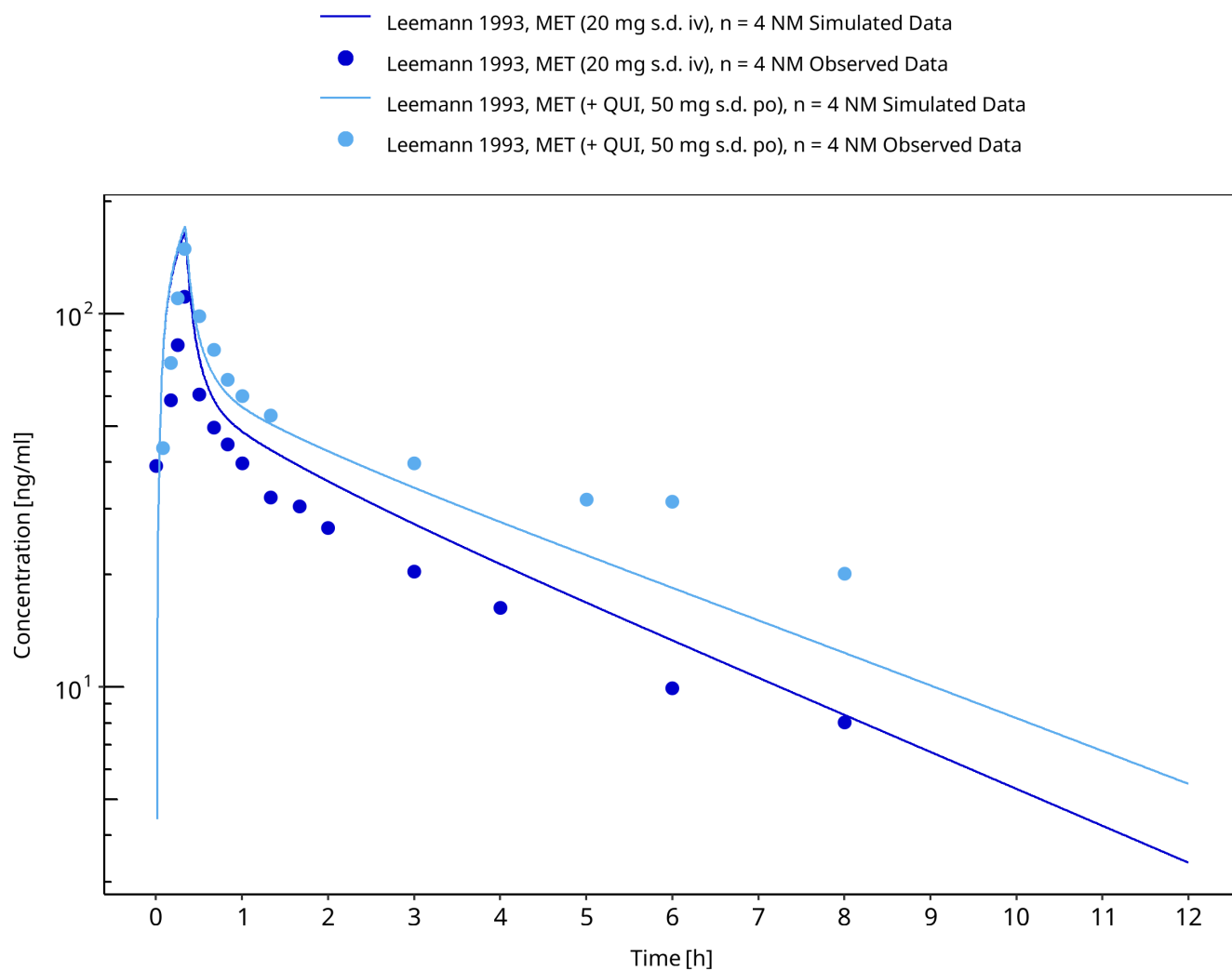


Figure 3-72: Quinidine-Metoprolol-DDGI

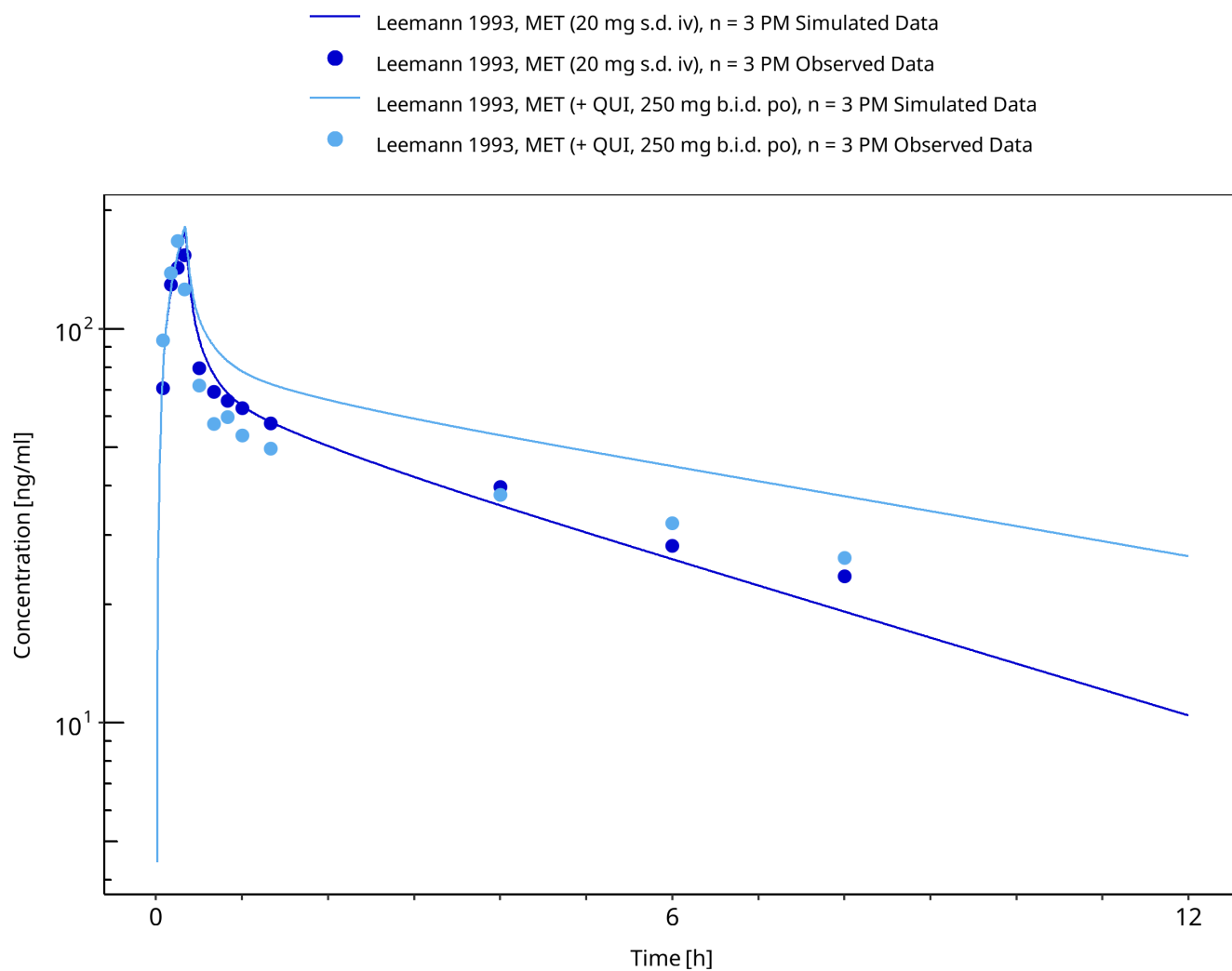


Figure 3-73: Quinidine-Metoprolol-DDGI

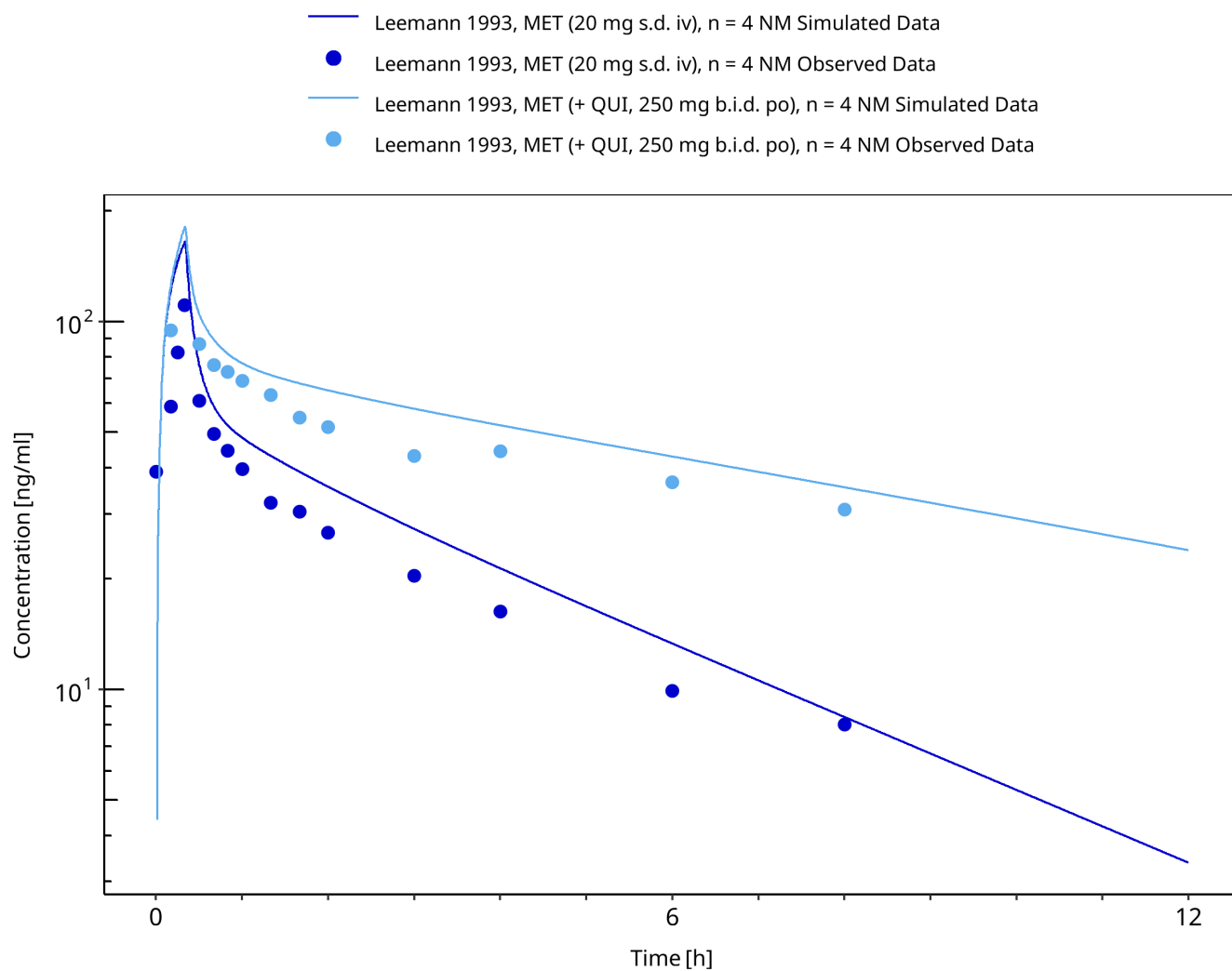


Figure 3-74: Quinidine-Metoprolol-DDGI

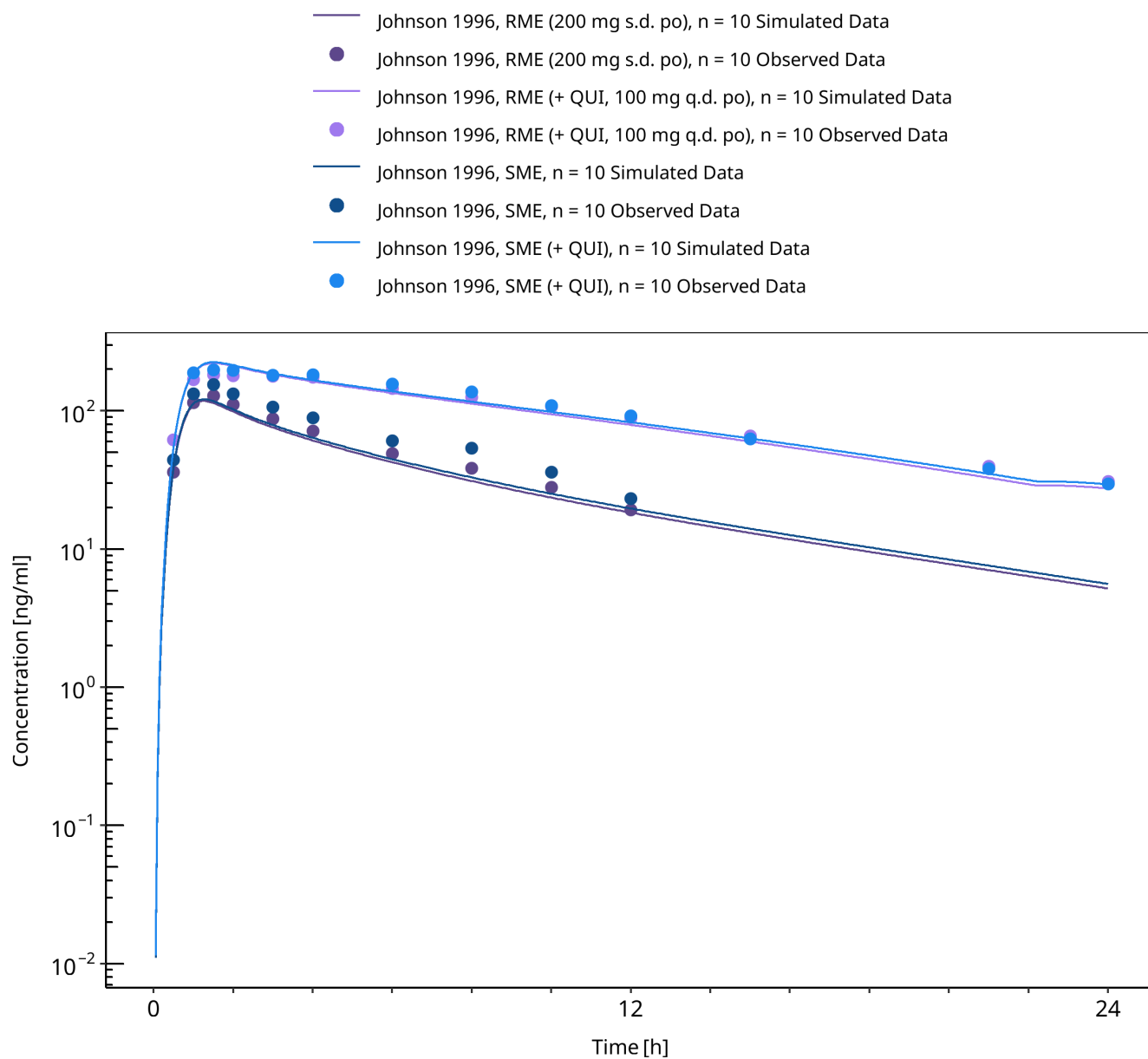


Figure 3-75: Quinidine-Metoprolol-DDI

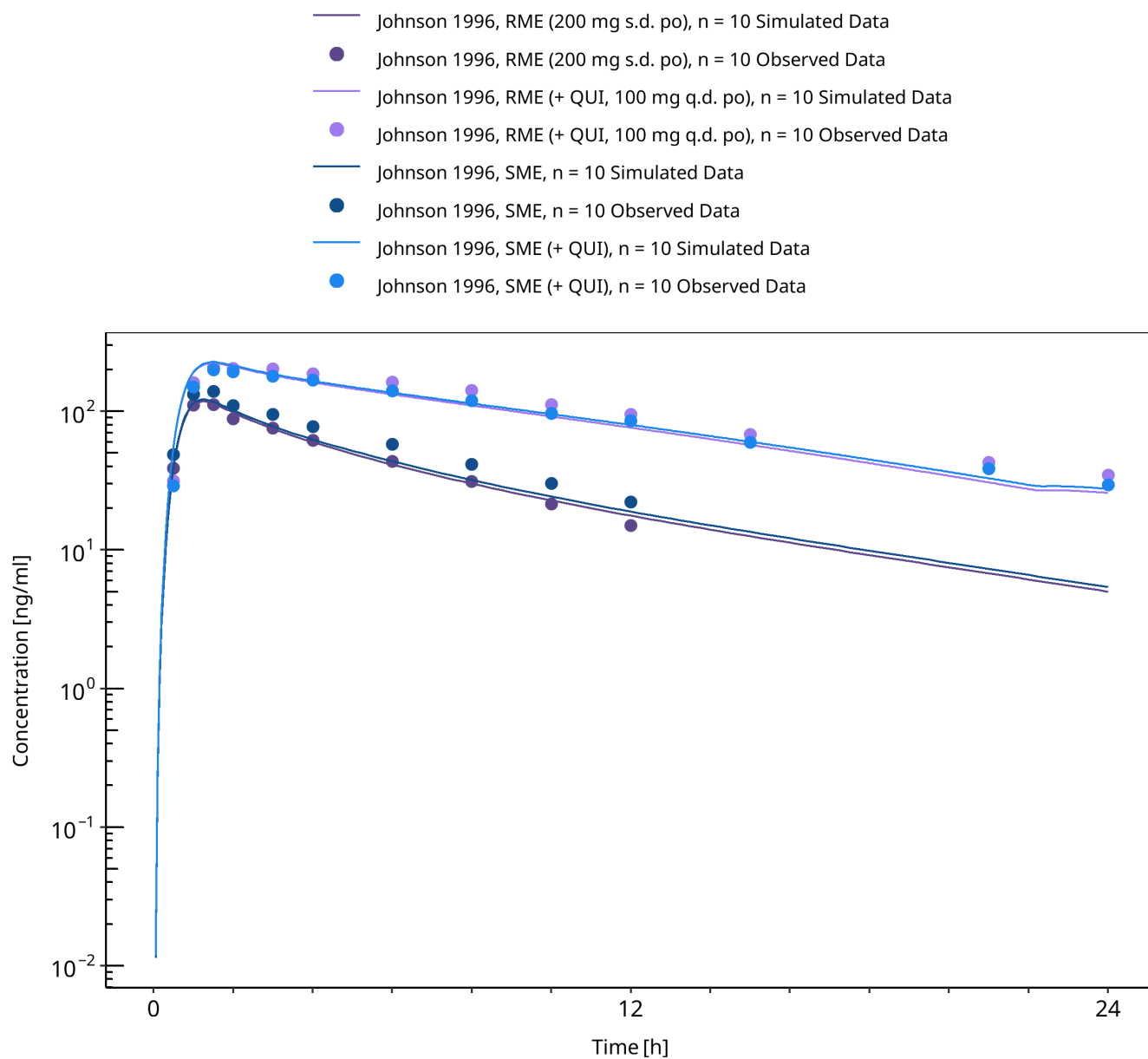


Figure 3-76: Quinidine-Metoprolol-DDGI

3.15 Rifampicin-Metoprolol-DDI

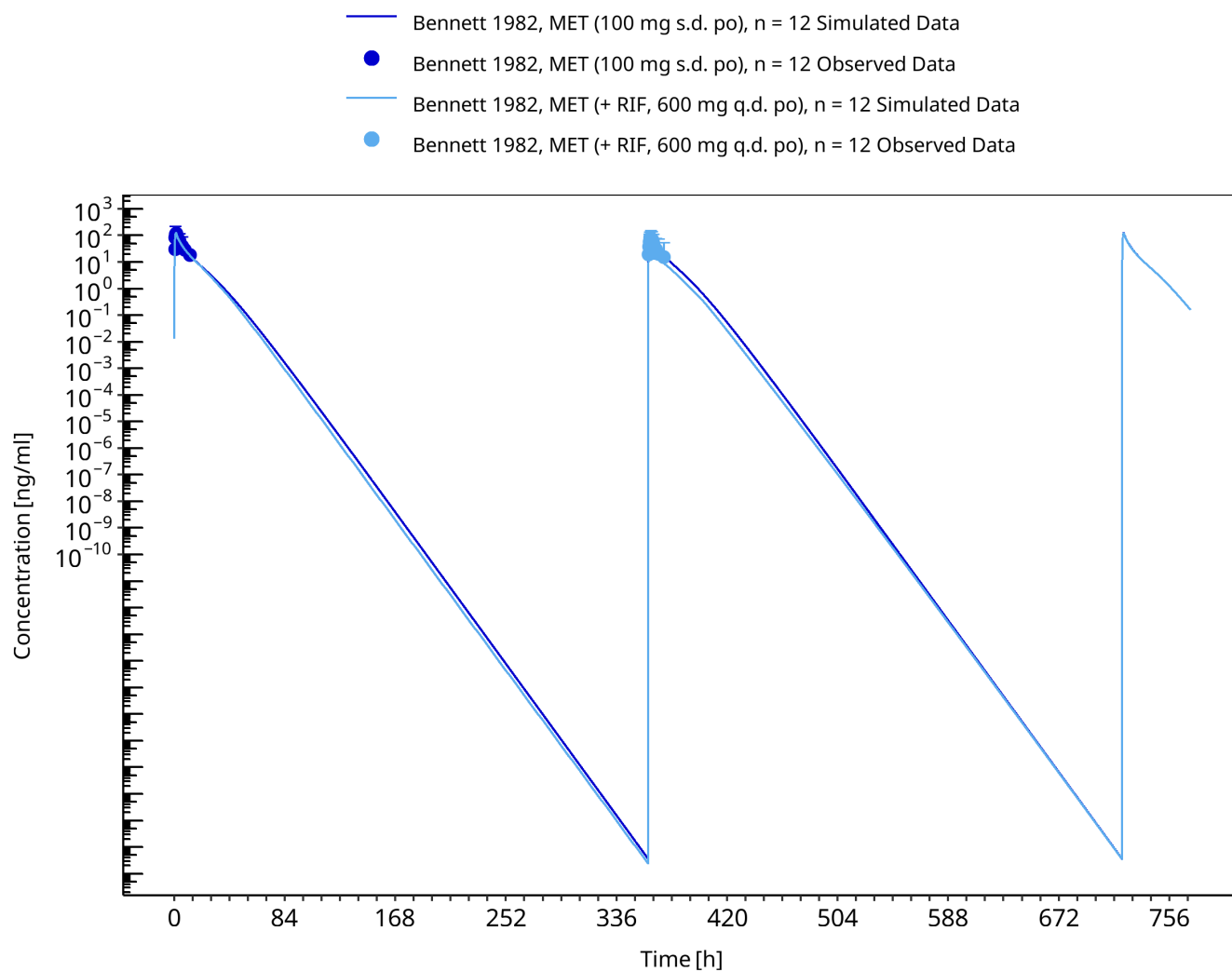


Figure 3-77: Rifampicin-Metoprolol-DDI

3.16 Quinidine-Mexiletine-DDGI

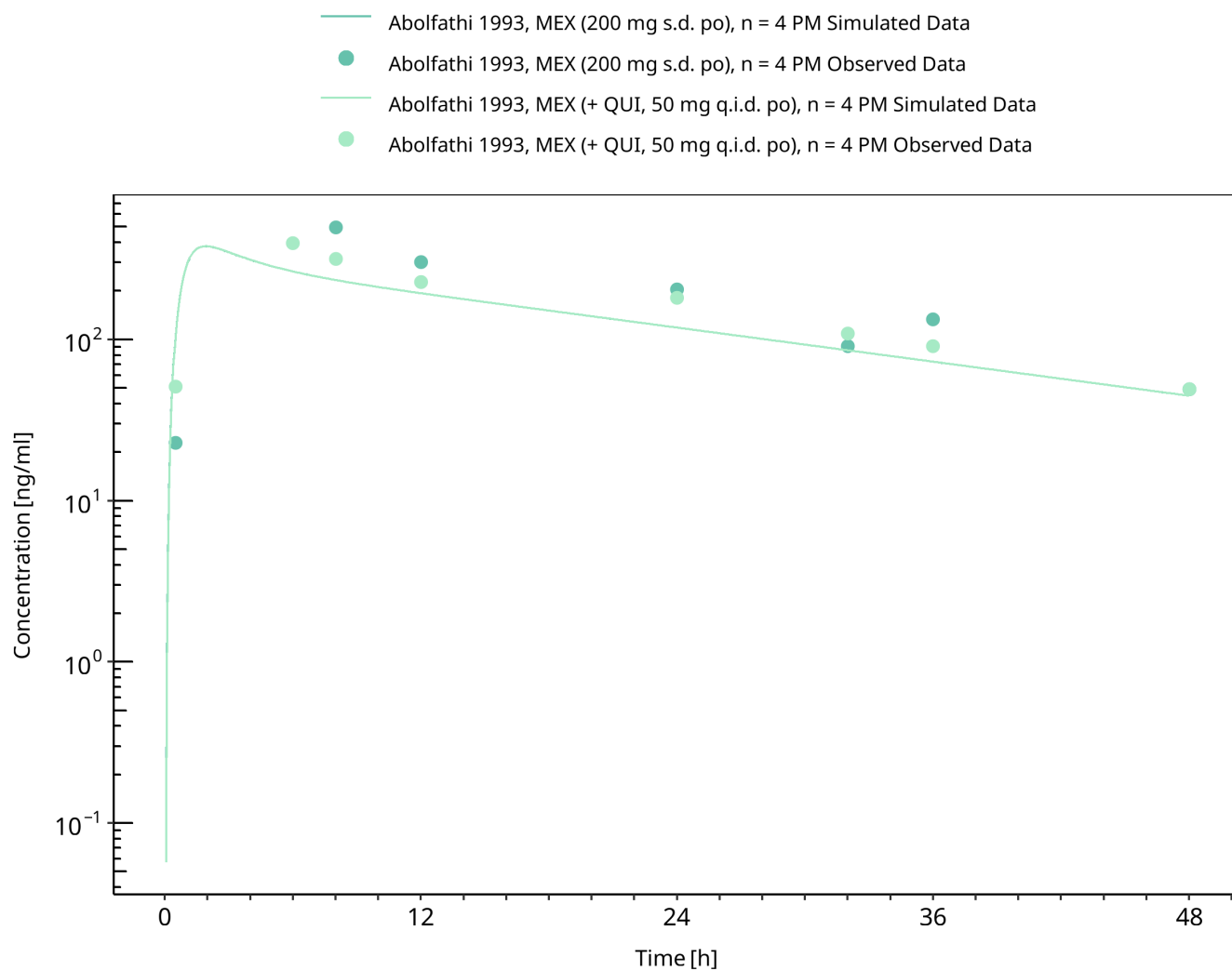


Figure 3-78: Quinidine-Mexiletine-DDGI

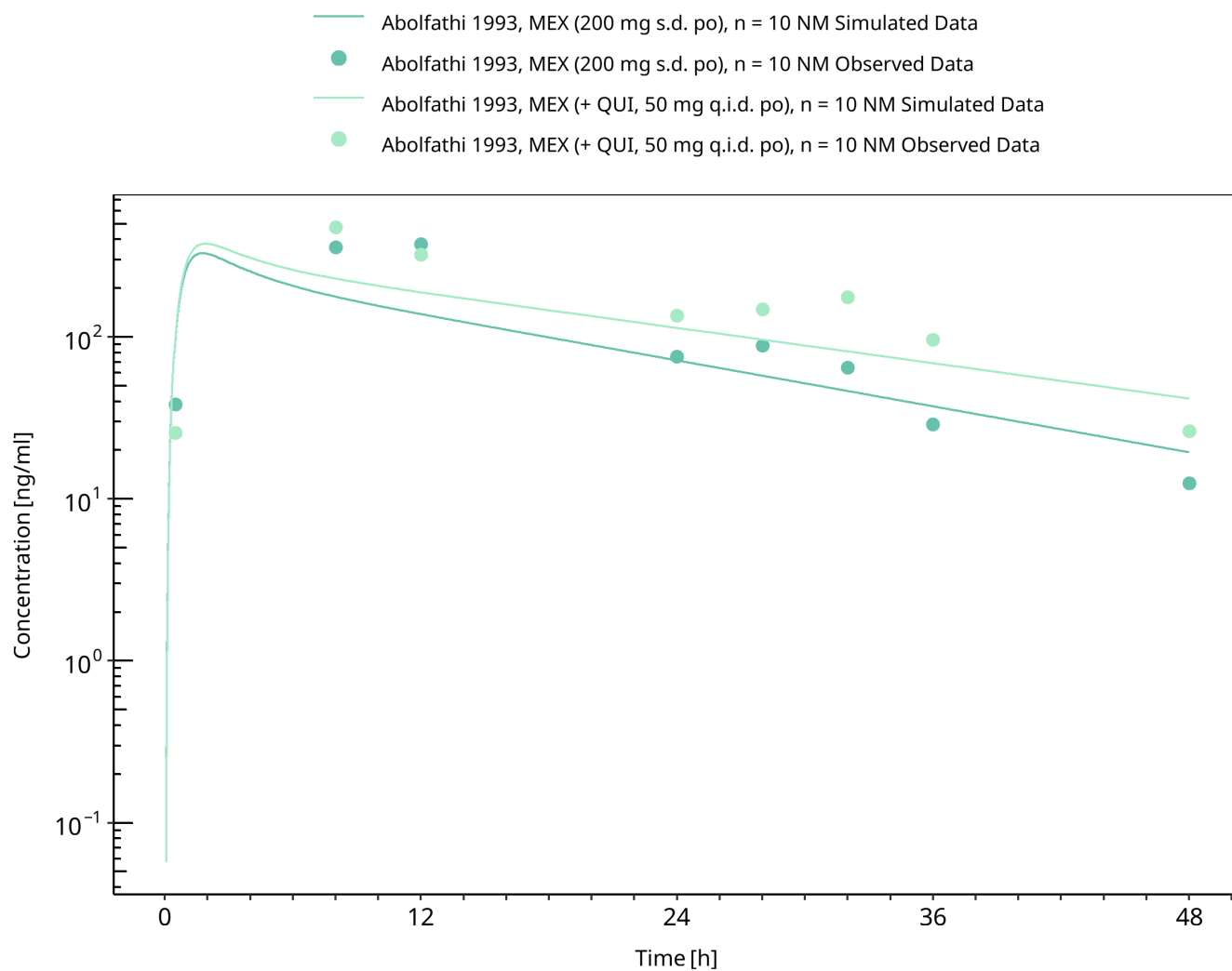


Figure 3-79: Quinidine-Mexiletine-DDGI

3.17 Atomoxetine-Midazolam-DDI

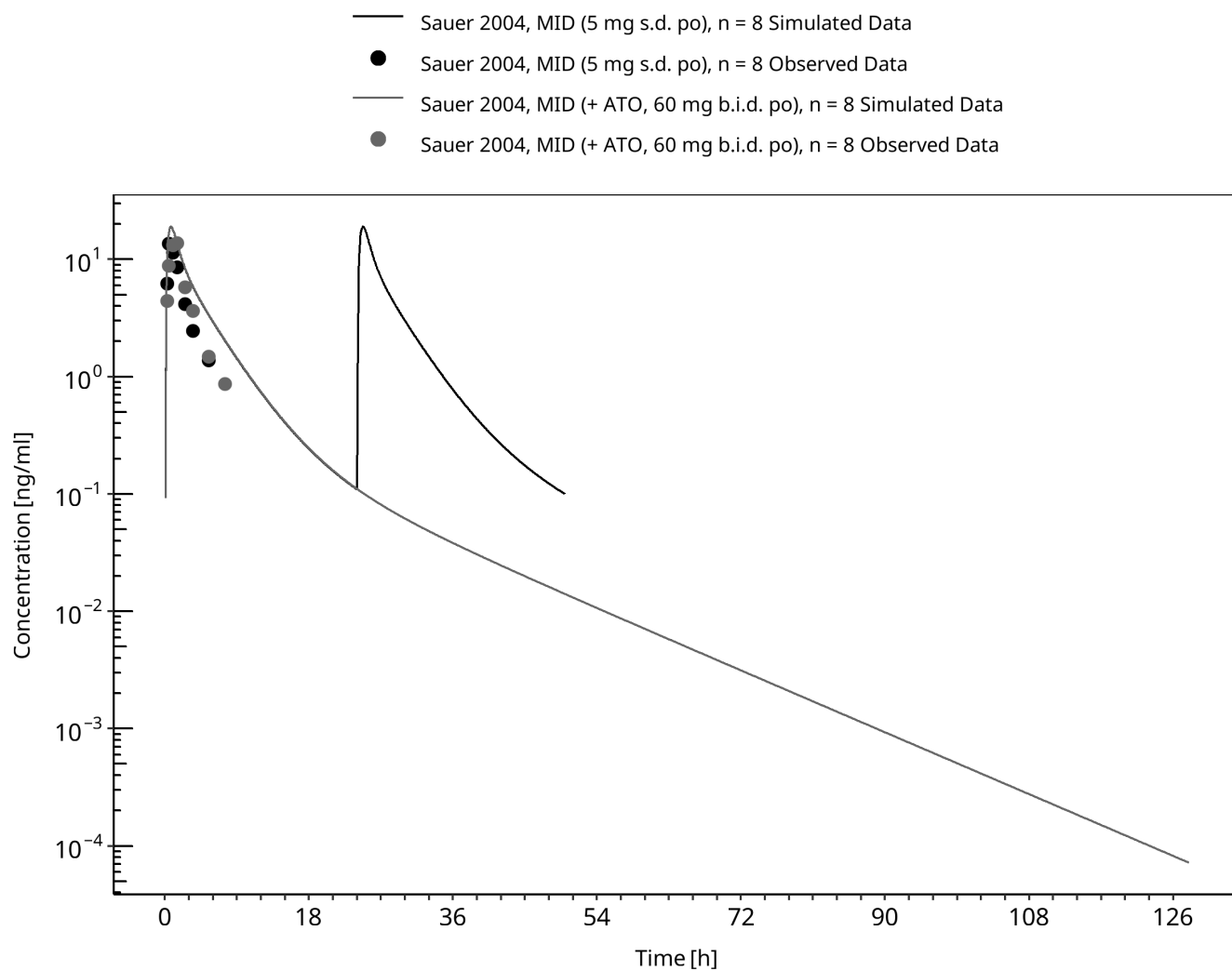


Figure 3-80: Atomoxetine-Midazolam-DDI

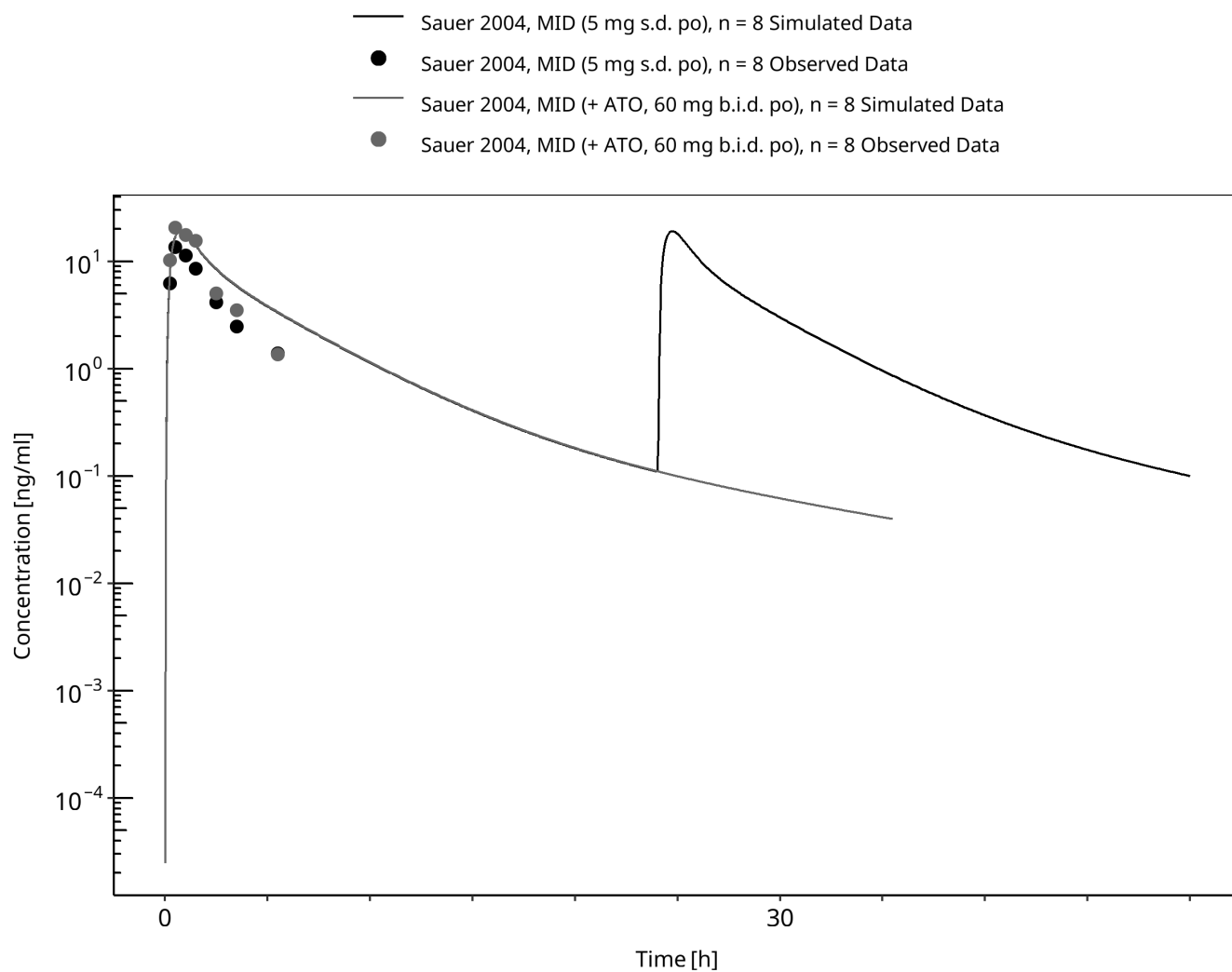


Figure 3-81: Atomoxetine-Midazolam-DDI

3.18 Itraconazole-Paroxetine-DDI

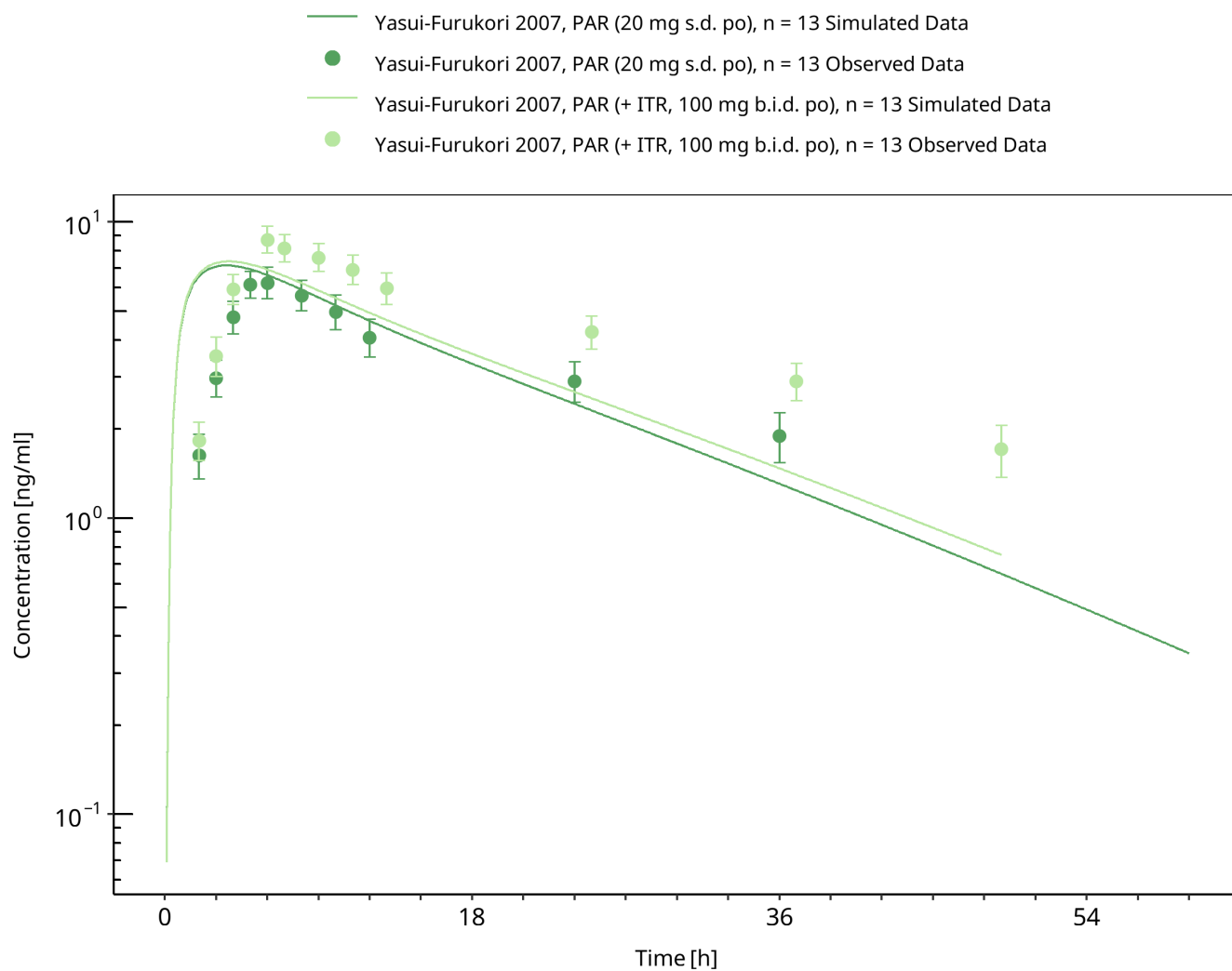


Figure 3-82: Itraconazole-Paroxetine-DDI

3.19 Quinidine-Paroxetine-DDI

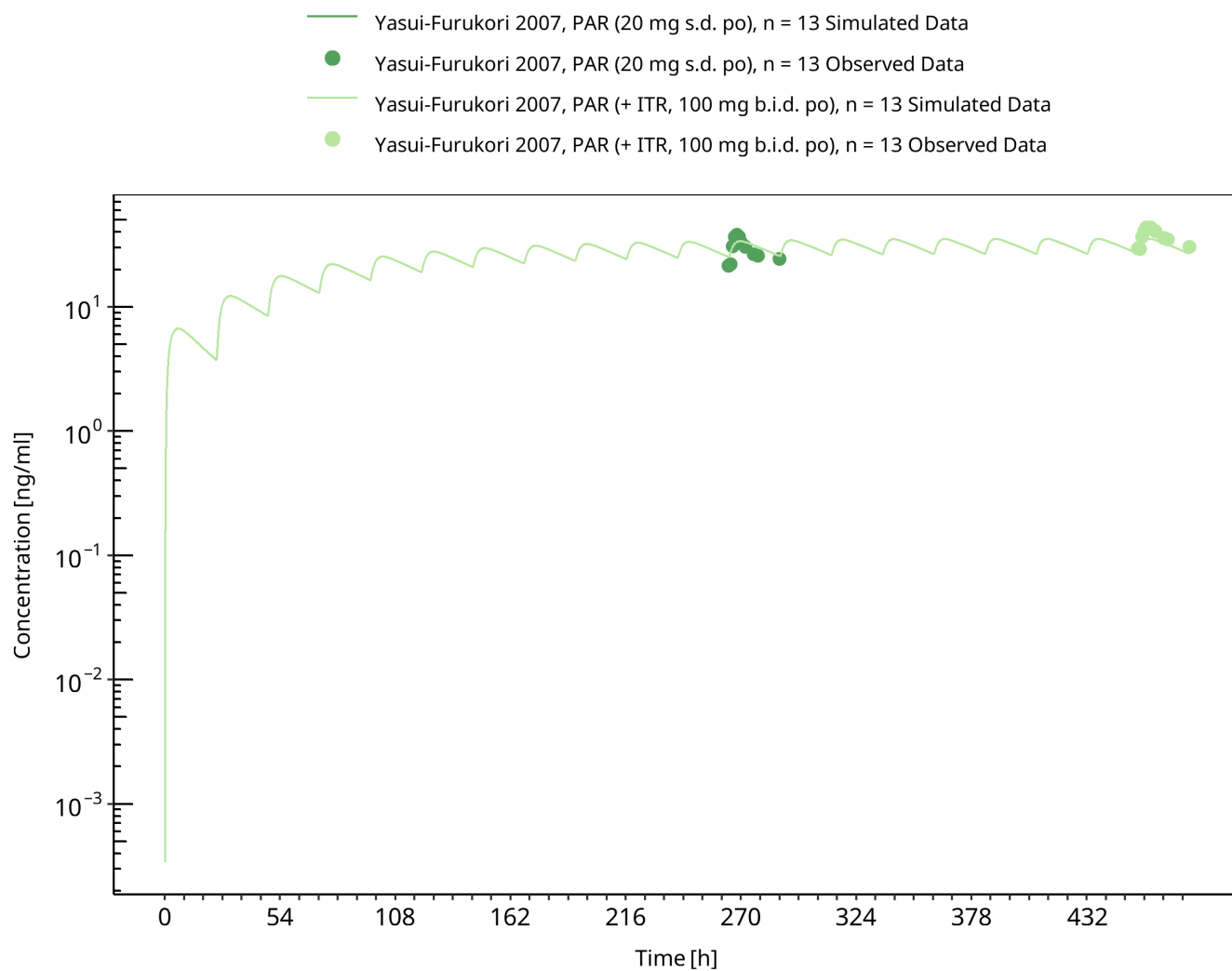


Figure 3-83: Quinidine-Paroxetine-DDI

3.20 Carbamazepine-Quinidine-DDI

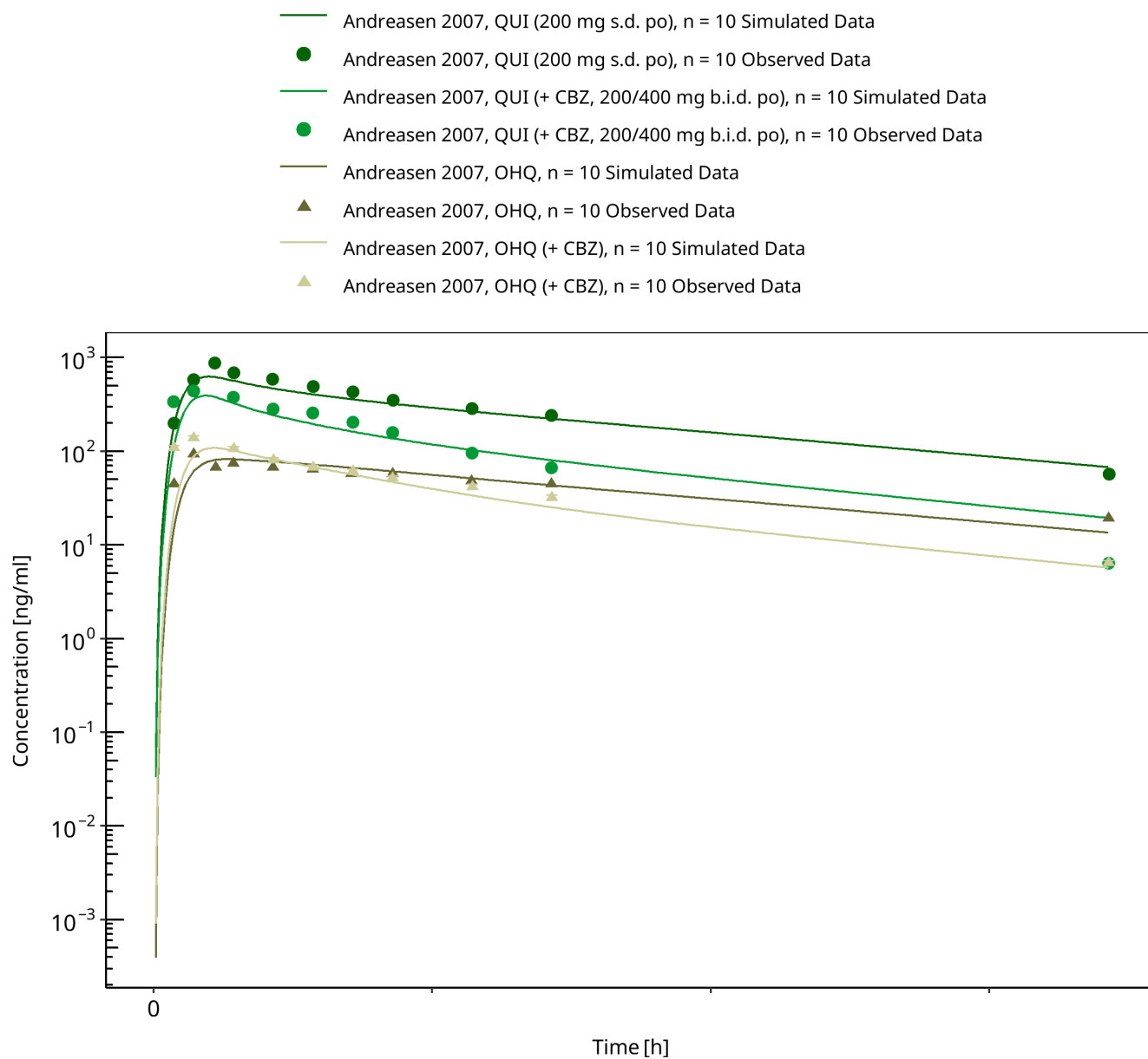


Figure 3-84: Carbamazepine-Quinidine-DDI

3.21 Cimetidine-Quinidine-DDI

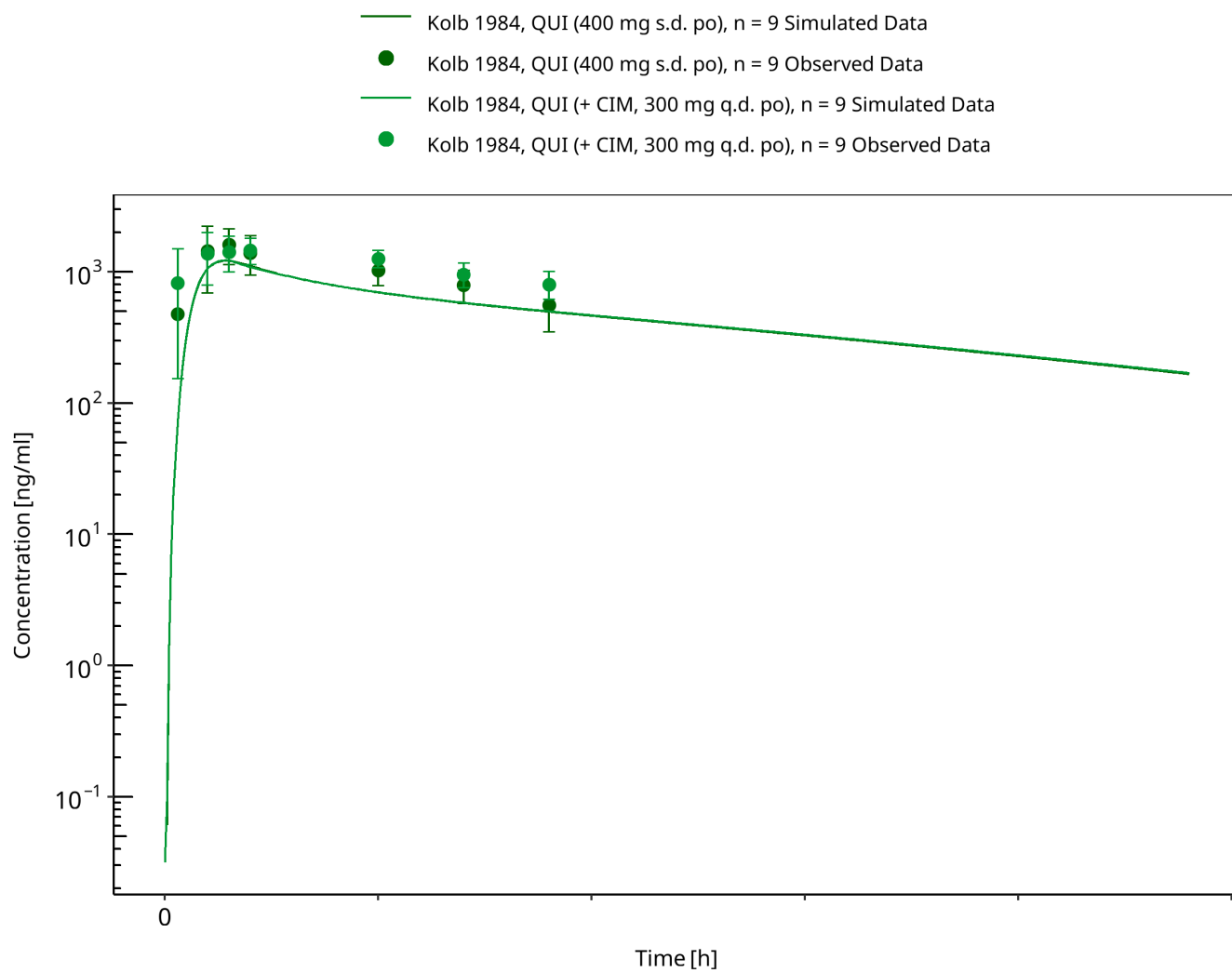


Figure 3-85: Cimetidine-Quinidine-DDI

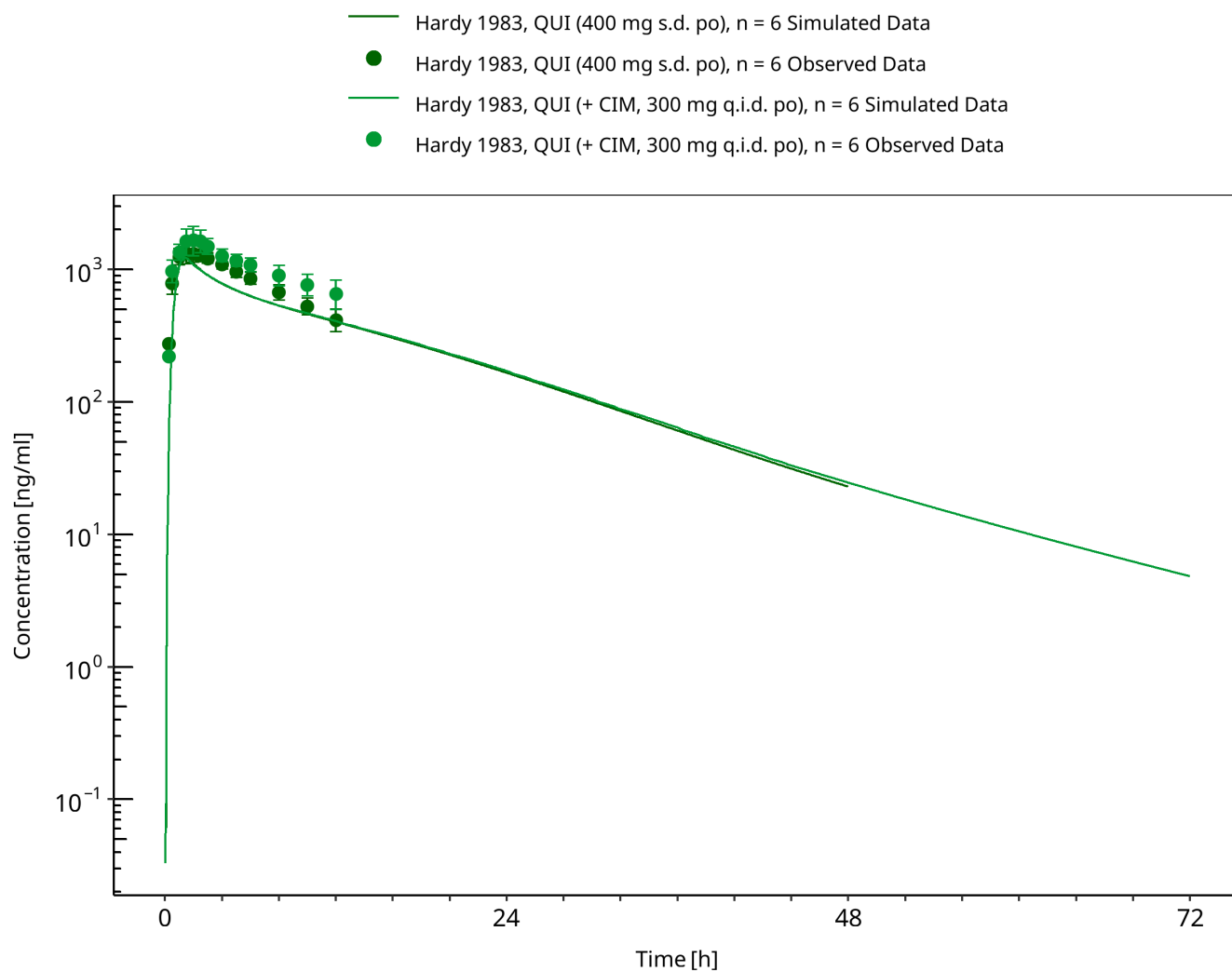


Figure 3-86: Cimetidine-Quinidine-DDI

3.22 Erythromycin-Quinidine-DDI

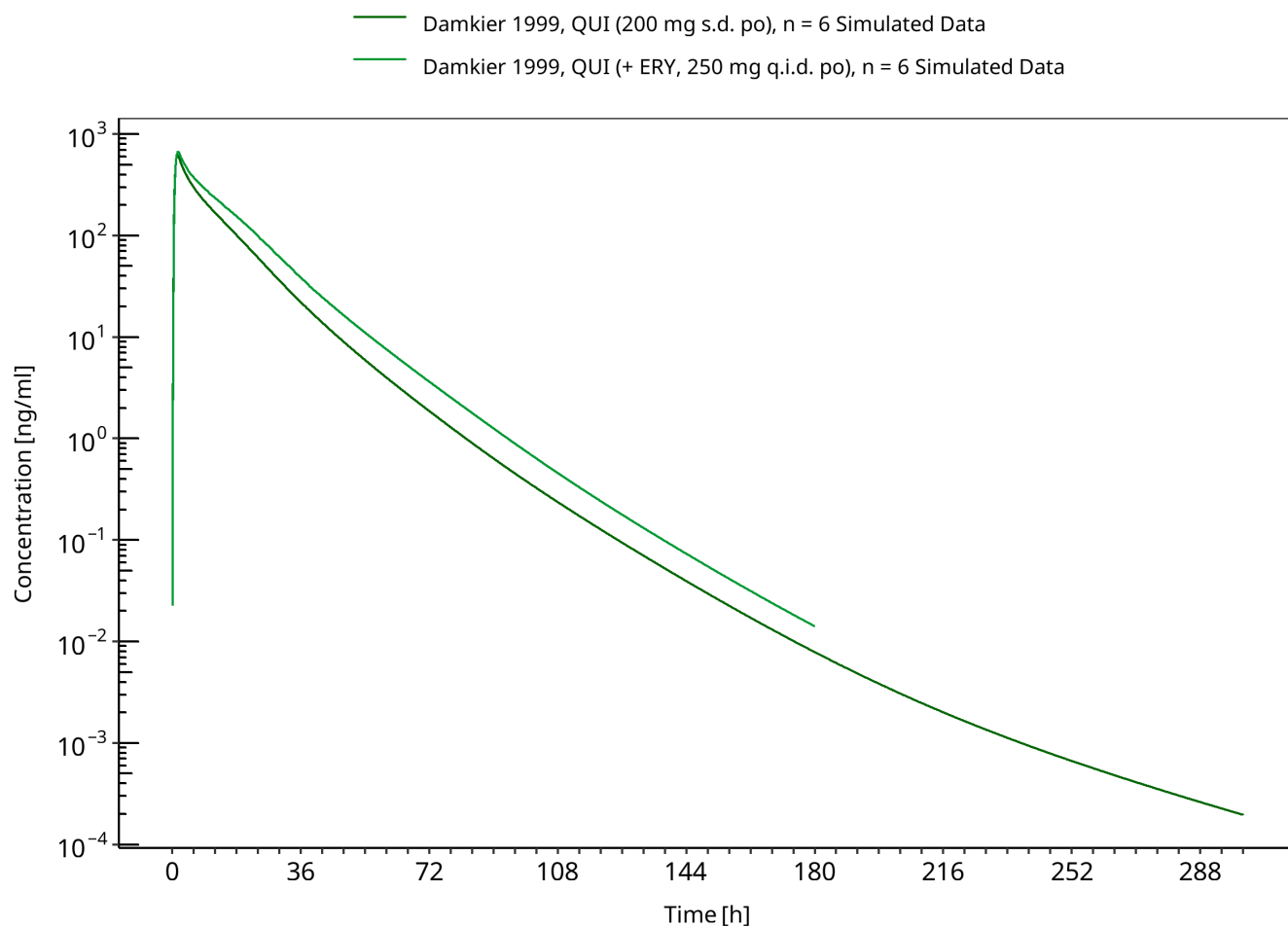


Figure 3-87: Erythromycin-Quinidine-DDI

3.23 Fluvoxamine-Quinidine-DDI

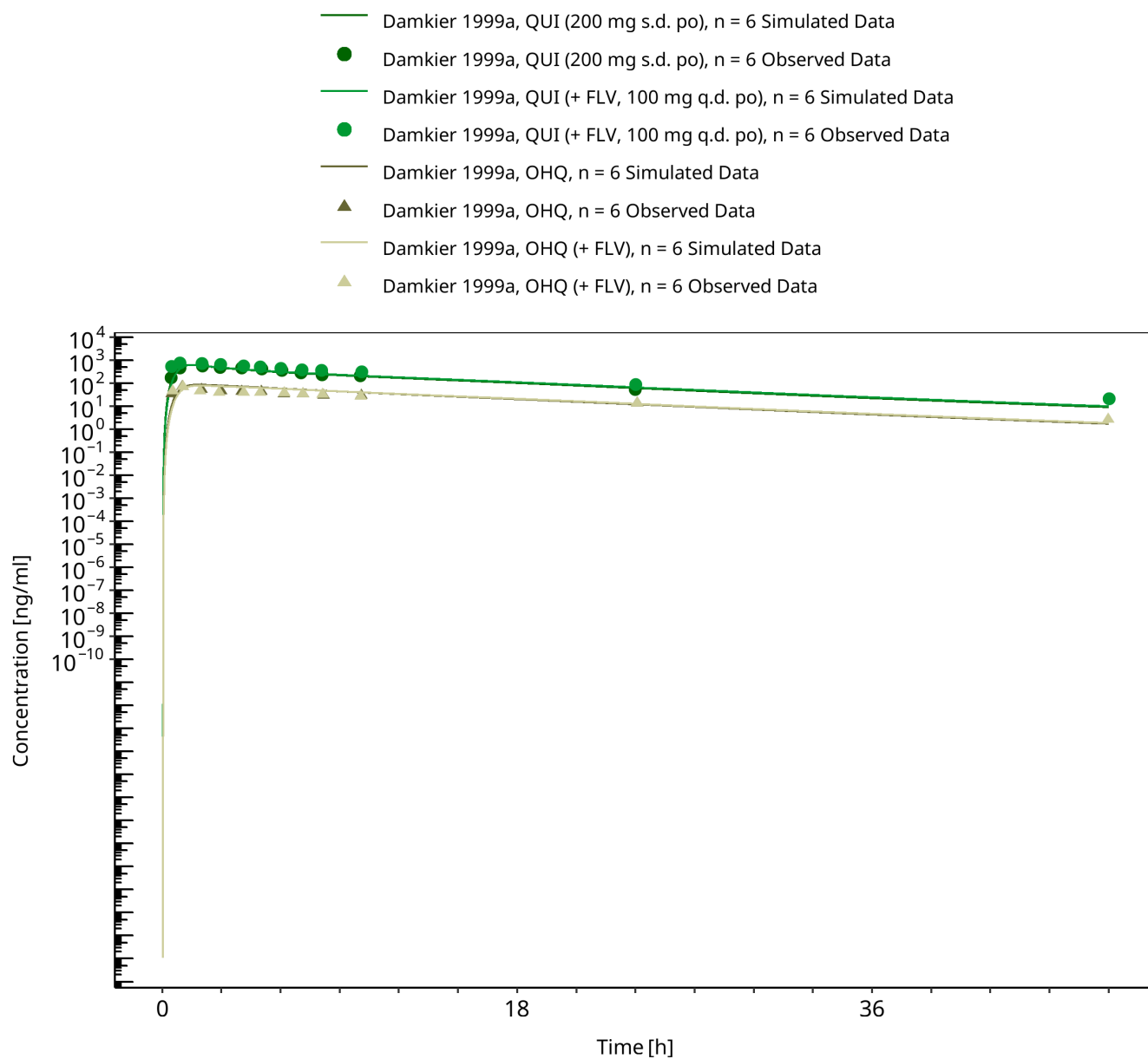


Figure 3-88: Fluvoxamine-Quinidine-DDI

3.24 Itraconazole-Quinidine-DDI

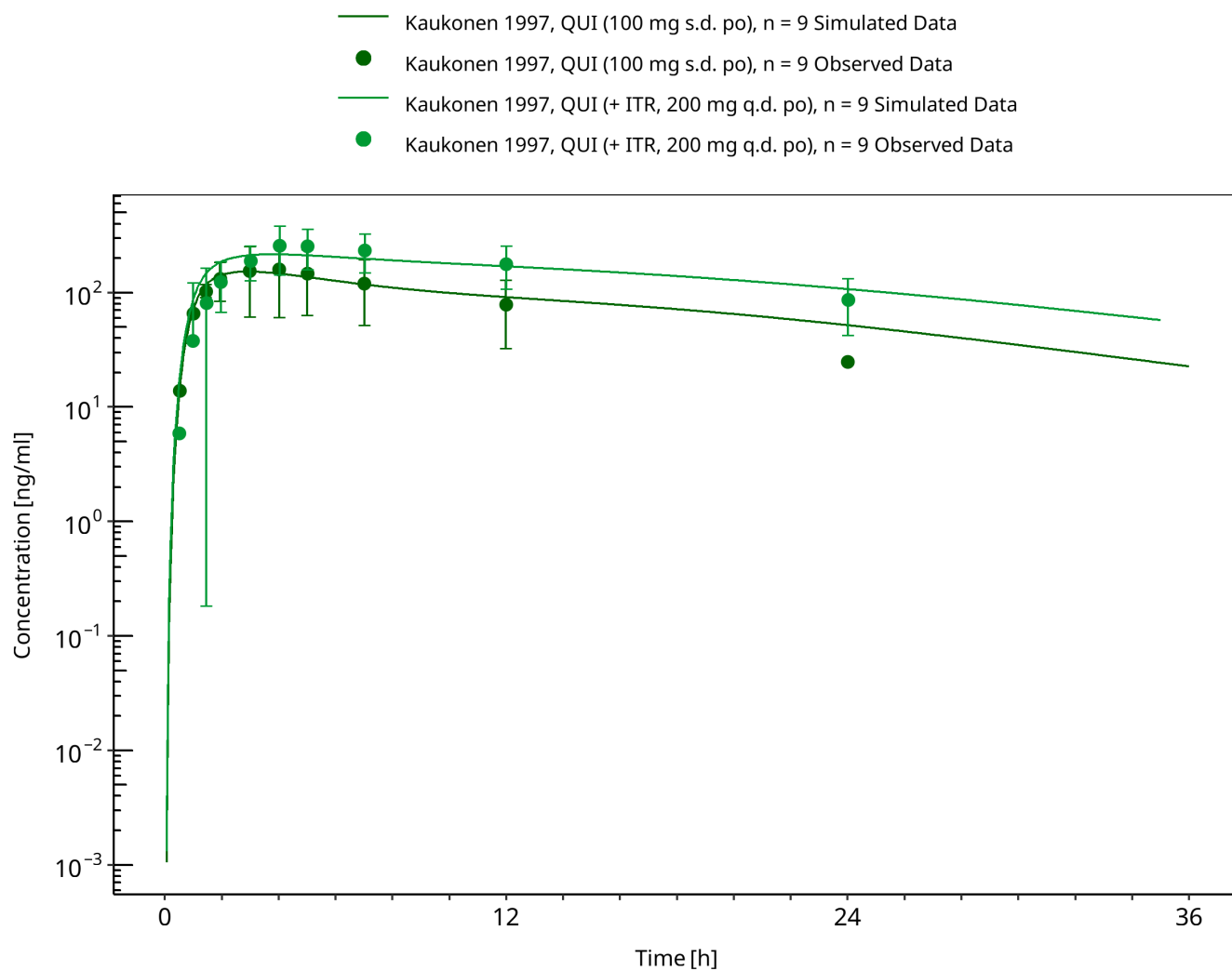


Figure 3-89: Itraconazole-Quinidine-DDI

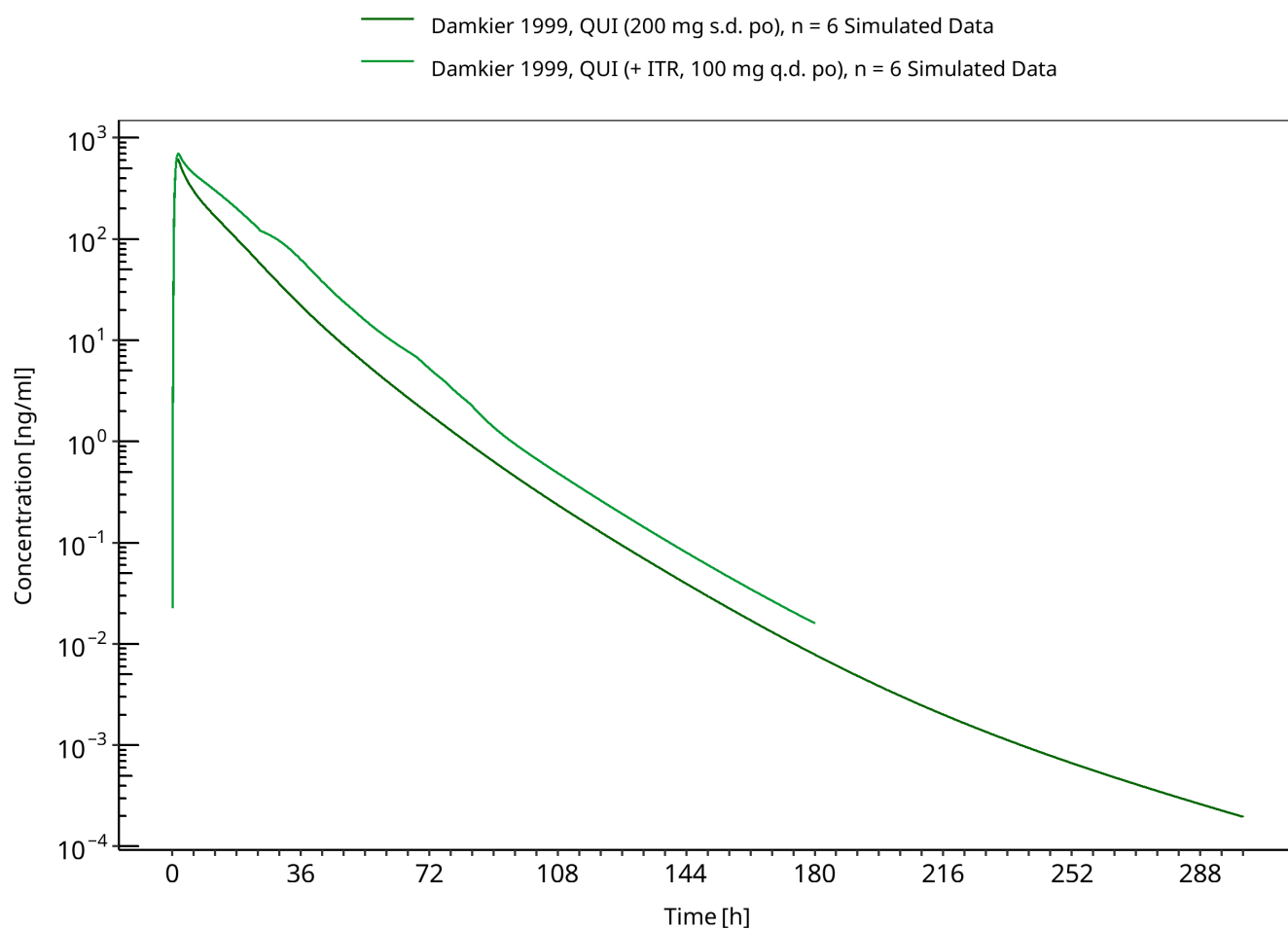


Figure 3-90: Itraconazole-Quinidine-DDI

3.25 Omeprazole-Quinidine-DDI

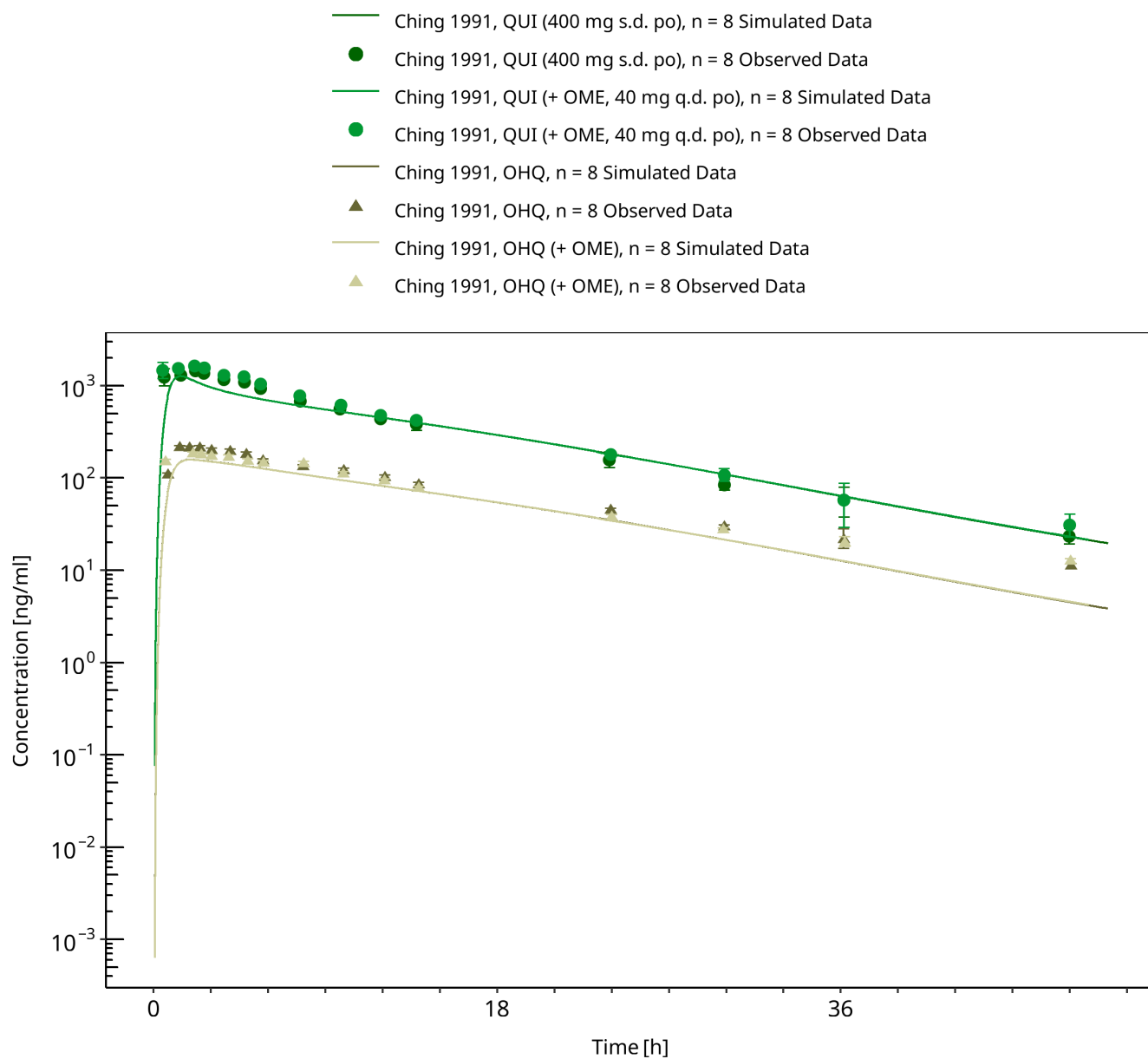


Figure 3-91: Omeprazole-Quinidine-DDI

3.26 Rifampicin-Quinidine-DDI

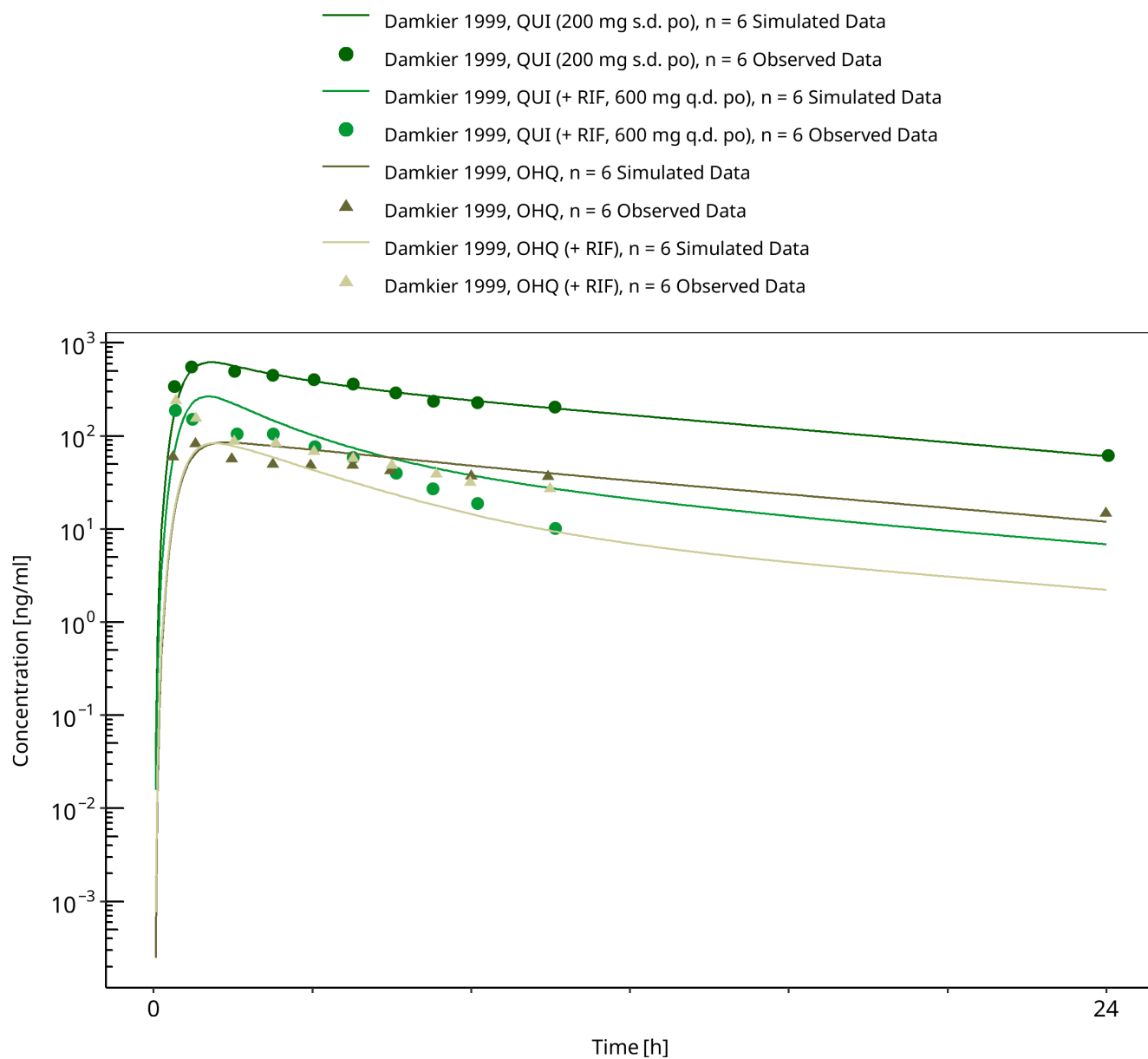


Figure 3-92: Rifampicin-Quinidine-DDI

3.27 Verapamil-Quinidine-DDI

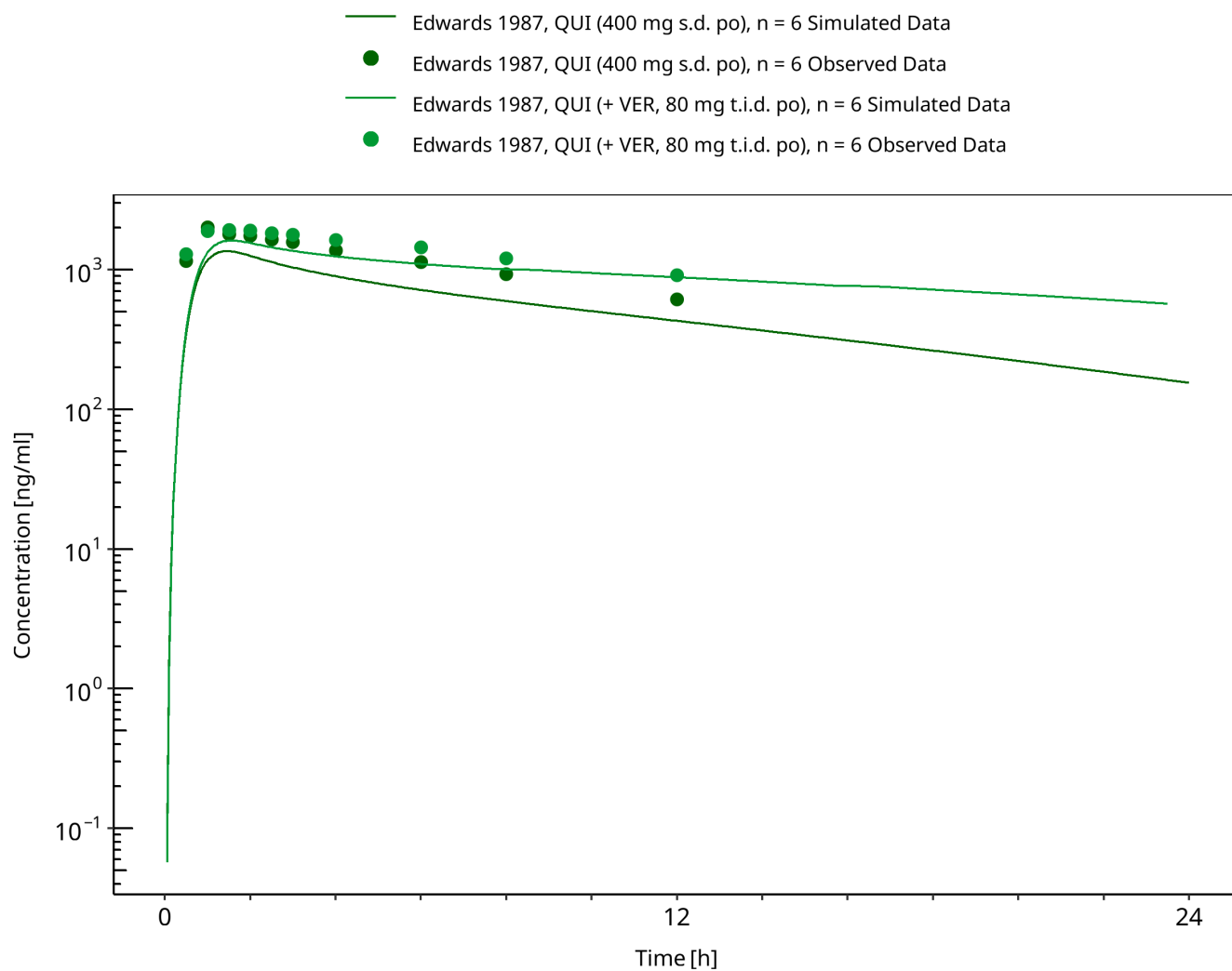


Figure 3-93: Verapamil-Quinidine-DDI

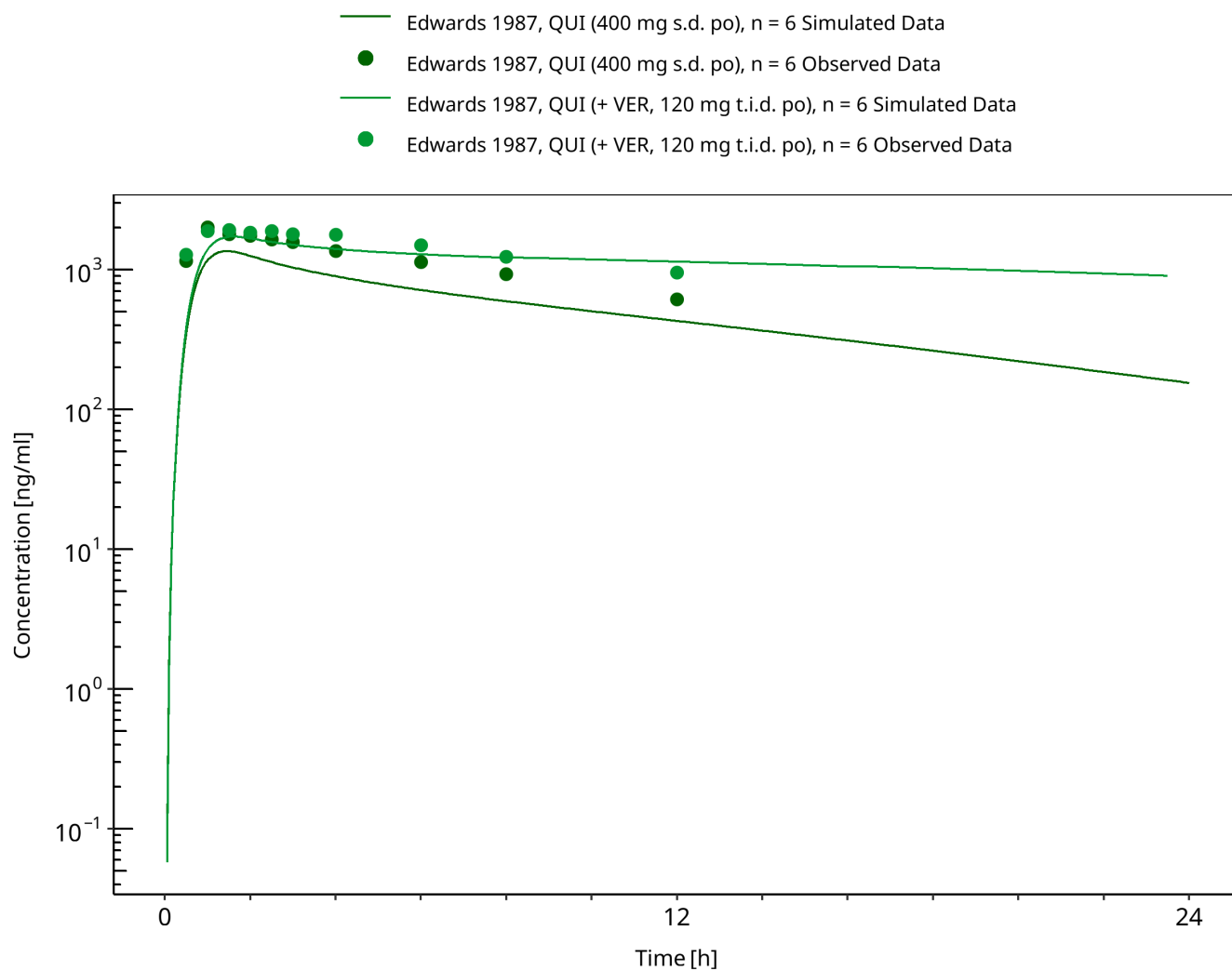


Figure 3-94: Verapamil-Quinidine-DDI

3.28 Ketoconazole-Risperidone-DDI

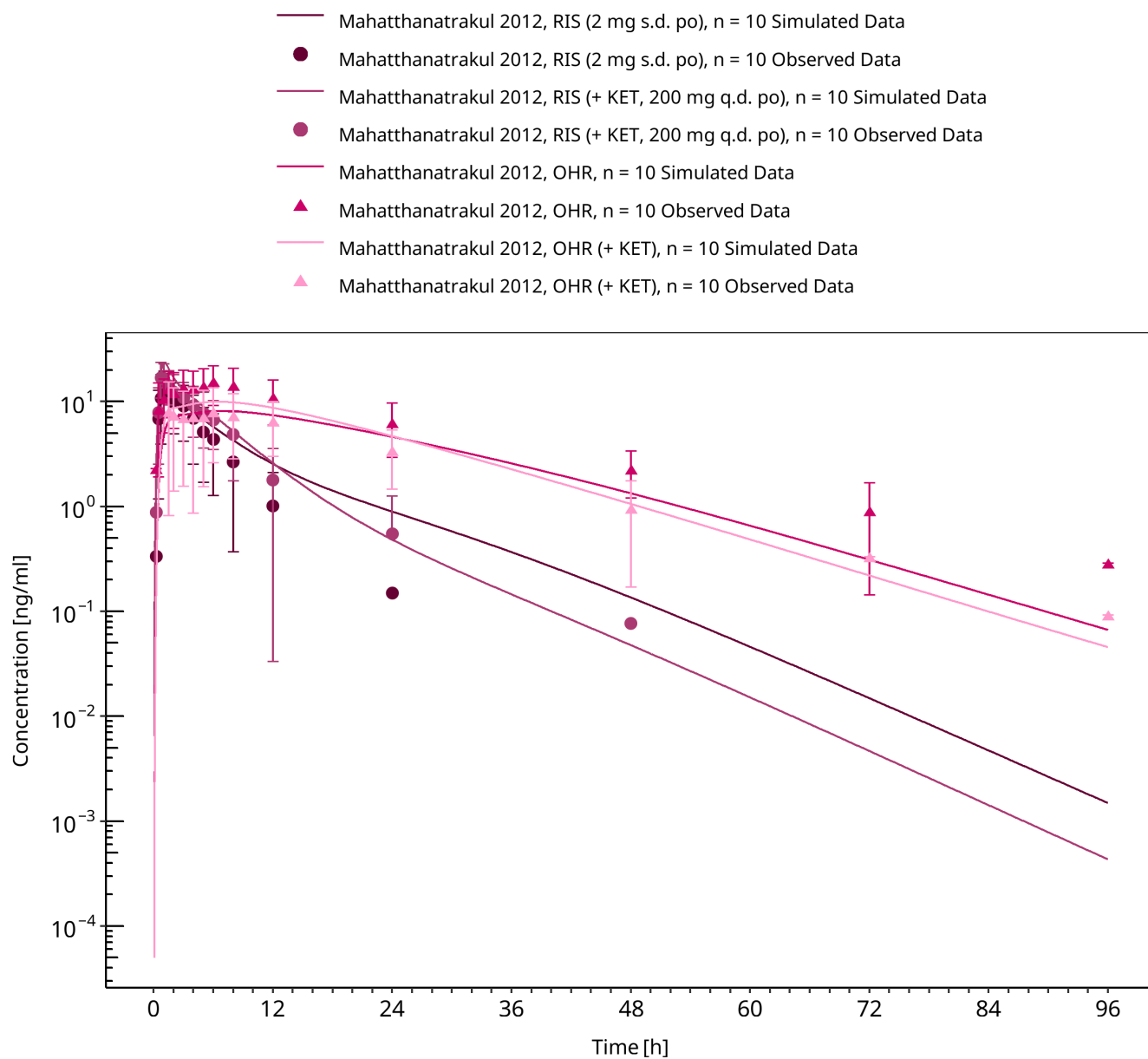


Figure 3-95: Ketoconazole-Risperidone-DDI

3.29 Rifampicin-Risperidone-DDI

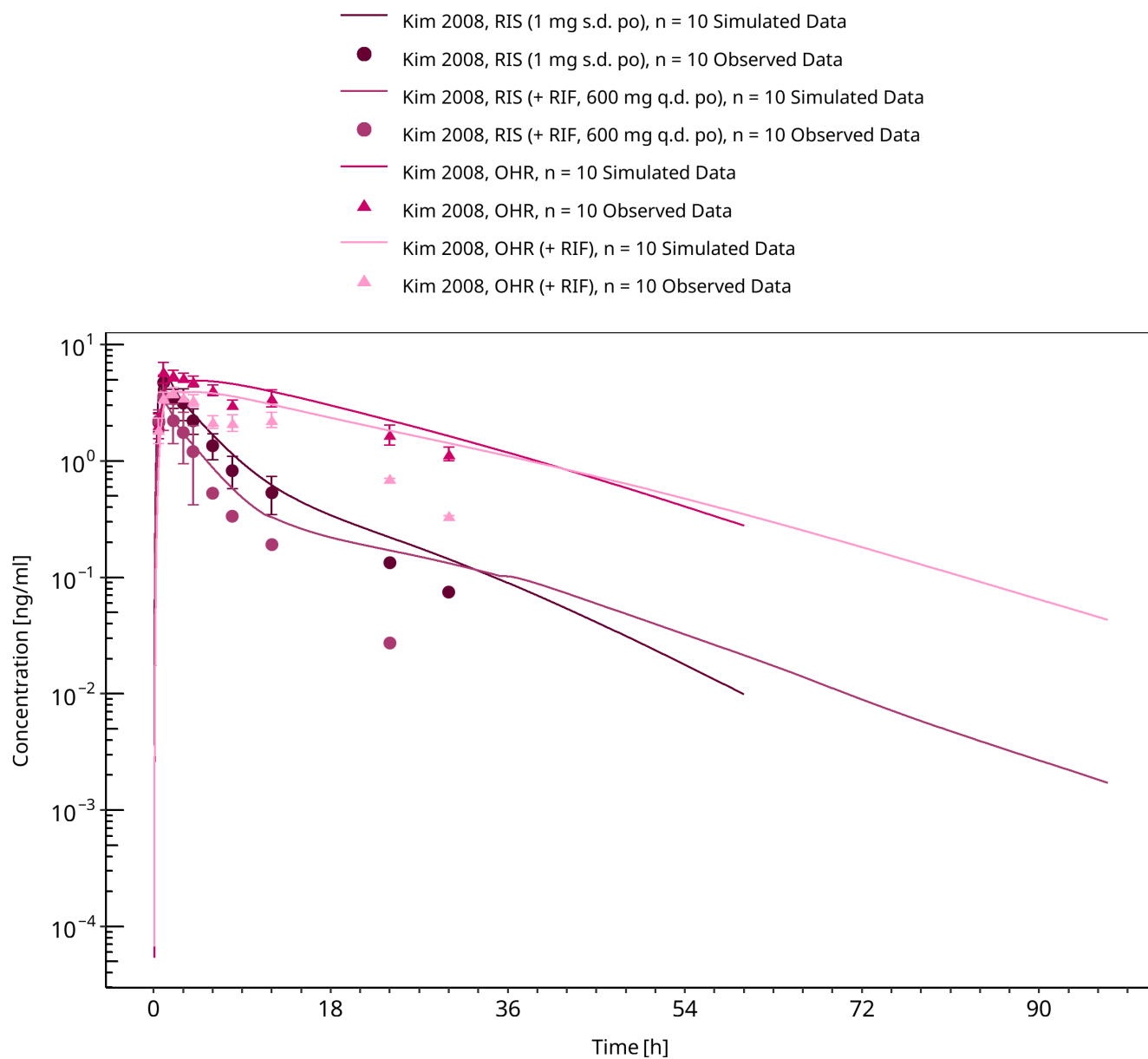


Figure 3-96: Rifampicin-Risperidone-DDI

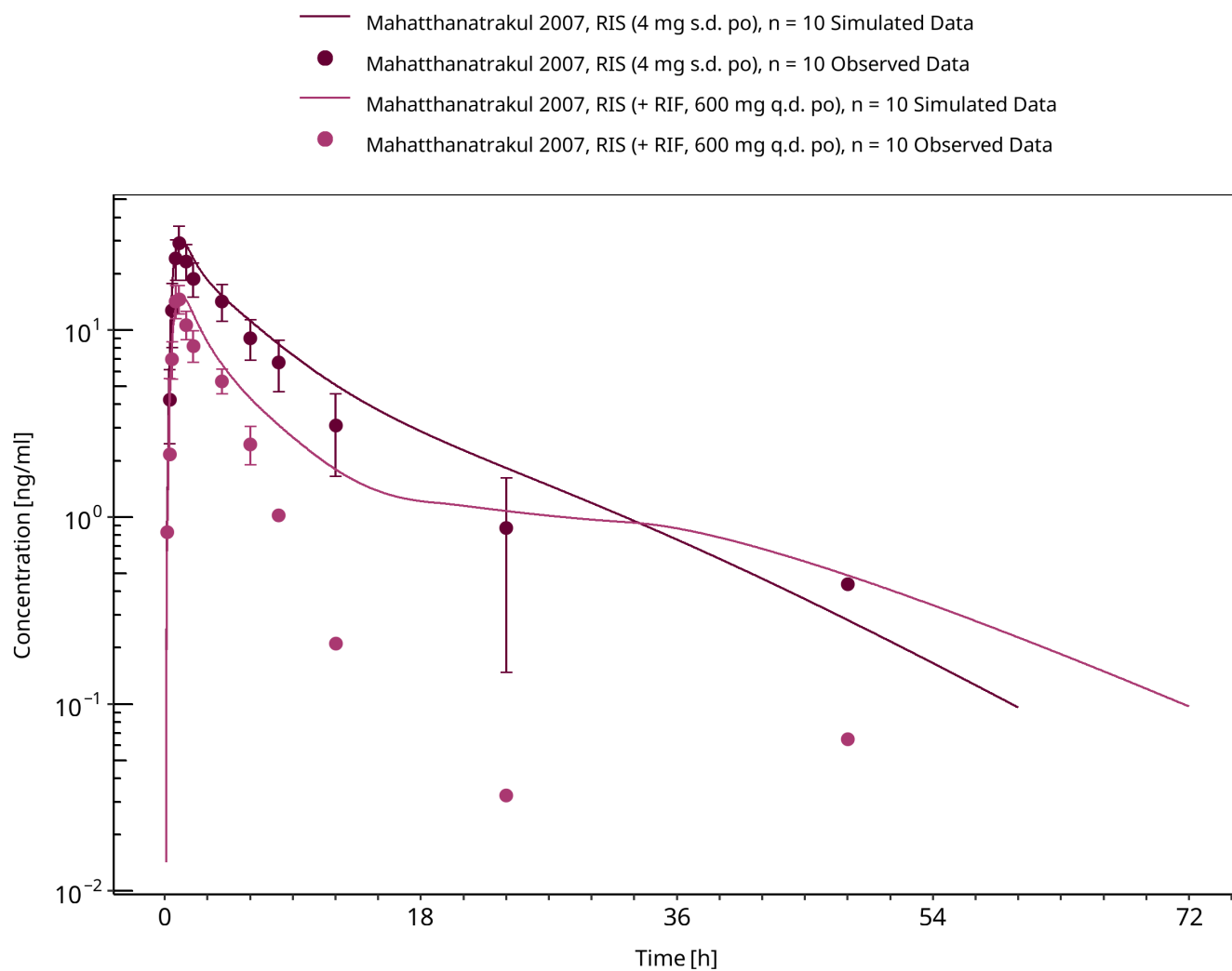


Figure 3-97: Rifampicin-Risperidone-DDI

3.30 Verapamil-Risperidone-DDI

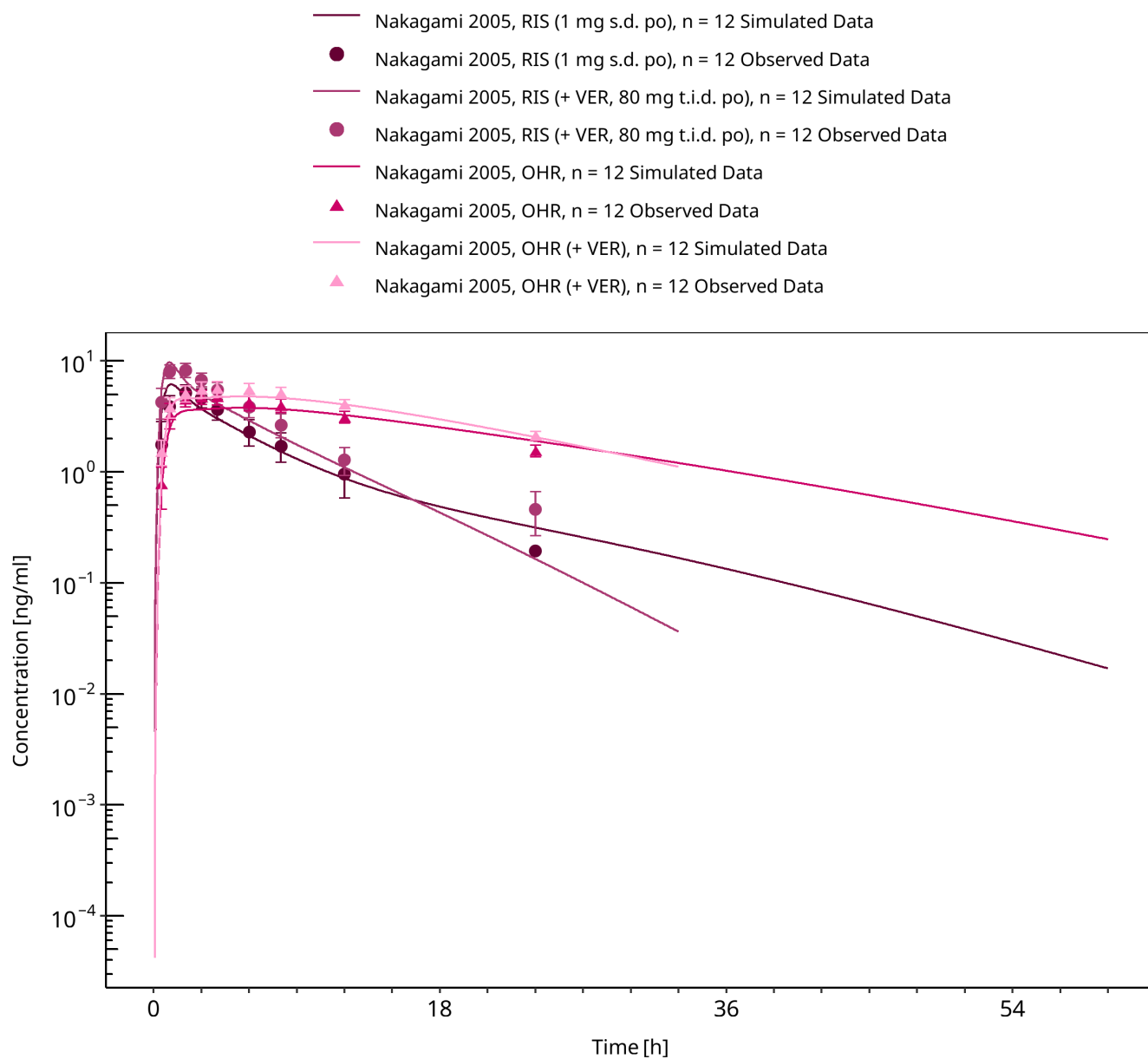


Figure 3-98: Verapamil-Risperidone-DDI

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5 Appendix

5.1 Open Systems Pharmacology Suite (OSPS) Introduction

5.2 Mathematical Implementation of Drug-Drug Interactions

5.3 Automatic (re)-qualification workflow

6 Glossary

| ALP | alprazolam | | AS | CYP2D6 activity score | | ATO | atomoxetine | | AUC | area under the plasma concentration-time curve |
| AUClast | AUC calculated between first and last concentration measurement | | AUCR | AUC ratio | | b.i.d. | twice daily | | BUP |
bupropion | | CBZ | carbamazepine | | CIM | cimetidine | | CLA | clarithromycin | | CLO | (E)-clomiphen | | Cmax | maximum
plasma concentration | | CmaxR | Cmax ratio | | CYP | cytochrome P450 | | DDI | drug-drug interaction | | DDGI | Drug-drug-gene
interaction | | DES | desipramine | | DEX | dextromethorphan | | DIG | digoxin | | DTT | total dextrophan | | DXG | dextrophan-O-
glucuronide | | DXT | dextrophan | | ERY | erythromycin | | FLV | fluvoxamine | | HDC | (E)-4-hydroxy-N-desethylclomiphen | | IM |
CYP2D6 intermediate metabolizer | | ITR | itraconazole | | iv | intravenous | | KET | ketoconazole | | MET | metoprolol | | MEX |
mexiletine | | MID | midazolam | | N | number of study participants | | NDC | (E)-N-desethylclomiphen | | NM | CYP2D6 normal
metabolizer | | OHC | (E)-4-hydroxyclophen | | OHD | 2-hydroxydesipramine | | OHQ | 3-hydroxyquinidine | | OHR | 9-
hydroxyrisperidone | | OME | omeprazole | | OSP | Open Systems Pharmacology | | PAR | paroxetine | | PBPK | physiologically
based pharmacokinetic | | P-gp | P-glycoprotein | | PM | CYP2D6 poor metabolizer | | po | oral | | q.d. | once daily | | q.i.d. | four
times daily | | QUI | quinidine | | RIF | rifampicin | | RIS | risperidone | | RME | (R)-metoprolol | | s.d. | single dose | | SME | (S)-
metoprolol | | t.i.d. | three times daily | | VER | verapamil |